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AIRPORT COOPERATIVE RESEARCH PROGRAM

ACRP RESEARCH REPORT 233

Airport Biometrics

A PRIMER

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2021

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AIRPORT COOPERATIVE RESEARCH PROGRAM

Airports are vital national resources. They serve a key role in transportation of people and goods and in regional, national, and international commerce. They are where the nation's aviation system connects with other modes of transportation and where federal responsibility for managing and regulating air traffic operations intersects with the role of state and local governments that own and operate most airports. Research is necessary to solve common operating problems, to adapt appropriate new technologies from other industries, and to introduce innovations into the airport industry. The Airport Cooperative Research Program (ACRP) serves as one of the principal means by which the airport industry can develop innovative near-term solutions to meet demands placed on it.

The need for ACRP was identified in *TRB Special Report 272: Airport Research Needs: Cooperative Solutions* in 2003, based on a study sponsored by the Federal Aviation Administration (FAA). ACRP carries out applied research on problems that are shared by airport operating agencies and not being adequately addressed by existing federal research programs. ACRP is modeled after the successful National Cooperative Highway Research Program (NCHRP) and Transit Cooperative Research Program (TCRP). ACRP undertakes research and other technical activities in various airport subject areas, including design, construction, legal, maintenance, operations, safety, policy, planning, human resources, and administration. ACRP provides a forum where airport operators can cooperatively address common operational problems.

ACRP was authorized in December 2003 as part of the Vision 100—Century of Aviation Reauthorization Act. The primary participants in the ACRP are (1) an independent governing board, the ACRP Oversight Committee (AOC), appointed by the Secretary of the U.S. Department of Transportation with representation from airport operating agencies, other stakeholders, and relevant industry organizations such as the Airports Council International-North America (ACI-NA), the American Association of Airport Executives (AAAE), the National Association of State Aviation Officials (NASAO), Airlines for America (A4A), and the Airport Consultants Council (ACC) as vital links to the airport community; (2) TRB as program manager and secretariat for the governing board; and (3) the FAA as program sponsor. In October 2005, the FAA executed a contract with the National Academy of Sciences formally initiating the program.

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Primary emphasis is placed on disseminating ACRP results to the intended users of the research: airport operating agencies, service providers, and academic institutions. ACRP produces a series of research reports for use by airport operators, local agencies, the FAA, and other interested parties; industry associations may arrange for workshops, training aids, field visits, webinars, and other activities to ensure that results are implemented by airport industry practitioners.

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FOREWORD

By Theresia H. Schatz
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ACRP Research Report 233: Airport Biometrics: A Primer defines and describes the landscape of biometric processes and applications for airports and their respective stakeholders.

This primer includes benefits and considerations of implementing biometrics in an airport environment and addresses many elements, including broad educational information on types and applications of biometrics and the benefits and challenges of applications beyond the government-mandated biometric entry/exit program. It discusses facial recognition biometrics, including opportunities for and benefits to the airport, and jurisdictional issues related to governmental requirements, statutory authorities, and commercial interests. There is a section on legal and privacy implications, along with industry- and government-driven initiatives, system design options, and interoperability at all identification points in the passenger journey and airport employee processing, as well as considerations for planning and implementation and future trends in the use of biometrics at airports.

This timely report presents solutions and additional suggestions for the growth of this technology and addresses safe airport operations during the COVID-19 pandemic and beyond.

Biometric recognition refers to identification and authentication of an individual based on physical traits. Biometric solutions are complex, multifaceted, and quickly changing, and include technical, operational, financial, legal, and programmatic considerations, among others.

Currently, some airports use biometrics for employee access control. U.S. Customs and Border Protection (CBP) and the Transportation Security Administration (TSA) are working together with airport operators and airlines to pilot the use of biometrics at airports for passenger processing. As airport operators see greater use of biometrics and as they consider exploring the technology, research was needed to produce a primer to help airport operators and their stakeholders understand the uses and considerations of biometric technology.

ACRP Project 03-55 was led by Inter*VISTAS* with advisory support from Baldwin Liaison Consulting and Dr. Ann Cavoukian. The research undertook vision and use cases; reviewed the legal and policy considerations, planning and processes, and system architecture and design elements; and collected input in a series of one-hour webinars hosted by the project team.

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CHAPTER 1

Introducing Biometrics

Biometrics are among the most powerful yet misunderstood technologies used at airports today. The ability to increase the speed of individual processes, as well as offer a touch-free experience throughout an entire journey, is a revolution that has been decades in the making. Biometrics can be a fundamental building block for key aspects of airport infrastructure. However, like many foundations, it must be built on a range of considerations beyond technology—process, human factors, policies, to name a few.

Considering biometrics in airports alone would be inappropriate: the technology must also be understood against the backdrop of the use and misuse of emerging technologies. From unlocking a smartphone to the automated recognition of individuals from a library of personal photos, the power of biometric recognition is widespread in our digital lives. At the same time, abuse of powers and flaws in recognition technologies have been widely publicized. Database breaches that enable the misuse of identity information as well as the limitations of certain technology types have given rise to public anxiety about facial recognition. Oversimplification and misunderstanding of technology uses have created an environment where several public entities have elected to outright ban the use of biometrics. Technology firms have reacted by pausing activities leveraging facial recognition conducted in concert with law enforcement agencies as society at large determines some of the parameters to better manage the potential misuse of biometrics.

Biometrics in Aviation

So where does this leave aviation and the use of biometrics at airports? *ACRP Research Report 233: Airport Biometrics: A Primer* was commissioned to encapsulate the range of issues and opportunities associated with the use of different forms of biometric identification. The primer is prepared by experts with a deep understanding of technical issues, but also the range of policy and business case aspects that require consideration by those seeking to introduce the use of biometrics for their operation.

Three overarching clarifications regarding the primer should be noted:

1. This study is not about mass surveillance: The word "biometrics" is often conflated with mass surveillance. In other words, some in the media and public conflate the use of biometrics in the context of personal identification for individuals who opt in with a model used typically in law enforcement to use video to surveil individuals contrary to their privacy interests.
2. This report could be adapted into a living primer or could be updated: biometrics technology has accelerated significantly over 25 years at airports in the United States and continues to evolve. The concepts described herein are directional in nature, but a living primer could be considered to incorporate overall trends/changes in the future. This report would benefit greatly from an update in due time.

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2 Airport Biometrics: A Primer

3. There are no absolutes: biometrics are also often misunderstood due to the complexity and the number of choices required for an individual deployment. The overall system architecture requires much consideration before effective deployment, which extends beyond details such as the choice of biometrics technology to be employed (e.g., fingerprint, iris, or hand geometry).

Objective of This Primer

The objective of this primer is to help aviation stakeholders, especially airport operators, understand the range of issues and choices available when considering and deciding on a scalable and effective set of solutions using biometrics. These solutions may serve as a platform to accommodate growth and to address the near-term focus regarding safe operations during the COVID-19 pandemic.

Fundamentals of Biometrics

Definition of Biometrics

Biometrics are biological and behavioral characteristics that can be used in a system to recognize someone or verify someone's identity. Thus, the main purpose of biometrics is to accurately identify individuals, typically in relation to enabling access to a controlled environment.

Types of Biometrics

The earliest use of biometrics dates from the 1860s, with unique patterns (i.e., dash and dot signals demonstrating behavioral characteristics) used by telegraph operators to identify senders/authenticate messages. In the past three decades, with the advent of sensor technologies and computing power, automated devices used to identify a person based on certain unique characteristics have proliferated.

Biometric identification consists of verifying the identity of a person. One needs to capture an item of biometric data from a person. It can be a photo of their face, a record of their voice, or an image of their fingerprint. This digital biometric is now in a data format that can be stored for use by algorithms. However, biometric data are often stored as a biometric template, where the biometric data are encoded, for instance, as a random sequence of numbers and letters.

There are many different types of biometrics that can be used to create unique templates for each individual. Some of the common types of biometrics are fingerprint, iris, facial, and voiceprint recognition. Many of these can increasingly be found in consumer electronics such as smartphones or laptops and are used to unlock devices or to confirm mobile payments. Emerging forms of biometrics include DNA matching, patterns of walking, and other forms of site-specific applications to capture unique traits/behaviors of individuals. A non-exhaustive selection of biometrics is explained in the following.

Iris Recognition

Iris recognition is an automated method of biometric identification that uses mathematical pattern-recognition techniques on video images of one or both of an individual's irises, which have complex patterns that are unique, stable, and can be seen from some distance.

Facial Recognition

Facial-recognition technology can identify or verify a person from a digital image or a video frame from a video source. There are multiple methods by which facial-recognition systems

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work, but in general, they work by comparing selected facial features from given images of faces within a database. Facial-recognition technology is also described as a biometric artificial intelligence–based application that can identify a person by analyzing patterns based on facial textures and shape.

3D Hand and 3D Finger Geometry

Three-dimensional (3D) hand and finger geometry exploits discriminatory information provided by the 3D structure of the hand, or more specifically the fingers, as captured by a 3D sensor. The advantages of 3D hand and finger biometrics over traditional two-dimensional (2D) hand and finger geometry (more typically know as fingerprints) are authentication techniques, improved accuracy, the ability to work in a contact-free mode, and the ability to combine this form of authentication with facial recognition using the same sensor.

Hand Geometry

Hand geometry is a biometric that identifies users by the shape of their hands. Hand geometry readers measure a user's hand along many dimensions and compare those measurements to measurements stored in a file.

Voice Recognition

Voice biometric technology works by digitizing a profile of a person's speech to produce a stored model voiceprint or template. The tones collectively identify the speaker's unique voiceprint. Voiceprints are stored in databases in a manner similar to the storing of fingerprints or other biometric data.

DNA Matching

DNA matching is a complex technique that can be used to identify someone or verify someone's identity. DNA matching requires highly specialized laboratory equipment and is not readily (or quickly) achieved. Further, techniques that use DNA may offer much deeper and far-reaching capabilities—with extensive research being done to map the human genome stored in our DNA—because someone's DNA can provide much more information about that person than they are often comfortable sharing, even on a personal basis. DNA can store information about heritage and physical characteristics but also elements related to one's health such as elevated risks to diseases. For these reasons, DNA matching is not (yet) applied in the airport environment and is not a candidate for biometric applications that require speed and efficiency.

Nonetheless, DNA matching is mentioned here because it is used in other areas, such as (forensic) research applications in criminal justice cases, challenges in paternity cases, and in some other cases where it may be used either as a last resort or because of the high accuracy of identification it can provide. Use of DNA for biometric technologies in aviation would most likely occur only in the distant future, partially because the storage of one's DNA in a database (in either the private or public sector) would likely be subject to legal challenges.

Practically, there are various methods for analyzing DNA to establish if two samples are the same or different. These methods are often referred to as DNA fingerprinting. For example, two sample pieces of DNA can be studied in the laboratory to determine if they have portions in common and thus overlap with one another.

Combinations of Biometrics

At times, one biometric is insufficient, and a combination of is used. This combination raises the level of security and is sometimes used to control access to high-security buildings or areas such as vaults.

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Understanding the Differences Between Detection, Matching, Identification, Verification, and Authentication

When dealing with biometrics, there are many ways to describe the expected outcomes of the technology. From detection to authentication, each term used has different connotations, from the concept of operations to the application of law. The following terms are described in relation to their use in biometrics:

- "Detection" answers the question, "Is a biometric being sensed?" Often, a specific part of the biometric technology is in place only to process from sensors whether a biometric is presented (i.e., is there a finger on the fingerprint scanner? Is there a face in front of the camera? If so, the biometric can be digitally recorded.) Detection does not specify whether a biometric is scanned, used, stored, or retained.
- "Matching" is the step that may follow detection of a biometric. In this case, the biometric is scanned, stored at least temporarily (as a template or not), and used for the purpose of comparing with a second (similar type of) biometric. The second biometric must be present or stored in order to be compared and matched. A positive match is when the two biometrics exhibit sufficient likeness to pass a specific matching threshold.
- "Identification" answers the question, "Who are you?" The identification process implies using a process of matching a scanned biometric to a set of stored biometrics, with the intent of discovering the identity of the person to whom the scanned biometric belongs.
- "Verification" answers the question, "Are you really who you say you are?" The process also implies the matching of two biometrics, one scanned and the other either on file or presented by the same person in the form of a token (e.g., an e-passport). Because both biometrics are presented, verification only aims to prove a positive match between the two, rather than to discover the identity of the person to whom the biometric belongs. In this case, the actual biographic information does not have to be linked to the biometric or stored with the biometric. It should be noted that facial recognition is a common type of biometric verification. The difference between facial recognition and facial verification is exemplified for the facial biometric in the following:
  - Facial recognition is a software-driven application that looks at facial points and contours to analyze and compare two face biometrics and make an attestation of likeness.
  - Facial verification is considered a form of biometric verification by definition and looks at unique biological characteristics so one's identity can be verified.
- "Authentication" answers the question, "Is this identity credential/document authentic?" This implies the checking or verifying of the authenticity of an identity document or credential; for example, is the passport fake or real? Passports, much like many currencies, have elements and attributes that allow an expert to prove the authenticity of the document. E-passports also have a digital attribute that allows for automated checking of the document's authenticity (i.e., whether the document is indeed issued by the right issuing party or government). Biometrics are typically not used in the authentication process.
- "Validation" is at times confused with verification (or authentication). A slight distinction is important to clarify here: validation looks at whether the specifications that are evaluated are appropriate to accomplish the goal, while verification checks whether the specifications have been met. For example, in the case where a passport is used to verify identity, one checks if the specifications (e.g., the name printed in the passport) match what the person claims his/her/their name to be. In validation, one may conclude that a passport is an adequate tool to verify one's identity because the characteristics of the passport (printed name in the document) can be used to accomplish the goal of identity verification.

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Using Biometrics: What Is the Purpose or Objective?

There are three main modes of using biometrics for matching (see also Figure 1-1):

- One-to-one (1:1) matching: verifying a biometric based on a previously registered identification document (i.e., is this really the same biometric as on the identification document?).
- One-to-many (1:N) matching: verifying a biometric or identifying an individual out of a large database of biometrics/people (i.e., in case of verification, is the biometric presented also in the database? In the case of identification, to whom does the biometric presented belong?).
- One-to-few (1:few) matching: verifying a biometric or identifying an individual out of a subset of a database of people.

There are substantial differences between the three types of matching. One-to-one biometrics involves, for example, an individual approaching a door or checkpoint and uses a mechanism such as a previously registered template that is compared to a photograph to verify access rights.

On the other hand, one-to-many is exemplified through mass surveillance techniques such as those integrated in closed-circuit television cameras for law enforcement and national security. This mode works through leveraging camera technologies to identify individuals who are in view of a camera, often without their knowledge.

One-to-few is different from one-to-many because it uses a subset of a larger database to verify or identify individuals. This database is smaller due to a selection process prior to the comparison.

While each type of biometric matching has privacy implications, each is rooted in a different technology type. One-to-many requires a substantial amount of data storage and computation power. India and China's drive toward a mandatory national biometric system will require large amounts of information to be collected and protected to enable recognition of billions of individuals.

One-to-few is less computationally intensive because it uses a subset of a larger database. Instead of comparing an image to the whole database, a selection of profiles can be used when it is known ahead of time who will be expected. At an airport, most passengers have booked their flights well in advance, allowing for the potential comparison to the passenger manifest as the population of the "few."

One-to-one biometrics is the least computationally intensive of the modes and has a range of options for data storage, including databases, smartcards, and the microchip on an e-passport.



Figure 1-1. The three types of biometric matching commonly seen today.

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How Are Biometrics Collected and Stored?

Biometrics are typically captured by a camera, a microphone, a scanner, or some other sensing device and turned into a digital signal. This signal is then transmitted to a computer chip that can process the signal with software to create a digital code called the biometric template. The biometric template is not a copy of the biometric scan itself, but rather is converted by an algorithm to a unique data file. The critical issue regarding biometric data security relates to the location of the stored biometric templates (see Figure 1-2). It is considered best practice for actual sensor data to be discarded upon creation of the biometric template. The biometric templates are typically stored in a database or on a mobile device or a token. "Token" is the generic term used to describe an item that one can carry with them on which a biometric template is stored, such as a passport, identity (ID) card, or memory stick.

This guidebook discusses four distinct types of storage:

1. Locally on a mobile device,
2. On a portable token such as a chip on an identity pass,
3. Centralized on a database server, and
4. Distributed data storage.

A locally stored biometric template can only be used on that device (e.g., using a fingerprint to unlock a phone). Biometric templates can also be stored on a database server in a data center. This method is typical for large corporations using biometrics to grant their employees access in office buildings or secured facilities. Storage on database servers comes with a higher risk since multiple templates are stored in one location. Biometric templates can also be stored on a portable token such as a USB drive, an access card, or an e-passport. Finally, biometric templates can be stored in distributed data storage. This method stores biometric templates on a local device and on a server, both of which must be accessed concurrently for authentication. Because of the split nature of this biometric template storage method, hacking is nearly impossible and, therefore, this method is highly secure.

Although encryption of the stored data significantly improves security, determining who has access to the encrypted data and how the data are used is a critical issue. With respect to the privacy retention and sharing of personal information, civil liberties groups and the Electronic Frontier Foundation have challenged the need to collect and use facial information. The legal, policy, and privacy aspects related to the collection, use, and retention of biometrics are discussed in Chapter 3.

Current State of Biometrics

We are in an era of mass adoption of consumer biometrics, with most common devices providing the option of password-based unlocking mechanisms or the use of facial recognition or other biometrics for unlocking. At the same time, several governments have elected to use



Figure 1-2. Where a system stores the biometric data is critical in the design, whether it be on a token, on local storage, on a personal device, or in the cloud.

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biometric technologies to enforce social order. For example, in Shenzhen, China, facial recognition is used to issue jaywalking citations as well as collect fines—all taking place before the individual reaches the other side of the street.

Some entities and individuals seek an outright ban on the use of biometrics, which others consider analogous to banning the use of passwords for computers. Accordingly, it is important for organizations to define the range of biometric use cases, respecting laws, regulations, and best practices for privacy-protected interface with passengers, employees, and consumers to ensure that potential risks are proportional to benefits. The privacy discourse to date has focused, in part, on the retention and sharing of personal information. Chapter 2 explores U.S. use cases, and Chapter 3 addresses the legal, policy, and privacy issues.

Vision for Biometrics at Airports

With the fundamentals of biometrics described in the previous section, this section narrows the scope to the application of biometrics within the airport environment. The current vision for biometrics at airports dates from the late 1990s and accelerated in different phases centered on e-passport adoption in the 2000s as well as alternate forms of storing biometric information in the 2010s. Through this time period, multiple initiatives advanced common interests across stakeholders, including:

- The Simplified Passenger Travel Interest Group initiated a discussion among airlines, airports, and governments to explore mechanisms for the implementation of biometrics.
- OneID from the International Air Transport Association (IATA) is aimed at providing passengers with a seamless experience, enhanced security for governments, and reduction of costs for airlines and airport operators.
- New Experience Travel Technologies (NEXTT) from IATA and Airport Council International (ACI) extends the biometric vision to other use cases for baggage handling on and off airport as well as to applications of artificial intelligence.
- The Digital Travel Credential (DTC) standard from the International Civil Aviation Organization (ICAO) envisions new opportunities from its passport standards documents.
- The World Travel and Tourism Council's Safe and Seamless Traveler Journey initiative aims to enable a seamless, safe, and secure end-to-end traveler journey through systematic, biometric-verified identification at each stage of the journey by replacing manual verifications.

Broadly, there are also different visions for biometrics among entities in the United States. The U.S. Chamber of Commerce Facial Recognition Policy Principles, from December 2019, point to "facial recognition technology as an enormous potential to enhance security and safety and enable innovation across a wide variety of sectors including transportation, retail, hospitality, and financial services" (Furlow 2019). The DHS and Customs and Border Protection (CBP) have also articulated a long-range vision for biometrics, as have individual airlines.

This vision has four components:

1. Collection of tools: The use of biometric technologies is not simply one mode of operation; it is a family of different tools/devices/storage mechanisms, each with different implications.
2. Trust in identity: From credit card fraud to intercepting imposters, biometrics can provide more trust. This indirectly connects to improved security and safety in the aviation system.
3. Seamless flows: From the promise of a single form of identification, biometrics can ensure that the amount of document fumbling is minimized (passport, visa, health forms, etc.).

To simplify the vision of biometrics specific to aviation, the following vision is offered:

Biometric technologies are the collection of tools to establish trust in identity, seamless flows, and a touchless airport experience.

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4. Touchless airport experience: From the status of COVID-19 immunization, to other ways of enabling airport retail to function, biometrics can augment minimized contact with disease-transmission media, surfaces, or staff.

Interoperability and Scalability

With the overall trend of biometrics moving to multi-stakeholder solutions, "interoperability" means a single biometric system that can be used by airlines, governments, or third parties through an airport environment and the ability for identity verification to be used throughout the journey.

Other industries can provide interoperability examples to the aviation industry. In banking, for example, there was a point in the 1980s when every bank had a separate system for every ATM. Over time, interoperability became available so that you could use an ATM card from a U.S. bank at a different bank in France.

Biometrics are similar but perhaps more complicated due to the potential for future use cases to be added. For example, if an identity verification system is used for airline check-in, the TSA security check, and flight boarding, this is a deployment that could be in place for several years before adding additional phases. Scalability planning dictates a pathway to enable more modules to be added, for example for rental cars, airport retail, or even other uses outside airports. A private third-party that has demonstrated scalability is CLEAR, which offers identity verification as a service through a paid membership.

Related to interoperability is the common use of processes, installed systems, and facilities at airports. Common use is often characterized as passengers being able to use self-service kiosks and check-in desks for all airlines at an airport, but the concept of common use applies to a much broader set of facilities and sets a common standard for the design and operation of these facilities. The benefits of common use are as relevant to biometric technologies and solutions as they are to traditional facilities and include spatial savings, time savings (e.g., leading to shorter minimum connection times), and improved passenger experiences. For more on the design, implementation, and benefits of common-use systems and facilities at airports, refer to the *ACRP Report 30: Reference Guide on Understanding Common Use at Airports* (Belliotti et al. 2010) and to the forthcoming report from ACRP Project 03-52, "Guidelines for Adapting and Managing Airport Common Use Programs."

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CHAPTER 2

How Advanced Is the Employment of Biometrics at Present?

Summary

This chapter describes ways that biometrics are employed in airports: process facilitation, access control to secured areas, tailored customer services to the individual user, commercial use by retail/concessions vendors, and fraud identification and risk mitigation to minimize loss.

Ten case studies illustrate the variety of biometric technology use in airports today. These case studies showcase uses at five U.S. airports and five airports worldwide. The case studies highlight (1) a potential template for current and future models for touchless and cashier-free retail at airports; (2) the VeriFLY initiative, which combines the idea of reservation systems with quick response (QR) codes to provide queue access and other possible passenger processing uses; (3) a trusted traveler model for true identification of low-risk, pre-approved travelers who have been vetted in advance; (4) an example of a public–private partnership in handling biometric/biographic data for employee vetting for TSA approval of secure identification display area (SIDA) badges; (5) an example of the power of end-to-end facial recognition in one system that offers benefits for border processing, check-in, TSA, and other processors; (6) a Canadian–Netherlands pilot illustrating the potential of the DTC for global travel; (7) implementation at multiple terminals at London Heathrow (LHR) for a more seamless experience and offering scalability for international application; (8) Easy Airport, which illustrates how biometric uses can be designed and refined to meet airlines' business needs, including during COVID-19; (9) an example of an effective communication strategy with passengers and the general public about biometric use; and (10) one of the earliest examples of biometric data use in Aruba's Happy Flow.

The 10 case studies illustrate specific and, in most cases, unique applications and benefits of biometrics, as well as lessons learned, identification of challenges, and key trends, which include the following:

- Trend 1: The deployment of integrated and multi-stakeholder biometric solutions is increasing because of the potential benefits of biometrics.
- Trend 2: Digital transparency and privacy concerns are adding to the complexity of implementation, particularly as the legal landscape changes.
- Trend 3: A focus on identity verification solutions is evident, in part to distinguish from mass surveillance programs that also leverage biometrics.
- Trend 4: Global biometrics and standards are emerging from a variety of governmental and nongovernmental entities that are addressing privacy, security, ethical, and technological concerns.
- Trend 5: Smartphones are expected to enable more use of biometrics through the on-device storage capability for biometrics and the promise of the transmission of digital travel credentials.

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Key Takeaway

The increased use of biometrics in everyday life can generally be assigned to five practical uses:

- Facilitating and simplifying processes
- Granting access for certain people to certain areas
- Personalizing interaction with technology
- Expediting commercial exchanges
- Guarding against fraud and improving security and safety

Five Primary Use Cases for Biometrics at Airports

Five primary use cases were identified in the aviation sector—or more specifically, were implemented at airports around the world (see Figure 2-1). These five use cases are each described in the subsequent sections and are followed by 10 case studies that provide examples of one or more of the use cases identified.

Process Facilitation

The aviation system involves a series of steps and processes. Passengers, employees, and suppliers need to record and demonstrate identity information several times during a workday or journey. While a document check requires on average 3 to 5 seconds, it is the multiplication of transaction times to present paper documents that can lead to difficulties, especially in a group environment. Adding 5 seconds per passenger may sound insignificant, but the practical effect can create a queue of hundreds of people if there is not a corresponding reduction in time elsewhere in the overall process.

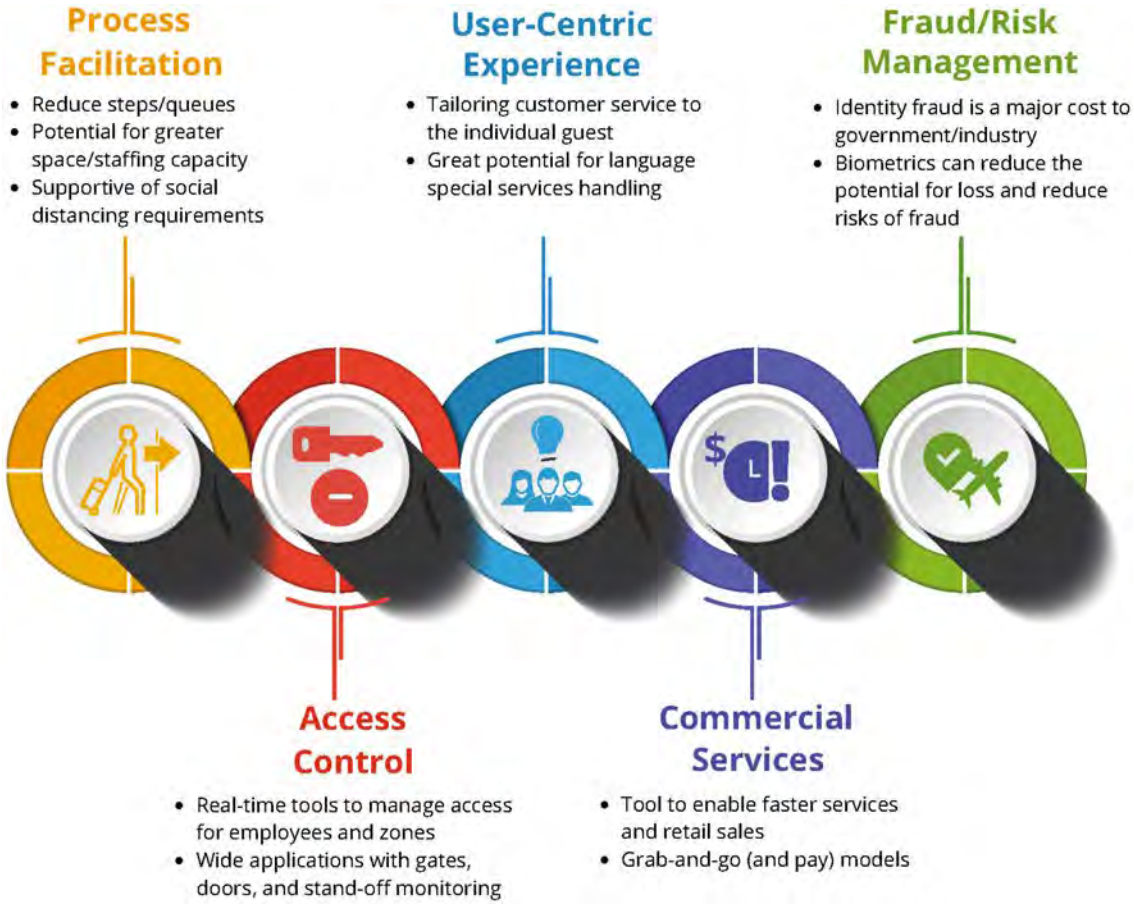


Figure 2-1. Overview of primary use cases of biometrics.

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During the COVID-19 pandemic and the recovery phase, these considerations have been even more important in addressing social distancing and health screening requirements, which can further elongate queues. Airport operators must also be mindful of the much greater level of handling of exceptions. In other words, as the travel environment adjusts to new requirements or steps, there will invariably be missing forms or other steps that passengers are unaware of that will require special handling, space, and time to resolve. The aim must be to have the vast majority of the population (95%+) able to proceed through airport processes without delay.

To unlock the potential for process improvements, there is a move to reduce sequential steps. Biometric use cases offer the potential to combine several process steps for an individual into one, reducing the number of sequential steps (and exceptions). The net result could be the elimination of passenger queues and converting these steps to a more automated process.

Access Control

Security is about safeguarding that which needs protection from harm, but it is also about allowing that which will do no harm. Determining one from the other is typically based on one of three things: who you are, what you have, and what you know. The latter two are best described as having a key (what you have) and having a password or PIN code (what you know). Biometrics can be an added layer, leveraging the physical features of an individual for identification or authentication purposes.

One of the most mature uses of biometrics at airports is access control—in other words, the automated authorization to proceed through a door to a secure area of an airport. A combination of a key card and a biometric is the dominant use case. While this use case may be conventionally thought of in the context of airport employees, there are a number of emerging use cases geared toward airline passengers.

Passenger access to a boarding gate or managing access to a premium lounge on the basis of biometrics are two prominent examples of passenger-facing access control.

User-Centric Experience

While identity management and process facilitation are geared toward confirming identity, an emerging use case for biometrics is to customize the user experience.

Aviation markets are increasingly segmented into different market types and preferences. From millennials to older travelers, to different travel segments (leisure, business, visiting family/friends, passengers with reduced mobility), airports are increasingly offering customized experiences. Providing identity management to help differentiate the user experience, from brand loyalty to personalized experiences, is a use case to make the experience more streamlined and welcoming to travelers.

Some tangible examples are interaction with passengers to only display information relevant to the individual pertaining to flights, weather, and preferred services, as well as customizing other elements of direct communication (language, font size, etc.) toward individuals. Vendors have also demonstrated new screen technologies that can ensure that on-site wayfinding signage is customized and directed to specific individuals only—all powered with a voluntary biometric identification algorithm.

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Commercial Services

Airports around the world obtain roughly 45% of their revenues from non-aeronautical sources. This share can be higher for some airports (e.g., Las Vegas) that have a larger concessions program. While some world regions (Europe, Asia) have traditionally had a higher amount of retail/concessions/duty-free revenue, there is potential for biometrics to increase revenue per passenger.

Increased revenues may be achieved through biometric-enabled use cases that convert lost time into potential spending time by eliminating queues or waiting. The use of cashierless airport retail has accelerated to an extent due to the COVID-19 pandemic. Increasingly, between flights, passengers can simply grab items and go without needing to queue for payment. Biometrics and other tap-to-pay technologies can enable this process.

Other groups are also treating dwell time in airports as opportunities to sell products for pickup at an arriving airport, and this may be fueled by biometrics.

The realignment of commercial services in airports is necessitating a touchless experience for any passenger-facing service, and there are many biometric use cases for this.

Fraud/Risk Management

Finally, the dominant area of biometric solutions for governments is the interdiction of criminal behaviors, including those related to travel. From CBP's ability to catch 300 imposters through facial-recognition software come a range of use cases related to defeating criminal activity (U.S. Customs and Border Protection 2021).

Hundreds of millions of dollars are lost annually by airport retailers and the airline industry due to fraudulent activities. According to IATA, the estimated cost of payment fraud to the airline industry each year is about $858 million (IATA 2016). Whether it is credit card fraud at a store inside an airport or ticket fraud, there is a cost to the industry that the use of biometrics may mitigate.

In addition to combating criminal behavior, there is also the added aspect of risk management. Risk-based security initiatives for TSA, for example, have the ability for biometrics to aid in linking different systems. Results of passenger security screening checkpoint scans can help to inform the level of baggage screening scans—a platform that may be supported by biometrics tied to departure control systems.

Key Takeaway

The main takeaways from five U.S. and five international biometric case studies are presented, ranging from complex multi-stakeholder implementations to simpler and more local solutions. Implemented solutions inform:

- Granting enhanced retail experiences
- Fast-tracking and improved security solutions
- Health-related separation
- Complete biometrics-enabled journeys, from the curb to the gate
- Returning data ownership to the passenger
- Seamless passenger flows and less intrusive technology in the journey
- Governments implementing a nationwide biometric identity

U.S. and Worldwide Lessons from Deployments

Table 2-1 presents an overview of the key highlights and lessons learned from 10 case studies, in which the application and importance of biometric technologies are illustrated. These case studies provide a representative cross-section of the efforts and developments that have taken place in the United States and the rest of the world. Further details regarding each case study can be found in Appendices A through J.

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Table 2-1. Case study overview.

Case Study Number	Case Study Name	Case Study Highlights
1	Amazon Go Cashierless Retail Experience	• Potential template for current and future models for touchless and cashier-free retail at airports • Some enrollment challenges may be resolved with simple two-step palm enrollment.
2	Denver–Daon Biometric Partnership	• Combines the idea of reservation systems with QR codes to provide queue access • One of several platforms with the potential to link passengers to health passport information • Remote passenger screening pilot can also help meter the peaked nature of operations, even during a reduced-traffic period.
3	CBP Trusted Traveler Programs at U.S. Airports	• Ability to facilitate not just U.S. citizens, but a range of country nationals (S. Korea, Canada, UK, etc.) • Model for true identification of low-risk, pre-approved travelers that have been vetted in advance • Added benefits to facilitate these individuals through TSA via PreCheck
4	Seattle–Tacoma International Airport and Designated Aviation Channeling	• The designated aviation channeling (DAC) model is an example of a public–private partnership in handling biometric/biographic data. • Evolution of protocols for handling personally identifiable information offers a good set of protocols.
5	Curb-to-Gate Program by CBP and Delta Air Lines at Hartsfield–Jackson Atlanta International Airport	• A good example of the power of end-to-end facial recognition in one system • Demonstrated capability of leveraging biometrics without the need for re-enrollment • Specific measurements of benefits for border processing, check-in, TSA, and other processors
6	Known Traveller Digital Identity at Aéroport International Montreal–Pierre Elliot Trudeau, Montreal, Canada	• Lessons learned on privacy of information and transmission to foreign governments, especially regarding legal frameworks between the project stakeholders
7	The Seamless Passenger Journey at London Heathrow	• Scalability of platforms could also potentially work with transatlantic flights to the United States. • Integration quality and deployment across multiple terminals is a significant model to emulate for smooth delivery.

(continued on next page)

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Table 2-1. (Continued).

Case Study Number	Case Study Name	Case Study Highlights
8	Risk Management During COVID-19 Using Biometrics at Carrasco International Airport, Montevideo, Uruguay	• Ability to meet different airline business cases; some saving of resources/re-dedication of resources through the biometric system • Add-on data portals provide added value for project partners. • Used for the risk profiling of passengers as part of efforts to control the spread of COVID-19
9	Digi Yatra and the Seamless Passenger Journey at Kempegowda International Airport, Bengaluru, India	• Focus groups and simple communication; "using your face as boarding pass" • Model notable also for an opt-out pathway for those who do not elect to use biometrics
10	Happy Flow at Aruba's Queen Beatrix International Airport	• Early implementation of biometrics with Happy Flow model to anticipate transatlantic preclearance • Some improvements on the mechanisms to conduct identity checks for participants • Broader vision for enrollments at any location may not be supportable by all stakeholders.

Case Study 1: Amazon Go Cashierless Retail Experience

Summary

Initiated by Amazon, the Amazon Go retail experience uses a combination of cameras, sensors, computer vision techniques, machine learning, and artificial intelligence techniques to create a checkout-free retail experience for its customers. The technology is also available for other retailers and is referred to as "Just Walk Out." The customer's identity is verified when signing up through the Amazon Go app, and entry to the store is gained by showing a QR code, one's registered palm biometric (for an Amazon Go store), or one's credit card (for other retailers equipped with the Just Walk Out technology). See Figure 2-2 for an overview of the Amazon Go steps, and Table 2-2 for key points of the case study.



Source: Adapted from Amazon Go 2020.

Figure 2-2. Overview of the steps to shop at an Amazon Go store.

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Table 2-2. Key points for Amazon Go case study.

Criteria	Overview
What	• Touchless checkout-free retail
Where	• 21 stores across the United States
Passenger process	• Sign up through a mobile app and create Amazon account • Scan QR code, palm, or credit card when you walk into store • Take items and just walk out
Who	• Amazon Go (Amazon)
Why	• Revolutionize the retail experience: to make it safer and more efficient
How	• Movement tracking and smart sensors in store track items customers take. Once they leave, their accounts are then charged.
Enrollment	• Sign up through a mobile app and register palm biometric (where available)
Verification of identity	• Identity is verified through mobile app when customer signs up for an account.
For	• Customers with Amazon Go accounts

Program Concept

By leveraging computer vision and machine learning to track customers and distinguish between the items they pick, customers can be charged for their items without stopping at a cashier. The methodology does not rely on facial recognition but rather on movement tracking and, where available, a biometric palm scanner.

Each customer, upon entry, is associated with a name, an account, and a consumer profile reflecting all interactions with items on the shelves (e.g., which items they view, pick up, and buy). Future use of the customer's data is still undefined. However, the initial patent application included some examples where the customer's purchase history could be used to confirm which items were being picked up from the shelves by the user.

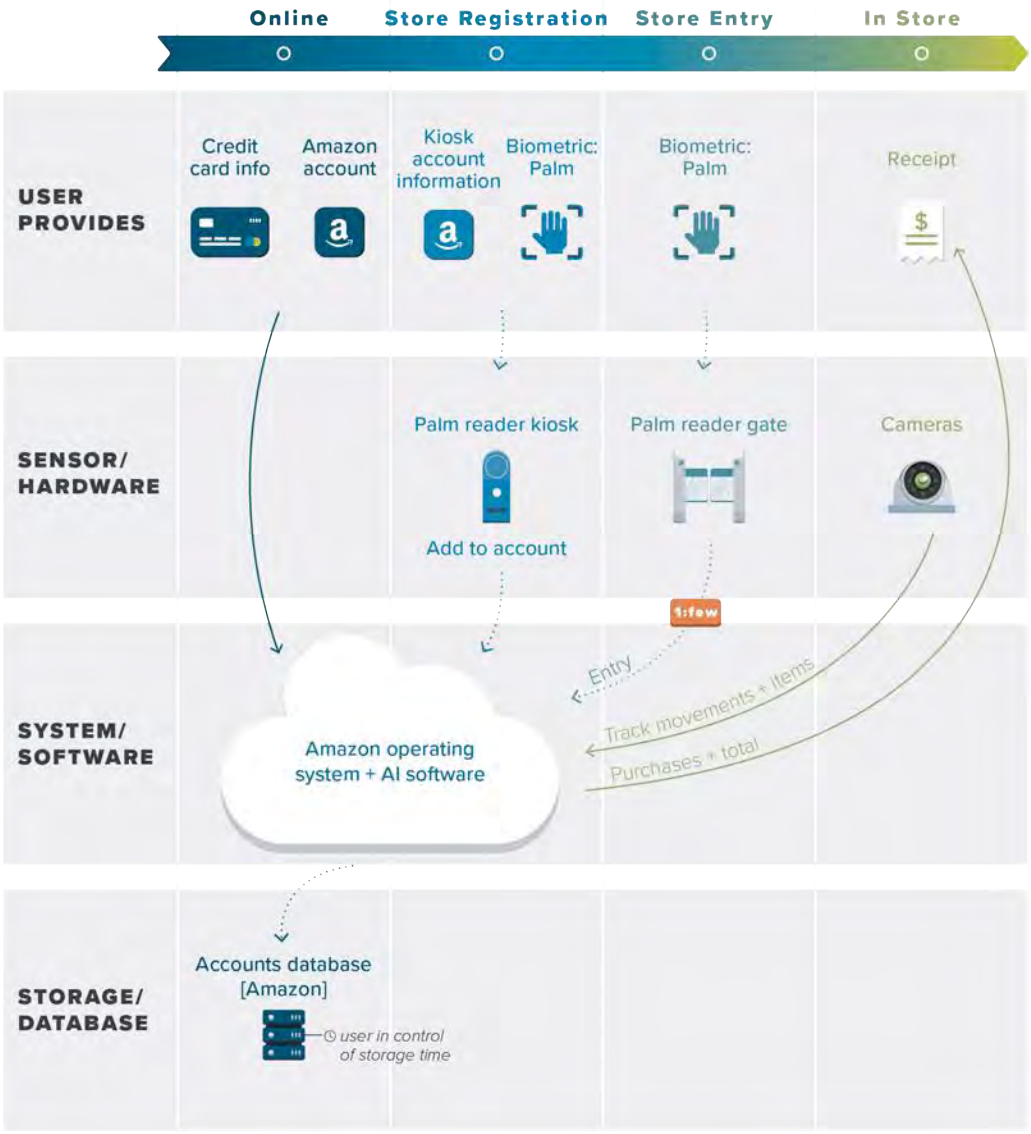
Program Key Takeaways

The main benefits behind using this type of technology are significant time savings for customers and financial savings for retailers through lower operational expenditures on employees (e.g., labor wages and benefits). Some enrollment challenges may be resolved with simple two-step palm enrollment, which is interesting because some may feel this methodology is less invasive compared to facial recognition. Furthermore, this type of software presents a potential alternative to current and future models for touchless retail for both airports and airlines.

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Case Study #1

Amazon Go Cashierless Retail Experience



Case Study 2: Denver–Daon Biometric Partnership

Summary

To balance the needs of passenger health and safety, Denver International Airport partnered with public health stakeholders and biometric technology firm Daon to develop a biometric authentication system known as VeriFLY, which is aimed at making passenger journeys safer and more predictable. A second pilot for remote screening would involve a security checkpoint located remotely from the passenger terminal. See Table 2-3 for key points of the case study.

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Table 2-3. Key points for Denver–Daon case study.

Criteria	Overview
What	• Biometric authentication system for security lane reservations and remote security screening
Where	• Denver International Airport
Passenger process	• Mobile app sign up to a reservation for the security checkpoint • Health form completion • Arrive at airport for designated slot
Who	• Denver International Airport and Daon
Why	• Speed the processing of passengers • Create health-conscious and more predictable journey from security checkpoint to concourse
How	• VeriFLY: Facial recognition to verify identity in the app to bring up a QR code for airport touchpoints • Remote security screening
Enrollment	• Voluntary; appointments on the app can be booked 2 weeks in advance
Verification of identity	• Facial recognition via the mobile app
For	• Departing passengers on domestic and international flights

Program Concept

With the intention of putting high-risk and health-conscious passengers first, passengers can use a reservation system on a mobile app to access a dedicated security lane. At the airport, after a temperature check, the passenger can enter the security lane by scanning a QR code shown in the app at an e-gate. A health questionnaire must also be completed within 24 hours prior to their flight. Passengers access a reserved train car with other VeriFLY travelers, which ensures a socially distanced and contactless concourse ride to their respective gates.

At its initial implementation, the program was a stand-alone system that was not tied into other databases. Future development may automate more steps of the process by including facial recognition and third-party health information.

Remote screening: In the future, the airport may offer a security screening checkpoint remote from the passenger terminal, adding capacity and offering a unique experience for segments of the traveling population.

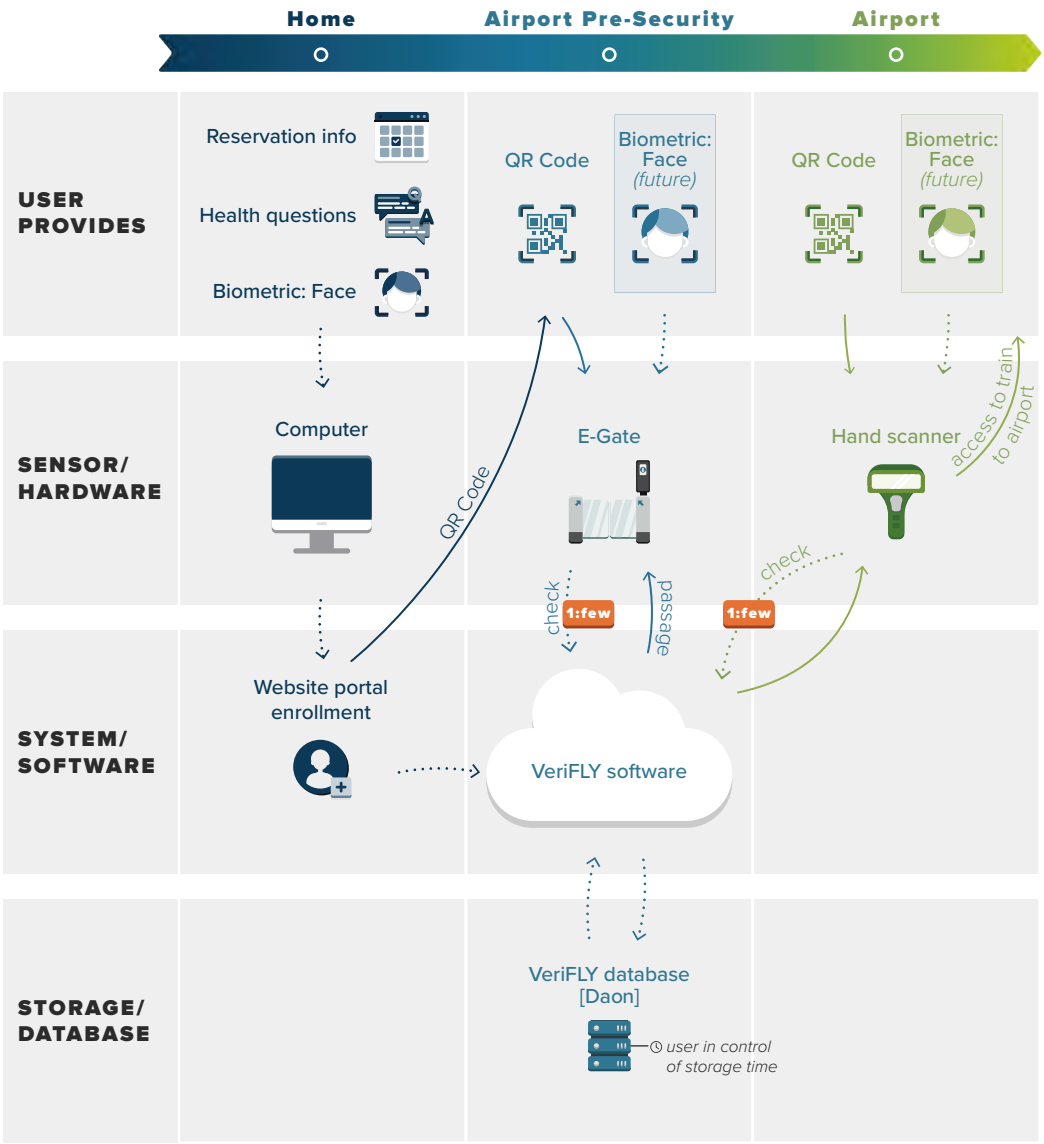
Program Key Takeaways

The VeriFLY initiative presents an opportunity to add reliability and confidence to the passenger travel journey by reducing customer interactions and the potential for congestion. Furthermore, VeriFLY combines the idea of reservation systems with QR codes and can help reduce the strain on airport operations, even during reduced-traffic periods as a result of the COVID-19 pandemic. It is one of several platforms that could leverage health passport information, many of which are under development. Interestingly, the remote screening pilot can also help meter the peaked nature of operations, even during a reduced-traffic period.

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Case Study #2

Denver-Daon Biometric Partnership



Case Study 3: CBP Trusted Traveler Programs at U.S. Airports – Global Entry

Summary

Designed to provide quick, easy, and convenient processing, the Global Entry program from CBP allows for expedited clearance for pre-approved, low-risk travelers on arrival in the United States. Global Entry is designed to reduce the time required for travelers in the passenger verification process when entering the United States. Similar programs include SENTRI and NEXUS. The roots for Global Entry date back to the U.S. Immigration and Naturalization Service (INS) Passenger Accelerated Service System (INSPASS) in the 1990s; the current

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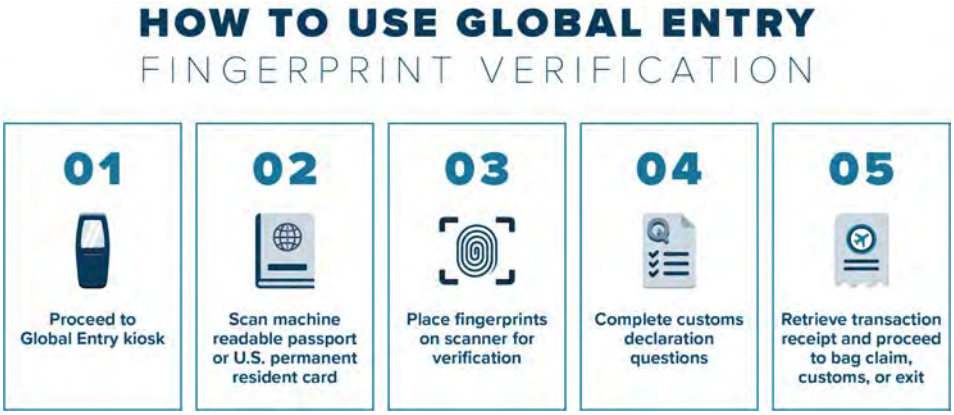


Figure 2-3. Overview of Global Entry program procedures.

iteration was piloted by CBP starting in 2008. See Figure 2-3 for an overview of the Global Entry program procedures and Table 2-4 for key points of the case study.

Program Concept

At airports, program members proceed to Global Entry kiosks, have their picture taken, and are matched against a gallery pre-compiled from a CBP database. The kiosk then issues the traveler a transaction receipt and directs the traveler to the baggage claim and the exit.

Travelers must be pre-approved for the Global Entry program. All applicants undergo a rigorous background check and in-person interview before they are approved for enrollment. Applications are reviewed by CBP, and information is processed through various government databases. Once conditionally approved, the individual goes to one of over 100 enrollment centers to provide fingerprints and a facial photograph along with native documents (e.g., passport, proof of residency) for review by CBP. The biometrics collected at enrollment are used for facial matching.

Table 2-4. Key points for CBP Trusted Traveler/Global Entry case study.

Criteria	Overview
What	• Expedited clearance for pre-approved, low-risk travelers
Where	• 75 U.S. ports of entry, including preclearance airports
Passenger process	• Individuals have their pictures taken at the automated border kiosk
Who	• CBP • DHS • Participating airports
Why	• To fast-track border crossing procedures for individual travelers who are deemed low risk
How	• Verification of identity through facial recognition
Enrollment	• Voluntary program with multistep application process, including an in-person interview
Verification of identity	• Facial recognition • At enrollment: passport, photograph, fingerprints
For	• U.S. citizens and lawful permanent residents, as well as citizens of certain countries

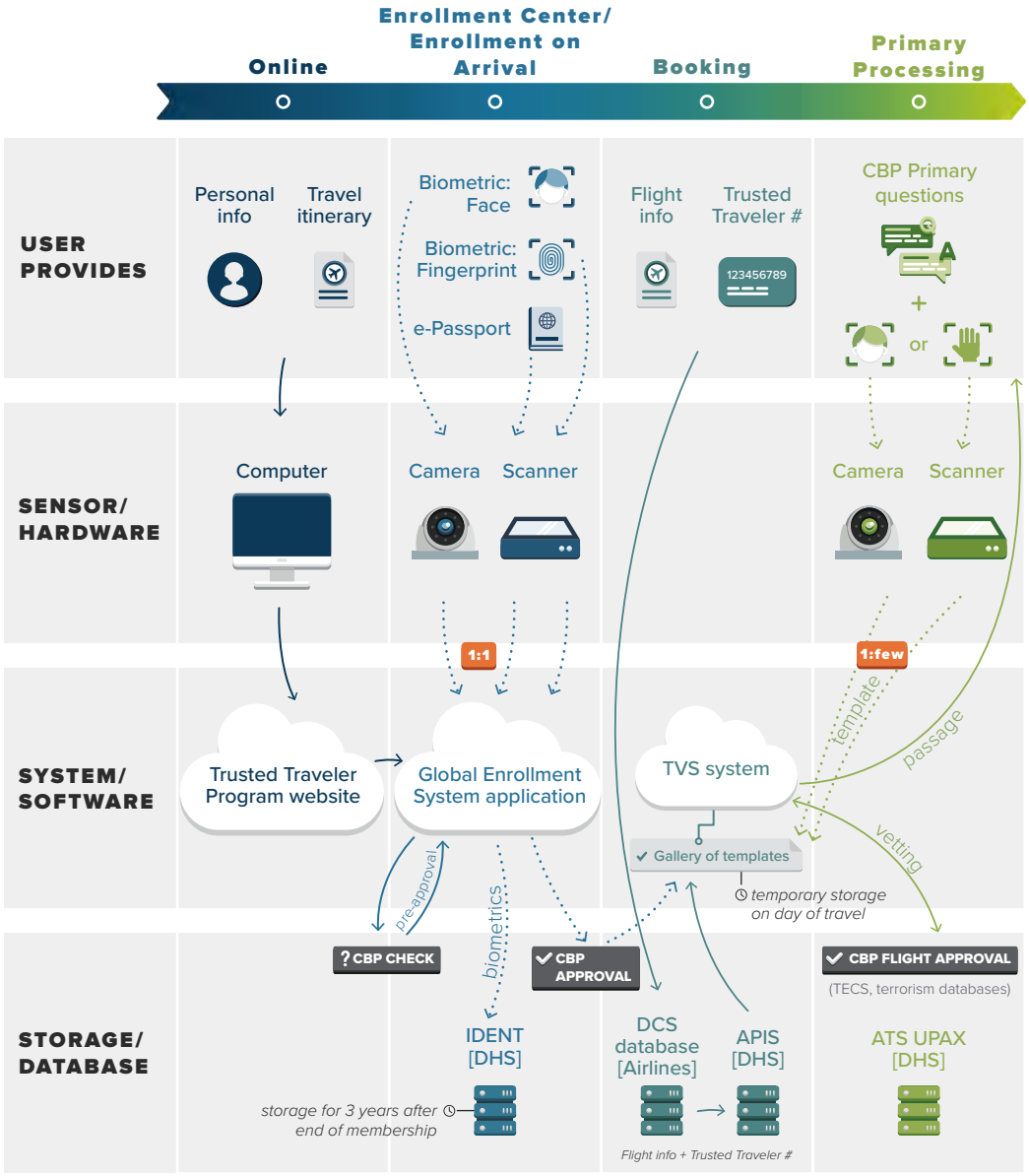
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Program Key Takeaways

The Global Entry program provides airports the ability to safely and securely facilitate international air travel for U.S. citizens and a range of nationals from other countries (e.g., South Korea, Canada, the United Kingdom). The program provides a method for identifying low-risk passengers by vetting pre-approved travelers in advance of their trips, thereby streamlining the clearance and screening processes. An added benefit is that members are granted clearance to use TSA PreCheck.

Case Study #3

CBP Trusted Traveler Programs at U.S. Airports



Note: TVS = Traveler Verification System; DCS = departure control system; APIS = Advanced Passenger Information System; UPAX = unified passenger module; IDENT = DHS Automated Biometric Identification System; TECS = Treasury Enforcement Communications System; ATS = Automated Targeting System.

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Case Study 4: Seattle–Tacoma International Airport and Designated Aviation Channeling

Summary

Pursuant to the Aviation Transportation Security Act, TSA requires all employees of airport authorities and airline carriers, as well as other airport stakeholder employees who require unescorted access to secured areas of an airport, to submit an application and be approved for a SIDA badge. The intent of doing so is to screen an applicant's information against federal criminal and immigration databases to determine whether the applicant may be a threat to transportation or national security. See Table 2-5 for key points of the case study.

Program Concept

For airports using designated aviation channeling (DAC) vendor services, the process of obtaining a SIDA badge begins when the airport employee submits the application, supported by authorized signatories (or endorsers), to the DAC vendor. Specifically, the application consists of obtained employee biometric and biographic information and, in some cases, additional information (e.g., I-9 form). After the application is completed, a DAC vendor ensures that the application is complete and properly formatted and verifies the employment and education information provided.

The application is accompanied by fingerprints and a photo, which are electronically submitted to TSA for a security threat assessment (STA) and to the Federal Bureau of Investigation (FBI) for a criminal history records check (CHRC). On approval of the application by TSA, the results are provided to the airport operator, and a badge is issued. The badge authorizes access to secured areas of an airport and, depending on the biometric technology in use at the airport and by the employer, could be scanned for timekeeping purposes.

Program Key Takeaways

The DAC model is a prime example of a public–private partnership in the handling of biometric and biographic data. The use of DAC also enhances current protocols and represents a set of comprehensive procedures for improved handling of personally identifiable information.

Table 2-5. Key points for SeaTac DAC case study.

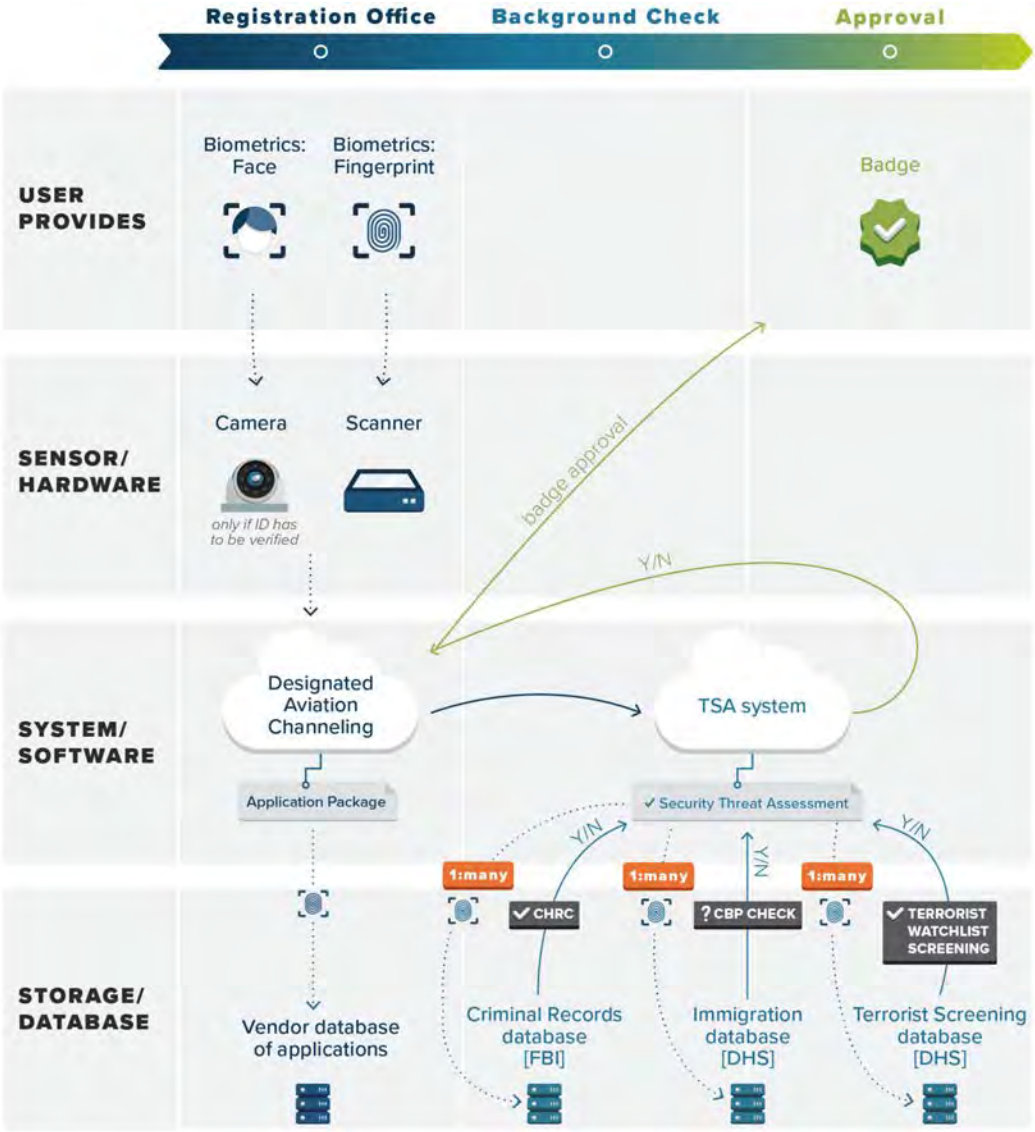
Criteria	Overview
What	• DAC for aviation employee background checks
Where	• Multiple airports in the United States (e.g., Seattle–Tacoma International Airport)
Passenger process	• Application-based process
Who	• Airports across the United States
Why	• Mandatory requirement for all airport and airline employees seeking unescorted access to secured areas, except federal state or local government employees • Ensure safety and security to general public and nation • Requirement by TSA
How	• Fingerprints and in some cases a photo (biometrics) • Criminal history background check • Biographical information
Enrollment	• DAC screening application
Verification of identity	• Vendor representative ensures that all information is correct and accurate during application.
For	• Airport employees, airline carrier employees, and select aviation stakeholder employees

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Case Study #4

Seattle-Tacoma International Airport and Designated Aviation Channeling



Case Study 5: Curb-to-Gate Program by CBP and Delta Air Lines at Hartsfield–Jackson Atlanta International Airport

Summary

CBP's Curb-to-Gate program is an application of facial recognition with the intent of reducing the number of times passengers must present their form of identification and boarding pass, as well as to increase the reliability and efficiency of both airport CBP and security officers. See Table 2-6 for an overview of the case study.

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Table 2-6. Key points for CBP's TVS with Delta at Atlanta case study.

Criteria	Overview
What	• Seamless flow through facial biometric matching
Where	• Hartsfield–Jackson Atlanta International Airport (first pilot)
Passenger process	• Opt in at check-in/bag-drop • Have photo taken at other touchpoints
Who	• Partnership among CBP, Delta Air Lines, Hartsfield–Jackson Atlanta International Airport, and TSA • NEC (technology and software) • DHS, Department of State
Why	• Speed passenger processing
How	• Cameras, including those in tablets • Matching live photo taken at passenger touchpoint with preloaded library of biometric templates
Enrollment	• No formal enrollment; collected when applying for a passport/VISA or at entry/exit from the United States
Verification of identity	• Facial recognition and 1:few matching
For	• International flights with Delta and its partners • Expansion planned to U.S. domestic flights

Program Concept

The Curb-to-Gate program led by CBP and Delta Air Lines allows passengers to verify their identity in a reliable and more efficient manner through leveraging CBP's Traveler Verification System (TVS):

- At check-in/bag-drop, a real-time photo is sent securely and encrypted to the TVS.
- The incoming photo is matched to a biometric template in a pre-compiled library based on the airline's passenger manifest for a specific flight.
- In case of a positive match, the passenger's ID is verified, and the use of facial recognition for the rest of the journey is enabled.
- At bag drop, security, and boarding, the passengers' identity and right to entry/passage are verified using facial recognition.
- For border crossings, an entry/exit record is made in the Arrival and Departure Information System (ADIS).

Program Key Takeaways

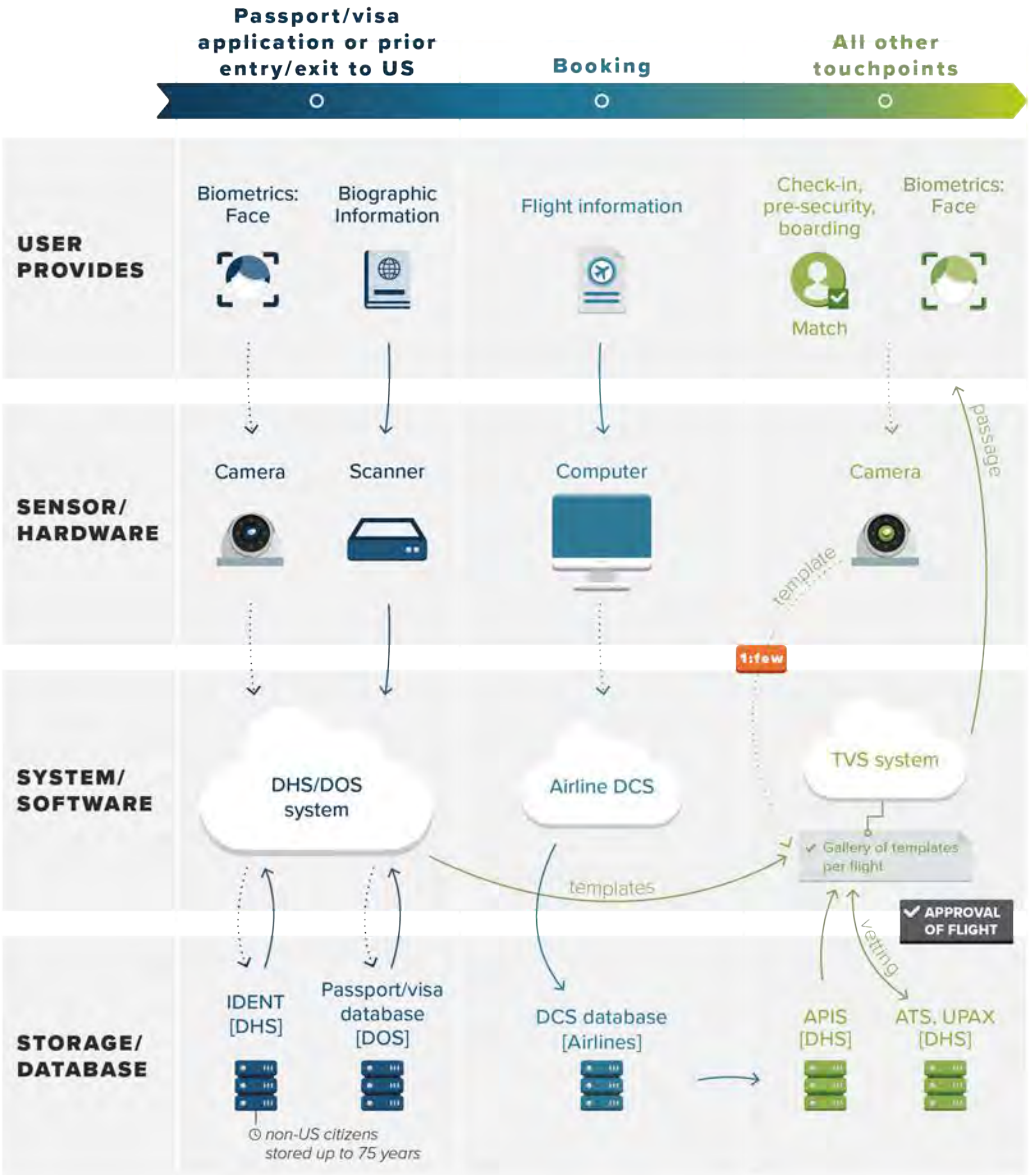
The benefits of using curb-to-gate are specific and measurable time savings for passengers and airport personnel at border processes, check-in, TSA, and other processes; lower operational expenditures for airports; optimization of the use of space throughout the airport; and an opportunity for airports to revolutionize the passenger travel experience. Additionally, curb-to-gate significantly reduces the number of physical interactions among airport stakeholders, thereby lowering the probability of transmitting diseases and hazardous pathogens (e.g., COVID-19). Furthermore, this represents a significant milestone in terms of the power of end-to-end facial recognition and demonstrates the capabilities of biometric software. This case study demonstrates the capability of leveraging biometrics without the need for re-enrollment.

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Case Study #5

Curb-to-Gate Program by CBP and Delta Air Lines at Atlanta International Airport



Note: DOS = Department of State; APIS = Advanced Passenger Information System; IDENT = DHS Automated Biometric Identification System.

Case Study 6: Known Traveller Digital Identity at Aéroport International Montreal–Pierre Elliot Trudeau, Montreal, Canada

Summary

To respond to the increasing demand for a more efficient and streamlined passenger process, the World Economic Forum (WEF) has launched a program for the development of a trusted digital identity based on biometric technology. The goal of this initiative is to make it possible

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for all industry partners to use this digital identity securely and easily while protecting the passenger's privacy. See Table 2-7 for key points of the case study.

Program Concept

The program is still in the early development phase and relies on the self-sovereign concept, where passengers are the owners of their personal data and can elect to share the data (or not) with different parties (e.g., airport, airlines, border authority) at different passenger touchpoints throughout their journey. The digital identity should be created by an identity-issuing authority (third party or government). This authority would create a biometric template of the passenger as well as a unique security feature. With the use of distributed-ledger technology, the immutability of the digital identity can be secured.

The Known Traveller Digital Identity (KTDI) concept aims to use the distributed ledger to register every successful identity claim at various authentication touchpoints in order to build up trust in the passenger. In 2018, the governments of Canada and the Netherlands established a pilot group to advance the KTDI efforts.

Table 2-7. Key points for KTDI and DTC case study.

Criteria	Overview
What	• KTDI concept, developed by WEF and partners
Where	• Amsterdam Airport Schiphol (AMS) • Montreal–Pierre Elliott Trudeau International Airport (YUL) • Toronto Pearson International Airport (YYZ)
Passenger process	• Create digital identity at issuing authority • Passenger manages own identity and information sharing to various airport stakeholders • Digital identity can be used for all processing steps at the airport (like seamless flow)
Who	• WEF • Airports: AMS, YUL, YYZ • Airlines: KLM, Air Canada • Governments of Canada and the Netherlands • Private partners: Vision-Box, Accenture, Idemia
Why	• Create a trusted interoperable ID verification system that can be used around the globe • Avoid the creation of multiple proprietary solutions that make the passenger process more confusing • Create a solid solution for data privacy protection • Create a framework that enables self-sovereign sharing of data required for the destination of the passenger [health information, Advanced Passenger Information System (APIS)]
How	• Facial-recognition technology, biometric-enabled processing points (e.g., e-gates), blockchain technology
Enrollment	• Voluntary
Verification of identity	• On enrollment by matching the face with the image that is stored on the e-passport chip • At authentication touchpoint by face recognition, when required with additional credential (smartphone or e-passport)
For	• Dutch and Canadian citizens who enrolled in the program

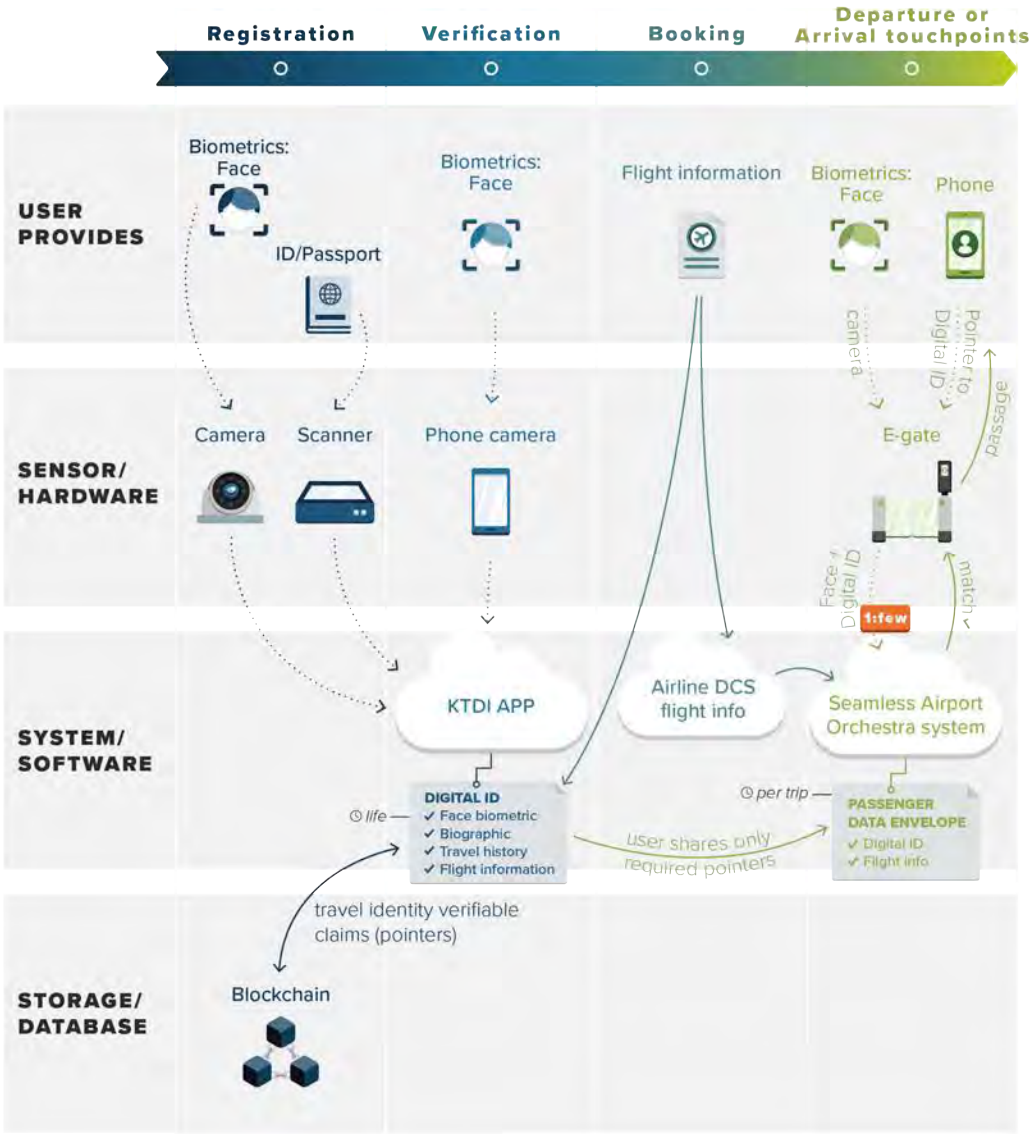
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Program Key Takeaways

In the KTDI project, the challenge for the large group of stakeholders from two different countries, including their two respective governments, was the agreement of sharing, transmission, and storage of information and biometric data. To comply with all Canadian and EU privacy laws, mitigate cybersecurity risks, and satisfy the private and public interests among the group, a lengthy process to fine tune the legal framework was required.

Case Study #6

Known Traveller Digital Identity at Aéroport International Montreal-Pierre Elliot Trudeau, Montreal, Canada



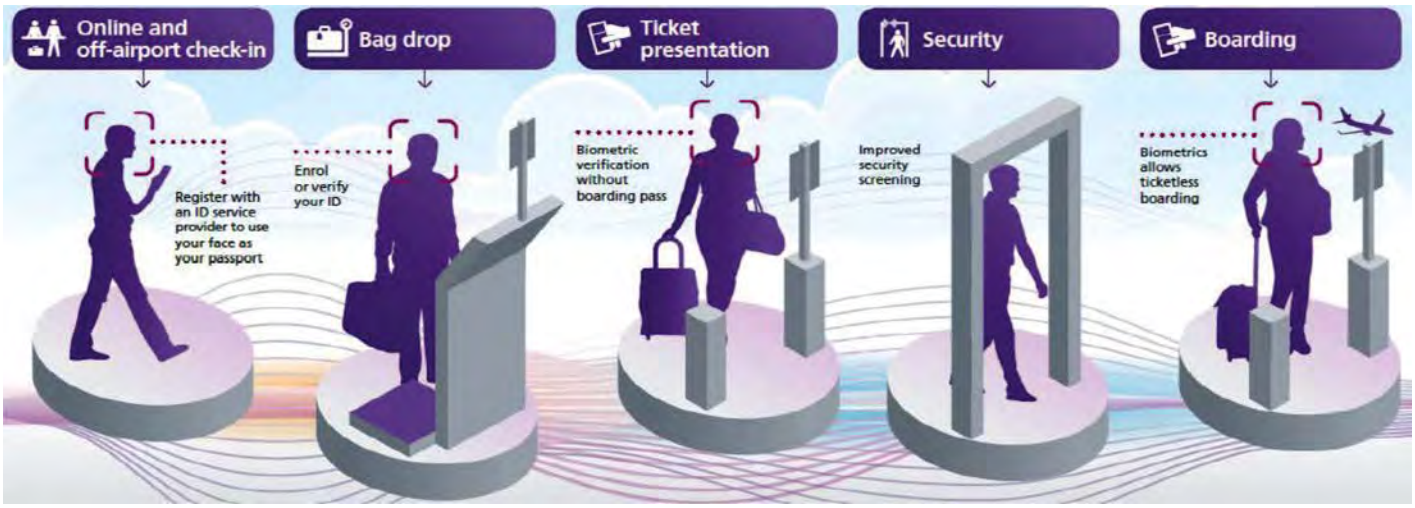
Note: DCS = departure control system.

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Case Study 7: The Seamless Passenger Journey at London Heathrow

Summary

In the Seamless Passenger Journey, Heathrow Airport launched its first end-to-end biometrics program, which is built on the premise of using facial-recognition technology at the check-point of the departing passengers' journey. Part of the journey will involve biometric self-boarding gates, which will be installed across multiple terminals throughout the airport. See Figure 2-4 for an overview of the Seamless Passenger Journey, and Table 2-8 for key points of the case study.



Source: Wilcox 2019a.

Figure 2-4. Overview of the Seamless Passenger Journey.

Table 2-8. Key points for the Seamless Passenger Journey case study.

Criteria	Overview
What	• Seamless Passenger Journey using end-to-end biometrics
Where	• Heathrow Airport, London
Passenger process	• Processes vary by type of flight • Passengers present facial recognition at multiple airport touchpoints
Who	• Heathrow Airport, Atkins, Dormakaba, ICM, CEIA, Rockwell Collins, Arora
Why	• Streamline the passenger travel experience • Accommodate and prepare for anticipated increases in passenger demand levels
How	• Facial recognition at multiple check-in points throughout the airport
Enrollment	• In person at self-service airport kiosks • Mobile enrollment to come later
Verification of identity	• Matching an image of the day taken of the passenger to that of their e-passport
For	• Domestic and international departing flights

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Program Concept

Biographic and biometric data are captured at the first touchpoint (self-service check-in kiosk, self-service bag drop, or ticket presentation gate) by reading the passenger's e-passport chip, scanning their boarding pass, and taking their photo, which is referred to as the "image of the day." All information is then collected and stored in what is referred to as the "Passenger Data Envelope."

The biometric data stored in the passenger's e-passport is compared to the image of the day and, if there is a positive match, the passenger's ID is verified and, from that point onward, the image of the day is used to recognize the passenger throughout the remaining journey. This approach eliminates the need for any additional identification measures.

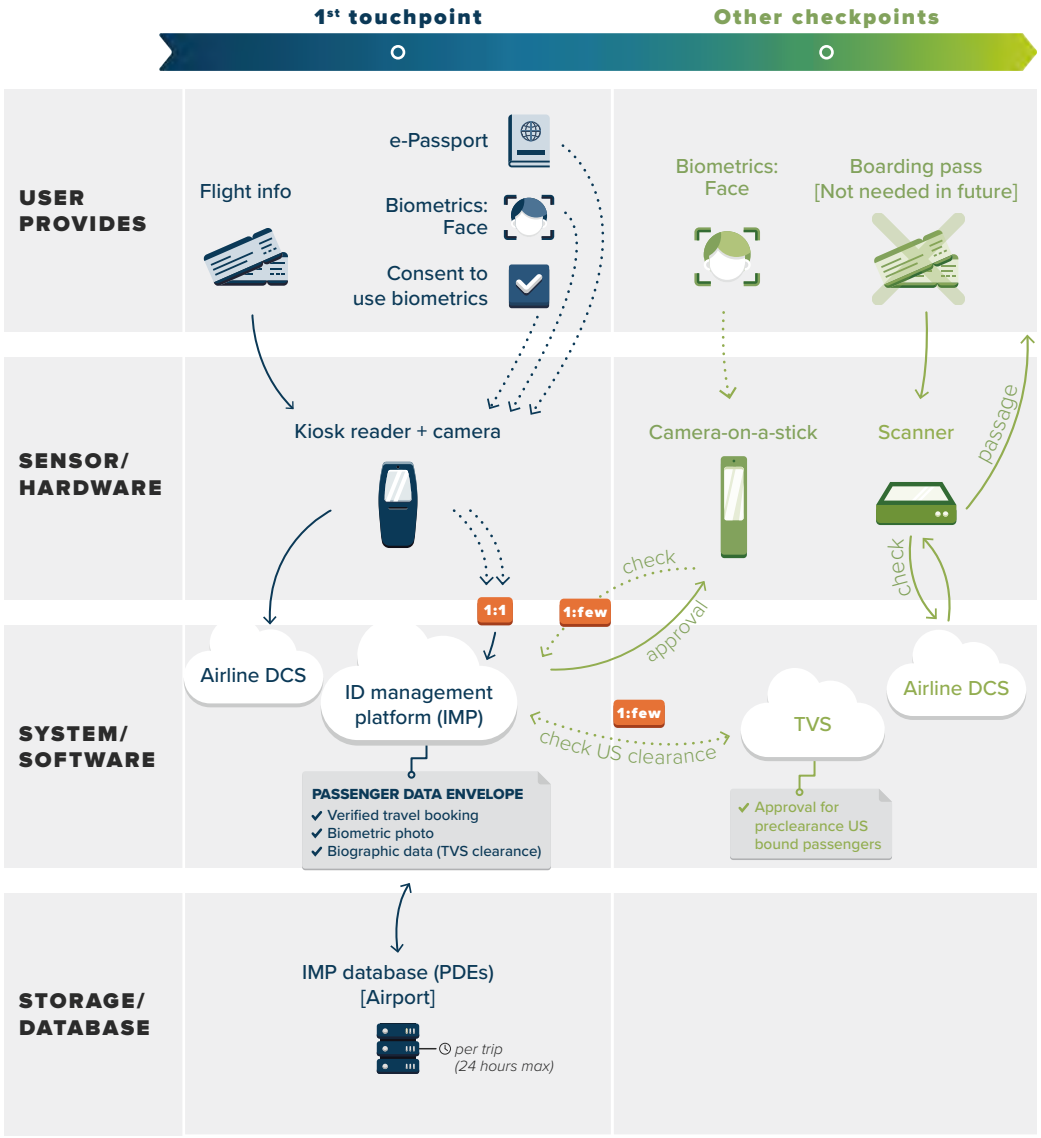
Program Key Takeaways

The Seamless Passenger Journey at Heathrow Airport provides a scenario where the use of biometric technology is scalable to large international airports, both in the short and long term. The quality and deployment across multiple terminals are a significant undertaking that can provide a seamless process. For passengers traveling to the United States, the Seamless Passenger Journey technology also provides ID cross-validation with CBP's TVS.

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Case Study #7

The Seamless Passenger Journey at London Heathrow



Note: DCS = departure control system; PDE = passenger data envelope.

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Case Study 8: Risk Management During COVID-19 Using Biometrics, Carrasco Airport, Montevideo, Uruguay

Summary

In 2016, Carrasco International Airport implemented a fully biometric boarding procedure as well as a biometrics border and customs checkpoint for arriving passengers. The program, referred to as Easy Airport, is the first self-service boarding system within the region based completely on facial recognition. The use of such technology has enabled border agents to initiate risk profiling during the arrivals process, specifically related to issues stemming from the COVID-19 pandemic. Passengers arriving from low-risk countries are processed quickly, while passengers from higher-risk countries are required to undergo additional screening. See Figure 2-5 for a photograph of an Easy Airport e-gate and Table 2-9 for key points of the case study.



Source: Meleiro 2016.

Figure 2-5. Easy Airport e-gate at Carrasco.

Table 2-9. Key points for COVID-19 risk management case study.

Criteria	Overview
What	• Biometric border crossing and boarding by facial recognition; risk profiling arriving passengers
Where	• Carrasco International Airport in Montevideo, Uruguay
Passenger process	• Border e-gates to verify passport authenticity, passenger identity (and link to flight information) • Automated boarding e-gates via facial recognition
Who	• Corporacion America (airport authority), border and customs authorities, and vendor Vision-Box
Why	• Enhanced control and security • Facilitate passengers along their journeys
How	• Biometric e-gates
Enrollment	• No enrollment necessary; all is done at the e-gate
Verification of identity	• Facial-recognition matching to e-passport
For	• Passengers from Uruguay, Argentina, Brazil, United States, and Europe with an e-passport/biometric ID card

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Program Concept

Departing passengers go through an e-gate for identity verification by facial recognition and are asked to select their flight on a touchscreen display to link the passenger information to the correct flight. Boarding requires a positive facial-recognition match.

Arriving passengers enter a similar e-gate for passport authentication, background checks, and biometric identity verification. Significant estimated reductions in per-passenger processing times were from 40 to 50 seconds down to 15 seconds.

Easy Airport was specifically designed for international travelers who are Uruguayan, Argentinean, Brazilian, American, or European citizens; are 18 years of age or older; and have an e-passport.

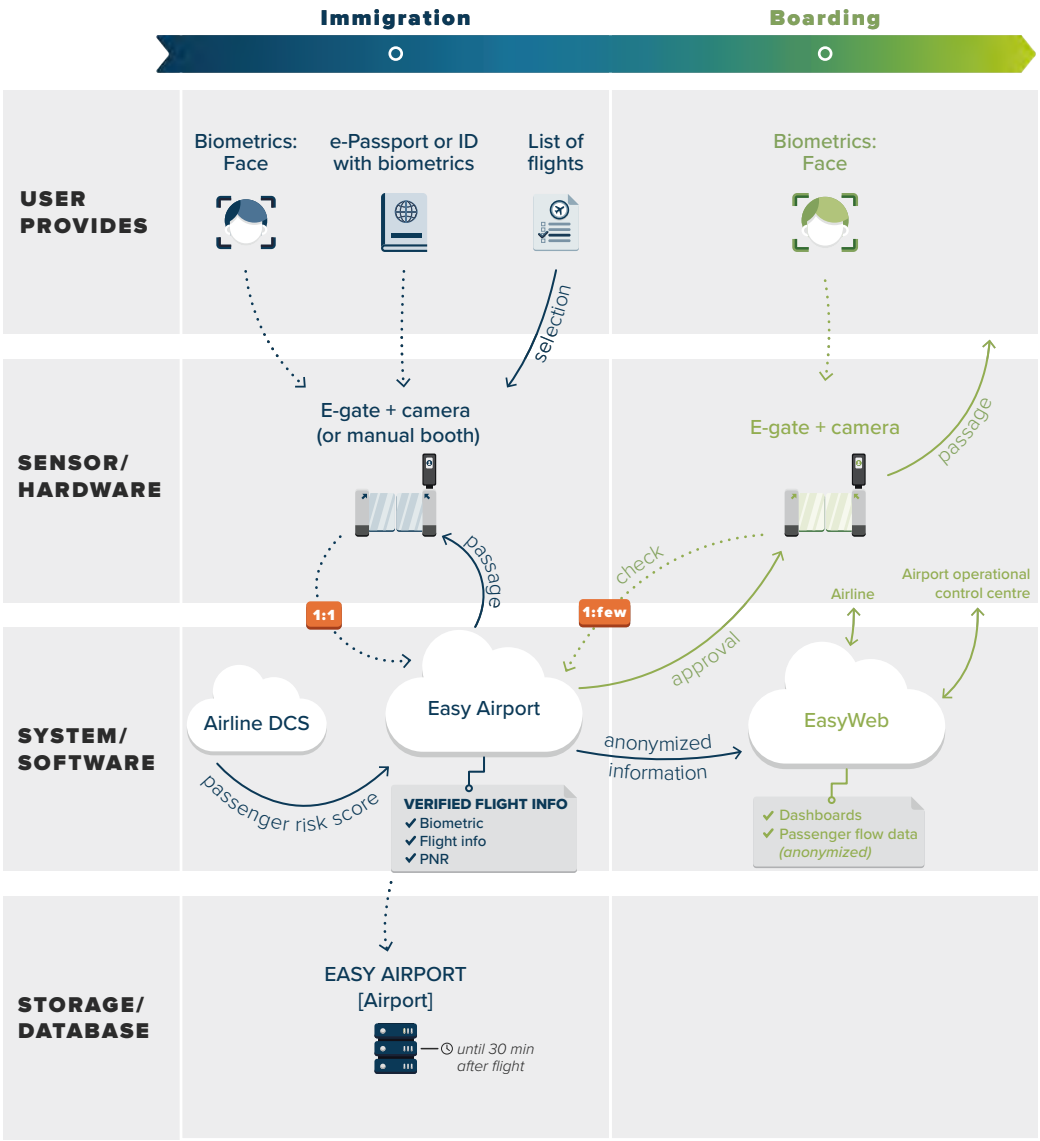
Program Key Takeaways

Easy Airport provides the ability to meet different airline business cases and strategic objectives, including those related to resource and time savings. It also provides the opportunity to install state-of-the-art technology and infrastructure to meet the airport's future demand. The added data portals for airlines and other stakeholders provide additional value and deliver anonymized data streams. Also, Easy Airport was used during the COVID-19 pandemic in risk profiling passengers into specific categories automatically, and in notifying border agents of passengers entering the country from specific origins. Those passengers underwent a different (health) screening process.

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Case Study #8

Risk Management During COVID-19 Using Biometrics, Carrasco Airport, Montevideo



Note: PNR = passenger name record; APIS = Advanced Passenger Information System; DCS = departure control system.

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Case Study 9: Digi Yatra and the Seamless Passenger Journey at Kempegowda International Airport, Bengaluru, India

Summary

Aviation in India has been growing at a considerable pace, and India is expected to become the world's largest domestic civil aviation market over the next 10 to 15 years. As a result, more emphasis has been placed on investing in innovation and digitalization to accommodate expected growth. To this effect, the Ministry of Civil Aviation has begun to implement an initiative known as Digi Yatra, which is designed with the intention of providing a seamless, hassle-free, and paperless experience to all air travelers in India. See Figure 2-6 for an overview of Digi Yatra and Table 2-10 for key points of the case study.

Program Concept

Under Digi Yatra, travelers in India are no longer required to show their boarding passes or proof of identity at touchpoints throughout airports within the country. Rather, the Digi Yatra biometric boarding system boarding pass will be integrated with a passenger identification document, ensuring faster and simpler processing at touchpoints such as the terminal entry gate, check-in/bag-drop, security checkpoint, and boarding gate.

Program Key Takeaways

The primary benefits of the Digi Yatra program are that it removes redundancies at various touchpoints throughout airports by using the passenger's face as a boarding pass, enhances airport resource utilization rates, and improves security and airport system performance. Likewise, the Digi Yatra program results in lower operational costs for airports across India and allows them to achieve better throughput through existing infrastructures using a digital framework. Moreover, the program has a relatively straightforward enrollment/registration process, which is easy to opt out of in case passengers prefer to manually check in.



Source: Bhavan 2018a.
Note: PAX = passengers; PESC = pre-embarkation security check.

Figure 2-6. Overview of Digi Yatra.

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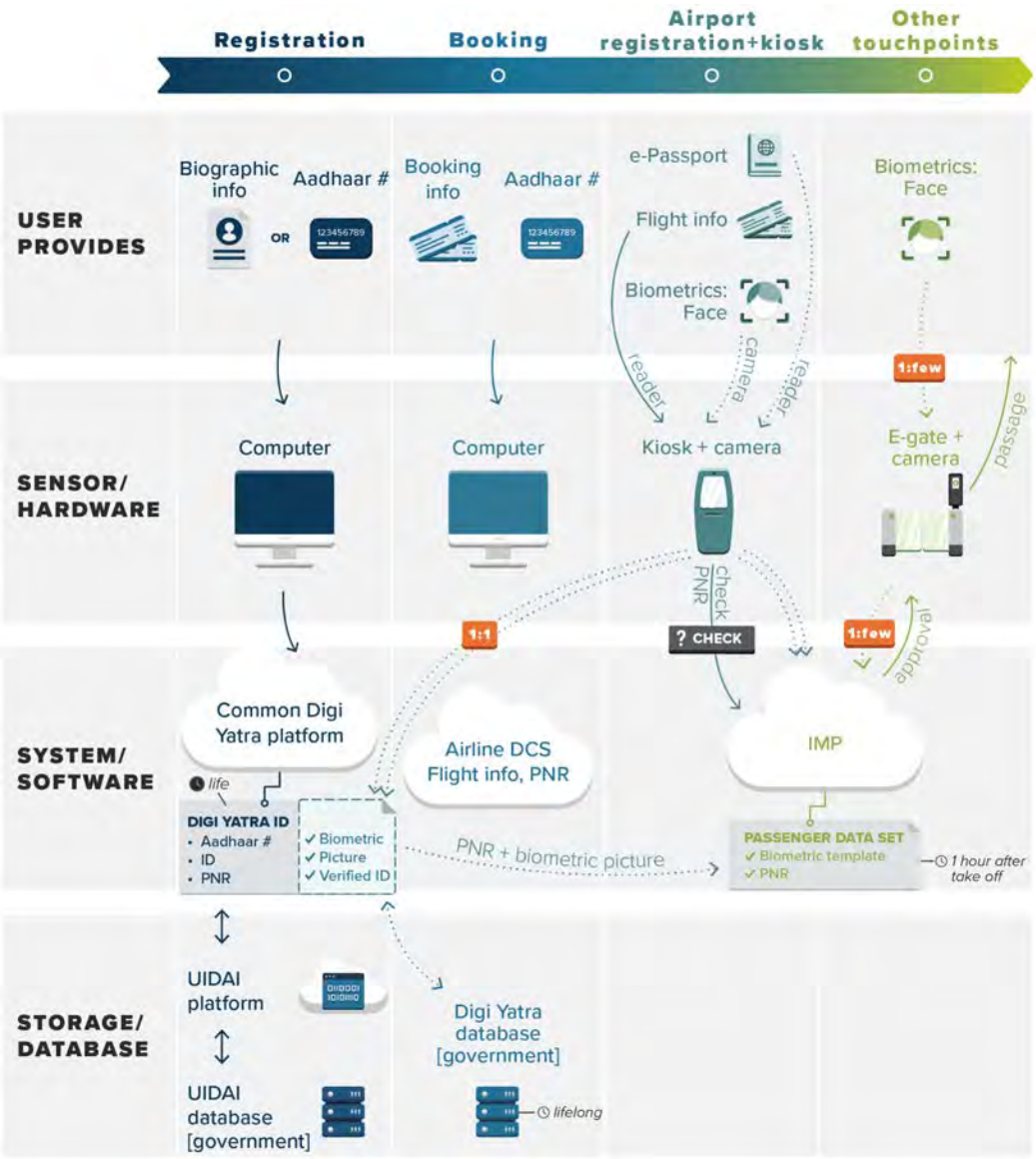
Table 2-10. Key points for Digi Yatra case study.

Criteria	Overview
What	Digi Yatra facial-recognition boarding system
Where	Airports across India Kempegowda International Airport (BLR) (first pilot)
Passenger process	Register at unmanned registration kiosks at airports across India
Who	Airports throughout India Indian Civil Aviation Industry Airports Authority of India
Why	Remove redundancies, increase resource utilization, and improve security
How	Registration kiosks at airports
Enrollment	Voluntary Digi Yatra ID enrollment
Verification of identity	Facial-recognition matching with government of India–issued IDs (e.g., driver's license, passport, voter card, student ID)
For	Indian citizens and foreigners who travel in and out of India

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Case Study #9

Digi Yatra and the Seamless Passenger Journey at Kempegowda International Airport, Bengaluru India (BLR)



Note: PNR = passenger name record; UIDAI = Unique Identification Authority of India; IMP = ID management platform.

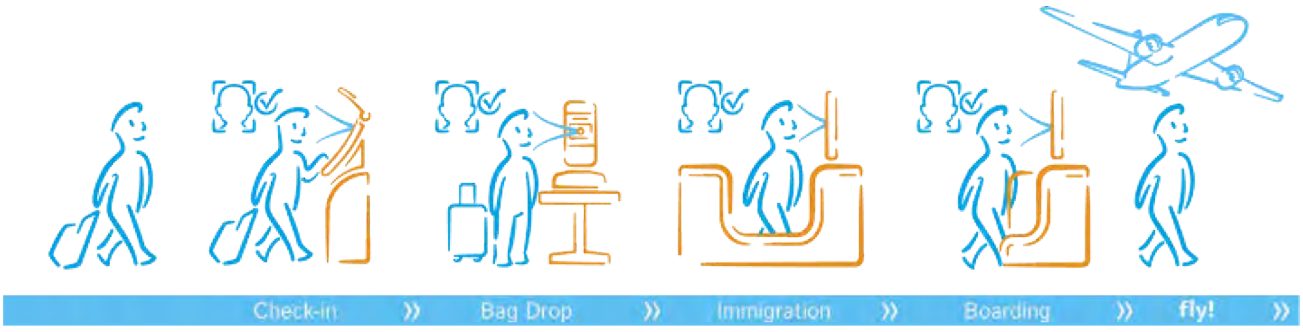
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Case Study 10: Happy Flow at Aruba's Queen Beatrix International Airport

Summary

Happy Flow is aimed at improving the departure process at Aruba International Airport by smoothing the flow of passengers and creating a unique and seamless passenger experience. The program adopted protocols and standards that would benefit authorities, the airport, and airlines.

Using facial recognition at multiple passenger touchpoints, passenger identity and right to travel are verified more securely. Plans are underway to include off-airport elements in the passenger journey (hotel check-in and car rental). During the initial launch of Happy Flow, there were also ideas to preclear into the European Schengen Area. See Figure 2-7 for an overview of the Happy Flow concept and Table 2-11 for key points of the case study.



Source: Aruba Happy Flow 2020.

Figure 2-7. Overview of Happy Flow.

Table 2-11. Key points for Happy Flow case study.

Criteria	Overview
What	• Happy Flow biometric process for preclearance at border control crossing
Where	• Aeropuerto Internacional Reina Beatrix, Aruba International Airport
Passenger process	• Enrollment • Baggage drop-off with biometric assistance • Biometric self-service border control checkpoint • Boarding e-gate with separate lane
Who	• Aruba Airport Authority, KLM Royal Dutch Airlines, Amsterdam Airport Schiphol, and vendor Vision-Box
Why	• Enhance the passenger travel experience • Streamlined, touchless airport journey for security and safety • Increase efficiency of operational processes of the stakeholders
How	• Facial biometric matching
Enrollment	• At a kiosk, passengers scan passport and boarding card, verify identity with data on passport, and create a facial ID
Verification of identity	• ID verification at kiosk • ID authorization at border control
For	• KLM departing passengers to Amsterdam Airport Schiphol with e-passport

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Program Concept

In the Happy Flow process, travel documents are only required at the enrollment station. There, the passenger's identity is securely checked under government-set standards, the passenger's biometric is checked against the one on the travel document, and a (temporary) virtual identity is created. After enrollment, the passenger goes through self-service passenger touchpoints (bag drop, security, border control, and boarding), where the passenger's face is matched to a secured database, and only authorized passengers are allowed to pass.

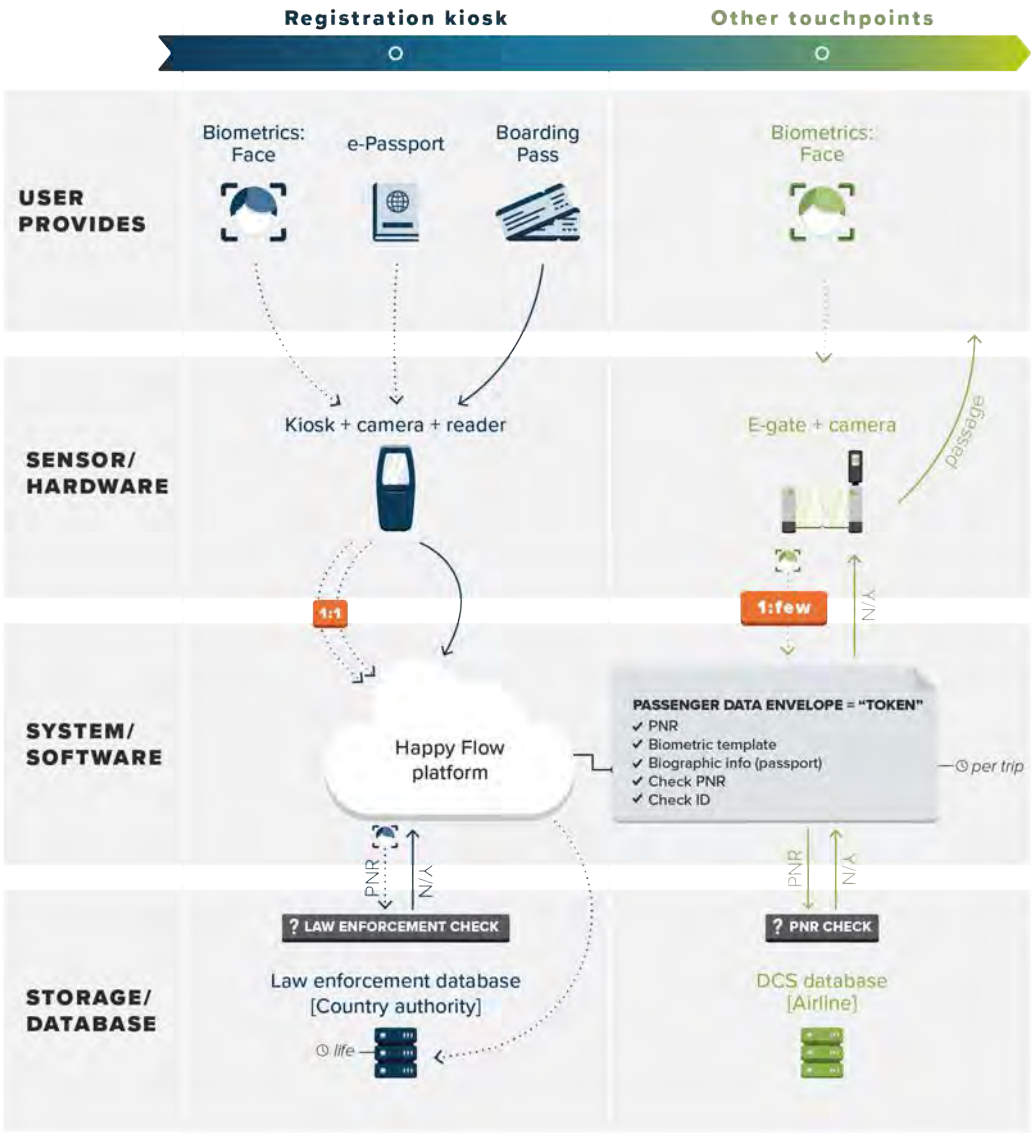
Program Key Takeaways

As with other facial-recognition initiatives and programs, the intent of Happy Flow is to promote a more streamlined departure process while improving overall security. It has been labeled a visionary model that anticipates transatlantic preclearance. Enrollment is the most time-consuming step, while processing is efficient after the biometric token is created. The system may be further optimized by introducing mobile enrollment using the ICAO DTC as well as measures to counter identity fraud from off-site/mobile enrollment. Nonetheless, a broader vision for enrollments at any location, which would be possible within the program, may not be supportable by all stakeholders.

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Case Study #10

Happy Flow at Aruba International Airport



Note: PNR = passenger name record; DCS = departure control system.

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Key Trends in Airport Biometrics

Based on the 10 case studies of recent deployments, there are five trends that are important to consider for the future of biometrics at U.S. airports (see also Figure 2-7):

- Trend 1: The deployment of integrated and multi-stakeholder biometric solutions is increasing because of the potential benefits of biometrics.
- Trend 2: Digital transparency and privacy concerns are adding to the complexity of implementation, particularly as the legal landscape changes.
- Trend 3: A focus on identity verification solutions is evident, in part to distinguish from mass surveillance programs that also leverage biometrics.
- Trend 4: Global biometrics and standards are emerging from a variety of governmental and nongovernmental entities that are addressing privacy, security, ethical, and technological concerns.
- Trend 5: Smartphones are expected to enable greater use of biometrics through the on-device storage capability for biometrics and the promise for the transmission of digital travel credentials.

Key Takeaway

These trends portend the development of solutions that may be more integrated, transparent, and privacy conscious. Solutions may offer more secure identification, within globally accepted practices, with the enabling technologies leveraging smart devices linked to our biometrics.



Figure 2-8. Overview of five key trends on airport biometrics.

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Trend 1: Deployment of Integrated and Multi-Stakeholder Biometrics Is Increasing

Where We Were

Airport biometrics were previously constructed around single-purpose ideas. Access control, for example, created an environment where a biometric clearinghouse was compiled to create a relationship with credentialing systems.

Where We Are Going Now

We are now moving into a world where a single enrollment could have multiple stakeholders using the system. The premise is that overall costs could be reduced by having five separate enrollment centers for five different stakeholders combined together. Known as interoperability, there are multiple ways to enable a single biometric to be used across airlines, government(s), and airports.

Typically, stakeholders have depended on a face-to-face customer-service model. Physical interaction with customer service representatives may increasingly be the exception and possibly available as a premium service, especially during the COVID-19 recovery period. One cannot efficiently accommodate the multiple groups, from ride share, car rental, airlines, government agencies, and other parties, each having their own enrollment stations and different biometric technologies.

As a result, a multi-stakeholder model is the preferred direction to reduce the number of in-person contact points and provide the greatest overall system benefit.

Key Examples

- CLEAR: Can be used at the TSA checkpoint, airline lounges, flight boarding, and for access to some professional sporting events.
- DHS Office of Biometric Identification Management: Since 2003, the image collected with a visa application has been used to confirm identity at port of entry (POE), as well as for applications involving the TVS.

Trend 2: Digital Transparency and Privacy Concerns Increase Complexity

Where We Were

There are privacy laws and policies used for public- and private-sector deployment of biometric solutions. Many of these have strict requirements governing the use, retention, and disposal of personal information. On occasion, lapses of data ownership have occurred.

Where We Are Going Now

There is heightened attention being paid to the threat of cybersecurity attacks, theft of personal information, and vulnerabilities associated with privacy. Confusion and, in some cases, a lack of awareness of risks exist within the traveling public.

It is not good enough to simply tell the public "we will keep your information private" or provide an opportunity for an opt out. Privacy advocates are calling for more sophisticated solutions to ensure that easy-to-use tools are available, especially when accountability is spread among airline, governmental, and third parties.

Key Examples

- Data breaches: Lessons learned from recent data breaches can inform the need for all stakeholders to manage policies and governance to limit vulnerabilities from unauthorized data access.

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- Self-sovereign biometrics: A different model for the use of biometrics that reframes the individuals' control over their identity information and that can be used across multiple countries, governments, airports, and airlines.
- Digital Trust in Places and Routines: A project that creates standard communications methods, signage symbols, and accountability notices to make it easier to communicate to passengers.

Trend 3: Identification Verification Solutions (Not Mass Surveillance)

Where We Were

The word "biometrics" equally represents all forms of algorithms and techniques used to manage identity. An analogy can be made between the word "password" and biometrics to the general public—there are strong secure passwords, and there are poor, ineffective passwords.

There are a number of emerging mass surveillance solutions that use public and private databases. Clearview AI, for example, leverages 3 billion images taken from social networking platforms to supply clients with identity information. Although many governments have invested heavily in national biometric registration and identity systems, only a few of these systems have some features constructed with mass surveillance in mind.

Where We Are Going Now

Current and proposed uses of biometrics at U.S. airports are significantly narrower in purpose. They are related to an expressed purpose (facilitate process, manage identity) and are formulated to verify identity, not conduct mass surveillance.

However, due to the high-profile nature of Clearview AI and the rapid advancement of alternate biometric models, there can easily be confusion and a lack of consumer confidence about proposed solutions. A key trend emerging involves airlines, airports, and governments adopting identity verification as the description of the program being deployed. In so doing, the rollout of biometric solutions can be distinguished from mass surveillance applications that tend to inflame the user base.

In other words, instead of referring to solutions as "facial recognition" or "biometrics," which imprecisely implies a set of unfettered processes, airport operators may more precisely refer to processes as "identity verification solutions." In addition to appropriate terminology, it is imperative that airport operators and others wishing to deploy biometrics-enabled initiatives do so within a framework consistent with best practices with respect to data security and privacy considerations.

Key Examples

- CBP's TVS may be powered by biometrics but is accurately described as an identity verification solution that differs from mass surveillance.
- "My face is my boarding pass": Plain-speaking solutions are being used within the Digi Yatra program in Bangalore, India; these phrases clearly communicate the focused nature of the solution.

Trend 4: Global Biometrics and Standards Taking Hold

Where We Were

It was only in the 1980s when passports became standardized by ICAO. Since 1998, the ICAO standard included a chip within the passport for e-passports. Yet the work is not complete for the original vision of e-passport rollouts. Visas were originally meant to be stored in a larger-capacity chip. Other data, including more advanced encryption mechanisms using key public infrastructure, were also meant to be augmented to reduce the potential for document fraud.

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Where We Are Going Now

DTC represents the first major change in document standards since ICAO's Document 9303 (ICAO 2015) was modified to include e-passports. There are a number of options relevant to biometrics, including the mechanism to store information (blockchain, mobile phone, e-passport) as well as to grow capabilities to better manage digital identities.

The key aspect of the DTC platform is to enable interoperability in a standard fashion. In other words, there is the ability to ensure that there is guidance to solve for incompatibilities [e.g., Video Home System (VHS) versus Betamax video tapes] so that a common standard is established for exchanging digital identity information.

Key Examples

- Passport-free travel: A number of countries (such as Canada) and multilateral regions (such as Mercosur) are evaluating DTC as a mechanism to contain traveler information.
- Health information: While there are a number of advancements contemplated associated with COVID-19 recovery, one in particular is the development of applications that include certain health information rooted on the ICAO DTC platform. The so-called "health passport" containing personal health information and immunization records may be an augmentation that different countries adopt in the future.

Trend 5: Leveraging Smartphones

Where We Were

When biometrics first started being used broadly at airports in the 1990s, it was exclusively for voluntary products such as INSPASS trusted traveler models. There is still strong value in the face-to-face interaction between an officer and an individual when going through certain regulated processes, such as clearing through a POE into the United States.

Where We Are Going Now

Many applications for biometrics exist on mobile smartphones, but these mostly relate to device unlocking, mobile payments, and some identity information to be transferred across the Internet (such as a passport photo page).

Increasingly, there are mobile applications that leverage still photos or live video interviews in order to integrate voice, video, and data. Such applications can thus use a broader range (or multiple) biometric characteristics for identity verification remotely and seamlessly.

Where there are doubts about the identity of an individual, there could be a triaged process to help with identity verification during a trip through a home-, hotel-, or office-based interaction that could create a provisional biometric record that would then fully vetted on-site at an airport on the next journey.

Key Examples

- The proliferation of health information passes, such as Commons Project CommonPass and IATA TravelPass, that are built around a smartphone app to house personal information such as a COVID-19 test result, vaccination status, or other information such as photographs.
- Mobile passport control, offered by a third party or, beginning in 2021, by CBP directly, that includes the ability to scan the machine-readable zone of a passport picture page through a smartphone camera.

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CHAPTER 3

Legal, Policy, and Privacy Review

Summary

To pursue biometrics as potential solutions to challenges in aviation, industry stakeholders need to understand the legal, policy, and privacy issues associated with the use of biometric data. The two predominant policy and legal issues associated with the use of biometric data (including facial recognition) are protection of privacy rights and the inconsistent treatment of biometrics under a myriad of international, federal, and state laws.

This chapter explains basic constitutional principles associated with the collection of biometric data; the few federal laws applicable to the use of biometric technology and the proliferation of inconsistent state laws; ongoing federal government activities and the role of airport operators, carriers, and tenants, including any litigation risks; privacy principles applicable to the aviation industry; significant court decisions, particularly those eroding traditional precedents because of the use of heightened or pervasive biometric technology; penalties and costs of litigation for violations of law; increasing collaboration of international organizations' initiatives; best practices, applicable standards, and privacy-by-design principles; and international regulations that offer airport operators and stakeholders the safest course of action in their roles as airport operators, transportation industry providers, and retailers.

While the predominant concern under the U.S. Constitution is the right to privacy, another concern is the fact that in the absence of a comprehensive national law, individual states have passed laws, a handful of which limit/prohibit the use of biometric data specifically, some of which address sensitive personal data, and all of which require action and notices upon breach of the data storage and retention systems. Many of these laws impose sanctions for failing to meet requirements pertaining to collecting and protecting biometric data.

Another emerging trend is the development of laws and restrictions (some with a broad reach) by other countries, the EU, and international organizations that may affect airport operators and airlines in the United States and include the assessment of penalties for noncompliance.

There are many sources of best practices for the protection of personal privacy. These include privacy by design, 2019 EU Data Protection Guidelines, and fair information practice principles, which offer guidance on transparency, adoption of policies and practices that incorporate privacy protections, communication tips, and similar practices.

Introduction

To pursue biometrics as potential solutions to challenges in the aviation industry, it is first necessary to understand the legal, policy, and privacy issues associated with the use of biometric data. The two predominant policy and legal issues associated with the use of biometric

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data (including facial recognition) are protection of privacy rights and the inconsistent treatment of biometrics under a myriad of international, federal, and state laws.

Protection of privacy rights is the most prevalent concern surrounding the use of facial recognition. Some contend that the use of facial recognition is a violation of privacy rights, that such use may infringe on constitutional rights, and that systems where biometric data are stored provide inadequate legal and security protections from breaches, abuses, and criminal activities (Wong 2020). As discussed in the subsection on state laws, the majority of recent lawsuits assert violations of state law by private-sector businesses (Kugler 2019).

Additionally, notwithstanding the increased efficiencies and security benefits associated with facial-recognition technology, media reports have highlighted concerns over the effectiveness of the technology with claims that facial recognition has a high rate of misidentification and bias (see, e.g., Singer and Metz 2019; Harwell 2019). Further, federal studies report that differing algorithms result in differing degrees of misidentification and bias (National Institute of Standards and Technology 2019). In a recent wave of criticism associated with allegations of racially based policing and the potential abuse by law enforcement engaged in surveillance activities of crowd demonstrations, Amazon, Microsoft, and IBM have either suspended or terminated production and sale of the technology (Greene 2020).

A recent legal and policy trend in the area of biometrics has been a proliferation of state laws that address a wide range of biometric technology issues such as data breaches, use restrictions, and bans on police use. The legal landscape with respect to state laws is non-uniform and in flux.

Another area of legal divide reflected in both federal and state laws is whether the entity collecting and using the biometric data is a federal or state agency or a private-sector or commercial entity. There are no uniform legal requirements for the protection of biometric data, which parties are responsible for protecting data, and what types of data are entitled to protection and under what circumstances. Thus, the state of the law with respect to the collection, use, protection, retention, and disclosure of biometric data is unsettled and evolving.

Notwithstanding the state of the law, changes and advancements in the technology itself arguably serve to mitigate litigation risks. For example, in the past, facial images were retained in mass storage as photographs, whereas now facial templates are stored as numbers using sophisticated algorithms and, in most cases, are encrypted. With the use of facial templates and encryption, some concerns about data breaches have been addressed. Additionally, the courts have begun to debate/distinguish between capturing a photograph versus application of a biometric template to a facial image, bringing the image within the scope of a state biometrics law.[1]

In recent years, state legislatures, privacy advocates, and free Internet proponents have focused on cybercrime (i.e., hacking and misuse of data) and debated the need for (or opposition to) the application of additional security measures such as encrypting personal data to lessen the risk of access to and ease of illegal use of the data in the mass storage of data or jpeg files (Marks 2020). As a general rule, use of biometric data for the management of employees and for commercial operations is principally governed by state law (Carrero 2018).

There is no comprehensive federal law governing the collection and use of biometric data or industry employees' and consumers' privacy.[2] Under a "patchwork"[3] of federal laws, there are provisions aimed at providing privacy protections for individuals' data in specific industry sectors, such as limitations on federal agencies' collection of personal information (if relevant and necessary to comply with a legal mandate),[4] protections for medical and financial data,[5] and limited enforcement action for unfair and deceptive practices.[6] In the absence of any overarching federal framework, states have been enacting or amending laws to protect their residents' privacy regarding biometric data and, particularly, facial-recognition data.[7]

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Airport operators and stakeholders, such as airlines and vendors, not only must comply with applicable federal and state laws that apply to biometric collection and use for employees and consumers but, in certain circumstances, must confront conflicting laws from multiple states and other countries or international organizations. Airport operators face differing risks and requirements depending on whether they are using biometric data collection for employee timekeeping, access to restricted areas, or other airport operational purposes, or are dealing with passengers to facilitate travelers' experience or retail purchases.

What follows in this chapter is a summary of key legal and policy issues and trends associated with the use of biometrics in the aviation environment. (Additional detailed discussion can be found in Appendices K and L). In particular, the chapter highlights:

- Basic constitutional principles associated with the collection of biometric data;
- The few federal laws applicable to the use of biometric technology and the proliferation of inconsistent state laws;
- Ongoing federal government activities and the role of airport operators, carriers, and tenants, including any litigation risks associated with governmental and commercial activities using biometrics;
- Privacy principles applicable to the aviation industry;
- Significant court decisions, particularly those eroding traditional precedents because of the use of heightened or pervasive biometric technology;
- Penalties and costs of litigation for violations of law;
- Increasing collaboration of international organizations' initiatives;
- Best practices, applicable standards, and privacy-by-design principles, and international regulations that offer airport operators and stakeholders the safest course of action in their roles as airport operators, transportation industry providers, and retailers; and
- Guidance on mitigation of risk of litigation and findings that support a responsible framework for use of biometrics in the airport environment.

Interplay of U.S. Constitution, Federal Laws, and State Laws

The legal concern most frequently raised in connection with the use of biometrics, and in particular facial recognition, is the right to privacy. The body of law defining the right to privacy is an amalgam of concepts evolving from multiple sources: common law; the Fourth, First, and Fifth Amendments to the U.S. Constitution; and applicable federal and state laws.[8] In other words, there are many sources (and types) of privacy rights that may be implicated depending on the particular use of facial recognition, who is using it, and the nature of the violation asserted (Intille 1999). What follows is a discussion of those legal considerations in relation to the use of biometrics (especially facial recognition) in an airport environment.

Key Takeaway

This section provides a basic outline about the patchwork of federal and state laws and touches on areas relevant to aviation stakeholders.

The Right to Privacy Under the U.S. Constitution

The U.S. Constitution protects privacy from *governmental*[9] invasion, and this includes state and local authorities operating in the airport environment. Protection of a person's right to be left alone by private parties is left largely to the laws of the individual states.[10]

While the Constitution does not explicitly mention any right of privacy, the Supreme Court has declared that the right of privacy is a fundamental right guaranteed by the Constitution.[11]

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Privacy advocates have challenged the collection and disclosure of biometric data under the provisions of the Fourth and Fifth Amendments of the Constitution.[12] The Fourth Amendment to the U.S. Constitution states, "The right of the people to be secure in their persons, houses, papers, and effects, against unreasonable searches and seizures, shall not be violated, and no Warrants shall issue, but upon probable cause . . ." and the Fifth Amendment states, "No person . . . shall be compelled in any criminal case to be a witness against himself. . . ."

The question of whether the collection of facial biometrics constitutes a search for purposes of the Fourth Amendment would have been answered, until recently, by the application of historical precedent that no person has a reasonable expectation of privacy in his or her face.[13] However, recent lower-court decisions suggest this may not be true for the pervasive or intrusive use of facial biometrics technology for identification or authentication purposes.[14]

The use of facial-recognition technology by government actors can be analyzed in three distinct aspects:

- The initial collection of biometric data,
- The request for identification, and
- The comparison between the stored data[15] and data presented by an individual on a particular occasion.

The first is likely consistent with the Fourth Amendment. The second and third aspects are settled areas of law: A request for identification does not, by itself, constitute a Fourth Amendment seizure,[16] and "the process of matching one piece of personal information against government records does not implicate the Fourth Amendment."[17]

"The ultimate touchstone of the Fourth Amendment is reasonableness."[18] Generally, reasonableness requires that, whenever a government actor conducts a search or seizure of a person or his or her property, that official must have a warrant based on probable cause and issued by a judge or magistrate.[19]

To determine whether a search occurred under the scope of the Fourth Amendment, a court initially will examine the search under tests formulated by the Supreme Court.[20] One test looks at whether the government physically intrudes (i.e., trespasses),[21] and the other whether the intrusion violates a person's reasonable expectation of privacy.[22,23] If a court determines that the intrusion falls within the scope of Fourth Amendment protection, it examines whether the search complied with the scope of the terms of the warrant, if one was obtained, or whether a recognized exception to the warrant requirement applies (e.g., administrative search, border search, exigent circumstances).[24]

Public Spaces and the Fourth Amendment

Historically, the Supreme Court has determined that certain activities are not searches or protected by a constitutional right to privacy: those related to persons on public streets and beaches,[25] conveyances on public roads,[26] prisoners in jail,[27] and information on the exterior of mails.[28] Further, the Supreme Court has recognized that a temporary detention of a person for police questioning in connection with an airport search was reviewable under a lower standard than the rule announced in Terry v. Ohio[29] (requiring reasonable suspicion that the person is involved in criminal activity for a temporary detention by a law enforcement officer).[30] The Court found the temporary detention, although constituting a seizure under the Fourth Amendment, was constitutional because of the officer's articulable suspicion and the public interest involved in the suppression of serious crime.[31]

In recent years, however, the courts and, in some cases, the Supreme Court, have wrestled with the traditional framework about public spaces when faced with advances in technology

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that pose a potential for unreasonable intrusion into a person's privacy.[32] The Supreme Court noted that technological advances provide "access to a category of information otherwise unknowable,"[33] and "implicate privacy concerns" in a manner as different from traditional intrusions as "a ride on horseback" is different from "a flight to the moon."[34]

The trend from recent Supreme Court cases focused on GPS and cell phone tracking appears to signal that, where the use of heightened technology is more intrusive than traditional forms of observation, courts may find the tracking and identification of persons to be unreasonable under the Fourth Amendment.[35]

Exceptions to the Fourth Amendment Warrant Requirement

Even where the Fourth Amendment text would otherwise suggest its application, the Supreme Court has recognized certain governmental activities as reasonable per se under the Fourth Amendment. Because these activities are deemed reasonable per se, they are regarded as exceptions to the warrant requirement of the Fourth Amendment. For example, routine searches of persons and items at the border by duly authorized officers [e.g., CBP and Immigration and Customs Enforcement (ICE)] are deemed reasonable and may be conducted without suspicion.[36] The border search exception is not limited to searches that occur at the border itself but includes searches that occur at the "functional equivalent" of the border (e.g., at an airport inspecting and clearing arriving international passengers).[37]

Courts have noted that there is a diminished expectation of privacy at airports.[38] In that context, the courts have held that routine airport screening searches are reasonable under the special needs exception and do not require a warrant, individualized suspicion, or consent,[39] provided that they are properly calibrated to their purpose and not in furtherance of traditional law enforcement goals.[40] Under the special needs doctrine, courts assess the constitutionality of the challenged conduct by weighing the government conduct—in light of the special need and against the privacy interest advanced—through an examination of three factors: (1) the nature of the privacy interest involved, (2) the character and degree of the governmental intrusion, and (3) the nature and immediacy of the government's needs and the efficacy of its policy in addressing those needs.[41]

Liability

For alleged violations of the Fourth Amendment, persons can sue state and local government actors under 42 U.S.C. §1983, which provides a private cause of action against state officers for violations of federal civil rights.[42] With respect to alleged unlawful searches or seizures by federal officers, the Supreme Court in Bivens v. Six Unknown Agents held that an individual could sue for monetary damages for a violation of the Fourth Amendment.[43]

Private Party Acting as an Agent of the Government

Traditionally, searches by private parties do not trigger the Fourth Amendment unless the party is acting at the direction of a government actor.[44] In other words, a search conducted by private individuals at the instigation of a government officer or authority constitutes a governmental search for purposes of the Fourth Amendment. Where a private party assists or acts at the direction of a federal agency in performing an activity lawful under an exception to the Fourth Amendment warrant requirement, such as the border search exception (i.e., CBP or ICE) or special needs exception (i.e., TSA), the private party's assistance would fall within the protected scope of the exception.[45] For example, courts have indicated that there is a diminished expectation of privacy at airports,[46] and thus, searches conducted by airport personnel in

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accordance with a federally approved security plan are deemed to fall within the special needs exception to the Fourth Amendment.[47]

Recently, there has been increased attention and scrutiny where private parties are engaged in voluntary data sharing with law enforcement. Most cases decided in this area focus on the degree of governmental influence over the initial collection (Brennan-Marquez 2018).

At least one commentator has suggested that the analysis for determining whether a private party engaged in voluntary data sharing with law enforcement is sufficient to make that party an agent of the government should include consideration of the following four factors:

- First, whether data sharing is repeated or, instead, spontaneous,
- Second, whether the data transfer was aimed to assist law enforcement,
- Third, how powerfully equipped the private actor is to perform data surveillance, and
- Fourth, whether law enforcement practice has evolved to reflect the availability of privately collected data (Brennan-Marquez 2018).

DNA Identification as a Case Study

Although courts have not yet had the opportunity to examine the *governmental* use of facial-recognition technology,[48] recent case law on the collection of DNA and creation of DNA profiles provides a useful framework for understanding how courts will likely address such conduct in the future. Ultimately, the similarities and differences between DNA profiling and the use of facial-recognition technology arguably demonstrate the latter's legality. (See Appendix K for a case study analyzing courts' treatment of DNA.)

Federal and State Privacy Laws Relevant to Airport Operators and Stakeholders

Key Takeaway

As discussed in this section on federal and state laws, aviation stakeholders should note that if you are doing business in a state that does not have a law expressly limiting the collection and use of biometric data, such as facial recognition, but you collect and use biometric data on a resident from one of those states, you may be required to implement changes and follow the requirements of the resident's state or potentially face stiff penalties.

Generally, federal privacy laws (1) govern a federal agency's collection and use of biometric data, or (2) protect consumers' privacy with respect to the collection and use of their biometric data and provide them a cause of action. There are a number of federal privacy laws applicable to the collection and use of biometric data *by federal agencies* as well as specific activities by various industries in distinct business sectors, some of which are relevant to airport operators and stakeholders. Interestingly, a 2015 U.S. Government Accountability Office (GAO) report observes that there are no federal laws restricting the *capture* of facial images except with respect to matters associated with minors.[49]

With respect to legal protections pertaining to *commercial uses* of biometric data, including facial-recognition technology, federal laws can be divided into three broad categories that address privacy and consumer protection for: "(1) the capture of facial images; (2) the collection, use, and sharing of personal data; and (3) unfair or deceptive acts or practices, such as failure to comply with a company's stated privacy policies."[50]

The overarching federal law governing federal agencies' collection, use, and disclosure of personal identifiable information (PII), such as biometric data, is the Privacy Act of 1974.[51] Other federal laws pertinent to or regulating the collection and use of biometric data are the Federal Trade Commission Act (addressing unfair or deceptive acts) and the Public Health

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Service Act (setting forth authorities to control public health emergencies such as COVID-19). (See Appendix K for a detailed discussion of federal laws governing privacy in specific sectors.)

U.S. Congress Legislative Activity

On January 1, 2019, the 116th Congress commenced, and it ended on January 3, 2021. During the 2-year period that Congress is in session, a bill may be introduced at any point, and it remains eligible for consideration until the Congress ends or adjourns *sine die* (i.e., adjourning without specifying a return date).[52] Since January 2019, there have been approximately 70 bills introduced that propose to protect consumers' privacy or impose limits on the private sector's or government agencies' collection and use of biometric data.[53] Critics of the collection and use of biometric data and privacy advocates have voiced the need for national legislation to establish a comprehensive framework for government and commercial use and collection of biometric data (Schubert 2020; Cunningham 2019). By the end of the session, except for a handful, most of the bills were not voted out of committee, and only a handful of generic provisions applicable to Department of Defense (DoD) activities within defense authorization acts (FY21) were enacted.[54] It is worth monitoring activity in the 177th Congress, which commenced in January 2021.

Survey of State Laws

In the absence of a comprehensive federal law regulating the collection and use of biometric data, states have enacted basic laws protecting citizens' and consumers' privacy, most of which do not expressly specify coverage for biometric data. Distinct from specific provisions governing biometric data, there are many state laws protecting *personal* data. In some cases, where there is no express mention of biometrics, biometric data are regarded as a subset of personal data (Mandler et al. 2017). In other cases, however, the law distinguishes between personal identifying information and biometric data by considering the latter as a unique category of sensitive personal information.[55] Even among the handful of states that have enacted laws expressly governing the use of biometric data, the definitions of biometric data are not uniform, and requirements imposed on companies' use of biometric data are inconsistent, which adds to the uncertainty for private companies seeking to ensure compliance with applicable laws.[56] A discussion of state laws (from Texas, Washington, California, New York, and Nevada) that expressly govern biometric data can be found in Appendix K.

State laws are particularly relevant in circumstances where airport operators and stakeholders are collecting personal data and using biometrics, such as facial recognition, to manage their workforces and for commercial or transportation services for travelers. Four types of state laws are as described in the following (as well as in Figure 3-1):

- Some 19 states regulate the collection and use of facial-recognition data (and an even larger number protect such data under a broad definition of personal information) (National Conference of State Legislatures 2015);
- All 50 states, plus the District of Columbia, Puerto Rico, Guam, and the U.S. Virgin Islands, require private companies or governmental entities to notify residents of a data breach (National Conference of State Legislatures n.d.; Ramirez 2019);[57]
- Approximately 35 states, plus the District of Columbia and Puerto Rico, mandate that entities destroy (or otherwise make inaccessible) personal information that is no longer of use or after a specified period of time (National Conference of State Legislatures 2019a); and
- A handful of states prohibit state and local agencies from collecting, using, sharing, and retaining biometric data, particularly facial-recognition data, except in limited circumstances (National Conference of State Legislatures 2019a).

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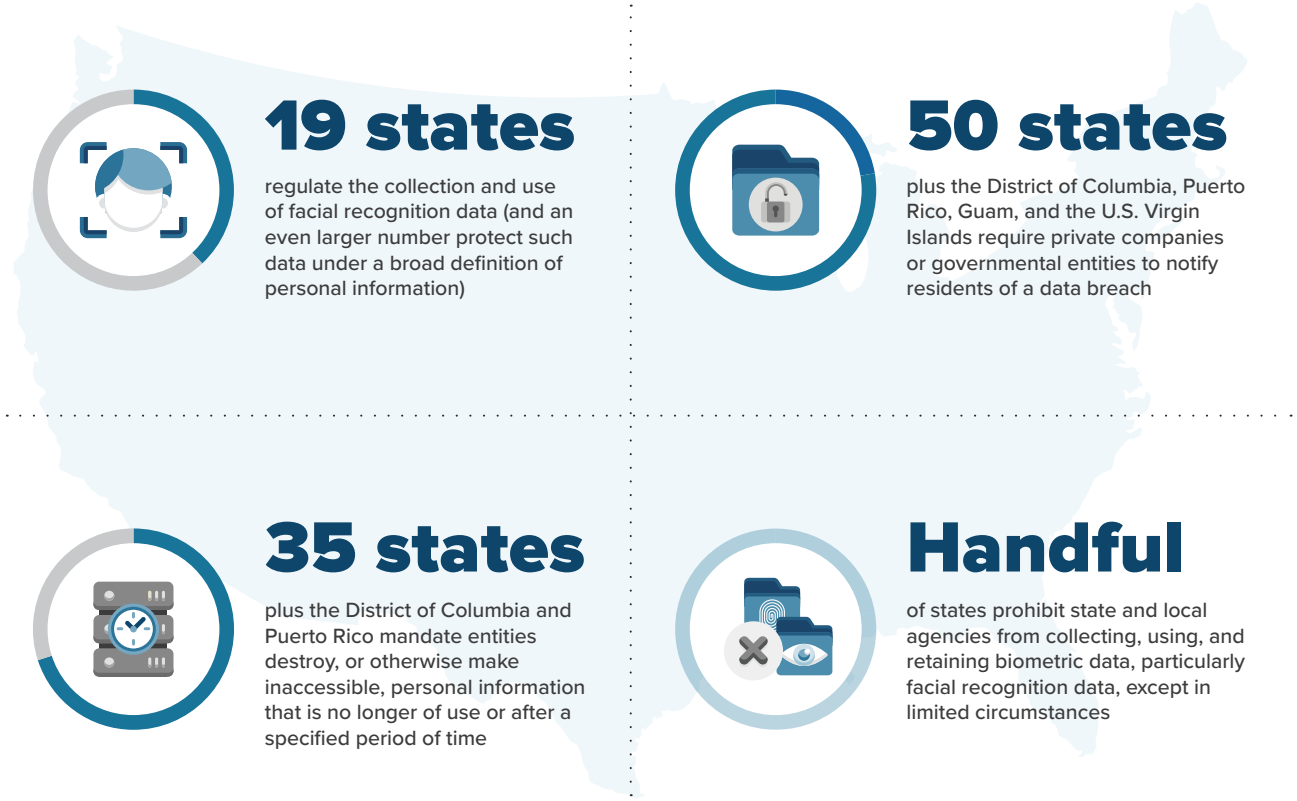


Figure 3-1. Summary of state laws related to the collection and use of biometric data.

State laws widely vary in their protections, penalties for violations, and requirements for businesses.[58] According to at least one article, as of May 2020, it is legal in 44 states to identify an individual using images taken without consent while they are in public, while New York, California, Washington, Illinois, Nevada, and Texas do not allow it for commercial use (Thales 2020) (see Figure 3-2). Several states, however, either expressly define personal information to include biometric data or their laws have been interpreted to provide such protection.[59] Several states have determined that the definition of personal information excludes biometric data.[60] In 2019 alone, legislatures in 25 states considered biometric privacy bills (National Conference of State Legislatures 2015).

Illinois

Enacted in 2008, the Illinois Biometric Information Privacy Act (BIPA) is the first law to expressly address biometric data.[61] BIPA imposes requirements on employers and private entities to make written privacy policies publicly available and provide written notice to, and obtain written consent from, employees on whom the biometric information is being collected and retained, and also imposes limitations associated with the purpose of the collection and the period of storage. It also prohibits profiting off biometric data, allows only a limited right to disclose collected data, and sets forth data protection obligations for business (Hodges and Mennemeier 2020).

Significantly, BIPA requires companies to adhere to reasonable standards while possessing the data and creates a private cause of action and authorizes enforcement by the state's attorney general.[62] Of note, BIPA carries strong penalties: $1,000 for each negligent violation and $5,000 for each willful or reckless violation.[63] Pursuant to a private cause of action, plaintiffs can recover injunctive relief and any actual damages (that exceed the prescribed penalties).

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Figure 3-2. Six states in the U.S. regulate commercial use of biometric data.

Over the period of 2018 to 2019, over 200 lawsuits were filed for alleged violations of BIPA, and reportedly this is on the rise (Prescott 2020). Illinois state courts rendered decisions on whether a technical violation of BIPA was sufficient to pursue a case since federal courts and state courts define injury differently for purposes of determining standing to bring a suit in the respective courts.[64]

To date, most of the cases under BIPA have been class actions targeting employers' use of biometric technology at work. These lawsuits are not only on the rise but are expensive and difficult to defend against. For example, in 2020, Facebook offered to settle a class action lawsuit (in California) for alleged violations of BIPA for $650 million (after the trial judge initially rejected the proposed $550 million settlement) (Shibu 2020). In May of 2020, plaintiffs filed lawsuits against Facebook in Texas and Arizona, adding to the many other suits pending nationwide, and the company is reportedly facing billions in damages (Justia n.d.).

There is also uncertainty with respect to application of BIPA (and other states' biometric privacy laws) under circumstances where airport stakeholders are partnering with government agencies, such as CBP under the TVS program. Factors bearing on applicability, compliance, and potential liability include whether the state law addresses only commercial uses of biometric data and excludes government agencies from its scope, as is the case with all of the biometric state laws discussed in this section (with the exception of the recent Washington law); and whether the airport stakeholder's partnership can be viewed as acting as an agent of the government by collecting facial images and transmitting them to CBP (and not retaining them), but this becomes complicated if the stakeholder collects the data for a dual business purpose as well, such as facilitating the boarding process. (See Appendix K for a discussion of other state laws.)

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Most recently, similar legislation has been introduced in several other states (e.g., Arizona, Florida, and Massachusetts).[65] States have also enacted laws to protect the privacy of minors and students.[66] In addition to the proliferation of state laws enacted or under consideration, local jurisdictions have approved ordinances banning the use of facial-recognition technology by city government agencies and, in particular, police forces (Hudgins 2019). It is anticipated that this trend will continue as privacy concerns increase.

Supremacy Clause and Conflict of Laws

The Supremacy Clause in the U.S. Constitution provides that even if a state statute is enacted in the execution of acknowledged state powers, state laws that "interfere with, or are contrary to the laws of Congress" must yield to federal law.[67] In the event of a conflict between existing/future federal law governing the use of facial-recognition technology, federal law is deemed to preempt state law in three situations (Vogel 2015): (1) when the federal law specifically states that it preempts conflicting state laws;[68] (2) when the federal law purports to "regulate conduct in a field that Congress intended the federal government to occupy exclusively";[69] and (3) when a state law "actually conflicts with federal law . . . the [Supreme] Court has found pre-emption where it is impossible for a private party to comply with both state and federal requirements. . . ."[70,71]

By way of example, this issue has been at the heart of lawsuits and commentaries challenging the Government's Secure Communities program seeking to mandate the sharing of fingerprints collected by state and local authorities as a means of checking immigration status and enforcing federal immigration laws. State and local authorities enacted policies and laws seeking to defy the mandate. While Congress has the power to preempt state law, it is not always clear where to draw the line between immigration enforcement and the exercise of general police powers (McCauley 2018).

With respect to a conflict between state laws, federal courts analyzing the issue "ordinarily must follow the choice-of-law rules of the state in which it sits."[72] Where a commercial entity includes within its terms of service that any litigation will be conducted in a chosen state, the court will uphold that term if the chosen state has a "substantial relationship" to the parties and transaction unless the decision would be contrary to a fundamental policy of the alternate state.[73]

Numerous commentaries have recommended enactment of comprehensive national legislation regulating privacy protections in connection with the collection and use of biometric data, particularly in the face of conflicting state laws on data privacy (Schubert 2020; Cunningham 2019). Several challenges render this a difficult task. Should national legislation preempt all the state laws in effect and set the bar for minimal requirements? How would the law make it clear to all stakeholders what the specific requirements and obligations are, and would it distinguish between personal information and sensitive personal data? Should national legislation incorporate state law provisions pertaining to consent, opting out, correction of data, enforcement, and so forth? Thus, while there has been a call for passage of a federal law governing privacy and personal information, the drafting, scope, and construction of a comprehensive framework addressing these and other related issues pose a daunting task.

Additional Legal and Policy Considerations

Surveillance Versus Identification

As noted earlier, there is a long-standing history of the use of stationary security cameras at airports, retail establishments, subways, and public streets.[74] The courts have generally found that individuals do not have a reasonable expectation of privacy in public places and

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have tended not to view public surveillance as a search governed by the Fourth Amendment. In a recent decision regarding the use of four fixed automatic license plate readers to surveil the ends of a bridge, a state court stated that, while the limited use of the technology did not constitute an unreasonable search, it struggled with where to draw the line.[75] Part of the challenge the courts face is assessing society's reasonable expectation of privacy where heightened technology influences or shapes those expectations, particularly with concerns centered around pervasive police presence.[76] Legal and policy challenges associated with private-sector use are on the rise where security cameras are linked to some form of facial-recognition technology.[77]

Key Takeaway

States are considering and enacting laws limiting/governing the collection and use of biometric data at a rapid pace, so air industry stakeholders need to stay current with emerging privacy law trends and comply with the enactment of state laws and other nations' requirements if they collect and use or propose to collect and use biometric data about their customers.

Bias

While the intersection of facial-recognition technology and discrimination, misidentification, and bias has occasionally cropped up in criminal prosecutions, policing, or cases involving false arrest,[78] there has been little litigation against private employers applying the technology where the claim is one of bias or discrimination. The litigation risk would most likely arise in the context of contract constraints or violation of state laws (Bickel et al. 2003).

The issue, however, is front and center in assessments of the technology. Under a number of studies by the National Institute of Standards and Technology (NIST), the agency found that the accuracy of the facial technology depends on factors such as the specific vendor, whether the purpose is for verification or identification, and whether the algorithm is one-to-one or one-to-many (Grother et al. 2019a). Further, another NIST report acknowledged that the type of recognition algorithm used (i.e., one-to-many identification or one-to-one verification) in various facial technologies resulted in a higher incidence of false positives and false negatives for certain ethnic, gender, and age groups (Grother et al. 2019b; Grother et al. 2019c). The NIST study also reported that the impacts of false positives and false negatives varied greatly depending on the use or purpose (e.g., some capable of mitigation by a second attempt, some presenting a security concern and requiring additional follow-up, some leading to false accusation, and some inconveniencing commercial applications such as access to phone use) (Grother et al. 2019b; Grother et al. 2019c). That same study concluded that advances in the vendor technologies available have consistently improved upon mitigating bias and misidentification (Grother et al. 2019b; Grother et al. 2019c), and reinforced the finding in the NIST report that "massive gains in accuracy have been achieved in the last five years (2013–2018)."[79] In February 2020, NIST indicated that it is in the process of assessing CBP's facial-recognition algorithm (Nash 2020).

Title VII Discrimination

There may be potential litigation based on allegations that an employment practice, such as the collection and use of an employee's biometric data, may implicate and violate religious beliefs.[80] Courts may require employers to provide a reasonable religious accommodation, such as an alternative method of clocking in.[81] Other employment considerations may apply, such as the necessity of reasonable accommodations under the Americans with Disabilities Act.[82]

Foreign Laws and Regulations

A short discussion of key foreign laws and regulations is included because they may involve challenging restrictions and legal risks to the use of biometrics by parties arguably subject to both U.S. and foreign laws and regulations. Many countries have turned to the use of biometrics,

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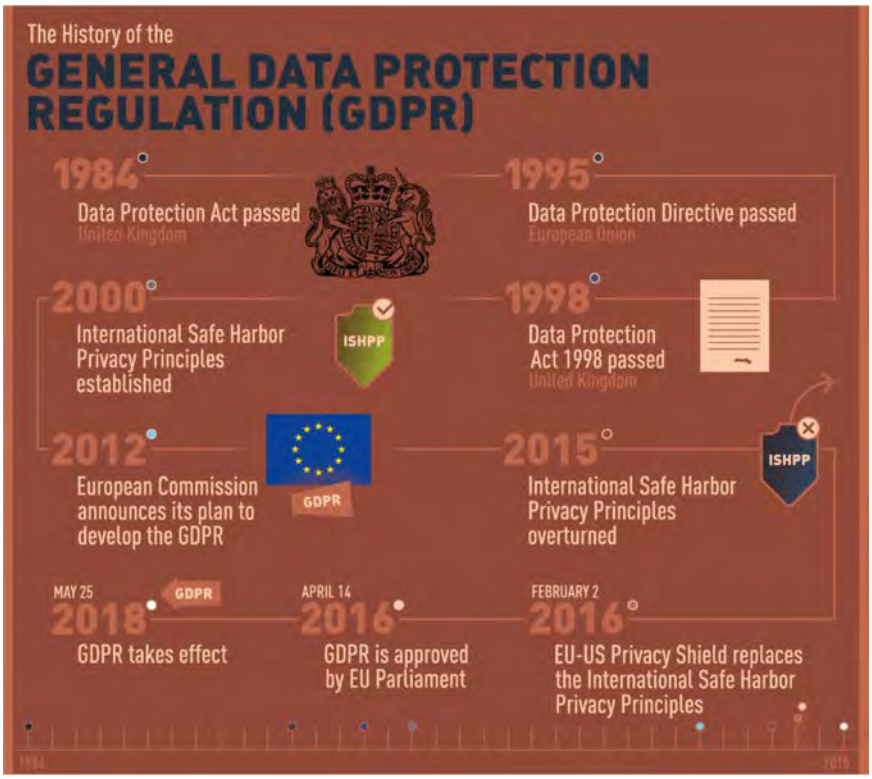
Key Takeaway

Although the focus of much of this report is on the current U.S. legal and policy landscape, this short section describes the EU privacy rules and their potential impacts for U.S. businesses.

particularly facial-recognition technology, to perform verification functions associated with traveler facilitation programs, criminal investigations, and passport administration.[83]

The European Union (EU) General Data Protection Regulation (GDPR)[84] was adopted officially on April 27, 2016, and entered into force on May 24, 2016. EU members were required to incorporate it into their national law by May 25, 2018 (see Figure 3-3). It offers insights into foreign trends and highlights potential conflicts with entities such as carriers subject to two sets of laws. The GDPR superseded the 1995 Data Protection Directive in the EU, but like that directive, the GDPR carries extra-jurisdictional ramifications by requiring countries or entities to prove compliance with the GDPR before allowing the transfer of EU-held personal data (Cunningham 2019).

The GDPR restrictions and requirements apply to entities processing personal data in three circumstances: (1) where the company processing the data is established in the EU, (2) where, regardless of where the company is located or where the processing occurs, the processing relates to an offer of goods or services to EU residents or monitoring their behavior that occurs within the EU, and (3) where the processing occurs and the GDPR applies as a matter of public international law (e.g., within an EU mission or consular post).[85] Article 2 provides four exceptions to the GDPR's applicability, including when government agencies and law enforcement collect and process data for the prevention, investigation, detection, or prosecution of criminal offenses or the execution of criminal penalties or for preventing threats to public safety.[86] Arguably, however, the GDPR may apply to companies that provide EU residents' data to national security



Source: Lord 2018.

Figure 3-3. A history of EU data protection regulations leading up to the GDPR.

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or law enforcement agencies. Further, it should be noted that the GDPR applies to personal data obtained from public sources (Heward-Mills 2020).

The GDPR is intended to provide safeguards for consumer data security rights. It regulates the private sector with respect to the collection, use, retention, storage, and sharing of automatically processed personal data, including biometric data, defined as "special categories of personal data."[87] Pursuant to Article 4 (14), biometric data are classified as a special category and defined as "personal data resulting from specific technical processing relating to the physical, physiological, or behavioral characteristics of a natural person, which allow or confirm the unique identification of that natural person, such as facial images or dactyloscopic data." The processing of sensitive personal data is otherwise prohibited except if one of any number of enumerated exceptions apply, including explicit consent, specified public interest considerations, and certain exemptions in the fields of employment and social protection law.[88]

In essence, there are seven key requirements for organizations that process[89] biometric data pertaining to EU residents, as shown in Figure 3-4.

Significantly, the GDPR limits the ability for cross-border transfers of personal data and generally requires a finding that the receiving country has adequate data protections in place (the so-called "adequacy decision").[90] The EU had issued the United States "only a limited adequacy decision with respect to companies that registered voluntarily under the EU-U.S. Privacy Shield program that the U.S. Department of Commerce administered and whose principles the Federal Trade Commission enforced."[91] This status recently changed.

On July 16, 2020, the European Court of Justice handed down a ruling that could have significant impacts for U.S. companies that collect personal data, including biometric data. "The Court of Justice invalidated Decision 2016/1250 on the adequacy of the protection provided by the EU-US Data Protection Shield."[92] The court found that information on EU citizens when transferred to U.S. servers was not adequately protected from "government surveillance."[93]

Essentially, the court found that the data were not adequately protected from U.S. government authorities and failed to provide EU citizens with certain rights guaranteed under the GDPR.[94] At a minimum, U.S. businesses that collect data on EU customers may need to assess their data protection policies and systems or risk severe sanctions.

As of February 26, 2020, Google has been fined approximately $55 million for GDPR violations relating to unclear terms and for failing to provide valid consent.[96] It should be noted that U.S. courts do not recognize nor are they required to enforce judgments for "the collection of taxes, fines, or penalties rendered by the courts of other [nation] states."[97]

On February 17, 2020, the EU's digital and competition chief told reporters that automated facial recognition breaches the GDPR because the technology fails to meet the regulation's requirement for consent (Macaulay 2020). Moreover, it was reported that the EU plans to regulate certain applications of facial-recognition technology in the future because it can violate EU subjects' privacy rights (Stupp 2020).

Although used in a majority of airports worldwide, biometric data systems such as fingerprint image and facial recognition technologies, are subject to very strict conditions under the European 2016 General Data Protection Regulation (GDPR). Due to its sensitivity, prohibition of biometric data processing is the rule, except if strong safeguards are provided to protect personal data (express consent, proportionality, necessity, limited storage time are some of the key principles). Thus, while automatization of border control (i.e., PARAFE E-Gates or a CLEAR® security check point) is legitimate, and even, to a certain extent, improvement of passenger journey experience (facilitation), mass surveillance to monitor passenger behavior along his airport journey is usually banned. In addition, given the high financial penalties at stake, the GDPR extra-territorial scope is another crucial issue a non-EU stakeholder shall consider, in particular if it has an establishment based in the EU or if it provides services to individuals based in the EU (no matter the individual nationality).

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Figure 3-4. Seven GDPR requirements.[95]

In 2018, the EU released the Regulation (EC) 2018/1725 of the European Parliament and of the Council of 23 October 2018 for the protection of natural persons with regard to the processing of personal data by the EU institutions, bodies, offices, and agencies and on the free movement of such data, and repealed Regulation (EC) No 45/2001 and Decision No 1247/2002/EC.[98] In other words, as a counterpart to the GDPR, the EU implemented rules to protect the privacy rights of its residents when dealing with *public* EU institutions.

In addition to the EU, approximately 28 countries have enacted legislation or promulgated regulations governing biometric data (Thales 2020).[99]

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International Organization Activities

There are a number of international organizations working on, coordinating, and collaborating with others to support initiatives to achieve a more seamless travel experience in the commercial aviation industry. Significantly, most advocate for global standards in an effort to incorporate uniformity in international aviation.[100] (See Appendix K for a discussion tracing the history of several international initiatives relying on the collection and use of biometrics, including among others the ICAO DTC, the UN High Commissioner for Refugees Population Registration and Identity Management and EcoSystem, IATA and ACI Smart Security and New Experience Travel Technologies, and WEF Known Traveller Digital Identity.)

Best Practices/Privacy by Design

There are many publications and organizations offering advice and recommendations on protecting the privacy of consumers' and employees' biometric data.[101] In the early 1970s, a set of principles, known as the Fair Information Practice Principles (FIPPs) (Cate 2006), were developed and evolved over the next several decades to provide a framework for the protection of personally identifiable information. (see Appendix L for the description of the FIPPs core principles).

During the mid-1990s, Ann Cavoukian introduced another widely accepted approach, privacy by design.[102] Privacy by design garnered international acceptance upon the unanimous passage of a resolution by the International Assembly of Privacy Commissioners and Data Protection Authorities (2010). The concept advocates the integration of privacy into data systems and technologies at every stage of the design process to reflect a "design thinking" perspective.[103]

Key Takeaway

With the expected increase in the use of biometric data, including facial recognition, and the anticipated benefits and efficiencies from its use in the aviation industry, stakeholders need to find the right balance of privacy protections and technological applications.

In essence, the seven principles of privacy by design promote incorporation of privacy protections into the design and build of technology and information technology (IT) systems, as well as applicable policies and practices, including those governing retention and destruction. The principles stress that inclusion of these protections can be made without sacrificing functionality or security. Further, they advocate business accountability to the public for privacy protections and providing data subjects with avenues of redress and an "active role" in the collection, use, and protection of their data.

The seven foundational principles for privacy by design are shown in Figure 3-5.

The seven principles are explained in the following.

1. **Proactive not reactive.** Builds privacy considerations into technologies, policies, practices, and so forth to prevent invasive events before they can occur.
2. **Privacy as the default setting.** Incorporates privacy protections automatically into IT systems or business practices to provide maximum safeguards by default.
3. **Privacy embedded into design.** Promotes the integration of privacy into IT systems and operations in a holistic way to ensure that privacy is a key foundational element without sacrificing functionality.
4. **End-to-end security – full life-cycle protection.** Extends privacy and security protections for the entire retention period until destruction.
5. **Respect for user privacy – keep it user-centric.** Aims to provide data subjects an active role in the collection, use, and protection of their data.
6. **Full functionality – positive-sum, not zero-sum.** Accommodates non-privacy objectives without trade-offs to achieve full functionality while protecting privacy; in other words,

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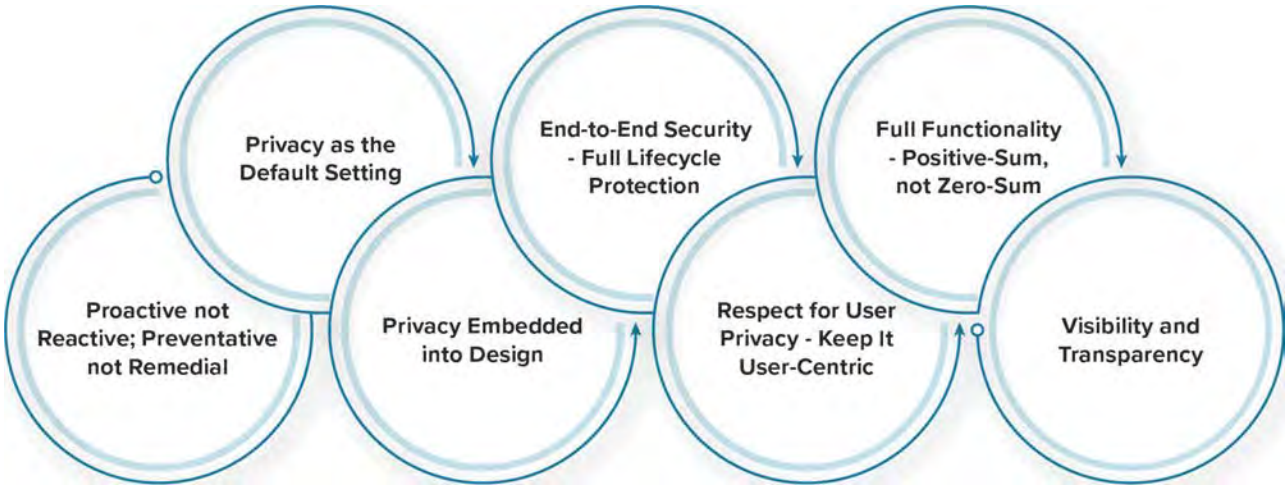


Figure 3-5. The seven principles of privacy by design.

this principle advocates the accommodation of all legitimate interests and objectives in a win–win manner, not through a dated either/or approach where unnecessary trade-offs are made, and supports the position that it is possible to have both privacy and security.

7. **Visibility and transparency – keep it open.** Assures users and providers that the collection and use of personal data comply with representations and objectives while providing for accountability through independent verification as well as avenues of redress (Cavoukian 2008).

Key Takeaway

Many of the principles of privacy protection (such as privacy by design) and best practices discussed in this section have been incorporated into U.S. law and internationally so that compliance with these principles may place stakeholders on surer footing with the law.

In 2012, the Federal Trade Commission (FTC) issued a report on consumer data privacy (FTC 2012a)[104] that recommends a number of best practices and addressed both substantive and procedural protections. Building on its 2010 preliminary report (FTC 2010) that had recommended that businesses build privacy protections into their operations as described by privacy by design, the 2012 FTC report proposed best practices that offer simplified choices that give consumers more meaningful control and would increase the transparency of their data collection and use practices (FTC 2012a; for a list of the 2012 FTC Report recommended practices see Appendix L). From a global perspective, it is worth noting recent EU developments, which include the 2019 EU Data Protection Guidelines (4/2019) on the GDPR requirements for Data Protection by Design and by Default (European Data Protection Board 2019).[105] The draft guidelines (which were open for comment until January 2020) proposed measures and guidance addressing data protection by design, data protection by default, data subjects' rights, safeguard requirements, practical guidance on the application of the principles, and certification. Thus, the guidelines propose direction regarding what data protection obligations mean in practice and how to implement the data protection principles effectively. Similarly, the EU published guidelines on consent describing the elements of valid consent [i.e., under GDPR article 4(11), European Data Protection Board 2019].

The EU also published Guidelines (2/2020) on GDPR provisions on transfers of personal data between EU and non-EU public authorities and bodies (European Data Protection Board 2020b).

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Commercial Developments

A recent initiative that focuses on transparency, consumer education, and the mitigation of privacy concerns is the Digital Trust in Places and Routines (DTPR).[106] DTPR is a collaborative project initiated by Sidewalk Labs, currently stewarded by Helpful Places, and joined by more than 100 participants to develop a standard that informs and advises individuals in simple language about complex forms of technology and data-collection activities. In particular, DTPR seeks to design transparent patterns and prototypes that entities can use to develop signage that conveys to members of the public information about the use and collection of digital technology.[107]

There are four components to DTPR: icons, a signage system (based on the icons), a "digital channel" (which, when you scan a QR code, connects you with a website for additional information), and a set of definitions of key concepts or taxonomy.[108] Different icons convey different information such as "the purpose of the technology; another, the logo of the entity responsible for the technology; and a third contains a QR code that takes the individual to a digital channel where they can learn more. In situations where identifying information is collected, a privacy-related, colored hexagon would also be displayed."[109]

The icons in Figure 3-6 are initial prototypes of a visual language for signage in the public realm that alerts the public to the presence of a digital technology. The black hexagons express the purpose of the technology, the blue and yellow hexagons show how identifiable information is used (see online version of report for color figure), and the white hexagons display the entity responsible for the technology. Another white hexagon with a QR code and URL enables people to learn more.

Findings

Significantly, the World Economic Forum (WEF) published a report in February 2020 in furtherance of an initiative launched in 2019 to design a governance framework for the responsible use of facial-recognition technology (World Economic Forum 2020c). The project applied a use-case approach focused on improving the air-passenger boarding experience. The WEF reports suggest a comprehensive approach to using not only facial-recognition technology but also other forms of biometric technology in the air environment (World Economic Forum 2020c). Components of the WEF framework include 10 principles for action,[110] best practices,



Source: Lord 2018.

Figure 3-6. Open-source icons aimed to facilitate digital transparency goals.

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an assessment questionnaire, and an audit framework (World Economic Forum 2020c). The two reports set forth a four-step approach to building the framework. Each step is intended to represent an additional level of commitment for achieving transparency and public trust. The December 2020 report focuses on auditing and certification not only to validate compliance with the principles for action, but also to build in accountability, monitoring, and opportunities for improving the flow management process (World Economic Forum 2020d).

WEF, while noting that the framework may serve as a "blueprint" for other applications, concludes that the next steps of the pilot or "journey" are to:

> Test the audit framework and certification scheme with industry actors, assess their relevance and the amount of work they create for actors seeking certification, and review them based on the observed results. If successful, this policy pilot will pave the way for the design of a standard for the responsible application of facial-recognition systems. Once the pilot project is completed, a multi-stakeholder coalition of actors committed to respecting and promoting this certification model will be formed (World Economic Forum 2020c).

The following are findings on actions drawn from legal research and best practices and designed in part to mitigate litigation risks as well as promote sound privacy practices. This list is intended to identify considerations that may be factored into decisions to use or support the use of biometric data technology:

- Determine and assess whether the application of a type of biometric technology is, as a matter of policy and operations, key to your business as an employer supervising a workforce (e.g., timekeeping, access to restricted areas) or a member of the aviation transport industry providing a service to customers;
- Consult with legal advisers and understand legal (both federal and state) requirements and restrictions (particularly those with broad reach and severe penalties) that govern the collection and use of personal data, especially biometric data, and appreciate that laws may impose different requirements depending on the type of collection and use of specific biometric data;
- Review existing company policies, establish a compliance plan, and designate a compliance officer responsible for ensuring that applicable policies, practices, and systems meet privacy principles and legal requirements;
- Make sure that policies address obligations and issues of (advance) notice, (affirmative/written) consent (and the ability to opt out), sharing with third parties, the ability to access and amend personal data, and why your company collects specific data and the purposes for which it uses it;
- Focus on and include policies and practices that allow for reasonable accommodation as the law may require (e.g., religious reasons, Americans with Disabilities Act);
- Identify and address any applicable union issues;
- Inform employees and the public in clear and plain terms about your company's privacy policies and practices, including measures to protect the data, since communication is critically important;
- Meet with privacy and civil liberties groups, as warranted, and legislative representatives to inform them of specific programs and protections, to appreciate any privacy concerns, and to avoid misunderstandings or unwarranted criticism;
- Ensure that IT systems safeguard personal data, limit others' access, and are governed by practices limiting the collection of data to that which is authorized and necessary, retaining data for only as long as necessary, and providing for appropriate means of storage and disposal of data;
- Prohibit the sale or transfer of personal data to third parties without the express consent of the data subject;

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- As a precaution, ensure that a mechanism is in place for notification of data breaches; and
- Stay current on changing laws (both domestic and international), try to anticipate foreseeable changes, and seek advice from counsel to appreciate and address any impacts on your company's use of biometric data.[111]

Endnotes

1. See, e.g., In re *Facebook Biometric Info. Privacy Litigation*, 326 F.R.D. 535, 540 (N.D. Cal. 2018); *Rivera v. Google Inc.*, 238 F. Supp. 3d 1088, 1090 (N. D. Ill. 2017).
2. *Id.*; see also Kosseff 2019.
3. This descriptive term has been used in over 400 legal articles. LEXIS NEXIS advanced search (5/19/2020).
4. Privacy Act of 1974, 5 U.S.C. § 552a (2018).
5. Health Insurance Portability and Accountability Act of 1996 (HIPAA) (codified at 42 U.S.C. § 1320d-6); Gramm-Leach-Bliley Act of 1999 (GLBA), 15 U.S.C. § 6805(a).
6. Federal Trade Commission Act, 15 U.S.C. §§ 41–58 (2018).
7. *See, e.g.,* Tex. Bus. & Com. Code Ann. § 503.001 (West 2017); 2017 Wa. ALS 299; 2017 Wa. Ch. 299; 2017 Wa. HB 1493.
8. 1 Data Privacy, Protection, and Security Law § 1.02 (2020).
9. *See Katz v. United States*, 389 U.S. 347, 350–51 (1967).
10. *Id.*
11. *Griswold v. Connecticut*, 381 U.S. 479, 484 (1965).
12. *See Griswold,* 381 U.S. at 484 [quoting *Boyd v. United States*, 116 U.S. 616, 630 (1886)] (describing the Fourth and Fifth Amendments as "protection against all governmental invasions 'of the sanctity of a man's home and the privacies of life.'").
13. *United States v. Dionisio*, 410 U.S. 1, 14 (1973) ("No person . . . can reasonably expect that his face will be a mystery to the world."); *see also Maryland v. King*, 569 U.S. 435, 476 (2013) (Scalia, J., dissenting) (noting that taking a person's photograph is a not a Fourth Amendment search).
14. *See, e.g., Patel v. Facebook, Inc.*, 932 F.3d 1264, 1273 (9th Cir. 2019) (recognizing a concrete privacy interest in one's biometric face template), *cert. denied*, 140 S. Ct. 937 (2020). Although *Patel* does not squarely address the reasonableness of the use of facial recognition technology under the Fourth Amendment, its Article III standing analysis necessarily involves an examination of the privacy interests of persons subject to facial recognition technology, and such interests logically reflect those same persons' expectations of privacy.
15. Note that there is no Fourth Amendment right to expunge government records of one's identity. *Johnson v. Quander*, 440 F.3d 489 (D.C. Cir. 2006), *cert. denied*, 549 U.S. 945 (2006). Although indefinite retention of data deeply concerns other countries, U.S. courts have generally not shown the same level of concern. *See, e.g., Gubala v. Time Warner Cable, Inc.*, 846 F.3d 909, 912 (7th Cir. 2017) (holding that the mere retention of data does not in itself cause an injury-in-fact absent a substantial risk of disclosure); *United States v. Hasbajrami*. 945 F.3d 641, 670, 670 n.20 (2d Cir. 2019) (noting that "[s]torage has little significance in its own right," but caveating that "[t]he considerations might be different if the storage involved data responsive to a warrant and retained for the purpose of a domestic criminal prosecution").
16. *INS v. Delgado*, 466 U.S. 210, 216 (1984).
17. *Quander*, 440 F.3d at 498 (citing *Arizona v. Hicks*, 480 U.S. 321 (1987)); *see also King*, 569 U.S. at 465 ("[O]nce respondent's DNA was lawfully collected the STR analysis of respondent's DNA . . . did not amount to a significant invasion of privacy that would render the DNA identification impermissible under the Fourth Amendment.")
18. *Riley v. California*, 573 U.S. 373, 381 (2014) (quoting *Brigham City v. Stuart*, 547 U.S. 398, 403 (2006).
19. *Id.* at 382.
20. *United States v. Moore,* 381 F. Supp. 3d 139 (D. Mass. 2019) (citing *United States v. Bain*, 874 F.3d 1, 11–12 (1st Cir. 2017)) (law enforcement's use of a pole camera to record the homeowners' comings and goings constituted a search).
21. *Florida v. Jardines*, 569 U.S. 1, 5 (2013).
22. *Katz v. United States*, 389 U.S. 347, 360 (1967).
23. Later cases analyzing *Katz* and drawing on Justice Harlan's concurrence in the case have explained that, under this approach, the Fourth Amendment protects legitimate or reasonable expectations of privacy where: (1) "the individual, by his conduct, has exhibited an actual (subjective) expectation of privacy," and (2) "the individual's subjective expectation of privacy is one that society is prepared to recognize as reasonable." *Smith v. Maryland*, 442 U.S. 735, 740 (1979) (internal quotation marks omitted) (quoting

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Katz, 389 U.S. at 361 (Harlan, J., concurring)); Mariko H., *Privacy in Public Spaces: The Reasonable Expectation of Privacy Against the Dragnet Use of Facial Recognition Technology*, 49 Conn. L. Rev. 1591 (2017).

24. *See* Howick, J. L., The Fourth Amendment and Airports (2016), ACRP 1101, https://www.nap.edu/catalog/23500/the-fourth-amendment-and-airports
25. *Kowalski v. Scott*, No. Civ.A.02-7197, 2004 WL 1240658, at *1 (E.D. Pa. May 26, 2004), *aff'd*, 126 Fed. App'x 558 (3d Cir. 2005).
26. *United States v. Knotts*, 460 U.S. 276, 281 (1983); *United States v. Jones*, 565 U.S. 400, 4430 (2012) (noting that monitoring of a person's movements is reasonable where short-term).
27. *Hudson v. Palmer*, 468 U.S. 517 (1984) (no reasonable expectation of privacy).
28. *United* States *v. Ramsey*, 431 U.S. 606, 619 (1977).
29. *Terry v. Ohio*, 392 U.S. 1 (1968).
30. *Florida v Rodriguez,* 469 U.S. 1 (1984).
31. *Id.*
32. *Kyllo v. United States*, 533 U.S. 27, 34 (2001); *United States v. Jones,* 565 U.S. 400, 416, 428 (2012) (GPS monitoring for extended periods of time); *Carpenter v. United States,* 138 S. Ct. 2206, 2215 (2018) (technological advances in tracking cell-site location information); *Riley v. California,* 573 U.S. 373, 386 (2014) (modern cell phone storage of "vast quantities of personal information).
33. *Carpenter*, 138 S. Ct. at 2218.
34. *Riley*, 573 U.S. at 393.
35. *But see United States v. Moore-Bush,* 963 F.3d 29 (1st Cir. 2020) (affirming the *Katz* principle held use of pole camera comports with rule that "a person does not have a reasonable expectation of privacy in the actions he or she exposes to the public view"); Koops, Bert-Jaap, et al. *Location Tracking by Police: The Regulation of 'Tireless and Absolute Surveillance',* 9 U.C. Irvine L. Rev. 635 (March 2019) (Suggesting that factors such as duration, intensity, use, active generation of data, etc. are relevant to an assessment of privacy intrusion by police tracking).
36. *United States v. Ramsey*, 431 U.S. 606 (1977).
37. *See Almeida-Sanchez v. United States*, 413 U.S. 266, 272–273 (1973); *Ramsey*, 431 U.S. at 606, 610 n.2.
38. *Cassidy v. Chertoff*, 471 F. 3d 67, 76 (2d Cir 2006) (noting that "society has long accepted a heightened level of security and privacy intrusion with regard to air travel"); *United States v. Herzbrun*, 723 F.2d 773, 775 (11th Cir. 1984); *United States v. Albarado*, 495 F.2d 799, 805 (2d Cir. 1974); *United States v. Skipwith,* 482 F.2d 1272, 1275 (5th Cir. 1973*); United States v. Davis*, 482 F.2d 893, 910 (9th Cir. 1973); *United States v. Hartwell*, 296 F. Supp. 2d 596, 602–05 (E.D. Pa. 2003); *People v. Hyde*, 524 P.2d 830 (Cal. 1974).
39. *United States v. Aukai*, 497 F. 3d 955, 960 (9th Cir. 2007) ("The constitutionality of an airport screening search, however, does not depend on consent. . . . Rather, where an airport screening search is otherwise reasonable and conducted pursuant to statutory authority, 49 U.S.C. § 44901, all that is required is the passenger's election to attempt entry into the secured area of an airport."); *United States v. Marquez*, 410 F.3d 612, 617 (9th Cir. 2005) (noting that airport searches are conducted for the parallel purposes of "prevent[ing] passengers from carrying weapons or explosives onto the aircraft" and "deter[ring] passengers from even attempting to do so").
40. Although there is no Supreme Court holding directly on point, Supreme Court *dicta* and lower courts are all in general agreement. *See, e.g., Chandler v. Miller*, 520 U.S. 305, 323 (1997) ("We reiterate, too, that where the risk to public safety is substantial and real, blanket suspicionless searches calibrated to the risk may rank as 'reasonable'—for example, searches now routine at airports and at entrances to courts and other official buildings."); *see also Corbett v. TSA*, 767 F.3d 1171, 1179 (11th Cir. 2014) (holding that suspicionless scans for explosives and pat-downs are reasonable administrative searches); *Ruskai v. Pistole*, 775 F.3d 61, 77 (1st Cir. 2014) (same, for just pat-downs); *Electronic Privacy Information Center v. DHS*, 653 F.3d 1 (D.C. Cir. 2011) (same, for just scans for explosives).
41. *Cassidy*, 471 F.3d at 75.
42. *Harlow v. Fitzgerald,* 457 U.S. 800 (1982).
43. *Bivens v. Six Unknown Named Agents of Fed. Bureau of Narcotics*, 403 U.S. 388, 389 (1971).
44. *See, e.g., United States v. Jacobsen*, 466 U.S. 109, 115 (1984) (explaining that private searches do not trigger the Fourth Amendment unless the private actor was operating as an agent or instrument of law enforcement at the time of the disputed conduct); *Skinner v. Ry. Labor Executives' Ass'n, 489 U.S. 602, 614 (1989)*; *Coolidge v. New Hampshire*, 403 U.S. 443, 487 (1971).
45. *See United States v. Momoh*, 427 F.3d 137 (1st Cir. 2005) (Court noting that even if private party was acting as a government agent, her search of defendant's package would have been permissible under the "border search" exception); *United States v. Rodriquez*, 596 F.2d 169 (6th Cir. 1979) (private search of package and subsequent examination of contents valid under plain view); *see, also Gonzales v. FedEx Ground Package Sys.*, No. 12-CV-80125-RYSKAMP/HOPKINS, 2013 WL 12080223, at *1 (S. D. Fla. Aug. 1, 2013) (holding that 19 U.S.C. § 507 protected Fed Ex acting reasonably to assist CBP officers).

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46. *Cassidy v. Chertoff*, 471 F.3d 67, 74 (2d Cir. 2006); *Brees v. HMS Global Mar. Inc.,* 431 F. Supp. 3d 1207 (W. D. Wash. 2020)
47. The Supreme Court has not directly addressed the application of the a special needs exception to airport administrative searches, but, in dicta, the Court signaled the likely validity of its reasonableness in two cases: *See Chandler v. Miller*, 520 U.S. 305, 323 (1997) (suspicionless searches "may rank as 'reasonable'—for example, searches now routine at airports"); *City of Indianapolis v. Edmond*, 531 U.S. 32, 47-48 (2000) ("holding also does not affect the validity of border searches or searches at places like airports and government buildings, where the need for such measures to ensure public safety can be particularly acute.").
48. In a review of case law to determine any suits filed against the FBI, CBP, or TSA regarding the use of facial recognition, the only cases located to date concerned actions filed under the Freedom of Information Act for agency records pertaining to CBP's use of facial recognition technology. *See, e.g.*, *ACLU v. DHS*, Case 1:20-cv-02213 (S.D.N.Y, March 12, 2020); *EPIC v. CBP*, No. 19-cv-689 (D.D.C. March 12, 2019) (settled on April 24, 2020).
49. U.S. Gov't Accountability Office, GAO-15-621, Facial Recognition Technology (2015), at 28 (referencing the Video Voyeurism Prevention Act of 2004, codified at 18 U.S.C. § 1801)); but also would include the Children's Online Privacy Protection Act in that list (15 U.S.C. §§ 6501–6506).
50. *Id.*
51. 5 U.S.C. § 552a (2018).
52. Constitutional Topic: How a Bill Becomes a Law, https://www.usconstitution.net/consttop_law.html (last visited Aug. 3, 2020).
53. *See* www.congress.gov (last visited July 24, 2020). *See, e.g.,* Privacy Bill of Rights, S. 1214, 116th Cong. (2019); Consumer Data Privacy and Security Act, S. 3456, 116th Cong. (2020); American Data Dissemination Act, S. 142, 116th Cong. (2019); Data Care Act of 2019, S. 2961, 116th Cong. (2019); Digital Accountability and Transparency to Advance Privacy Act, S. 583, 116th Cong. (2019); H.R. 3900, 116th Cong. (2019); Privacy Score Act of 2020, H.R. 6227, 116th Cong. (2020); Office of Biometric Identity Management Authorization Act of 2019, H.R. 1729, 116th Cong. (2019).
54. www.congress.gov. (last visited 03/21/2021).
55. *Mini-Symposium on Comprehensive Data Privacy Reform Legislation in the United States*: *Data Privacy and the Financial Services Industry: A Federal Approach to Consumer Protection*, 24 N.C. Banking Inst. 527 (March 2020); General Data Protection Regulation (GDPR), Commission Regulation 2016/679, 2016 O.J. (L 119) 1, Arts. 4 and 9.
56. *Id.,* at fn. 4.
57. *See also* Oregon's Consumer Information Protection Act (OCIPA, ORS 646A-600, *et seq*.) effective 01/01/2020.
58. For list of state laws, see Ramirez 2019.
59. *See, e.g*., Ark. Code Ann. §4-110-103(7) (effective July 2019, Arkansas revised its breach notification law to include biometric data within its definition of personal information); Louisiana Revised Statutes 51:3071, *et seq*. (2018) (amending its definition of personal information under its breach law to extend protections to biometric data and include a private cause of action); N.C. Gen. Stat. §14-113.20(B); Iowa Code Ann. § 715C.1(11)(a)(5) (West 2014) (defining personal information to include "unique biometric data, such as a fingerprint, retina or iris image, or other unique physical representation or digital representation of biometric data"); Neb. Rev. Stat. Ann. § 87-802(5)(a)(v) (West 2016) (defining personal information as including "unique biometric data, such as a fingerprint, voice print, or retina or iris image, or other unique physical representation"); Wis. Stat. Ann. § 943.201(1)(b)(13) (West 2017) (defining biometric data as "including fingerprint, voice print, retina or iris image, or any other unique physical representation").
60. *See* National Conference of State Legislatures 2019b. List of state bills introduced including where proposed amendments to definition of personal information failed to pass (e.g., Florida FL HB 1153).
61. 740 Ill. Comp. Stat. 14/15 (2013) (BIPA).
62. *Id.*
63. *Id.*
64. *See Bryant v. Compass Grp. USA, Inc*., 958 F. 3d 617 (7th Cir. May 5, 2020) for a discussion of standing requirements and decisions (standing found in case where plaintiff alleged violation of BIPA for the failure to comply with statutory requirements associated with an account to biometrically access a vending machine).
65. H.R. 2478, 50th Leg., Second Reg. Sess. (Ariz. 2012); H.R. 1153, 2019 Sess. (Fla. 2019); S. 120, 191st General Court (Mass. 2019).
66. *See, e.g.,* National Conference of State Legislatures 2019b.
67. *Gibbons v. Ogden*, 22 U.S. 1, 211 (1824)
68. *See English v. Gen. Elec. Co.,* 496 U.S. 72, 78–79 (1990*).*
69. *Id*. at 79.
70. *Id.* (citing *Fla. Lime & Avocado Growers, Inc. v. Paul*, 373 U.S. 132, 142-43 (1963)).

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71. For an extensive discussion of the Supremacy Clause and the doctrine of preemptions, see *State v. Martinez*, 896 N.W.2d 737 (Iowa 2017)
72. *Palomino v. Facebook*, 2017 U.S. LEXIS 2971 (N. D. Cal. 2017) (quoting from *Atl. Marine Const. Co. v. U.S. Dist. Court for W. Dist. of Texas*, 134 S. Ct. 568, 582, 187 L. Ed. 2d 487 (2013).
73. *Id.,* at 7–8.
74. *See, e.g.,* Iraola 2003.
75. *Commonwealth v. McCarthy*, 142 N.E.3d 1090, 1106 (Mass. 2020) (determining that it could not say precisely how detailed a picture of the defendant's movements must be revealed to invoke constitutional protections).
76. *Id.*; *see also* Blitz 2004 (comprehensive discussion of the evolution of privacy protections in the law).
77. *Bah v. Apple Inc.,* No. 19-cv-3539 (PKC), 2020 WL 614932, at *1 (S.D.N.Y. Feb. 10, 2020). In a nightmare of a case, Ousmane Bah, a teenager, lost his learner's permit, which was subsequently used by person/persons unknown to commit thefts from Apple stores in four states. Bah was arrested for the thefts based on facial recognition misidentification information from Apple "linking" him to the thefts provided to law enforcement by Apple. Bah has sued for false arrest, defamation, and malicious prosecution.
78. *Id.; see, e.g.*, Hamann and Smith 2019.
79. NIST.IR 8271, at 2.
80. *See, e.g., EEOC v. Consol. Energy, Inc*., 860 F.3d 131 (4th Cir. 2017), *cert. denied*, 138 S. Ct. 976 (2018).
81. *Id.* at 143.
82. *Nat'l Fed'n. of the Blind v. United Airlines, Inc.,* 813 F.3d 718 (9th Cir. 2016) (referencing 14 CFR § 382.57 providing that generally biometrics may not be the only means of user identification or control); *see also EEOC v. Orion Energy Systems,* 208 F. Supp. 3d 989 (E.D. Wis. 2016) (no violation of the ADA since biometric health screening assessment was voluntary).
83. *See* Countries Applying Biometrics, https://en.wikipedia.org/wiki/Countries_applying_biometrics (list of various countries' biometric laws); Carrero 2018 (citing examples involving Canada, Brazil, Australia, and Argentina); World Economic Forum 2017 (detailing various country programs using biometrics).
84. *See* Commission Regulation 2016/679, 2016 O.J. (L 119) 1, Art. 3.
85. GDPR, art. 3(1)-(3); Recital 25; Determann 2020, pp. 236–237; GDPR, Art. 3.
86. GDPR, art. 23
87. GDPR, arts. 2 and 4; *see also* art. 5 (setting out the principles governing the processing of personal data).
88. GDPR, art. 9(2)-(4).
89. GDPR, art.4(2) (Processing means collecting, recording, organizing, structuring, storing, adapting, using, sharing, etc.); *Id*., art. 33; *Id.,* art. 15; *Id.,* art. 7; *Id*., art. 20; *Id*., art. 25; *Id.,* arts. 37–39; *Biometric data and data protection regulations (GDPR and CCPA)*, Thales (June 27, 2020), https://www.thalesgroup.com/en/markets/digital-identity-and-security/government/biometrics/biometric-data.
90. *See* GDPR, art. 45(1).
91. European Commission, Commission Implementing Decision (EU) 2016/1250, 2016 O.J. (L 207) 11. *See also* Determann 2020, p. 238.
92. Court of Justice of the European Union Press Release No. 91/20 (July 16, 2020), https://curia.europa.eu/jcms/upload/docs/application/pdf/2020-07/cp200091en.pdf.
93. *Id.*
94. *Id.*
95. GDPR, art.4(2) (Processing means collecting, recording, organizing, structuring, storing, adapting, using, sharing, etc.); art. 33; art. 15; art. 7; art. 20; art. 25; arts. 37–39; Biometric data and data protection regulations (GDPR and CCPA), Thales (June 27, 2020), https://www.thalesgroup.com/en/markets/digital-identity-and-security/government/biometrics/biometric-data.
96. Shyy 2020. (Concluding that "the GDPR is both *ineffective* in protecting consumer privacy and *burdensome* on businesses of all sizes," at 157).
97. Restatement (Third) of the Foreign Relations Law of the United States, §483 (Am. Law. Inst. 1987); See Wallace 2019 for an extensive discussion of comity and the principles associated with the enforcement of the GDPR in the United States.
98. See Parliament and Council Regulation 2018/1725, 2018 O.J. (L 295) 39, available at https://eur-lex.europa.eu/legal-content/EN/TXT/PDF/?uri=CELEX:32018R1725&from=EN.
99. For comprehensive list of foreign data privacy laws, see Greenleaf 2019.
100. *See, e.g.*, ICAO n.d.-a.
101. *See, e.g.,* National Institute of Standards and Technology 2020; U.S. Department of Homeland Security 2008a; Mandler et al. 2017; Amelung 2019.
102. Cavoukian is credited with developing this approach while Information and Privacy Commissioner of Ontario (Cavoukian 2009).
103. *See* Cavoukian 2009; Wong 2020.

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104. S*ee also* Federal Trade Commission 2012b.
105. *See also* European Data Protection Board 2020a. For a comprehensive summary of the elements of data protection by design and by default, see www.twobirds.com/en/news/articles/2019/global/edpb-publishes-guidelines-on-data-protection-by-design-and-by-default.
106. See Designing for Digital Transparency in the Public Realm, https://sidewalklabs.com/dtpr/.
107. *Id.*
108. *Id.*
109. *Id.*
110. Those 10 principles for action are (1) Proportional use of facial recognition systems, (2) Risk assessment, (3) Bias and discrimination, (4) Privacy by design, (5) Performance, (6) Right to information, (7) Consent, (8) Information display, (9) Right of access to vulnerable groups, and (10) Alternative option and human presence. World Economic Forum 2020c, p. 16.
111. For examples of other checklists, *see, e.g.*, Prescott 2020; Information Commissioner's Office n.d.

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CHAPTER 4

Planning and Process Considerations

Summary

This chapter highlights the role of technology (such as the use of biometrics) in passenger terminal area operations and planning. More specifically, in considering the use of biometrics in process improvements, airport stakeholders are advised to ultimately balance the demand and capacity of a facility to achieve certain operational requirements and the desired level of customer service.

The chapter provides a discussion of numerous factors for determining the applicability of biometrics in the airport setting, such as passenger demographics, airport size and geography, and the operational profile (e.g., primarily serving international or domestic traffic). Airports that are capacity constrained in some manner will likely be the early adopters with regard to the use of biometrics given the imperative that any incremental benefit, regardless of magnitude, may offer a meaningful overall improvement. While discussed in more detail within the chapter, critical factors to be considered are:

- Would passenger or facility security (e.g., identity assurance and accuracy) be enhanced?
- Would a small increase in capacity or level of service be meaningful to the overall experience of the population affected?
- Are there reasons besides capacity enhancement and security to implement biometrics, such as the desire to reduce staffing costs or improve the reliability of certain processes?
- Are the stakeholders required to implement the changes supportive of the goals?

Other questions that airport decision makers may address include:

- In what parts of the passenger or employee journey should the use of biometrics be considered?
- What are the potential benefits to the airport and associated stakeholders from implementing biometrics?
- Given the potential costs associated with making enhancements through the use of biometrics, where will it have the most impact for the least amount of investment?
- How are biometrics successfully implemented in an airport environment?
- Do the passengers using the airport have a high rate of passport ownership or Trusted Traveler program membership?
- Do the airlines wish to implement biometrics at the airport, and who is paying for the up-front costs of capital investment and the ongoing operations and maintenance of the equipment?

Airport operators are advised to review the pros and cons of specific biometric implementations to decide whether an implementation is warranted. In most cases, the return on investment (ROI) for the airport operator goes beyond strictly financial considerations for the implementation of biometric solutions. Instead of making decisions based solely on the financial ROI, airports are evaluating several additional considerations such as customer service, airport technology goals, passenger safety, operational efficiency, and competitive concerns.

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Introduction: Biometrics Will Disrupt Traditional Airport Planning

Key Takeaway

Technologies such as biometrics are changing how terminals function, requiring that future terminal design delivers flexibility instead of simply physical purpose-built expansion.

Terminal planning in the airport environment is ultimately an exercise in balancing between the demand and capacity of a facility to achieve certain operational requirements and the desired level of customer service. Traditionally, capacity could be gained by adding new processing capability, which often involved physically expanding the size of facilities, with an overview of the different processing steps at an airport shown in Figure 4-1. While these static planning decisions were sometimes easier to make, building expansion requires a significant time and financial investment to implement. These expansions were also often purpose-built to serve a specific need in the terminal such as airline check-in or security screening and could not be repurposed as demand changed. More recently, technology has been integrated into facility planning to increase capacity and improve processes while lessening the need for more expensive and time-consuming expansions.

This shift toward technological solutions requires that future terminal design deliver flexibility instead of simply physical purpose-built expansion. Technology and innovation have changed and will continue to change how terminals function. Increased throughput at various passenger processing functions, for example, leads to less processing space required. This provides new opportunities to transform unused processing space into more valuable passenger dwell space, preferably airside rather than landside, as was formerly the case with many pre-September 11th terminal designs.

Biometrics are one of those recently deployed technological innovations that are influencing the way airport terminals are designed and operated. Biometric use has been a part of everyday life for many years, including in personal banking, building access, smart home devices, hands-free cars, and access control and single sign on. It is not uncommon to use a fingerprint scan to

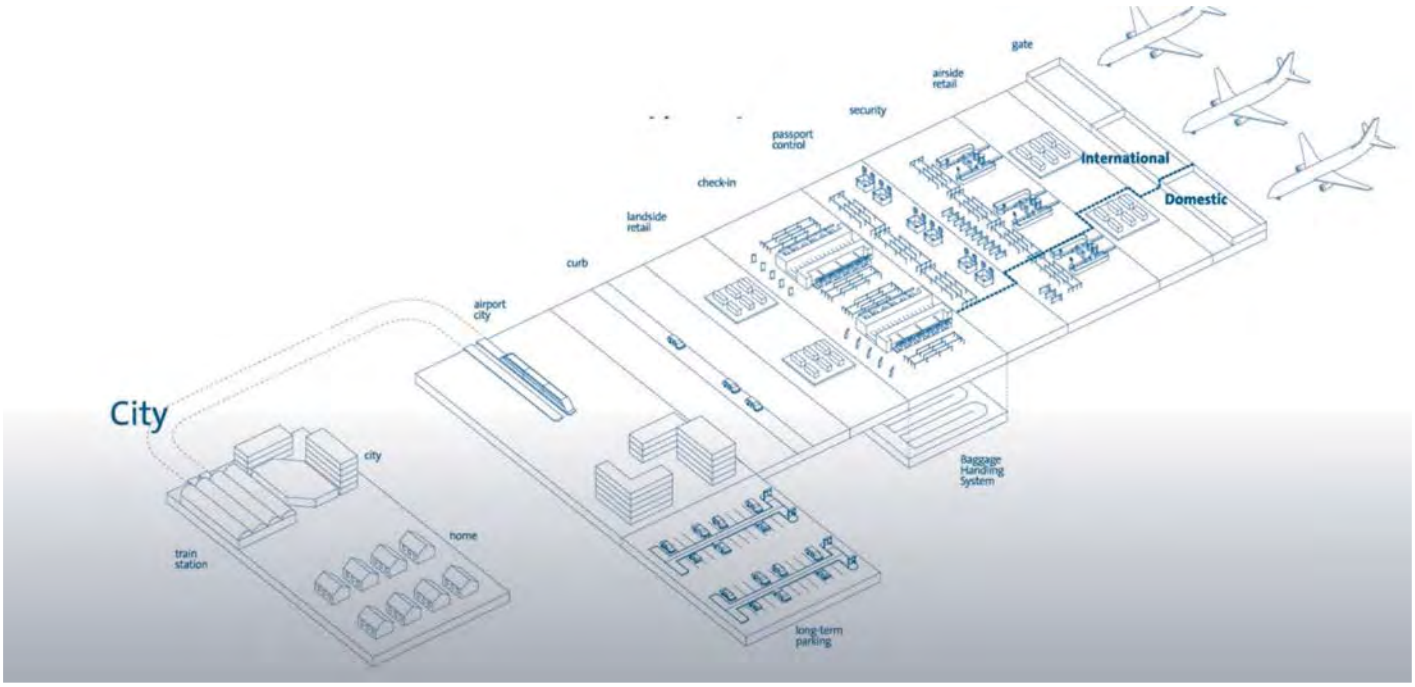


Figure 4-1. Airport overview of several passenger touchpoints in the departure process.

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access a secure building or to use your voice to unlock your car or make a Bluetooth phone call. These concepts are now being expanded in application to airport facilities.

As discussed in Chapter 2, there are a number of use cases worldwide and within the United States. Of particular note for the planning of terminal facilities:

- An example of recent biometric implementations, which have been driven primarily by airlines, is JetBlue's first fully integrated biometric self-boarding gate (see Figure 4-2) for use by international flights. Developed in partnership with CBP, JetBlue started trials on select international flights at Boston Logan International Airport (BOS) and Fort Lauderdale–Hollywood International Airport (FLL) in 2017, before permanent installation began at John F. Kennedy International (JFK) in late 2018.
- Delta Air Lines' biometric Terminal F at Hartsfield–Jackson Atlanta International Airport (ATL): In partnership with CBP, this curb-to-gate program uses facial recognition for all



Source: JetBlue 2017.

Figure 4-2. JetBlue self-boarding process.

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stages of the departure process, from check-in to security access to flight boarding, eliminating the need for passport or boarding-pass verification.
- Alaska Airlines' and Delta Air Lines' fingerprint access to airline lounges: For those members who opt in, both airlines partner with CLEAR and use fingerprints as a biometric token to grant access to their full network of domestic U.S. airline lounges.

These individual implementations of biometrics have proven beneficial to passengers, airlines, and airports and have resulted in reductions in queuing, space required, staffing, and human error. To achieve maximum impact, however, it is important to think about the end-to-end journey. This may provide for the opportunity to reduce disparate terminal processes as well as improve the integrity of operations and the passenger experience. To facilitate this biometric journey, it would be ideal for solutions to be developed with an awareness of trends and standards addressed in Chapter 2 of this primer.

As with most technological advances that change the functionality of a process or task, biometrics introduce new considerations for the airport facility. Some examples include the need to understand how to accommodate opt-out passengers, passengers who might require additional support or screening, and the premium passenger experience. Accommodating and troubleshooting passenger match errors, system errors, equipment failure, or other technology down-time scenarios are other new challenges that must be considered as part of the implementation. Privacy concerns also need to be addressed early and incorporated into every aspect of the design and implementation of biometric solutions.

It is important to note that airports are different in terms of size, airline market shares, passenger demographics, and the role each plays in the broader aviation system. For example, while the U.S. Department of State (DOS) reports that over 44% of Americans hold a passport, which is a key document for enabling biometrics without additional enrollment requirements, the rates of passport ownership are not consistent from state to state. High-population-density states like California and New Jersey have high rates of passport ownership, while states like Alabama and West Virginia have much lower rates. There is not a one-size-fits-all solution for biometrics implementation in every airport, but this primer is intended to inform airport decision makers on how the use of biometrics is evolving inside and outside the airport landscape and the associated impacts and benefits of biometric implementation at each stage in the passenger and employee journey. Important considerations for evaluating biometrics will be reviewed and used to evaluate potential biometric solutions in individual terminal processes along the passenger journey.

Considerations for Evaluating Biometrics in the Airport Environment

As with most emerging technologies, implementation of biometrics is a moving target that is constantly evolving and changing. In some airports, biometrics have been deployed or trialed in piecemeal implementations by a diverse set of stakeholders, including CBP, airlines, and ground transportation providers. This reality can make it challenging to consider biometrics in a terminal operational solution because biometrics affect critical design decisions and may quickly be rendered obsolete if implementation is not planned properly. To better understand whether biometrics should be considered, airport decision makers may ask the following questions:

- In what parts of the passenger or employee journey should the use of biometrics be considered?

Key Takeaway

Biometrics are an evolving and changing technology, so it is important to think about biometric implementations at each terminal process from varying perspectives, including space, security, staffing, and capacity, to evaluate the appropriate use case.

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- What are the potential benefits to the airport and associated stakeholders to implementing biometric technology?
- Given the potential costs associated with making enhancements through the use of biometrics, where will they have the greatest impact for the least amount of investment?
- How are biometrics successfully implemented in an airport environment?

Since every airport is physically, geographically, and demographically different, it is important to think about biometric implementation at each terminal process in the passenger journey from several perspectives to evaluate the appropriate business case. The following questions have been developed to serve as a checklist when evaluating the business case for a biometric solution.

- Who are the stakeholders that are involved with this terminal process? Are they willing to participate with airport management to make an investment of time and attention to enable the improvement to be realized or even piloted?
- What are the space implications for a biometric solution? Consideration should be given to the type of furniture and equipment required as well as wayfinding to account for added passenger segmentation, such as non-enrollers, premium passengers, and mobility-impaired passengers.
- Does the biometric solution reduce staffing expenditures and, if so, for which stakeholders?
- Are there customer service benefits to the biometric solution, such as increased airside dwell time or fewer touchpoints?
- Can a biometric solution improve the security or reliability of certain processes?
- Does the biometric solution increase the capacity of the terminal to accommodate more passengers at improved levels of service?
- Are the benefits of biometric solutions offset by the associated costs that are allocated to the affected airport stakeholder?
- Are there governmental, regulatory, or other external considerations that necessitate a biometric implementation?

This primer explores the opportunities and implications of implementing biometric technology at each major terminal process in the passenger journey to provide a comprehensive understanding around these considerations.

Key Takeaway

Biometrics can play a role in almost all terminal processes, and in many cases, already does. This section identifies the stakeholders, space and capacity implications, and other benefits of biometric implementations at each terminal process in the passenger journey.

Applying Biometrics in Terminal Processes

The airport terminal contains several functional areas that support departing and arriving passengers in their journeys as well as the employees that work at the airport. The functional areas evaluated for biometrics in this primer are check-in, health screening verification, security checkpoint access, employee access and screening, airline lounges, concessions and retail spaces, self-boarding, immigration and customs, baggage claim, and ground transportation facilities and modes. The applicability of biometrics for each of these functional areas is summarized in Table 4-1.

Biometrics for Airline Check-In

The airline check-in process can be disjointed since some parts of the process may occur remotely from the airport (e.g., at home or work) while others may occur inside the terminal. Additionally, unlike many terminal functions, check-in differs by airline and passenger profile (e.g., need to check a bag, obtain a boarding pass). The four primary check-in processes can be

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Table 4-1. Considerations for processing of passengers and employees.

Terminal Process	Can Biometrics Play a Role?	How Are Biometrics Currently Accounted for or Potentially Accounted for?	Stakeholders	Space and Capacity Benefits	Other Benefits (e.g., staffing, security, customer service)
Check-in	Yes, now	Two models are being employed: • Remote biometric enrollment – passenger arrives at the airport ready to fly as the biometric token is already created and is recognized at each check-in stage • On-airport biometric enrollment – passenger has not enrolled in the biometric seamless journey in advance so enrollment must occur at kiosk or other check-in touchpoint	Passengers, airport, airlines, government agencies (e.g., CBP)	• Remote biometric enrollment reduces requirements for check-in facilities in the lobby by shortening the processing times. • On-airport biometric enrollment does not reduce processing time at enrollment touchpoint but does at all check-in touchpoints downstream. • Spatial implications have to be analyzed case by case based on passenger mix and check-in channel distributions.	• Reduces airline staff, but reductions may be higher in some cases than others • Enhanced passenger experience • Shorter processing times at the airport (in most cases) • Enhanced security with more accurate ID matching • Postponement of capital investment in capacity enhancement
Health screening verification	Yes, future for integration with biometrics	• Programs in development to collect health information on passengers, with potential to leverage biometrics • VeriFLY (see Case Study 2 in Chapter 2) passengers enroll their facial biometric and confirm their health status to receive personalized processing; potential to connect to passenger health records • Airports and airlines providing medical testing resources in airport facilities	Passenger, airport, airlines, government agencies, medical labs	Smaller facilities dedicated to health screening are required if passengers are prescreened and approved.	• Reduced burden for health screening to occur on airport • Enhanced passenger experience • Reduced curb-to-gate time for passengers who do not require manual health check to be performed • Enhanced verification process and increased accuracy

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Table 4-1. (Continued).

Terminal Process	Can Biometrics Play a Role?	How Are Biometrics Currently Accounted for or Potentially Accounted for?	Stakeholders	Space and Capacity Benefits	Other Benefits (e.g., staffing, security, customer service)
Security checkpoint access	Yes, now	• Biometric token used to verify ID at travel document checker, prior to entering screening lane, as modeled at ATL Terminal F • Pre-enrollment programs such as CLEAR and VeriFLY provide expedited access through checkpoint queue. • Checkpoint access could be incorporated in the seamless journey using single biometric token.	Passengers, airport, government agencies (CBP and TSA)	• Processing times will be reduced, resulting in fewer access points required, but flow will still be metered by the checkpoint lanes themselves. • Queue area could potentially be reduced.	• Reduced security staffing • Enhanced passenger experience (seamless and touchless) • Enhanced security with more accurate ID matching
Employee access/ screening	Yes, now	• Biometric access control is already in place through SIDA badging. • Programs like VeriFLY expand employee screening facilities to members of traveling party using biometric token.	Airport, employees, DAC vendor services	Negligible, though some space savings possible if programs like VeriFLY can divert some traveling employees to other facilities	• Additional staff reduction if biometric access is extended to additional access points • Enhanced security with more accurate ID matching
Airline lounges	Yes, now	• Airlines (Delta, Alaska) partner with CLEAR to use fingerprint biometric to validate lounge access. • Third-party operators like American Express are getting closer to biometric implementation by using QR code but still require physical ID check. • Lounge access could be incorporated in the seamless journey using single biometric token.	Airlines or third-party lounge operators, passengers	Negligible	• Reduces airline or third-party lounge staff • Faster check-in process, no need to provide membership cards and IDs • Enhanced verification process and increased accuracy

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Concessions/ retail	Yes, future for integration with biometrics	• Amazon Just Walk Out technology is shifting from supermarket application to airport application. • Payments and remote ordering or delivery-to-gate apps can be expanded to include biometric token. • Lines blurred between concession areas and holdrooms	Airports, passengers, concessionaires	• Ability to create more open-space concessions not bound by physical barriers • More shared common space between airside terminal functions could reduce overall space needs.	• Reduces concessionaire staffing • Seamless shopping without having to queue at checkout; fewer touchpoints • Better tracking of food/retail purchase trends • Push coupons or advertising based on traveler behavior
Self-boarding	Yes, now	• Biometric boarding currently for international flights with both U.S. and foreign-flag airlines • E-gate or camera-on-a-stick approach to validating credentials • Location of gate podium, security around boarding door, and electrical connections require consideration.	Airports, airlines, passengers, government agencies (CBP)	• Could require more space than the traditional boarding to account for exception processing and mobility considerations as well as barriers to prevent passengers from bypassing. • Faster boarding process could reduce space required for queue (9 minutes saved on wide-body boarding).	• Reduces airline staff required for a flight departure, allowing them to focus on customer-service issues • Faster boarding times allow passengers to spend more time relaxing or spending money in the terminal. • Turning aircraft more quickly could allow for gates to be available for additional flights. • Enhanced verification process and security
Immigration and customs	Yes, now	• CBP Simplified Arrivals process uses photographs to match against biometric token to complete passenger inspection. • Traditional static booths can transition to podia and potentially to mobile units. • Trusted traveler programs such as Global Entry and NEXUS process low-risk travelers using biometrics.	Airports, airlines, governmental agencies (CBP), passengers	• Shift from kiosks and counters to podia and mobile units will decrease overall size of immigration halls. • Increased capacity expected since CBP can clear arriving flights an average of nearly 12 minutes faster than through using traditional processing methods. • Secondary screening area for customs is still required.	• Staff spend less time on low-risk passengers and more time on higher-risk ones; could present staff savings. • Faster processing time for passengers, especially those who are low risk • Enhanced passenger experience (seamless and touchless) • Standardized process with increased identity assurance and accuracy that reduces the imposter threat

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Table 4-1. (Continued).

Terminal Process	Can Biometrics Play a Role?	How Are Biometrics Currently Accounted for or Potentially Accounted for?	Stakeholders	Space and Capacity Benefits	Other Benefits (e.g., staffing, security, customer service)
Baggage claim	Maybe, but benefits may not outweigh capital costs	• Matching passengers with bags upon arrival has traditionally been a manual process. • A passenger's biometric token would be required to call for the specific bag to be delivered to bag claim, requiring replacement of traditional claim units with more of a baggage storage and recall system. • A less-intensive biometric implementation could simply identify stolen or mismatched bags.	Airport, airlines, passengers	• Incorporating biometrics could require significant physical modifications that may replace the area currently allocated to baggage-claim belts. • Batching of arriving passengers may drive up requirements for bag retrieval points in baggage claim.	• Staffing at bag claim is not significant (bag service counter, tug operator, bag porters), so minimal savings are expected. • Improves the security of the process because it is less likely for passengers to take the wrong bag or for bags to be stolen, but these percentages are likely fairly low.
Ground transportation	Yes, with more integration possible	• Hertz partnership with CLEAR to expedite vehicle rental process trialed • Biometric token could be used to bypass transit ticket purchase or eliminate pay gates in an on-airport parking facility, expanding beyond toll tags.	Airport, passengers, concessionaires (rental cars, parking, transportation network companies, transit operators)	• Smaller footprint for passenger processing in ticket lobbies • Reduced size exit plazas at parking and rental car facilities	• Staff reduction at parking exit plazas, for example • Faster and enhanced (touchless) passenger processing • Enhanced verification process and increased accuracy

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summarized as (1) online/remote check-in, (2) self-service kiosk, (3) self-service bag drop, and (4) full-service counter. Biometrics can streamline this complex, multistep process while providing a faster and more secure experience for passengers (see Figure 4-3).

Currently, there are two high-level scenarios for enabling a biometrics component to improve the check-in process: remote biometric enrollment and on-airport biometric enrollment. Under the remote scenario, biometric check-in begins at home with online check-in. At Delta Air Lines' biometric check-in for international departures from ATL, for example, passengers confirm their passport information remotely during online check-in (see Figure 4-4). This enables the passenger to be ready to fly on arrival at the airport. The passenger can approach a self-service kiosk at the airport and be immediately recognized, so boarding passes and checked baggage tags can be seamlessly obtained without interaction with airline customer service representatives and without the presentation of any documentation.

The biometric token is used again when passengers drop their bags at baggage drop. Under the on-airport scenario, for passengers who did not check in online to confirm their biometric token, the passport can be scanned at the self-service kiosk, and the passengers can continue the biometric check-in experience from that point forward. This process requires a longer transaction time at the self-service kiosk, however, to account for the document verification process. At any step in the check-in journey, passengers can opt out of the biometric check-in process and continue to use the self-service kiosks and counter bag drops using the traditional process. The traditional process at check-in involves physical interaction with airline customer service representatives to verify identification rather than the use of biometrics to obviate the need for such interaction.

While biometric check-in in the United States has predominantly been applied to international flights because of the use of passport data as the biometric check, developments to allow biometrics for domestic flights are underway. Delta Air Lines is trialing curb-to-gate biometrics for domestic passengers at Detroit Metropolitan Wayne County Airport (DTW). Passengers with a passport and Known Traveler Number from enrollment in Global Entry or TSA PreCheck can store these identification documents in their SkyMiles profile and opt in at check-in on the Delta mobile app. Delta is not planning to save the biometric data used. This trial is limited in scope since only domestic passengers who have both a passport and Known Traveler Number can participate.

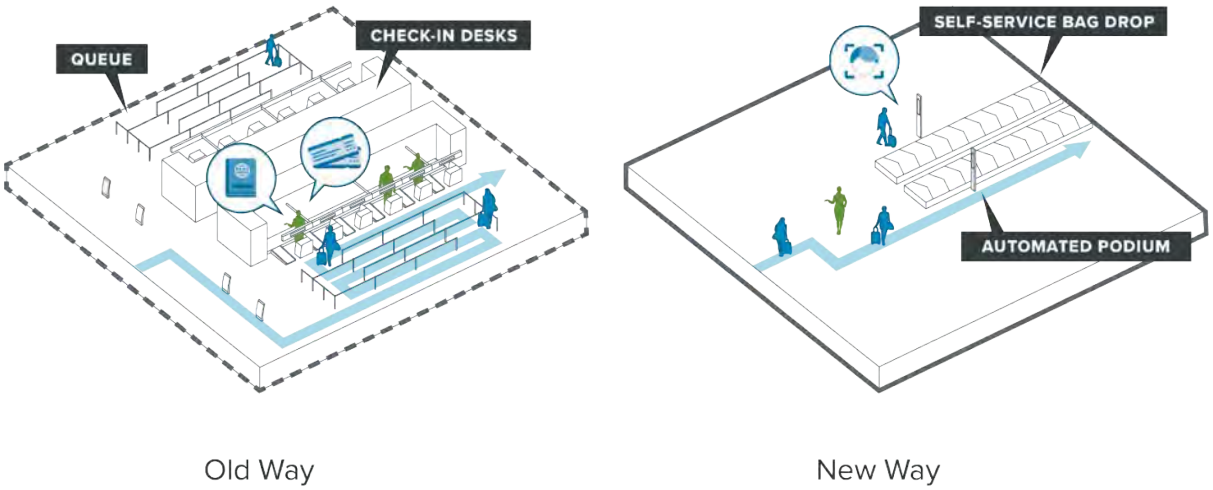


Figure 4-3. Traditional check-in (left) and modernized check-in (right).

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Source: Van Wert 2018.

Figure 4-4. Biometric check-in kiosk at ATL Terminal F.

The use of biometrics for domestic flights on a larger scale without a separate enrollment process would likely require access to additional databases of biometric information (such as state-issued drivers' licenses) to be able to make the facial biometric match. Leveraging drivers' license data could expand the numbers of stakeholders, which currently include passengers, the airport, airlines, and government agencies (e.g., CBP, TSA), to include state agencies as well.

There are three primary benefits that could accrue though the use of biometrics in the check-in process:

1. Biometric use at terminal check-in can reduce the space required for the queuing of passengers. There are benefits to both remote biometric enrollment and on-airport biometric enrollment for check-in, but remote biometric enrollment provides the most benefit to the overall check-in process because it affects more steps in the process. Reduced transaction times at kiosks, bag drops, and counters should reduce the overall space requirements to accommodate these processors in the check-in lobby. Even if passengers need to register their biometric on-airport at the kiosk, the remaining downstream processors can benefit from reduced transaction times. Regardless of the scenario, the spatial requirement of the check-in hall should decrease, which can postpone costly capital expansion or reconfiguration of facilities.
2. Biometric use at terminal check-in can reduce the number of staff in the check-in area. From a staffing perspective, biometrics in the check-in lobby should reduce airline staffing requirements. While some airline agents would still be required to roam the lobby and assist passengers who require exception handling, the number of passengers requiring direct agent interaction should be reduced. For passengers, the shorter processing times at the airport provide for an enhanced customer experience and reduced stress. In severely congested terminals, airports may also be able to reduce operational staff that are deployed to manage overflowing passenger queues.
3. Reducing the reliance on human interactions in the form of ID checks can create enhanced security for all users by providing for more accurate identification matching.

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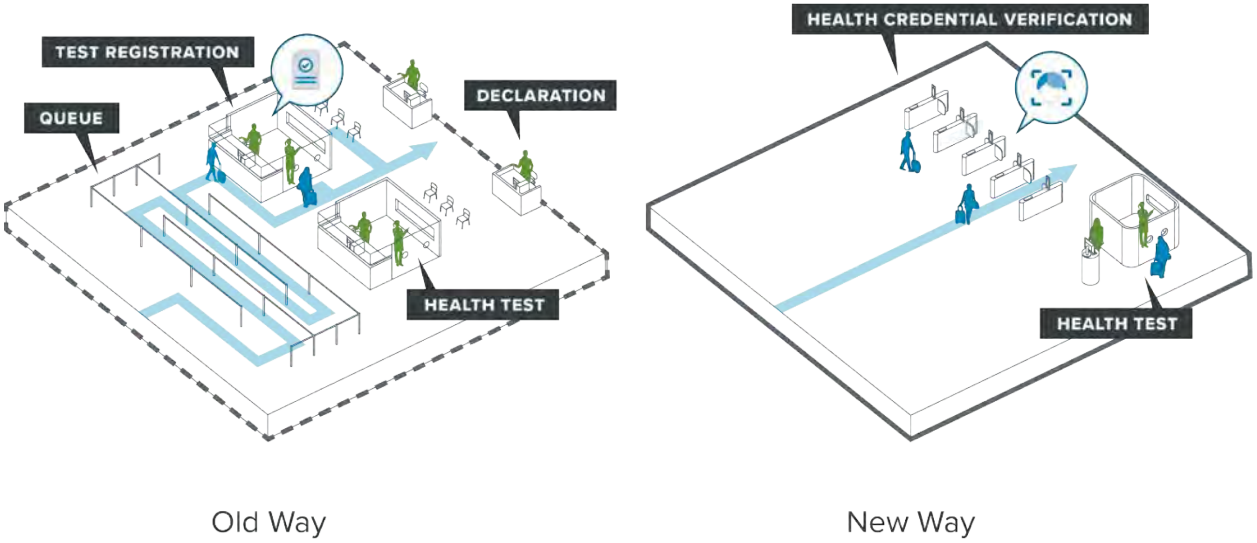


Figure 4-5. Health screening touchpoint that may leverage biometrics.

Biometrics and Health Screening

Health screening has received heightened attention given the COVID-19 pandemic, and biometrics can play a role in this emerging function. While solutions are in early development, health screening now plays a role in the passenger travel experience. Currently, most U.S. airlines ask passengers to self-certify their health status during the check-in process. This process involves answering a few questions on the airline mobile phone application or at the check-in counter. While not time-intensive, this self-certification does take longer and depends, without external verification, on the passenger providing accurate and honest responses. As the process of health screening is better understood and more widely adopted, it is reasonable to assume that biometrics will eventually contribute to the process (see Figure 4-5).

Programs such as VeriFLY at Denver International Airport (see Figure 4-6) have taken the first step of identifying passengers who are willing to enroll. The program uses their facial



Source: Denver International Airport 2020.

Figure 4-6. VeriFLY e-gate access to security queue.

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biometric template and enables confirmation of their health status on the day of travel to receive security screening and concourse train access isolated from the general traveling public.

Future capabilities of VeriFLY could include a connection to a passenger's health records to confirm testing certifications or vaccinations required to travel to certain destinations. Airports and airlines are also beginning to provide medical testing resources as well. As these solutions continue to evolve, biometrics could be used to link these results to passengers' profiles to allow them access to security or to board the aircraft.

Benefits in the area of staffing levels or terminal space are difficult to estimate because they would vary on the specifics of the implementation and the unique circumstances, but it is anticipated that biometrics may reduce the burden for health screening to occur on the airport since passenger records could be made available more easily through a biometric token. One potential benefit in times of severe outbreaks would be the ability to determine a passenger's identity through technological interfaces rather than through interaction with airport or airline personnel.

Biometrics and Security Checkpoint Access

Biometrics are currently being used to enable priority security checkpoint access by low-risk passengers to enhance the overall customer experience. Biometrics are deployed prior to the security checkpoint to verify a passenger's identification and acceptance into security screening (see Figure 4-7). In some cases, programs like CLEAR or VeriFLY go a step further and allow enrolled low-risk passengers to use their biometric token to bypass the queue and reach the checkpoint faster than other passengers. As with upstream processes such as check-in, biometric identification of passengers that are not part of a membership program such as CLEAR is currently limited to international departures and domestic trusted travelers. Expansion to additional domestic passengers is under development and is a TSA goal, as stated in its September 2018 *TSA Biometrics Roadmap: For Aviation Security and the Passenger Experience* (TSA 2018b). TSA and CBP have developed proofs of concept to explore expanded use of the TVS at the security checkpoint through data integration between the TVS and TSA Secure Flight. Additionally, in September 2018, TSA began collecting photos of travelers who renewed in person or were enrolling in TSA PreCheck for the first time. As an initial step, in 2020,

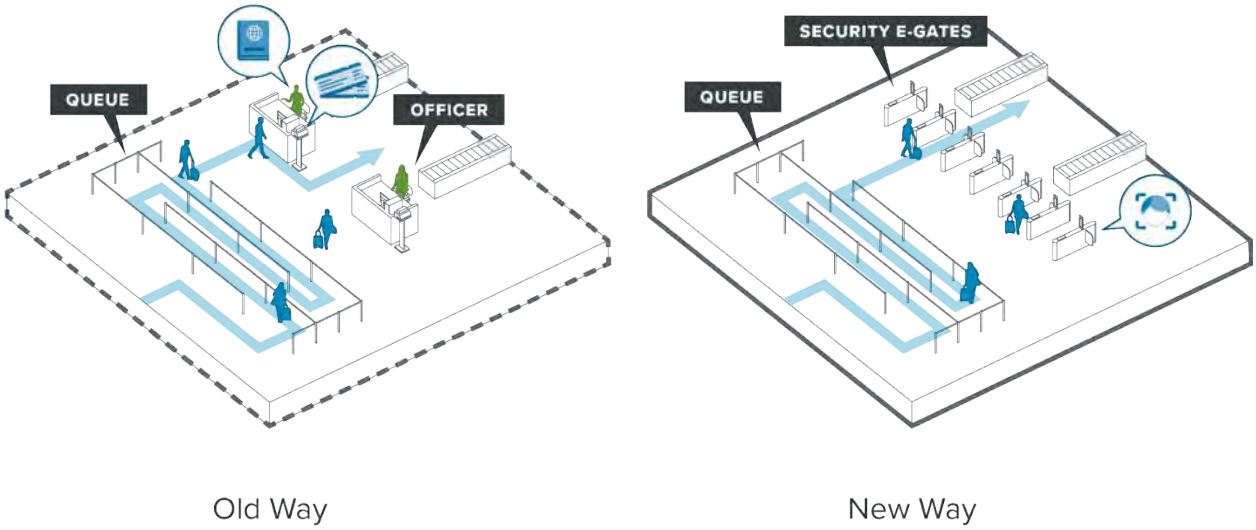


Figure 4-7. Security checkpoint touchpoint that may leverage biometrics.

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TSA piloted a touchless self-service technology at the checkpoint to match a traveler's real-time photo with the photo on their government-issued identification. These photographs are not being stored and are only used for identity verification.

Expanding to a larger pool of domestic passengers without requiring a separate enrollment process requires additional coordination and access to a biometric database larger than federally issued passports, such as state-issued drivers' licenses. Notably, the emergence of REAL ID, on its own, does not create a centralized database that enables the ability to match a person to their biometric data; rather, the purpose of REAL ID is to make identity documents more consistent and secure.

Stakeholders involved in the use of biometrics at security screening include the airport, passengers, TSA, and any governmental agencies that may be required to verify identification for entry/exit purposes (e.g., CBP). The benefits of deploying biometrics at the entry to security are primarily security driven and include:

- There is enhanced security and more accurate identification matching from biometrics than from manual review of a photo ID.
- Biometrics also reduce touchpoints to create a more seamless process and one that should verify a passenger's identity in less time.
- The number of staff required to verify passenger identification and documents may also be reduced because of the increased throughput. The increased throughput generated from reduced transaction times at ID check, however, is still metered by the capacity of the security checkpoint lanes themselves, so close coordination is required to ensure that the screening lanes are providing sufficient capacity to avoid queues between ID check and the lanes themselves. As a result, there may be limited space savings in the terminal.

Biometrics and Employee Access and Screening

Biometrics are already in place for employees working in an airport environment. All employees who are allowed access to the secured areas of an airport must be approved for a SIDA badge. While these badges are issued by the airport and require employees to meet certain TSA requirements, additional identification verification is required to prove the person's identity. In addition, fingerprints are taken as part of background checks.

Telos ID and the American Association of Airport Executives (AAAE), through the Transportation Security Clearinghouse, handle the biometric background checks for aviation workers at many airports. As of 2015, AAAE had processed more than 15.6 million fingerprint-based criminal history records and 6.8 million STAs (AAAE n.d.).

Doors and access points in the airport environment are controlled in several ways, depending on access purpose. Some access points, such as passenger boarding bridges, may require a badge scan and a multi-digit access code to allow for non-domicile airline crew to access. Others, however, may require a badge scan plus fingerprints or iris scan to gain entry, especially between non-secure and secured areas of the airport.

Given that airports are accustomed to rolling out access control for employees, operators may wish to trial equipment and processes associated with biometrics for enhanced employee access points. For one, the privacy concerns of passengers are different from those associated with employees who undergo background checks and documentation for the purposes of obtaining an airport badge. Secondly, airport operators can observe the performance of the equipment and the enhanced process to address (1) whether the enhanced process is faster, and if so, how much faster, (2) what the equipment's performance is in terms of maintenance, (3) what the backup process is when equipment fails, and (4) what process exceptions must be addressed. Finally,

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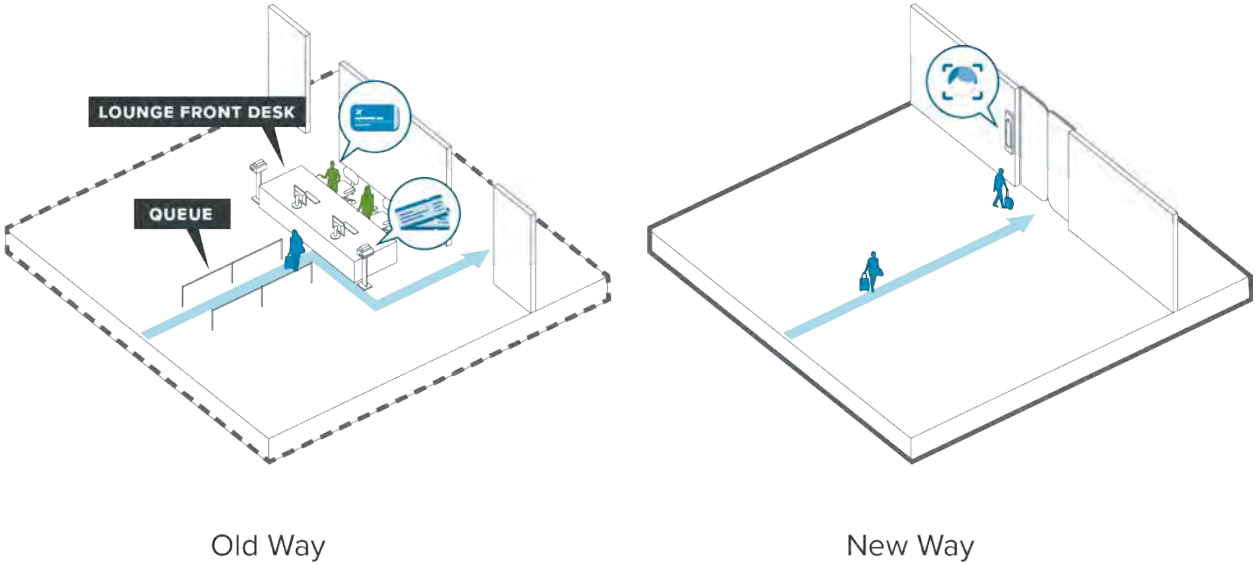


Figure 4-8. Airline lounge touchpoint that may leverage biometrics.

if airports are to gather biometric data beyond that already on file, airports will have a trial experience for a smaller population with respect to how much effort will go into enrollment, in terms of labor, equipment, and space requirements.

Biometrics and Airline Club Lounges

Biometrics have long been used by airlines to provide access to a smaller subset of passengers who use airline club lounges (see Figure 4-8). Starting in 2014, Alaska Airlines trialed opt-in fingerprint scanning to provide access to passengers at the airline's four Board Room airport lounges. In mid-2017, Delta Air Lines introduced biometric check-in for Delta Sky Club guests with CLEAR memberships at its Ronald Reagan Washington National (DCA) Sky Club (see Figure 4-9). The biometric check-in option rolled out to all 50 U.S. Sky Club locations in



Source: Future Travel Experience 2018.

Figure 4-9. Delta Sky Club fingerprint access scanner.

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early 2018 and is free for Delta Sky Club members who are U.S. citizens or permanent residents (those who are not already paying members of CLEAR). Other lounge providers have taken steps toward biometric access. These steps include providing QR codes as part of the access credential; however, a positive ID check is currently still required. Lounge access could eventually be incorporated into the biometric seamless journey, where one biometric token is used to verify access, including for lounge entry.

Lounges are unique compared to other terminal functional areas in that they affect a smaller number of passengers and are largely controlled by an airline or third-party operator. The stakeholders involved in the biometric deployment are primarily passengers, airlines, or third-party lounge operators. Unlike in other terminal functional areas, biometrics applied to the airline lounge environment are not likely to reduce the space required. There may be a staff savings, but this would only apply to the airline or lounge operator.

Passengers benefit from the use of biometrics at lounge check-in. The process is faster and does not require the passenger to produce an ID or membership card. Additionally, this provides an enhanced verification process and reduces any potential membership errors.

Biometrics and Airport Concessions and Retail

Biometrics have not yet been integrated into the airport concessions and retail environment and do not currently play a large role in this area. While self-checkout and app delivery services have been introduced in the airport environment, these services have not yet been tied to a biometric token. In Schiphol Plaza, located at Amsterdam Schiphol Airport (AMS), Dutch supermarket chain Albert Heijn trialed a digital store accessed by customers with their contactless debit card.

A future step could involve adding a biometric token to increase security and further expand mobile payments. Additionally, technologies, such as that used in Amazon Go convenience stores, are being made available for adaptation to other environments. For example, the Just Walk Out technology could shift from a supermarket application to an airport retail environment. As with Amazon Go stores, access could be controlled at an e-gate to restrict those travelers who do not have the biometric token (see Figure 4-10).

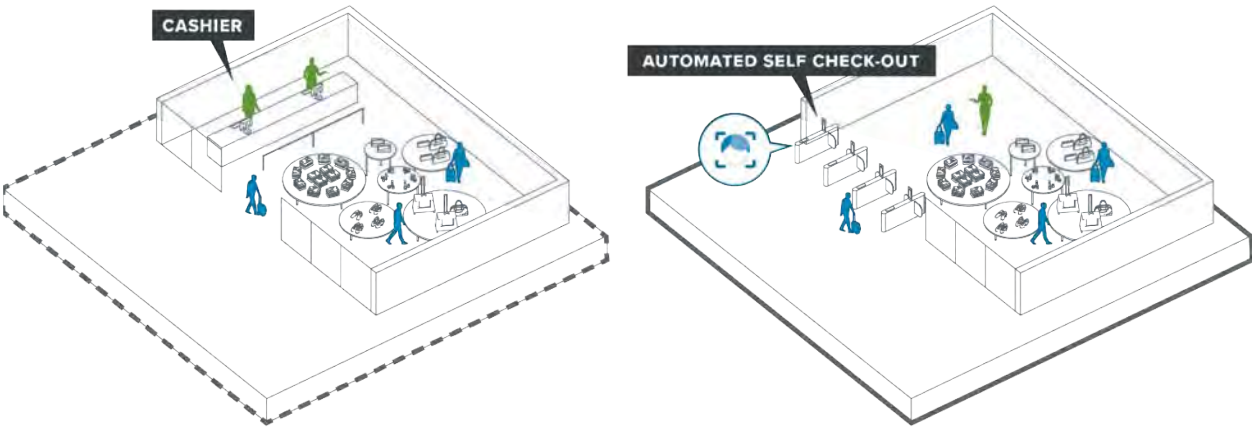


Figure 4-10. Concession and retail stores that may leverage biometrics.

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Additionally, third-party programs such as CLEAR have been using biometrics at concession stands in baseball and football stadiums. In Seattle, for example, fans can link their driver's license and a credit card to their CLEAR account and use their fingerprint to purchase food and drinks. At Citi Field in New York, concession items are placed on a self-checkout unit where they are scanned and charged to the CLEAR account associated with the fingerprint.

The primary stakeholders for integrating biometrics into concession and retail areas are passengers, concessionaires, and the airport.

Removing traditional cashier positions and payment systems provides the ability to break down many of the physical barriers separating concession spaces from other functional areas like the holdroom to create a more open-space and open-flow concept. Biometric payment systems would also reduce the staffing needs of the concessionaire and allow employees to focus their time assisting customers. As has been seen in Amazon Go convenience stores, a seamless, touchless shopping experience reduces queues, reduces stress, and increases the time passengers have available to spend money before departure.

Biometrics and Self-Boarding

Self-boarding has been using biometrics for several years, and its deployment may be accelerated. In recent years, biometrics have been applied to the aircraft boarding process with a focus on international departing flights. American Airlines, Delta Air Lines, and JetBlue are among the U.S.-based airlines using biometric boarding for international departures. In addition, there has been participation by foreign-flag carriers such as British Airways and Lufthansa. There are two primary biometric boarding systems being used at U.S. airports: (1) e-gates and (2) camera-on-a-stick.

In both models, passengers walk up to a camera, where their photo is compared against the biometric templates of other passengers on the flight. No passport or boarding pass is required. Once a match is made, either the e-gate opens, or an airline gate agent allows the passenger to board (see Figure 4-11). If a passenger chooses to opt out or if there is an error in the matching, the passenger must see an airline gate agent for manual processing of the boarding pass and passport.

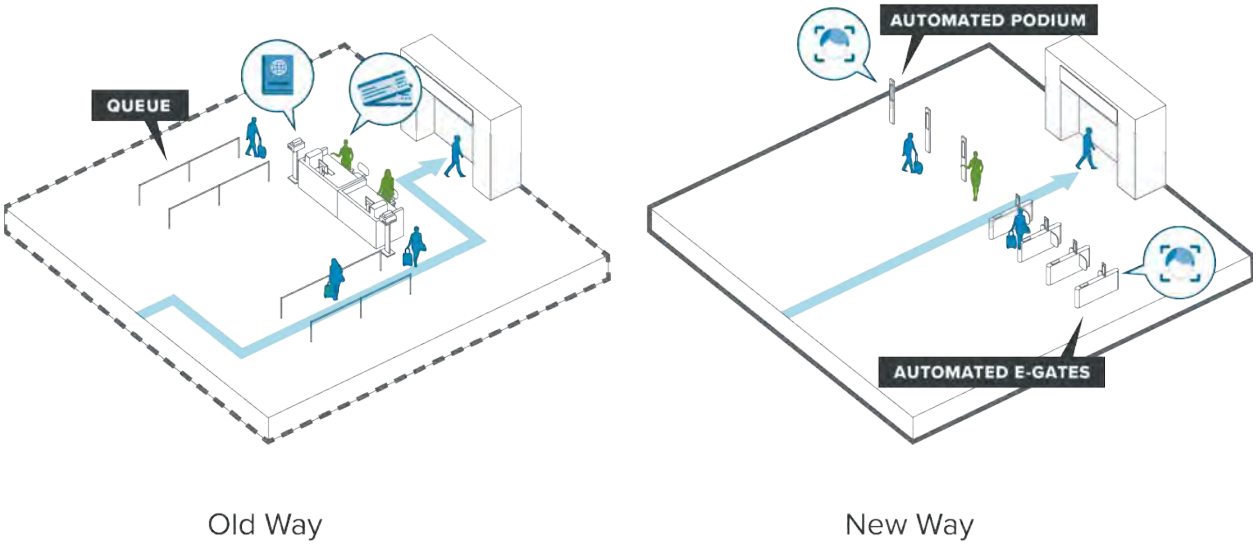


Figure 4-11. Boarding gates may leverage biometrics.

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It is important to understand these two self-boarding systems as they relate to exclusive-use and common-use terminal systems. Many airlines prefer to use their own equipment and have exclusive connections to their departure control systems (DCSs), limiting access to other airlines. Common-use facilities require connections to multiple DCSs and must accommodate different airline operating procedures. For the e-gates installed at Los Angeles International Airport (LAX) Tom Bradley International Terminal (see Figure 4-12), a common-use facility, airlines are enabling use of the system on an individual basis. While installation of the e-gates was funded by the airport authority as part of a terminal capital project, planned recovery of full projects costs would occur through landing fees, terminal rates and charges, and other non-aeronautical revenues.

As a result of the extra equipment required to accommodate biometric boarding, the location of the customer service podium may need to be relocated to ensure that there is sufficient clearance around the boarding door for the camera podium and possibly the e-gate. Electrical connections are also required in this area, although some may exist already to accommodate the boarding-pass scanner and agent computer. If e-gates are used as the preferred process, additional factors require consideration, including providing an Americans with Disabilities Act–compliant lane as well as barriers to prevent passengers from bypassing the e-gate.

Participation by several stakeholders is required to successfully implement biometric boarding: the airport, airlines, passengers, and CBP in the case of international departing flights.

There are three primary benefits that could accrue through the use of biometrics at boarding:

- Biometric boarding can improve the passenger experience and enhance security. Passengers are provided with a touchless experience without the need to present identification or boarding passes. There is also the added level of security and safety in ensuring that the correct passengers are boarding the aircraft.
- Biometric boarding can help airports with constrained gate facilities. According to initial feedback from Delta Air Lines, biometric boarding reduced each passenger's boarding time by 2 seconds, which resulted in 9 minutes of savings for a wide-body aircraft. For Lufthansa, its trials showed that 350 passengers could be boarded onto an Airbus A380 in 20 minutes. Reducing aircraft ground time and gate occupancy time could allow for additional flights to be accommodated at airports with constrained facilities.
- Biometric boarding can reduce holdroom congestion. Faster aircraft boarding times could reduce the space required for passenger queuing inside the holdroom, allowing that space to be used for seating or other purposes.



Source: Vision-Box Pressroom 2018.

Figure 4-12. Self-boarding lanes at LAX TBIT.

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Biometrics and Immigration and Customs

CBP has been a recent model for biometric adaptation through Simplified Arrival and Trusted Traveler programs (see Figure 4-13). At the time of this writing, over 50 million passengers have been processed on entry to the United States through what CBP refers to as the "Simplified Arrival" process. Using an airline manifest and a cloud-based matching service, CBP retrieves existing traveler images from government records and builds a photo gallery of expected travelers prior to a flight's arrival. As arriving passengers approach the primary inspection booth, CBP takes a photo of each traveler and compares it against the pre-assembled gallery to find a match. With a match, the CBP officer can complete the inspection in a few seconds. If a match cannot be made, the CBP officer reverts to the original process and manually inspects the passenger's travel documents.

Biometrics are also currently implemented as part of Global Entry, NEXUS, and other CBP trusted traveler programs as a way of processing low-risk travelers to the United States. Recent updates to Global Entry have created a more streamlined process where, in addition to shifting from fingerprint to facial biometrics, the typical customs-related questions put to passengers have been eliminated.

The primary stakeholders for biometric implementation at immigration and customs are the airport, airlines, passengers, and CBP. There are three primary benefits that could accrue though CBP's use of biometrics:

- Integrating biometrics into the immigration process will reduce the spatial needs within the terminal. Large, fixed immigration booths organized in a linear fashion can be replaced by smaller podia and eventually by mobile units. These touchpoints can shrink in size over time since less of the passenger processing will need to occur at the booth or podia.
- Biometrics increase throughput capacity and reduce passenger processing time. Under Simplified Arrival, CBP can clear arriving flights an average of nearly 12 minutes faster than using the traditional manual processing methods. In addition to the savings generated by Simplified Arrival, CBP says the new Global Entry kiosks have reduced the processing time of eligible passengers by nearly 90%, from 45 seconds to less than 6 seconds. Passengers benefit from faster processing times and an increased touchless experience.

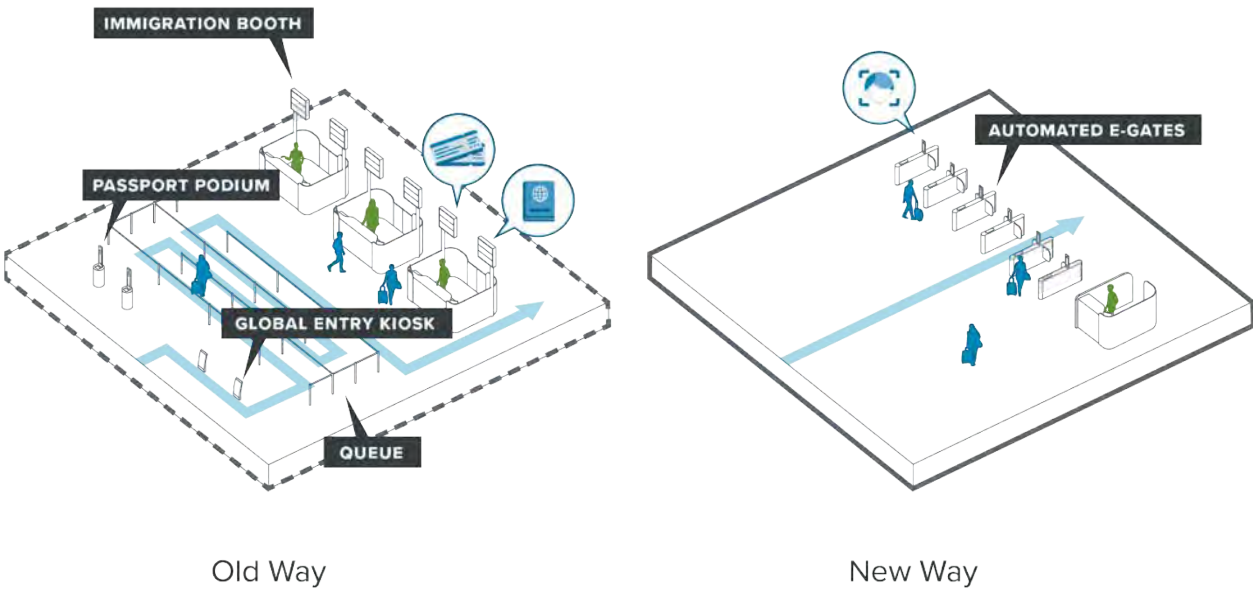


Figure 4-13. CBP leveraging biometrics.

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- CBP offices can spend more time on higher-risk passengers. Shifting away from the longer manual process allows CBP officers to refocus their time from low-risk passengers to higher-risk passengers. This streamlines the passport check process and provides a safety and security benefit because there is reduced risk of the imposter threat and increased identity assurance. These efficiencies may provide a staffing benefit to CBP as it can be more efficient in its resourcing.

The model CBP used for biometrics could serve as a basis for identity verification for TSA. Several airports already have systems deployed to use CBP's model to facilitate touchless identity checks through security screening checkpoints.

Biometrics in Baggage Claim

Biometrics in baggage claim (see Figure 4-14) will require the largest structural modification to the arrival process. Reuniting passengers with their checked baggage has remained a largely manual process, requiring passengers to self-identify their bags from a wide selection on an arriving flight. This honor-system process has traditionally been validated, when necessary, by matching the serial number on the bag tag against the passenger's receipt. For biometrics to be integrated into the baggage-claim process, a passenger's biometric token would be required to call for the specific bag to be delivered. This would likely require the replacement of traditional sloped- or flat-plate baggage-claim units with something resembling more of a back-of-house baggage storage or locker system. Bags would be retrieved one at a time once called for using the passenger's biometric token. While the elimination of baggage-claim belts would free up space in baggage claim, this space would possibly be replaced by the infrastructure required for baggage storage and retrieval systems. Passengers on arriving flights tend to approach baggage claim in a short time distribution, potentially driving up the baggage claim retrieval requirement.

Stakeholders that might consider a biometric implementation at baggage claim are the airport, passengers, and airlines or their ground handlers. Unlike many other functional areas in the airport, incorporating biometrics would require significant physical modifications, so increased coordination among stakeholders would be required to ensure a successful deployment.

Biometrics would improve the security of the baggage retrieval process because there would be fewer opportunities for bags to be stolen or picked up by mistake. Staff savings would not be

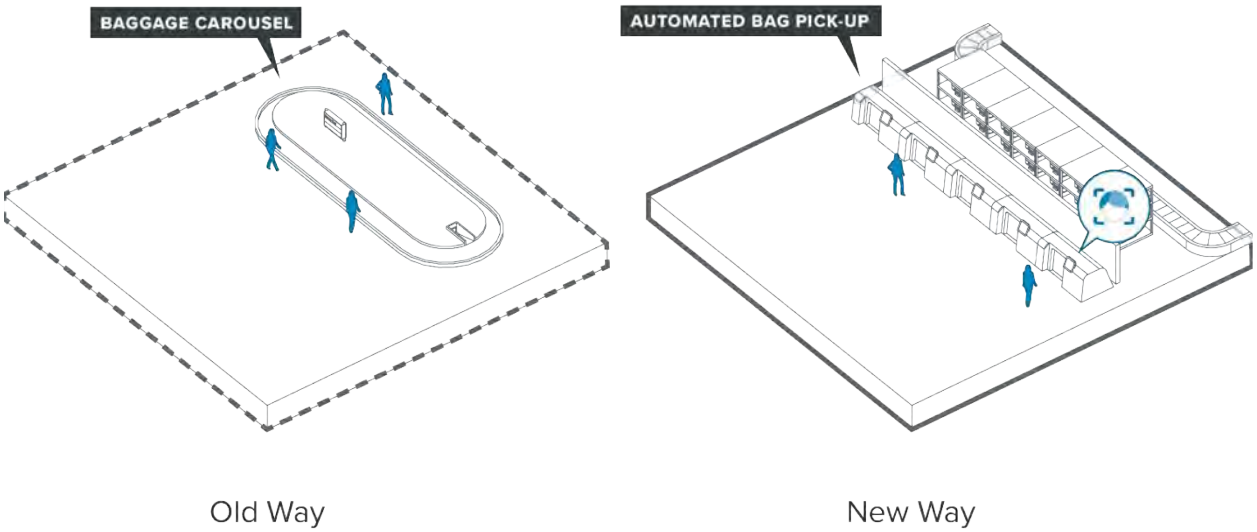


Figure 4-14. Baggage reclaim leveraging biometrics for self-service.

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expected to be significant since the baggage-claim process is generally less staff-intensive than other functional areas. Staff would still be required at the baggage service desk as tug operators and bag porters. Given that the number of bags stolen or lost in baggage claim is relatively low, and limited airline resources are available at baggage claim, the true benefits of implementing a full biometric reclaim process may not outweigh the costs of added security patrols and the airline liability associated with baggage replacement.

A less-intensive implementation of biometrics in the baggage-claim area could focus primarily on identifying bag thefts or mismatched bags as opposed to ensuring that the chain of custody between the bag and the passenger is maintained. Biometric readers and bag-tag scanners installed at the baggage-claim exits could be used to confirm that a bag that is claimed from the belt belongs to its owner. This concept would have significantly less capital cost and require less modification to the facility but would not completely ensure that the passenger receives the correct bag; it would simply identify mismatched or stolen bags. Facility modifications could include creating a controlled access to the baggage-claim area and ensuring that bags are scanned when leaving the baggage-claim area. The cost of the biometric implementation would have to be weighed against the liability incurred with lost or stolen bags.

Biometrics and Ground Transportation

Ground transportation biometrics are being trialed using some modes such as rental cars and rail (see Figure 4-15). Ground transportation is the first and last stage in the airport passenger journey. Residents may use on-airport parking facilities, for example, while visitors may rent a car. Biometrics have had limited use in airport ground transportation. The most prominent example is in the rental car industry. At the end of 2018, Hertz announced a partnership with CLEAR to launch "Hertz Fast Lane powered by CLEAR" (see Figure 4-16), a service that used biometrics to speed up the car rental and exit process to get customers on the road in 30 seconds or less. Customers used their facial biometric to confirm identity instead of showing the exit booth employee a driver's license. This was the first use of biometrics by a major rental car company. The program was expanded to several airports, but in July 2020, due to the Hertz bankruptcy, it was announced that the program would be discontinued.

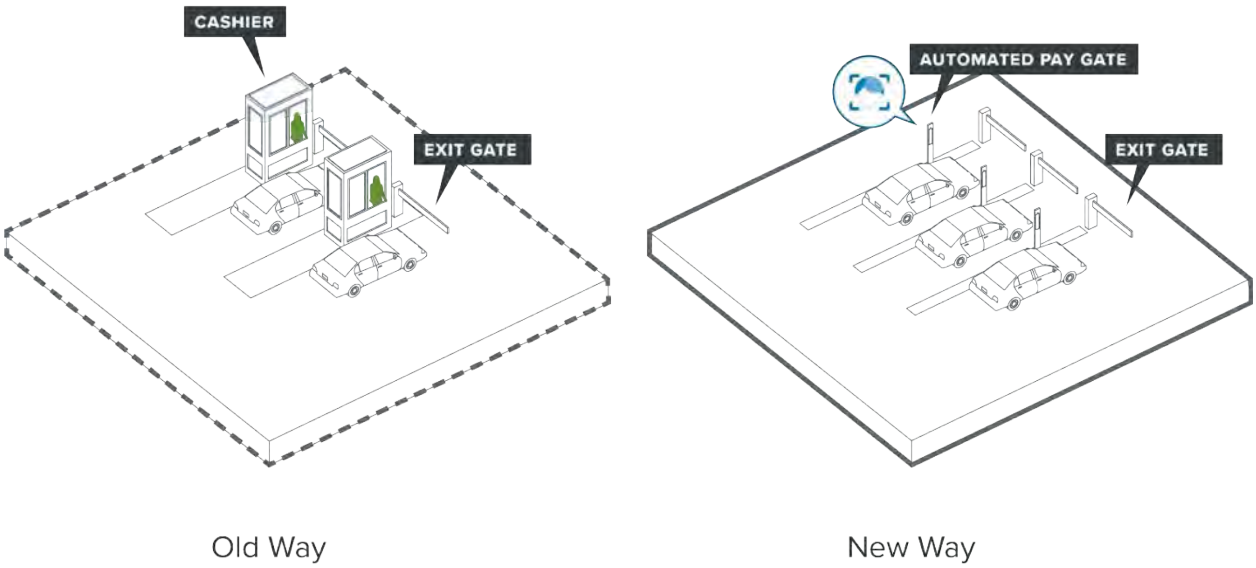
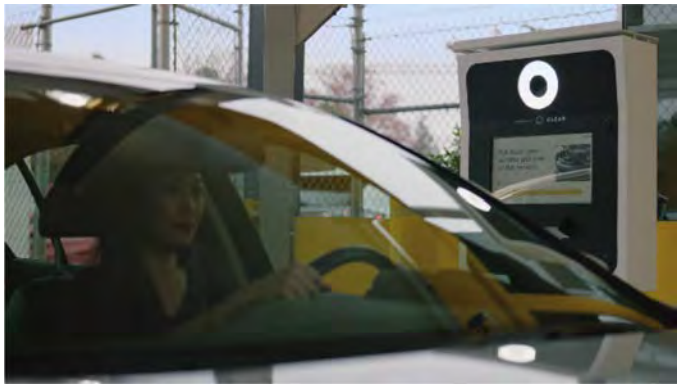


Figure 4-15. Ground transportation touchpoints leveraging biometrics.

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Source: Corum 2019.

Figure 4-16. Hertz Fast Lane exit.

It is anticipated that biometric tokens will soon allow passengers to pay for other ground transportation services such as parking transactions as well as train tickets. Biometric trials for train ticket checks and border exit processes are expected to start in March 2021 on the Eurostar departures from London St. Pancras. Biometrics could be expanded to parking facilities to allow passengers to pay for parking transactions without needing to stop at a pay station or pay gate.

Stakeholders would include passengers, the airport, ground transportation operators and concessionaires (e.g., parking concessionaire, transportation network companies, rental car companies), as well as local transit agencies (e.g., bus, rail).

Applying biometrics to ground transportation modes is not anticipated to provide a significant space savings, although the need for ticket lobbies could be reduced. The primary benefit would be felt by passengers, who would experience a faster, touchless, and more automated experience.

Evaluation of Biometric Implementations in the Airport Environment

Biometrics can be applied to several functional areas in the terminal and throughout the airport, but airport decision makers must decide if there is a strong business case to move forward with a biometric solution and how to best accomplish it. One way to evaluate these questions is to understand where biometrics provide the greatest impact and if there are trade-offs.

Key Takeaway

For airport decision makers, establishing a strong business case for biometrics is likely to extend beyond financial considerations to include customer service, airport technology goals, passenger safety, operational efficiency, and competitive concerns.

As discussed, biometrics should be treated as an emerging technology that is a moving target and constantly evolving. Biometrics are a leading-edge technology in that:

- There is not a complete U.S. database of biometrics since eligible passengers are currently limited to those with passports and Trusted Traveler program participation.
- There is a patchwork of laws that must be navigated governing the collection and use of biometric data, adding to the complexity.
- Biometrics are moving toward multi-stakeholder solutions, with the potential for future use cases to be added over time.

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Since every airport is unique, biometric implementations that work at some airports may not be viable at others. What works at an airport could depend on several factors, including passenger demographics, airport size and geography, and the airport's operational profile (e.g., primarily serving international or domestic traffic). Airports that are capacity constrained in some manner will likely be the early adopters of biometrics given the imperative that any incremental benefit, regardless of magnitude, may offer a meaningful overall improvement. Important factors to be considered are:

- Would passenger or facility security (e.g., identity assurance and accuracy) be enhanced?
- Would a small increase in capacity or level of service be meaningful to the overall experience of the population affected?
- Are there reasons besides capacity enhancement and security to implement biometrics, such as the desire to reduce staffing costs or improve the reliability of certain processes?
- Are the stakeholders required to implement the changes supportive of the goals?

The most mature uses of biometrics at the airport involve (1) airport employees (e.g., access to secured areas), (2) self-boarding of international flights, and (3) the international passenger arrivals process (e.g., CBP's Simplified Arrivals); these models are the most mature in that they have proven to provide greater benefits than other uses have. Many of these successful implementations have also been driven by airport stakeholders such as airlines and CBP rather than the airport operator. Accordingly, management seeking to implement changes at airport facilities may place a greater emphasis on these areas to derive the greatest benefits. Additionally, some airport operators may choose to simply participate in biometrics implementation rather than drive implementation because CBP, TSA, and the airlines control many of the facilities in the passenger terminal that would be affected by biometrics, and these stakeholders can access passenger biometric data via CBP/TVS and build an enrolled population of passengers more easily than airports can.

In most cases, the ROI for the airport operator goes beyond strictly financial considerations for the implementation of biometric solutions. Instead of making decisions based solely on the financial ROI, airports are evaluating several additional considerations, such as customer service, airport technology goals, passenger safety, operational efficiency, and competitive concerns. To better understand these considerations, airport operators may consider the questions discussed in the following when evaluating implementation of a biometrics solution.

Do the passengers using your airport have a high rate of passport ownership or Trusted Traveler program membership? Airports with high passport and Trusted Traveler program participation rates provide a significant population of passengers immediately eligible to participate in biometric programs. Further, no additional enrollment process is required for these passengers because the database of biometric data has already been developed and is being maintained. Airports with lower passport and Trusted Traveler participation can still consider biometrics, but operators should consider that an extensive effort to enroll the passenger population will be required. This is an important consideration for some states and may have a major impact on the ability of TSA or self-bag-drop processing (where biometrics are required) to use the TVS. If an airport operator cannot leverage the existing TVS, the case for a positive overall ROI will be more challenging since more labor, more space in the passenger terminal, and increased marketing of the program by the airport would likely be required.

Do the airlines wish to implement biometrics at your airport, and who is paying for the up-front capital investment and the ongoing operations and maintenance of the equipment? In some instances, airlines may be willing to fund biometric implementations, including paying for new equipment, operations, and maintenance, thereby reducing the financial burden on

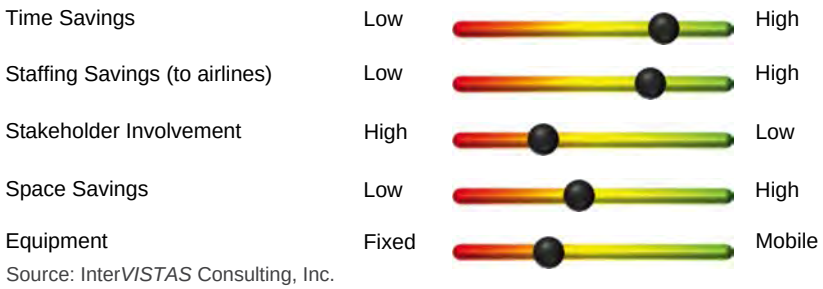
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the airport operator. If airlines are not willing to participate in the funding of the program, or if the potential biometric implementation occurs in a common-use facility, the ROI calculation is dramatically different. For example, if an airline wants to install biometric boarding at the gate, the airport may have to pay for the electrical infrastructure to enable the automated boarding gate equipment installation. If there are costs borne by the airport on either the capital or operations and management side, this should be a considered as part of the decision for implementation.

Do biometrics significantly reduce processing time, staffing, or space needs? Labor savings may not directly accrue to the airport operator, but savings to airport stakeholders such as the airlines, TSA, or CBP can indirectly benefit the airport operator. A lower cost structure for airlines affects their decisions on whether to introduce additional service or, in more challenging economic circumstances, maintain existing service. In terms of customer service, reduced processing times improve the passenger experience and decrease passenger stress levels. Calm passengers who have additional free time beyond the security checkpoint have been shown to spend more money at airport concession offerings. Finally, if significant reductions in space would accrue as a result of biometrics implementation, the airport operator may be able to defer capacity expansion plans or lease newly available space to interested tenants. As discussed in Chapter 1 and Chapter 2, biometric interoperability may provide for most gains in capacity and reductions in space needs. A biometric solution that is applicable to multiple processes along the passenger journey multiplies the benefits, however small they might be individually.

Airport operators should review the pros and cons of specific biometric implementations to decide whether implementation is warranted. A graphical example of such a comparison is provided in Figure 4-17 for biometric self-boarding gates. Five dimensions for ROI calculations are advanced that can help assess the overall benefits of costs of biometric implementation.

As described earlier, implementation of biometric self-boarding at the gate has been shown to reduce aircraft boarding time, which provides benefits to the airlines, since time is money, and also the airport, since expensive gate expansion projects might be deferred. Some airline customer-service staff savings are anticipated, or at a minimum, the reallocation of staff to serve other customer needs may be enabled. Depending on whether self-boarding is implemented in an exclusive-use or common-use facility, stakeholder involvement may be limited to one airline or may involve several airlines. The space required for self-boarding may increase, especially if e-gates are used, since additional space may be required to handle passenger exceptions. However, the faster boarding times may reduce passenger queue length in the holdroom, providing needed space for other functions (e.g., increased seating or standing space). Finally, regardless of whether e-gates or camera-on-a-stick is selected as the preferred implementation, the equipment is largely fixed, which means that it is not easily moved during changes in facility layout or facility modifications.



Source: Inter*VISTAS* Consulting, Inc.

Figure 4-17. Sample evaluation of biometric self-boarding.

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This more-focused evaluation of a specific biometric implementation will also bring to light some unforeseen capital costs that could be required to implement the solution. Examples of these costs are expansion of communication rooms or network bandwidth to accommodate additional data feeds, tinting of windows to help the resolution of camera or document scanners, and the relocation of existing infrastructure.

Reviewing the questions in this section and the spectrum of specific implementation considerations such those shown in the Figure 4-17 should help airport operators better understand whether to move forward with biometric implementations.

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CHAPTER 5

System Design and Information Technology Architecture

Summary

This chapter focuses on the IT architecture related to biometric technologies and solutions. First, the focus is on the variety of types of biometric credentials, choices for storage of biometric data, and different biometric interaction touchpoints such as e-gates, camera-on-a-stick, kiosks, and walk-through tunnels. Thereafter, the chapter identifies and explains five distinct IT architecture models to illustrate how these options tie together, with a real-world example of each to demonstrate its use case.

A secure digital biometric identity credential will bring many possibilities and advantages for passenger processing. Nonetheless, the supporting IT architecture design faces a larger challenge with more stakeholders because interoperability and scalability becomes more complex. With stakeholders in multiple countries, those countries' respective privacy laws also have an impact on the design. On the other hand, with ICAO DTC, an international standard is being developed to create common technical specifications that allow technologies to interact smoothly while abiding by multiple nations' (and states') laws and being software-, hardware-, and vendor-agnostic.

Although the U.S. market is big enough for the broad application of biometric systems such as the TVS, U.S. stakeholders might want to consider compliance with future global standards. U.S. stakeholders that invest in systems that are or can adapt to interoperable systems now may be challenged to change later when a global standard is adopted. In other words, where there is a development of a range of different systems, country-specific and extraterritorial application are the norms that will need to be contended with in the future.

Introduction

One of the challenges of a biometric implementation is assessing the IT architecture and associated business cases that can drive the ROI equation. Various drivers, such as the existing IT architecture at an airport, meeting performance standards and requirements, and the delivery of the desired objectives, may greatly affect that equation.

Many airports are challenged with implementing biometrics because of the scale of the capital investment and the technical know-how required, as well as the complexity to meet stakeholder requirements and achieving alignment and trust. This primer is not meant to be an all-encompassing standard guide for technicians; rather, it provides senior management and executives a comprehensive introduction to the most important concepts and acquaints them

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with the typical issues covering a range of business risks. By business risks in biometric rollouts, the authors mean:

- Will the capital program be right-sized for the objectives?
- Are performance specifications (false-accept/false-reject rates, speed) adequate to meet the desired customer experience?
- Are there risks/vulnerabilities that are introduced by the range of privacy and cybersecurity issues?
- To what extent is scalability built-in in the first year versus life-cycle reinvestment in future phases?

For implementation of biometric projects, it is important to pay attention to performance, risks, and scalability issues because of the different types of facilities. Age of existing assets, sunk costs that may form part of the business case, and localized makeup of the travel market can greatly influence the direction of biometric project definition. For instance, airports with a population of a low number of passport holders will require more emphasis in their IT strategy on the enrollment of passengers. Other airports using current technologies such as fiber-optic infrastructure backbones will have an opportunity to leverage existing investments.

This chapter discusses key lessons learned related to technology architecture and airport facilities. Invariably there will be more detailed technical expertise needed, and there will be local program requirements that will need to be addressed. These challenges are best solved by project teams that systematically create technology solutions, detailed designs, implementation programs, and rollouts. Use of this chapter will enable airport management to address some of the most important questions and considerations to evaluate what is required, understand the risks, and evaluate various investment scenarios.

What Is Information Technology Architecture?

Key Takeaway

IT architecture is composed of the IT software, IT platform, and IT infrastructure components, as well as the IT architectural principles that describe how these different components are designed to work together.

At any airport, the IT architecture describes the IT software, IT platform, and IT infrastructure components, as well as how these are designed to work together. The software is the applications, computer programs, and data that drive the airport's processes, passenger services, and operational systems. The IT platforms are the operating systems and middleware on which the software runs. As for the IT infrastructure, or hardware, these are the actual computers, network cabling, servers, switches, and other physical equipment that are needed for the IT system to function properly. But in addition to these software, platform, and infrastructure components, the IT architecture describes how these are designed to work together in a framework of principles. The IT architecture principles include, for example, technology standards, policies, and guidelines.

In order to have a well-functioning IT system that is able to meet the airport's needs, the IT architecture must be well-designed, maintained, and upgradable when needed. With airports developing rapidly over time, especially with more technology solutions being implemented in airport processes, services, and systems, the IT architecture should be aligned with the airport's business/strategic direction, supporting transitional requirements in support of future technology needs. With every airport being unique due to its demography, size, location, culture, and other factors, IT architectures are designed to best match the airport's needs.

When it comes to implementing biometric technologies and solutions, existing IT architectures are often redesigned to incorporate these new biometric hardware and software components and updated principles. For new airports, the IT architecture can be designed from the ground up.

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Summarizing:

- The IT architecture software is made up of the applications, computer programs, and data that drive the airports' processes, passenger services, and operational systems. Examples of biometric software are the mobile applications or programs used for access control with a fingerprint and check-in via facial recognition.
- IT architecture platforms include the operating systems and middleware. Operating systems such as Microsoft Windows, Apple OS, and Linux are common examples seen on today's laptops and personal computers. Larger IT systems might run on cloud-based platforms such as Amazon Web Services or Azure. When platforms host sensitive biometric data, it can be important where the platform is located since it might be in a different country. Middleware is the collection of software that is needed to run certain software or applications.
- IT architecture infrastructure is the hardware components such as computers, network cabling, servers, switches, and other physical equipment that are needed for the IT system to function properly. Biometric IT infrastructure components may be, for example, a fingerprint scanner, a camera that is used for facial recognition, or a self-service kiosk with biometric capabilities.
- IT architectural principles should be defined in a solution-free manner. This means that the principles are not prescribing any technology, software, hardware, or solutions but only act as criteria for the assessment of the different options when choosing specific solutions or technology. Examples of principles are (1) include privacy by design, (2) use modular components instead of monoliths, (3) performance is more important than cost (or vice versa), (4) stability is more important than innovation, and (5) user experience is most important.

Designing Airport IT Architecture for Biometrics

When designing airport IT architecture that incorporates biometric technologies or solutions, several generic architecture models can be chosen as a starting point. In considering the different architecture models, it is good to take note of several factors that influence the architecture model choice.

Governance of the IT Architecture

When designing and implementing IT architecture, the right governance model must be taken into account. A governance model appoints roles and responsibilities to enterprise architects, solutions architects, product managers, product owners, design thinkers, and procurement experts who orchestrate all the developments in line with the design and upgrading of the IT architecture.

Single-Party or Multi-Party Technology Solutions

One of the first main crossroads that the airport/airline/government finds itself at is whether to choose between one of generally two options:

- **A single-party** biometric technology solution is typically the best choice when the airport aims to create a system that services only the one airport, thus leading to IT architecture that supports the operations for a smaller user group with fewer stakeholders.
- **A multi-party** biometric technology solution provides interoperability between multiple stakeholders, often across multiple countries or locations (airports), which thus greatly increases the complexity of the IT architecture. The national Digi Yatra system, which uses the Indian government's Aadhaar database (See Chapter 2), is such a solution, where all airports and airline stakeholders are to be users of the countrywide solution.

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Stakeholders

Stakeholders are important in determining the feasibility of a concept in a real-world environment. Private- and public-sector stakeholders must collaborate to pilot a prototype, apply iterative development methodologies to demonstrate its value, and seek continuous feedback from each other to adapt accordingly. The use of an iterative approach encourages the necessary paradigm shift among stakeholders and establishes an environment for large-scale adoption. Each airport will have a unique mix of stakeholders affecting the IT architecture in a similarly unique way.

Existing IT Architecture

Implementations of biometric technologies and solutions need to align with the existing IT architecture. The existing IT architecture may not be ready to support biometric additions and, thus, should be assessed.

Airport Size

The airport size has a direct impact on the IT architecture and the type of (biometric) services to be considered. Due to some IT infrastructure components having a limited range, large airports will need a different and more extensive network topology to connect hardware over longer distances.

Funding Options

Throughout the world, there are major funding gaps and insufficient budgets to maintain assets and their IT architecture, which can lead to a higher risk to public safety. Therefore, airport operators may need to evaluate their mechanisms to fund capital and maintenance expenditures. Other mechanisms, beyond the traditional methods, include the following:

- Public–private partnerships can be an effective way of transferring life-cycle costs of infrastructure off public-sector budgets and simultaneously to create investable assets for the private sector.
- There should be a regional approach to infrastructure rather than direct funding from a country's government. This is especially true as urbanization transforms many of our global cities into mega-regions, requiring broad and interconnected infrastructure systems.
- Dynamic pricing is a usage charge policy that can increase revenues and ensure the efficient use of infrastructure assets. This policy can be linked to peak-hour congestion fees and passenger facility fees during peak travel times.

Depending on the funding, the IT architecture design may need to be selective with regard to what is hoped to be achieved, or it should be expandable in its design and incorporate more generous flexibility for future expansion.

Key Takeaway

There are various IT aspects that need to be considered when implementing biometric technologies and solutions, from the type of credential, how it is stored, and how the interaction at touchpoints is designed. These choices greatly affect the hardware components of the IT architecture.

Biometric IT Architecture Infrastructure Components

In the implementation of biometric technologies and solutions, there are three design choices that have the largest impact on the actual physical infrastructure components or the hardware of the IT architecture. These are:

1. What biometric *credential* to use?
2. How to *store* biometric data?
3. How do we design the *interaction* of the passenger at biometric touchpoints?

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Often, technology specifications of infrastructure, communication protocols, and other details such as how data are stored are left to the designer of the system, which in many cases is the vendor of the technology solution. Nonetheless, choices that an airport, airline, or government makes related to performance, envisioned use, and interoperability have a significant impact on which infrastructure components should be considered. Also, these choices can have significant capital impacts, such as upgrading existing infrastructure or replacing various components to address legacy issues and new technology standards and performance. For that reason, it is good to be familiar with the following considerations:

- Type of biometric credential: The credential is an object or data structure that authoritatively binds an identity to an authenticator. This can be done via one or multiple authenticators, such as a biometric. Examples of physiological biometrics are one's face, fingerprints, hands, and irises. Examples of behavioral biometrics are one's signature, voice, and keystroke. The selection of the biometric affects the choice of the type of credential and inherently affects the IT architecture.
- Type of storage for biometric data: The type of storage, in this case for the storage of the biometric data, opens the discussion of data ownership, location, and protection against data theft or misuse. Biometric data can, for instance, be stored in a database (locally or over the Internet in the cloud) or they can be stored on one's own smartphone or on an access card or token. The type of storage chosen affects how passengers can use their biometrics in the airport processes, services, and systems, and how the IT architecture is to support those uses.
- Type of interaction with passenger/customer/user at biometric touchpoints: There are several types of touchpoint hardware (e.g., kiosks, e-gates, camera-on-a-stick) that may facilitate the interaction of a passenger/staff member with the biometric technology and solutions. The choice of type of interaction affects which infrastructure components are used and, thus, the IT architecture.

Biometric Credentials

A biometric credential is an object or data structure that authoritatively binds an identity—via an identifier or multiple identifiers—to at least one authenticator owned and controlled by a subscriber (passenger or airport staff). The following credential types are discussed in this section:

- E-passports;
- Cards, tokens, or passes;
- Digital credentials (of three types); and
- Government-issued identity cards.

E-passports

The most evident globally accepted process for verifying a person's identity is a physical passport that is issued by the passport holder's home authority. The first standardization of the details on a passport were developed under the auspices of ICAO in 1980. Standardization included the person's image; a standard-format, machine-readable zone; and the passport holder's biographical information. The identity of a person can be verified by border officers manually by comparing the person to the image on the passport. In 1998, the format was upgraded with the introduction of an embedded electronic microprocessor chip (hence e-passport), which could contain both biographic and biometric information. The e-passport uses contactless smart-card technology, including an embedded radio-frequency identification chip. The passport's critical information is printed on the data page, repeated on the machine-readable lines, and stored on the chip with a country-specific digital signature.

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There are now more than 100 countries that issue e-passports (see Figure 5-1). The digital signatures are unique to each country and can be verified by using the originating country's respective certificates. A digital signature on an e-passport is authenticated by the issuing state's security certificates—the Country Signing Certification Authority (CSCA) Certificate and the Document Signer Certificate. Together, the signature and certificates form a trust chain wherein the CSCA certificate is securely anchored in the authority of the issuing state and the Document Signer Certificate is securely stored in the chip of the e-passport as the Document Security Object.

To validate an e-passport at an international border, the border control system retrieves the Document Security Object from the chip. The CSCA certificate can be derived from the ICAO Public Key Directory, which is a database maintained by the ICAO to facilitate the secure, online sharing of information between states. Certificates, however, can also be exchanged through a bilateral exchange process.

The data elements that are contained on the chip of the e-passport are the biographic details of the passport holder and an encoded digital photograph. An e-passport contains additional data fields for fingerprint and iris data. The actual storage of this biometric data on the e-passport is optional and depends on the individual country's legislation and policies.

Cards, Tokens, or Passes

A token system (portable) uses a smart card or a fob to store biometric data. This means that your fingerprint, once captured, is stored within the token. The benefits of storing biometric data on a portable token are that it need not be transferred over a network for verification purposes, and this reduces the risks that can come with network-related vulnerabilities. When using this method, users need to present their card or fob and then their biometric data as a two-step authentication process.

Digital Credential – Self-Sovereign Identity

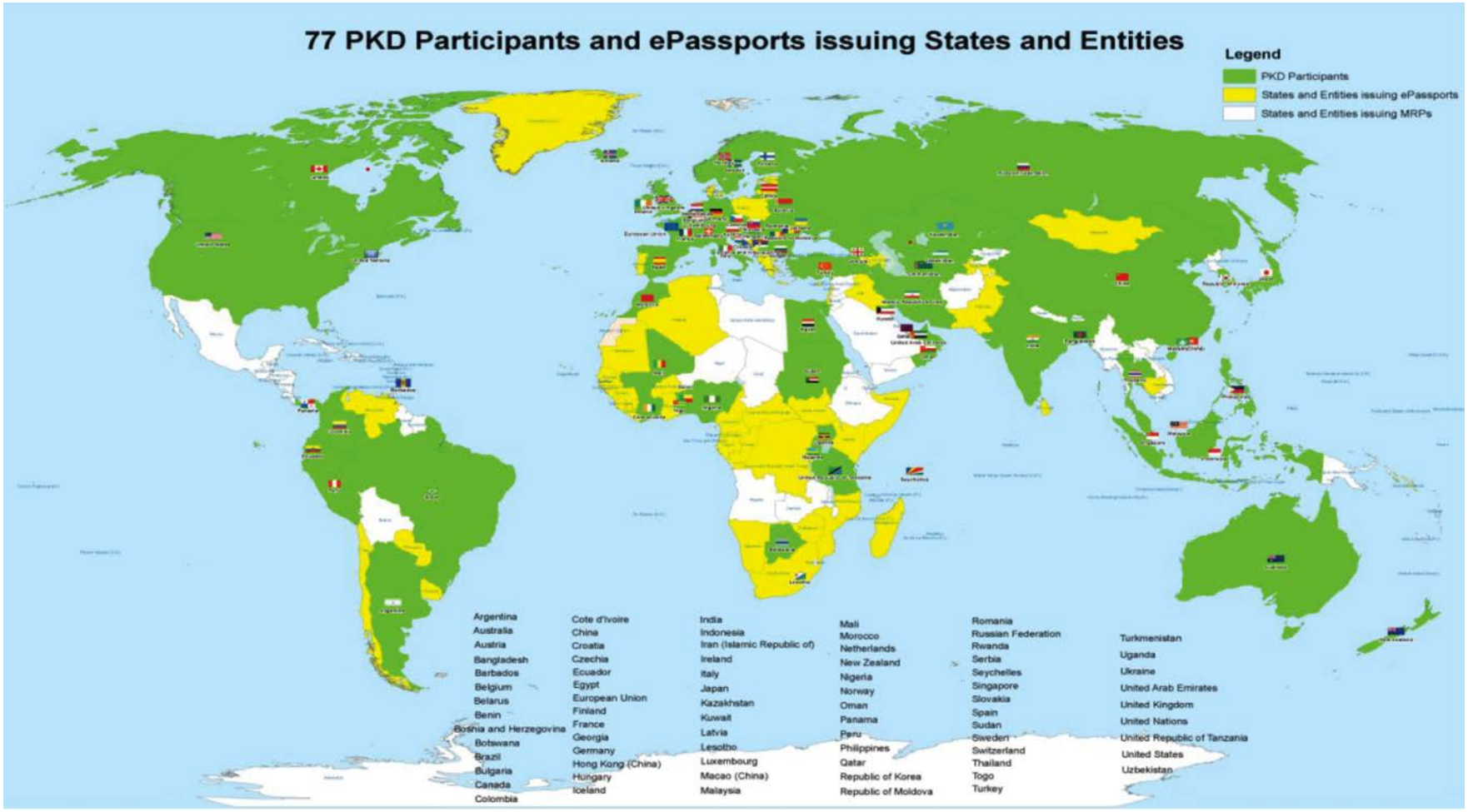
The most recent development in biometric passenger processing is the concept of self-sovereign identity (SSI). An SSI is owned by the individual just like a physical passport. As owner, the individual has access to, can refer to, and can share components of this identity at his/her discretion. While certain components of the identity are established by issuing authorities (e.g., passport number, bank details), the individual must consent to the sharing of his/her identity and any related data. This is achieved by individuals securely storing their own identity data on their own mobile devices and providing it on request to those who need to validate it.

Just as with the format of the e-passport, the New Technologies Working Group of ICAO has recently published guidelines for developing these digital travel credentials, titled the "Guiding Core Principles for the Development of a Digital Travel Credential" (ICAO 2020). The ICAO working group defined four basic criteria with which the credential needs to comply:

- It should be produced from a travel document issuing authority;
- It should be capable of being supplied unaltered to verifying entities in advance of the traveler's journey or arrival;
- It should be globally interoperable to ensure that it can be used for different use cases; and
- It needs to be easy enough to use and provide enough benefits for broad adoption by travelers (ICAO 2020).

The DTC workgroup is opting for a hybrid credential, which is a combination of a virtual credential (biometric template) that is linked to one or more physical credentials (authenticators). The credentials could be stored in a remote system, such as a database or web server, and the authenticator could be an e-passport, smart card, or mobile phone.

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Source: ICAO n.d.-c.
Note: PKD = public key directory; MRP = machine-readable passport.

Figure 5-1. Countries that issued biometric e-passports as of mid-2019.

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Any credential being developed will eventually need to be embraced by governments, industry, and citizens and accepted throughout the world, which is an inherently complex undertaking that will take time. With the recent COVID-19 experience, travel documents may need to incorporate health certificates, and consideration of this is needed in the development of the DTC. This requirement could increase the need to develop a global solution.

Digital Credential on a Mobile Device

Smart mobile devices provide a platform for individuals to carry and provide a digital credential that is trusted with a unique ID number that resides on the device. This allows the individual to provide a digital credential in place of a traditional physical credential. It also offers the same functionality of a physical credential when accessing a secure or controlled area.

Digital Credential in a Blockchain

Distributed ledgers, such as blockchains, are a developing technology that is proving to be adaptable across industries, providing cost-effective and secure real-time (data) transactions between two parties. The data stored are secure and immutable by design because the data transferred are recorded and verified by all the different participants or computers (called nodes) in the system (thus the term "distributed"). This creates a database of data records that offers transparency and traceability.

In many biometric identity solutions that utilize blockchain technology, it is not the actual credential that is stored in the blockchain, but rather a record of successful verification of that identity. These records are called "verified claims of identity" and can be shared as proof of verified and authentic identity.

These solutions often connect through an untrusted medium (e.g., the Internet) and thus require a secure connection and strong authentication (of the device, software, user, environment, and so forth). Due to their complexity, implementation is often achieved with the involvement of specialized technology partners.

Government-Issued Identity Cards

Identity cards are issued in many countries, while some countries are still discussing their implementation. These cards can also contain biometrics, much like the e-passport.

Biometric Data Storage

There are various forms of storage that can be used, and they will work with more than one mechanism to deliver the desired solution. In this chapter, the following storage mechanisms are discussed:

- Centralized server or in a local data center (on premise),
- Secure cloud in a remote data center,
- Mobile devices such as smartphones, and
- Distributed databases and ledgers.

Centralized Server or Data Center

A centralized server is one way to store data and allows biometric technologies and solutions to access data at any location connected to the server. In some cases, the centralized server only offers connections to the systems at the airport, not outside it. In cases where the server is connected to the Internet, it is susceptible to a cyberattack. To reduce the risk of the server being breached and the data stolen or copied, the data must be encrypted when stored or transferred

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over the network. The issue with encryption is deciding where encryption keys will be stored and who will be trusted with their access. With the implementation of data and privacy protection, there are increased responsibilities for managing and securing data.

Secure Cloud

The use of biometric security systems in cloud computing is progressively increasing. In general, storage in the cloud comes with a separation of responsibilities when it comes to maintenance of the IT systems. The cloud provider, such as Amazon Web Services, Microsoft Azure, or Google Cloud, is responsible for maintaining the underlying hardware, software, and security, and the entire cloud provider's organization is aimed at reaching the highest levels of security and performance in this. No single organization can reach the same amount of security and quality offered by these providers. The client organization, such as an airport or its supplier, is solely responsible for the configuration of the storage of the data, and the trouble of hardware, software, and network maintenance is taken out of its hands.

Biometric security systems have the potential to take cloud computing to the next level as they guarantee an extremely high level of security and ensure that the rendered services are accessible to only legal and authorized users and no one else.

There is a clear distinction between cloud service providers and the customer. Breaches of data in the cloud are not typically breaches of the underlying cloud provider's infrastructure. To make a cloud secure, it is vitally important that responsibility be shared. Customers are responsible for securing how they use the cloud services, including properly configuring identity and access management, storage and computation settings, threat analysis and defense, and the security of the application and data processed and stored in the cloud.

Cybersecurity for centralized server solutions and cloud-based services is the protection of IT-connected systems and protecting them from cyber threats.

Mobile Devices Such as Smartphones

Biometric data can also be stored on the end user's device. This is most common with smartphones that use touch ID fingerprint sensors. On-device storage can be used to store biometric data through a chip that holds the data separately on the device's network. When storing the data on the authentication device itself, the organization implementing the biometric verification process does not have control over it or the ability to copy the data and is also not liable for the theft of large databases with personal data.

Distributed Databases or Distributed Ledgers (Blockchain)

Public blockchains are a collaborative creation, with their goal being to create a world that is completely decentralized and where the ownership of digital assets is always protected and transferable. Blockchains can concurrently achieve high security, decentralization, and scalability. The core value of blockchain technology is not to provide rudimentary data services (like the decentralized database) but to build a new system of digitized data assets and automated trust services. The global blockchain updates its state autonomously, and data are traceable to their source. Data contained in a blockchain are immutable, which supports privacy concerns when personal data need to be stored.

In contrast, a distributed database is centrally managed by a service provider. The goal is to create a logical center that can provide efficient, low-cost services with great scalability.

The core value of distributed database is to supply data storage and access services to business systems, focusing on the analysis and retrieval processes.

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Biometric Touchpoint Interactions

Travelers are required at various points in their journeys to interact with various customer-facing biometric solutions. Some of these biometric interactions (e.g., e-gates, camera-on-a-stick, kiosks, and walking through gate or tunnel) are already commonplace in many countries. This section describes some system architecture implications related to the operations of these different solutions.

There is a major debate on the optimal technology types to use for flow control. E-gates are the prevailing methodology used outside the United States to control flows. Several U.S. airlines and CBP, on the other hand, prefer the camera-on-a-stick method. Both require staff intervention to some extent, such as when dealing with exceptions when an e-gate does not open and monitoring a successful passage through a visual/audible signal from the camera-on-a-stick model.

E-gates

Automated e-gates at checkpoints with the capability to use biometric technologies for the verification of identity or granting permission for entry by scanning a biometric (credential) are by now common solutions at airports. With an e-passport, the e-gate can be used worldwide at border control checkpoints for customs, immigration, and emigration, allowing for automated self-service border crossing.

The e-gate might use the data stored in the chip of a biometric passport to verify the user's identity. Travelers undergo biometric verification using face, fingerprint, or iris recognition, or a combination of modalities. In some locations, systems such as Global Entry (in North America) are used for verification. Functionality can be extended so that the e-gate can authenticate the signature on the e-passport against the origin country certificates, read the biometric image, and match it with the person in front of the barrier (1:1 matching).

After identification, the person's details can be digitally vetted according to the national border control checks and procedures. Typically, e-gate biometric identification and verification will provide a two-gate option, allowing the traveler to cross a border with no ability to return unless following and meeting the entry or exit criteria of the country.

Automated e-gates used for boarding usually have a single gate system where a traveler presents a boarding pass to board a flight. This type of e-gate is controlled by the team responsible for managing boarding issues and incorrect boarding procedures.

After positive identification and authentication, there is no need to store the passenger's data in a database. Since the data are stored on the passport and not in a database, passport bearers are the keepers of their own data.

The advantage of an e-gate is the ability to manage flows with a device that physically obstructs passage if there is no biometric match to confirm identity. In other words, e-gates serve as a mechanism similar to the payment system gates used for mass transit. The disadvantages are that e-gates are bulky and require more planning around wiring and data, as well as the life-cycle wear-and-tear of gates.

Camera-on-a-Stick – Gateless Interaction Point

The point of interaction is where biometric identification or identity verification can take place without the passenger needing to go through an automated gate. Currently, these points are, in many instances, manned by ground agents or airline representatives, but they still offer the opportunity to increase security performance and decrease processing times. Many different applications can be designed to be incorporated in a totem (a contactless interface panel that can use facial recognition and can be either free-standing or integrated into a gate or kiosk),

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the most common being the camera-on-a-stick method, which refers to the camera being used for facial recognition.

The advantages of the camera-on-a-stick, from the point of view of CBP and some airlines, are that the overall capital cost is low and the biometric processes are integrated with other customer interactions. For CBP officers, it is the questioning of passengers; for airline agents, it is a face-to-face customer-service driver to help with additional services. There are some clear disadvantages: a camera-on-a-stick model is less bulky physically but negates the full passenger self-service potential. In other words, to truly get hundreds of passengers through without the need for staff time is much harder if the concept of operations is built on front-facing passenger interaction. Moreover, with social distancing measures and reduction of face-to-face time for customer service, there is also a realignment of how to use methodologies that do not require a gate.

A hybrid model may be developed that uses a wayfinding approach to direct passengers to Point A if successful and Point B if a biometric match has not been achieved. In so doing, the bunching up of people is minimized, and there are methodologies to ensure that individuals are not routed to the wrong location.

Kiosks

The typical choices available for passenger check-in are via mobile phones, airline desks, and kiosks. Kiosks provide self-service devices that are simple to use and convenient, making it possible for passengers to check in any time on the day of departure.

The next generation of kiosks is being tested by various airports and airlines. These kiosks include sensors to help with touchless check-in using touchless infrared technology. Robotic biometric kiosks are also being used to alleviate congestion at check-in areas.

Contactless check-in via a smart device remotely and at the airport may be used to print boarding passes (if necessary) and baggage tags at kiosks.

Check-in with a smart device or website may be done to obtain a boarding pass and provide personal details, including passport information. There is an increased focus from airports and airlines on enhancing existing processes with touchless options using passenger smart devices, and the introduction of biometrics through smartphone cameras is also a possibility. Alternatively, using on-device biometrics based on Apple iOS or Android systems to confirm identity is a methodology of SSI that can limit the amount of transmission of information but still validate individual identity.

Walking Through a Gate or Tunnel

This technology is being tested in various airports, allowing passengers after enrollment to clear immigration in seconds by walking through a defined path (e.g., a touchless gate or tunnel). Iris biometric technology is used to authenticate each passenger, ensuring that queuing is eliminated.

IATA OneID or Seamless Flow

The IATA OneID concept envisions an end-to-end travel journey where passengers enroll with their biometric details prior to each journey and are then able to pass several processing steps (e.g., check-in, bag drop, security, border control, boarding, and even airline lounge entry and duty-free purchases) by simply showing their faces. The concept aims to include the departure and arrival processes at both the destination and origin airport. In addition to smooth processing of passengers, the system can also recognize and differentiate passenger personas and provide services tailored to those personas (for example, for priority or disabled passengers).

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IATA has been working on this initiative for 15 years, with the aspiration that its solution can be adopted initially in markets struggling with capacity. The rollout of OneID has seen several pilot projects; however, a wider adoption will take time. A more detailed description of the complete system can be found in Appendix M.

IT Architecture Models

Introduction to Five Models

Key Takeaway

The five IT architecture models discussed here differ predominantly in terms of (1) who owns the biometric data, (2) which entity is responsible for the verification of the passenger's identity, and (3) how long biometric data are stored.

This chapter identifies and compares five different models, focusing on the enrollment, verification, data facilitation, and authentication of the various credentials that were used for the categorization of the case studies of biometric implementations in Chapter 2. Literature often references at least two types of model: a per-trip model and a per-life model, where the differences are obvious. In the first, the system architecture aims not to retain information such as biometric data much longer than necessary past the duration of one completed trip. In the latter, enrollment or the retention of information is for a much longer period, in this case for life. A distinction added to these two main models, due the trends of focusing more on privacy protection, data storage, and the governance of the system, is whether the system architecture facilitates biometric data to be governed by the passengers themselves, by a third party, or by an authoritative party such as a local or national government. This results in the following five models (see also Table 5-1):

Model 1: Identity as a service – a centralized, government/authority model
Model 2: Identity as a service – a centralized, third-party model (not bound by national/state borders)
Model 3: Per trip – a semi-federalized model
Model 4: Per life – a federalized model
Model 5: Per life – a federalized model with SSI

Model 1: Identity as a Service – Centralized, Government/Authority Model

The government-driven model is a centralized approach to passenger and data facilitation. The government establishes and verifies the traveler's identity and serves as the identity management service provider to other stakeholders within the airport system.

This model does not depend on formal traveler enrollment or adoption because the government uses centralized databases of pre-verified own nationals' and foreigners' biometrics (facial) to authenticate travelers. No further data are collected, and no booking information is integrated to create a digital identity.

The core platform is a cloud-based biometric matching service. The government can extend this to private-sector providers as an identity as a service (IDaaS), which provides a scalable, secure, and seamless solution that easily integrates with providers' systems through web-based application programming interface (API) connections.

On arrival at any journey touchpoint, a traveler's image is captured by facial-recognition technology (e.g., e-gate, stand-alone facial-recognition cameras) and sent to the government verification system for authentication; a match result is then returned to the travel provider.

The government-driven model demonstrates applicability and implementation success at touchpoints across the airport system. Any traveler providing consent to participate

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Table 5-1. Comparison of models.

	Model 1 Government/ Authority	Model 2 Third Party	Model 3 Per Trip	Model 4 Per Life Federalized	Model 5 Per Life SSI
Example	e-passport*	CLEAR	Happy Flow	Digi Yatra	DTC on blockchain
Enrollment					
How to enroll	Every 5–10 years, at a government location outside the airport	Once, at airport	Every trip, prior to check-in at or outside airport	Once, at government location outside airport	Once, on a personal device, verified by a government agency
Ability to opt out	Yes	Yes	Yes	Yes	Yes
Data (biometrics/ biographic) used	Biographic, facial, fingerprint	Biographic, facial (iris)	Biographic, facial, flight information	Biographic, facial (iris)	Biographic, facial
Type of hardware used	Government enrollment office or online	Kiosk	Kiosk or on a personal device	Government enrollment office or airport kiosk	On a personal device
Type of software and architecture required	Government cloud platform	Cloud-based orchestration platform	Cloud-based orchestration platform	Government cloud platform	Distributed ledger
Enrollment process	Apply or renew before expiration	Online plus CLEAR enrollment center or just CLEAR enrollment center	Check-in kiosk or counter	Register with central system, validate ID at start of first trip at registration kiosk	Download app and upload information
Retention period	Period of validity	Life of the credential, until one's membership terminated	Length of a trip, or limited number of hours after departure	Per life	Per life
Integration of reservation data	Separate	Integrated	Separate	Integrated	Integrated
(Biometric) Verification of Identity					
Where in biometric journey it is required	Check-in, bag drop, security, border, e-gate	Security/ border/ boarding	Check-in, bag drop, security, border, e-gate	Check-in, bag drop, security, border, e-gate	Check-in, bag drop, security, border, e-gate
Verifying party	Airline, airport, government	Third party, government	Airline, airport, government	Airline, airport, government	Airline, airport, government
Data (biometrics/ biographic) used	Facial	Facial (iris)	Facial	Facial (iris)	Facial
Hardware used	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate
Software required	Facial recognition on orchestration platform	Facial recognition on cloud platform	Facial recognition on local template database	Facial recognition on cloud platform	Facial recognition on orchestration platform, mobile DTC application
Method of verification	Picture of face (live) matched	Picture of face (live)	Picture of face (live) matched	Picture of face (live)	Picture of face (live) matched

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Table 5-1. (Continued).

	Model 1 Government/ Authority	Model 2 Third Party	Model 3 Per Trip	Model 4 Per Life Federalized	Model 5 Per Life SSI
	to stored biometric on the physical credential	encrypted and sent as photo/template to third party for matching with on-file biometric	to stored biometric on personal device or temporarily on airport system	encrypted and sent as photo/template to government for matching with on-file biometric	to biometric on personal device (verified by issuer of digital credential)
Data facilitation					
Type of platform	Local or cloud-based orchestration platform	Private cloud-based orchestration platform	Local template database and orchestration platform	Cloud-based orchestration platform	Orchestration platform, distributed ledger, and mobile DTC app
Storage of data at location	Government, airline, and airport servers (temporary)	Private, airline, and airport servers	Temporary: airline, airport servers; records of immigration, emigration may be kept by authorities	Government server	Blockchain, records of immigration, emigration may be kept by authorities; all stored data are immutable.
Connectivity through	Application programming interface (API)	Platform	API	API	Ledger
Data and privacy protection	Laws governing government data retention and protection	Laws governing commercial companies' data retention, use, and protection	Privacy by design integrated	Laws governing government data retention and protection	Privacy by design integrated
Authentication of credential					
When in journey?	Check-in, bag drop, security, border, e-gate	Security/ border/ boarding	Check-in, bag drop, security, border, e-gate	Check-in, bag drop, security, border, e-gate	Check-in, bag drop, security, border, e-gate
Authenticating party	Government	Third party	Airline, airport, government	Government	Airline, airport, government
Data (biometrics/ biographic) used	E-passport, facial	E-passport, facial (iris)	E-passport, facial	E-passport, facial (iris), digital credential	Digital credential (on mobile device), facial
Hardware used	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate	Kiosk/e-gate
Software required	Orchestration platform connecting to government authentication service	Orchestration platform connecting to government authentication service	Orchestration platform connecting to government authentication service	Orchestration platform connecting to government authentication service	Orchestration platform connecting to blockchain API
Method of authentication	Scan of e-passport for check of authenticity of certificate of issuing authority	Third party checks authenticity of identity documentation upon registration	Authenticity of identity documentation (e-passport) checked upon enrollment	Government is issuer of digital identity and checks the authenticity of identity upon enrollment.	Identity documentation authenticity checked upon enrollment, stored on blockchain as verifiable claim

*Specific requirements may differ slightly by country.

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experiences a streamlined journey without having to enroll in advance. The core challenge with this model is legislative restrictions that prevent the extension of IDaaS beyond its current mandated realm. As a result, there may be limitations when attempting to integrate car rental services and hotels into a model that was developed and approved for use only in the airport/governmental system.

Example: Border Control Authorities

Various border control authorities across the world use an e-gate or kiosk that supports biometric microchip passports:

- France: Charles De Gaulle Airport allows citizens from the European Economic Area (EEA), Andorra, Monaco, San Marino, and Switzerland to use the PARAFE gates. These gates can only be used by citizens aged 18 or over and holding a valid biometric passports.
- United Kingdom (UK): E-gates are in place at 15 air and rail ports across the UK. Users must be 12 and over and a British citizen or a national of the EU, Australia, Canada, Iceland, Japan, Liechtenstein, New Zealand, Norway, Singapore, South Korea, Switzerland, or the United States.
- The Netherlands: E-gates can be used by those 16 years or older and holding a valid EU passport.
- United States: At the time of writing, CBP was in the midst of rolling out the Simplified Arrival program to speed clearance for international arrivals. Simplified Arrival uses facial biometrics to automate document checks and provide travelers with a touchless process.

Model 2: Identity as a Service – Centralized, Third-Party Model (not Bound by National or State Borders)

The European Commission is implementing a central biometric matching system (BMS) that will serve the biometric identity requirements of multiple applications that are essential to European security. This system is ground breaking in terms of scale, transactional support, and strategic approach, offering many valuable lessons in the use and adoption of biometric identity capabilities. At the time of writing, the system was in development and was expected to be live during 2022. The system aims to instantiate a new entry/exit system (EES), creating a unified information system for recording data on the entry and exit movements of short-stay third-country nationals crossing the external borders of the EU.

The key body of the EES is EU-LISA, the European agency for the operational management of large-scale IT systems, which is headquartered in Tallinn, Estonia, with an operational site in Strasbourg and a backup site in St. Johann Im Pongau (Austria). The agency is responsible for the following tasks:

- Development of the central system,
- Implementation of a national uniform interface in each member state,
- Secure communication between EES and Visitor Information System (VIS) central systems, and
- Communication infrastructure between the central system and the national uniform interface.

Each member state is responsible for the organization, management, operation, and maintenance of its existing national border infrastructure and its connection to the EES.

The EES will be a centralized system through which the member states cooperate—hence the need for a common architecture and operating rules. Secure Internet access to a web service hosted by EU-LISA will allow third-country nationals to check their remaining authorized length of stay at any time. Airlines will also be able to use this function to check whether their passengers are authorized to enter the EU.

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Example: Traveler Verification Service

A digital ID verification service in the United States is the TVS. This service, which is provided by CBP, is based on the concept of IDaaS. CBP uses a biometric matching service. The system compares a new photo taken at the time of departure with a DHS-stored subset of a larger DOS dataset (based on the flight manifest) that includes images from photographs taken by CBP during the entry inspection; photographs from U.S. passports, U.S. visas, and other travel documents; and photographs from previous DHS encounters.

The TVS can be applied at all entry/exit locations (air as well as sea) in the United States. For air travel, instead of the airline conducting a manual passport verification and a registration of the boarding card of the passenger in the flight manifest, the process can be automated by means of facial biometrics.

The TVS can be applied at security, check-in, bag drop, or the boarding gate and uses facial-recognition software; a live-captured image is matched with the person's identity credentials that are stored in the TVS database. On a match, the system returns the biographic data of the person together with a unique identifier that the airline can use to link the passenger to its DCS and allow access or passage through the touchpoint.

Model 3: Per Trip – Semi-Federalized Model

The per-trip model is a semi-federalized approach to traveler and data facilitation throughout the traveler journey. Unlike the government-driven model, the traveler has the choice to opt in to participate in per-trip travel experiences at the time of enrollment.

The enrollment process for per-trip travel experiences typically begins on arrival at the airport. Using a biometric check-in kiosk, passengers verify their identity with an e-passport, biographic information, and facial image (biometric token). The data orchestration platform creates a digital identity that lasts only for the duration of the journey.

In this model, data is stored, managed, and facilitated by an orchestration platform, which all stakeholders trust to supply verified traveler data. Connections between the orchestration platform and travel provider or government agency systems normally only require API integrations. These platforms are built to adhere to privacy-by-design principles and securely store and send encrypted traveler data to stakeholders on a need-to-know and authorized-to-know basis.

A traveler's identity is verified on arrival at a touchpoint using facial-recognition technology that captures the traveler's image, which is transmitted to the orchestration platform in order to generate an authenticated status; any other data required to complete the process is transmitted as well.

The per-trip model has emerged as one of the most prevalent models to be tested and implemented worldwide. This model is widely accessible to a broad set of travelers, including occasional travelers and non-nationals of the country using the technology. It is also relatively easy to implement from a technical perspective and easy to use from a customer experience perspective. Challenges to designing an end-to-end model will be extrapolation beyond the airport environment and ensuring that different parts of the journey (e.g., connecting flight at another airport) are included without the need for passengers to re-enroll.

Example: Happy Flow

The Happy Flow model is based on collaboration among public and private stakeholders, including the government of Aruba, the Aruba Airport Authority, the Netherlands, KLM, and

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the Schiphol Group. In the past 4 years leading up to 2020, this group piloted a streamlined, user-friendly, end-to-end experience at Aruba International Airport.

Enrollment after opting in creates a single biometric token that is only kept for 24 hours. When the biometrics token is created, the passenger's e-passport is used for authentication, after which the holder's identity is verified at each touchpoint using facial recognition.

Model 4: Per Life – Federalized Model

This federalized per-life model has a federalized approach to traveler and data facilitation throughout the traveler's journey. Travelers have full discretion over how much of their data are shared, to whom, and at what point. Travelers also keep data integrity by storing their digital identity on their mobile device.

The per-life travel experience typically begins with the creation of a digital identity on a traveler's mobile device using a digital identity management app. This initial enrollment allows travelers to upload and verify core pieces of their identity (e.g., passport biographic information, facial image) using the app's built-in e-verification capabilities, and also allows verification by government. On completion, the digital identity resides on the traveler's mobile device for life. The traveler may add as much additional information as desired and can perform one-time verification of this information with relevant, trusted stakeholders (which may be but are not required to be government agencies). Finally, the traveler easily integrates bookings into digital identity to allow for seamless data management and sharing.

Pre-journey, a traveler's mobile device pushes minimum required data to relevant travel providers or government officials. Data facilitation is managed by distributed-ledger technology and cryptography, which ensures the secure transfer of data. On receiving traveler data, stakeholders can perform a host of activities (for example, government officials can perform risk-based assessments to streamline security processes, and travel providers can leverage shared data to enhance the traveler experience). At touchpoint arrival, a traveler is authenticated via facial-recognition technology, which captures the traveler's image, authenticates it against received data, and receives an authentication status.

This per-life model offers numerous opportunities for secure, seamless travel experiences across a multitude of use cases. The key challenge for end-to-end model consideration will be ensuring stakeholder acceptance and trust in this level of federalized digital identity.

Example: Digi Yatra

Digi Yatra is an industry-led initiative overseen by the Indian government, which uses the link to the Aadhaar system (12-digit unique identity number of all Indian citizens) in partnership with airlines and other system players during the booking process, facilitating faster airport entry and automated check-in without the need for any paper-based interventions.

Dubai International Airport recently introduced a pilot where enrolled travelers walk through a biometric tunnel without stopping due to the advanced biometric security solutions. The biometric tunnel uses iris and facial biometrics for identification and verification.

Once the traveler has successfully cleared the tunnel, the traveler can receive real-time notifications about congestion and delays, as well as obtain greater visibility on the next step of journey, which includes navigation though the airport using a smartphone or via an interactive kiosk and various augmented reality applications. Returning passengers can receive alerts about their luggage and its arrival on the baggage belt, submit baggage claims, and provide customer feedback.

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Model 5: Per Life – Federalized Model with SSI

An SSI is owned by the individual. As owner, individuals have access to and can refer to and share components of this identity at their discretion. While certain components of the identity are set up by issuing authorities (e.g., passport number, bank details), individuals must consent to the sharing of their identities and any related data. This is achieved by individuals securely storing their own identity data on their own personal devices and providing it efficiently to those who need to confirm it, without relying on a central repository of identity data.

Example: Known Traveller Digital Identity

WEF has published a white paper describing in detail its proposed biometrics-backed paperless international travel concept, the KTDI, a limited pilot project involving travel between Canada and the Netherlands (World Economic Forum 2020b).

A KTDI would be an international traveler's digital profile—one that is detailed, secure, and shareable (at a traveler's discretion). The new profile model would be decentralized and facilitated by blockchain processes, biometrics, mobile devices, and cryptography.

The KTDI concept is a public–private endeavor that the pilot consortium launched midway through 2019. The pilot is being run by the WEF, government agencies of Canada and the Netherlands, KLM Royal Dutch Airlines, Air Canada, Amsterdam Airport Schiphol, the Greater Toronto Airport Authority, Aéroports de Montreal, and Accenture Plc.

An open-source blockchain called "Hyperledger Indy" has been chosen as the decentralized identity platform. Hyperledger is a global enterprise blockchain project that offers the necessary framework, standards, guidelines, and tools to build open-source blockchains and related applications for use across various industries. For the Hyperledger platform, a person's digital identity, authenticated by a national government, would be linked to the person through biometrics. The identity would be encrypted and stored on the traveler's phone. The KTDI application would keep an ongoing list of attestations, beginning with the government authentication and continuing with a running travel history that includes border crossings and transactions with trusted vendors. Information shared by the traveler is verified by checking trusted databases.

Evaluation of Architecture Models

Table 5-2 provides an evaluation of the five architectural models as well as their differentiators, benefits, and challenges. Each model needs to be carefully evaluated based on one's circumstances since one may be more appropriate than another. One consideration is the type of air services offered (domestic, regional, or international, or a combination of all). Other considerations include the airport type, airport size, and types of airlines providing services.

Key Takeaway

The IT architecture design faces a larger challenge with more stakeholders as interoperability and scalability become more complex. With stakeholders in multiple countries, the privacy laws of those countries also affect the design.

Stakeholder Challenges: Interoperability, Scalability, and Privacy Protection

There are multiple challenges to the design of an IT architecture, which often merits careful collaboration between the airport, airline, government, and the stakeholders to achieve the desired interoperability. IT architectures must be increasingly capable and scalable to incorporate the new technology needs of airports. Many are also dependent on local or national laws, especially when it comes to the protection of privacy and biometric data. On top of that, the designer will want to

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Table 5-2. Evaluation of five architecture models.

Metric	Notes	Model 1 Government/ Authority	Model 2 Third Party	Model 3 Per Trip	Model 4 Per Life Federalized	Model 5 Per Life SSI
	Example	e-passport	CLEAR	Happy Flow	Digi Yatra	DTC on blockchain
Enrollment						
Pre-journey enrollment is possible.		No	Yes	Yes	Yes	Yes
Accessibility for all passengers	Passengers with reduced mobility, families with children, the elderly, requirement for a passport?	Low	Low	Medium	High	Medium
Speed of enrollment		Low	High	High	Medium	Medium
Integration of booking data is possible.		No	Yes	Yes	Yes	Yes
Authentication/Verification						
Level of acceptance (quality of authentication)	Authentication of an identity; is the identity (digital) credential authentic and accepted by government/ country/ authority?	High	Medium	Low	High	High
Quality of verification (confluence of verifiers)	In case of facial recognition	High	High	Low	High	High
Potential for U.S. preclearance	U.S. preclearance: some models allow for data sharing that is required by United States.	Yes	If third party is a trusted partner of CBP	No	Yes	Yes
Government acceptance (likelihood)		High	Low	Medium	Low	Low
Data and Privacy						
Traveler ownership of data, ID	Does the traveler own the data?	No	No	?	No	Yes
Privacy protection		Low	Low	High	Medium	High
Data security (of the platform) from external manipulation	Difference between distributed ledger and central database	Medium	Medium	Medium	Medium	High

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Table 5-2. (Continued).

Metric	Notes	Model 1 Government/ Authority	Model 2 Third Party	Model 3 Per Trip	Model 4 Per Life Federalized	Model 5 Per Life SSI
Operational Efficiencies						
Time savings (speed at checkpoints)		Medium	Medium	High	High	Medium
Customer experience improvement	Assuming a faster, relaxed journey is delivered, disregarding preference for data ownership (considered in a different metric)	Medium	High	Low	High	High
Space and resource savings		Low	Medium	Medium	High	High
Applicability outside airport system	The passenger journey outside the airport (ride share/taxi)	No	Yes	No	Yes	Yes
Other considerations						
Availability of technology (now)		Yes	Yes	Yes	Pilot	Pilot
Sharing data with stakeholders	Airport, airline, government authorities	Yes	Yes	Yes	Yes	No
Sharing data with third parties (anonymized)	Retail, food and beverage, hotels, etc.	Yes	If third party shares with airport	Yes	Yes	No
Operational big data applications (passenger flow tracking, prediction)		Medium	If third party shares with airport	High	High	High

implement best practices and make the architecture secure and user friendly. With more stakeholders, this becomes increasingly complex.

On the other hand, global standards, although typically 2 to 3 years behind technology development, aim to create common technical specifications, allowing technologies to interact smoothly while abiding by multiple nations' (and states') laws and being software-, hardware-, and vendor-agnostic. More specific to biometrics, the ICAO DTC development is an example of an emerging global standard and is explained at the end of this section.

Single-Party Biometric Solution

In this scenario, every airport/airline/government has designed a system for themselves that applies to local conditions. This is a technology solution that can be implemented more quickly, easily, and with less capital investment than the multi-party biometric technology solution. A few models exist, and a broader range of possibilities exist for choosing infrastructure components because these generally do not have to be agreed on by multiple

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stakeholders. This option also has the advantage of offering more opportunities for a stricter set of guidelines and allows for more transparency and broader privacy-protection guidelines.

Around the world, governments, airlines, and airports have rolled out or are testing passenger processing systems based on, for example, biometric recognition systems. Many of those platforms and systems have been developed based on national law, international border agreements, stakeholder interests, and limitations of existing facilities, which leads to a varied implementation of solutions. Stakeholders focus on individual interests and not the interests of others, and thus, data sharing across borders and to other stakeholders is not realized.

Multi-Party Biometric Solution

Multi-party biometric solutions comprise all other systems using biometric technology; aim to allow functionality across multiple stakeholders, parties, and countries, or even aim to allow global implementation; and are to a certain extent interoperable and scalable. Technology solutions are typically much more complex in design to allow for interoperability. Ideally, infrastructure components are made compatible and are not vendor specific, although operational requirements need to be specified to guarantee functionality across different infrastructures. Technology guidelines are the greatest hurdle, with many stakeholders having to agree on issues like privacy protection, data sharing, data protection, and data retention.

Stakeholders

As highlighted previously, the direction taken for a single-party or multi-party biometric solution affects the number of stakeholders at the table. The list that follows presents an overview of the main stakeholders, each of which will have a different effect on addressing the challenges of interoperability, scalability, and privacy protection.

- Border control authority (CBP) is responsible for carrying out customs, immigration, and emigration procedures on airport premises and has multiple biometric-ready implementations linked to the use of the TVS.
- Security screening (TSA) screens passengers at the airport as part of the layered approach to security to get travelers safely to their destinations. TSA's screening procedures are intended to prevent prohibited items and other threats to transportation security from entering the sterile area of the airport and being introduced onto the aircraft; are developed in response to information on threats to transportation security; and provide expertise and specialist processes to protect passengers, staff, aircraft, and airport property from accidental/malicious harm, crime, terrorism, and other threats. Aviation security is a combination of human and material resources to safeguard civil aviation against unlawful interference. Unlawful interference could be acts of terrorism, sabotage, threat to life and property, communication of a false threat, bombing, and so forth.
- Airlines are responsible for the verification of valid travel documents of passengers for entry to the destination country. They are concerned about delivering an optimum passenger experience and are therefore a stakeholder in the entire passenger journey at all touchpoints.
- The airport (operator) is responsible for verifying that passengers carry a valid boarding pass on entering the security checkpoint and facilitates the passenger process in its terminal. In addition, airports have a special interest in improved efficiencies in space usage and improving the passenger experience.
- Technology providers are responsible for the development, delivery, and installation of the biometric systems applied at the airports.
- IATA, as representative of the airlines and initiator of the OneID concept, is responsible for guidance and facilitation of the biometric passenger processing developments. IATA advocates striving for efficiencies for all airlines and their passengers.

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- ACI is a representative of the airport industry responsible for guidance and facilitation of biometric passenger processing developments. ACI strives for efficiencies in the entire industry.
- ICAO is the collective body of member state representatives with the role of setting global aviation standards and any rules for aviation safety, security, efficiency, capacity, and environmental protection. ICAO strives for efficiencies in the entire industry, notably resulting in, for example, Doc 9303 (ICAO 2015). ICAO also has the Facilitation Panel established in 1995 that plays a critical role in the accomplishment of priorities and helps to ensure that ICAO Annex 9 (ICAO n.d.-b) is kept current.
- The passenger requires a seamless travel journey without the need to continuously present the same documents at each point. The passenger seeks travel safety and security and an easier process to facilitate entry to a country (visa, supporting documentation, validation).

ICAO Digital Travel Credential

A global standard that has been under development under ICAO is the DTC. Its policy paper "Guiding Core Principles for the Development of Digital Travel Credential (DTC)" was published in October 2020 and sets out a clear set of principles that lay the foundation for a digital credential standard, which would also apply to biometric credentials stored on tokens, on a mobile device, or in a database (ICAO 2020).

The policy paper relates much of the technical specifications to current standards for e-passports, which are an example of an electronic machine-readable travel document (eMRTD). Following the current security of the eMRTD, which results in the verification and consistency of data between physical and electronic documents, the intent is to validate to the DTC at the same level of the eMRTD process. More on the DTC specifications can be found in Appendix N. For airport management considering implementing biometric technologies and solutions, it is important to track the DTC standard developments and make decisions that guarantee the IT architecture is able to incorporate or adapt to such emerging global standards.

Key Takeaway

Supporting your biometric solution and taking into consideration your existing architecture and systems ensures the efficiency of existing and future investments and best practices. By learning from prior completed biometric implementations, much risk can be prevented going forward.

Lessons Learned

Large facilities such as airports may have existing IT installations that are not necessarily designed to support biometrics due to their current architecture, ability to scale, or inability to meet privacy-protection requirements. There are elements that need to be considered to support biometrics, such as network cabling, managing processing loads, balancing network traffic, and applying network standards that address communication between a large number of devices connected to the network. If these are not addressed in planning, network bottlenecks can lead to passengers waiting an unusually long time for a response from the system, leading to the creation of unnecessary queuing.

Cabling, Local Area Network

The main artery to the IT network in a large facility is the cabling, which will link all devices connected to the network. This needs to be maintained and managed at all times. Existing IT installations may not necessarily be designed to support biometric requirements. The network must have the ability to scale as well as to meet privacy-protection requirements. Before starting the biometric journey, the present network architecture needs to be evaluated, not only to ensure the best use of existing assets but, more importantly, to identify limitations and the required best practices to provide the relevant biometrics foundation.

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Network Processing Loads and Network Switching

Network processing loads and network switching refer to the concept of ensuring that the network is efficiently designed and properly balanced and that no item in the network is doing all the work, leading to reduced performance. All the devices that are connected should be sharing the required workloads to make sure that there are no delays, bottlenecks, or slow response times.

A vital part of a biometric network design is what is known as "data packet switching" (transfer of small pieces of data across various networks). For example, when at least three biometric devices are connected or when a biometric device is connected to three or more central servers, different network routes may be used to reach the destination biometric device.

Network Protocols

A network protocol can be defined as the format and the order of messages exchanged between two or more communicating entities (e.g., computers), including any actions taken in transmission or receipt of a message.

Biometric devices connected with each other must use certain network protocols to transmit their data packets back and forth.

A biometric enrollment kiosk includes, in most cases, a passport reader to verify the person's identity based on the image in the passport. Such a kiosk can also have full common-use self-service (CUSS) functionality to guide a passenger through the airline's boarding process and obtain a boarding card for the passenger in the same step.

Facial Recognition: Liveness Detection

Facial-recognition technology uses a biometric reference database to compare an individual's identity with a verified credential—the biometric template. Facial-recognition software performs a series of tasks before it stores a person's biometric template, and at a later time, when an image is captured at a biometric touchpoint, conducts a similar series of tasks to perform the facial-recognition operation.

An important element to these operations that should be performed in parallel is liveness detection. This detects people spoofing or bypassing the system by showing photos, videos, masks, or artifacts to the camera. There are many different techniques available to check the liveness of the person, and these have variable accuracies and response times. Which technique is needed may depend on the scale of the threat that can be expected and the actual location and layout of the touchpoint. Supervision of the touchpoint during operating hours may deter certain threats.

Facial Recognition: Lighting

Facial biometrics supply an easy-to-use method of verifying an individual's identity. However, the accuracy of the verification is significantly less than that of other biometric methods. One of the shortcomings is that many facial biometric algorithms are sensitive to even minor changes in lighting and thus require standardized lighting.

This can prove to be challenging as facial biometric kiosks are incorporated into existing buildings. Modulated light sources synchronized with video cameras can be used to supply images that are lighting independent for these situations. The amount of modulated light needed

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is minimally obtrusive. The lighting arrangement should be appropriate for the application. It should also be remembered that good lighting can suppress the background in closed and open spaces.

Facial Recognition: Face Capturing Hardware

The equipment used for capturing images of passengers' faces has developed over the years and no longer shows large variations in performance, but the hardware and physical layout of a touchpoint can influence the match results. With a high-quality image and the right composition, the face-match system will be more accurate. First, the quality of the camera (light sensitivity, number of pixels, and encoding type) can influence the performance.

In addition, the enrollment station and verification touchpoint can be designed in such a way that every person is lined up in front of the camera in the same way, thereby leading to the same composition. Accuracy can be further improved by ensuring that faces receive enough and equal lighting conditions at every touchpoint. Decent quality images will require less processing by the facial-recognition software for detection and normalization, which in turn will lead to better-quality biometric templates. Equal conditions for images taken at enrollment stations and images at verification touchpoints will lead to more reliable match results.

The following features can be provided at biometric touchpoints to improve the quality of face capturing:

- Stickers of feet on the ground that show the position for the passenger,
- A camera on a moveable stick that is manually pointed at the target person,
- A speed gate with barriers to position a person in front of a camera,
- A camera that can move vertically,
- A dynamic indicator or an avatar to attract attention, and
- Illumination of faces.

Facial Recognition: Accuracy

With the application of facial-recognition systems in aviation, it is important that all stakeholders be aware of how accurate these systems are. Inaccurate capturing equipment or facial-recognition technology leading to a false positive match must not result in a passenger gaining illegal passage at a border checkpoint or entering a secure area or a plane. On the other hand, the algorithm should not be too strict, causing a high number of falsely rejected passengers (false negatives). Finally, the time to render the facial recognition and matching must not take too long. A balance between accuracy and speed of computation needs to exist, as well as the need to abide by national, state, and local laws and security and facilitation. More detail on this subject is included in Appendix O.

Technology Will Fail and an Opt Out Will Need to Be Ever Present

Technology has revolutionized consumer experiences for the past 30 years; however, as airlines and the infrastructure that serve them become increasingly reliant on technology, vulnerabilities become clearer, and minor technical failures can cause catastrophic outcomes.

Insufficient investment in technology infrastructure in the past can result in many technical outages that can cripple airport and airline operations for days on end. As exposure to technology increases, so do the risks of an IT failure. Measures can be introduced to try to limit the risk, but it is obvious that air travel of the future will be prone to the risk of IT failure. Using legacy systems that are already overstretched will increase the likelihood of an IT failure.

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It is important to plan for the inevitable because the biggest risk areas are the loss of landside or airside activity and the loss of baggage systems that work between landside and airside, leading to major impacts on airports, airlines, and passengers. Contingency plans with a recognized structure and model need to be developed to address what happens when things go wrong.

The very nature of IT, seamlessly working in the background, means that it is often not obvious that a problem is just around the corner. The next best thing to preventing a problem is being ready when one occurs.

Findings

Implementing Biometrics

Biometric technology takes the role and the benefits of passenger automation in airports a significant step forward. With the innovations that are taking place, passengers will be able to move through the airport with limited human contact using technology to create their own seamless journey at their own pace.

Key Takeaway

Three areas derive value from a biometric implementation: the use of cloud security, data ownership, and the opportunities of digital travel credentials.

Airports are already using automation to improve the passenger experience, which allows the flexibility to scale up and down their activities due to seasonal changes, passenger growth, and major disruptions to the industry.

The introduction of biometric technology needs to be implemented carefully and without major interruptions to present operations or disruption to passengers, and needs to be integrated into present systems and technologies. Planning is important because the replacement of assets is intrusive and is linked to current technologies and the introduction of a multitude of solution providers and contractors as well as existing infrastructure that is operating in different types of building and terminal designs. One of the challenges is to develop an interoperable identity management system that can be used worldwide rather than having countries and stakeholders invest in localized solutions that are not interoperable or accepted by different authorities.

Biometric systems should be designed to anticipate development and easily adopt new technological advancements, and components that are likely to become obsolete, such as biometric sensors and matching systems, should be modularized. A life-cycle approach is needed that considers the capabilities and limitations of the technology and devices. This approach must also be flexible enough to manage the unexpected reactions of users, operators, and other stakeholders.

Cloud Data Security, Ownership, and Technology

Data security has consistently been a major issue in information technology. Biometrics use cloud computing environments primarily because the data can be stored in different locations across the globe. Data security and privacy protection are the two main user concerns about cloud technology. Data security and privacy-protection issues are relevant to both hardware and software in a cloud architecture.

Cloud computing allows processing on demand and convenient, ubiquitous network access to shared configurable computing resources such as storage, networks, servers, services, and applications. This type of solution is suitable for biometrics. Biometric identification management systems are multimodal, and they generate large amounts of data. Processing petabytes of data cannot single-handedly be addressed by sheer central processing unit (CPU) power

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and needs flexible and economical infrastructure that can manage variable processing and data requirements.

Data ownership in any environment relates to both the possession of and responsibility for information. In other forms of control of information, cloud services not only provide the ability to access, create, modify, package, derive benefit from, sell, and remove data, but also protect by applying rules relating to the assignment of privileges to others.

In a cloud environment, the use of blockchain technology is beneficial when low processing volumes are expected and the transactional records are located in a distributed-ledger database that links to numerous stakeholder databases. The records are unchangeable and signed cryptographically with a consensus protocol to confirm that the data are correct (or are not).

In order to deliver a biometric solution across different geographies, the distributed-ledger database, linked to other stakeholder databases, will need to be applied.

Digital Travel Credential and Self-Sovereign Identity

When booking or checking in, travelers could potentially send their virtual credential in advance from their mobile phone to the border authority or to an airline for advanced passenger information/passenger name record (PNR) purposes. As ICAO's DTC standard is implemented by different countries, it could simplify biometric enrollment or potentially augment legacy systems such as the Electronic System for Travel Authorization (ESTA).

For the DTC, ICAO has the role of coordinating and establishing the digital identity format between all nations. Border authorities are the direct stakeholders charged with the responsibility of ensuring that border processes comply with national legislation. Airports and airlines are indirect stakeholders that may benefit from more efficient passenger processes. Technology providers have an interest in developing systems that they can sell on a global scale.

The DTC can be used as part of the credential-issuing stage of a biometric process. This stage can be performed from a passenger's mobile phone, and the credentials can be sent to any system or database. Related to the DTC, where no specific facilities are required, this will depend on the use case that is to be applied at the airport. Theoretically, all passengers could be able to prove their identity, store their biometric details digitally, and have them certified with an identity verifier.

A secure digital biometric system will bring many opportunities and advantages for passenger processing and can be adopted internationally. A similar time frame to the adoption of e-passport standards could be foreseen.

Due to individual countries developing and adopting their own biometric solutions, a wide range of different systems per country and on an extraterritorial basis could be the norm in the future.

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CHAPTER 6

Future Directions

Suggestions for Further Study

Acknowledging that this primer has been developed at a time when biometric technologies, the related laws and policies, and, most importantly, our understanding of the technology are developing, the research team suggests a timely update of this research, ideally within 2 to 3 years.

The research team is also mindful that a primer has limited scope to address emerging issues, especially regarding the tools deployed for COVID-19 pandemic recovery that are driven by the need for a touchless environment. There are five core suggestions for further study, which may be additional research scopes, pilot projects, or proof-of-concept ideas that government and industry stakeholders could pursue.

Running Inventory of Airport Pilot Projects and In-the-Field Research

Regarding additional research scopes and especially pilot projects, the research team realizes the immense value that feedback from in-the-field research into biometrics, together with new applications of biometrics and (in some cases ongoing) pilot projects, have provided to the research and development of this primer. For future projects, a running inventory of commencing, ongoing, and recently completed in-the-field research and pilot projects testing new technologies, procedures, or ideas would be a valuable resource for principal investigators and their teams. Maintaining such an inventory (possibly through the IdeaHub platform) in a collaborative manner would benefit the ACRP community.

For Further Study

Five core suggestions for further study are discussed in the following.

1. Human Factors Research

There are core aspects of biometrics that are heavily driven toward technological solutions. The language of solution development largely concerns false-accept and false-reject rates, as well as the architecture and solutions to share information. While many solutions involved a holistic review of human factors, more research is needed to assess the acceptability and suitability of the solutions to the subject of biometrics: humans themselves.

Research could involve a range of tests around the solutions and devices or emerging areas in passenger communications. Digital transparency in the public realm, for example, is an area that requires more attention to uncover opportunities to improve passenger and employee understanding of the use of biometric data.

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2. Self-Sovereign Identity, Middleware, and Traveler Verification Service

SSI could provide passengers and staff greater ability to own identity data fully without intervention from outside administration. At the same time, systems that are geared toward admissibility to the United States are structured around longer (75-year) retention periods that challenge the control function and privacy concepts such as the right to forget.

As systems like the TVS are geared toward the ability to host identity information in a government-controlled cloud system, there is an emerging area of problem solving that requires more detailed use cases to bridge the gap between SSI and models such as the TVS. Middleware is likely needed as a way of defining the models for the ability for the TVS to work with emerging global standards such as ICAO DTC.

Whether the research effort is suitable for a hackathon or series of hackathons to deliver potential solutions, there is an aspect of creating an environment to maximize the strengths of SSI and the power of the existing TVS framework.

3. Biometrics Best Practice Designations

The current state of facial recognition and biometrics is one where the language is expansive and unclear to the end user. High-profile abuse and misuse of artificial intelligence, selling of consumer identity information, and other biometric practices have created major concerns among the general public. Moreover, the methodologies, however different from ones used by airlines, airports, and many government agencies, can be tainted with the same negative brush when in fact the technology/governance may be quite different.

Whether it is green buildings, genetically modified food product certification, or other applications that are driven by potentially risky issues, there is further research needed on codification of best practice designations in biometrics that cover the range of privacy, operational, technology, or other aspects of uses to differentiate the myriad of technology deployments at airports.

The research that led to the WEF publication of "A Framework for Responsible Limits on Facial Recognition Use Case" (World Economic Forum 2020c) should be regarded in the research, and, ideally, be implemented at airports, especially as the evolution of domestic TSA checkpoint biometrics accelerates.

4. Equity, Diversity, and Inclusion Reviews

High-profile cases of biometric systems that have performed differently by race are indicative of issues from a societal and technological point of view. Where one community is adversely affected compared to another, issues exist that can challenge the perception of biometrics at airports, especially with populations of passengers from all around the world. Moreover, there are also known issues for individuals with different mobility issues who can benefit from or be challenged by biometric solutions.

5. Digital Twins and Biometric Models

Another area of research relates to creating a digital model of facilities for airport operators and aviation industry participants to use to better understand the potential of biometric systems. In the past, the U.S. Commercial Aviation Partnership had an econometric model in place to assess the value of new measures at airports. This research effort was conducted under the Aviation Security Advisory Committee (Peterson et al. 2006). There may be a need to augment past efforts to emulate across airports, airlines, and government agencies a standing model based on digital twins that is able to predict the power of different modes of biometric implementation across major passenger and employee flows.

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Abbreviations, Acronyms, and Initialisms

AAAE	American Association of Airport Executives
ACI-NA	Airports Council International–North America
ADIS	Arrival and Departure Information System
AGTS	Automated Guideway Transit System
APC	Automated Passport Control
API	Application Programming Interface
APIS	Advanced Passenger Information System
ATL	Hartfield-Jackson Atlanta International Airport
ATS	Automated Targeting System
BIPA	Illinois Biometric Information Privacy Act
BOS	Boston Logan International Airport
CBP	(U.S.) Customs and Border Protection
CHRC	Criminal History Records Check
CSCA	Country Signing Certification Authority
CUSS	Common Use Self Service
DAC	Designated Aviation Channeling
DCA	Ronald Reagan Washington National Airport
DCS	Departure Control System
DEN	Denver International Airport
DHS	Department of Homeland Security
DOS	Department of State
DTC	Digital Travel Credential
DTPR	Digital Trust in Places and Routines
DTW	Detroit Metropolitan Wayne County Airport
EEA	European Economic Area
EES	Entry/Exit System
eMRTD	Electronic Machine-Readable Travel Document
ESTA	Electronic System for Travel Authorization
EU	European Union
EWR	Newark Liberty International Airport
FBI	Federal Bureau of Investigation
FIPPs	Fair Information Practice Principles
FIS	Federal Inspection Station
FLL	Fort Lauderdale–Hollywood International Airport
FTC	Federal Trade Commission
FTCA	Federal Trade Commission Act
GAO	Government Accountability Office
GDPR	General Data Protection Regulation

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GES	Global Enrollment System
HART	Homeland Advanced Recognition Technology
IATA	International Air Transport Association
ICAO	International Civil Aviation Organization
ICE	Immigration and Customs Enforcement
IDaaS	Identity as a Service
IDENT	DHS Automated Biometric Identification System
IdMS	Identity Management System
INSPASS	INS Passenger Accelerated Service System
IT	Information Technology
JFK	John F. Kennedy International Airport
KTDI	Known Traveller Digital Identity
LAX	Los Angeles International Airport
LDAP	Lightweight Directory Access Protocol
NEXTT	New Experience Travel Technologies
NGI	Next Generation Identification
NIST	National Institute of Standards and Technology
PDE	Passenger Data Envelope
PIA	Privacy Impact Assessment
PII	Personal Identifiable Information
PNR	Passenger Name Record
POE	Port of Entry
QR Code	Quick Response Code
Rap Back	Record of Arrest and Prosecution Background
ROI	Return on Investment
SEA	Seattle–Tacoma International Airport
SEN	Secure End Node
SIDA	Secure Identification Display Area
SORN	System of Records Notice
SPII	Sensitive Personally Identifiable Information
SSI	Self-Sovereign Identity
STA	Security Threat Assessment
TTP	Trusted Traveler Program
TVS	Traveler Verification System
UK	United Kingdom
UPAX	Unified Passenger Module
VIP	Very Important Person
VPC	Virtual Private Cloud
WEF	World Economic Forum

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Glossary

Specific Terms Explained in Relation to One Another

For the definitions and differences between the terms **Authentication**, **Detection**, **Identification**, **Matching**, **Validation**, and **Verification**, see the Fundamentals of Biometrics section of Chapter 1.

Relating to sensitive and personal information, the following classifications exist:

Personally identifiable information (PII) – Information that permits the identity of an individual to be directly or indirectly inferred, including any information that is linked or linkable to that individual, regardless of whether the individual is a U.S. citizen, legal permanent resident, visitor to the United States, or employee or contractor of the Department of Homeland Security.

Sensitive personally identifiable information (SPII) – Personally identifiable information that, if lost, compromised, or disclosed without authorization, could result in substantial harm, embarrassment, inconvenience, or unfairness to an individual.

Sensitive personal information (SPI) – Information that does not identify an individual, but is related to an individual, and communicates information that is private or could potentially harm an individual should it be made public.

General Terms

Artificial intelligence – A branch of computer science dealing with the simulation of intelligent behavior in computers.

Bias – Systematic error introduced into sampling or testing by selecting or encouraging one outcome or answer over others.

Biographic data – Information describing non-physical characteristics of a person (e.g., name, date of birth, and place of birth).

Biometric data – Information describing physical characteristics of a person (e.g., height, color of eyes, fingerprints, or a photo).

Biometrics – The measurement and analysis of unique physical or behavioral characteristics (such as fingerprint or voice patterns), especially as a means of verifying personal identity.

Biometric template – A record of data that is created by encrypting biometric information through cryptography, creating a random sequence of numbers and letters. A biometric template cannot be reverted to its original form; it is a one-way method of encoding data.

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Biometric touchpoint – A dedicated area where a person can get his/her identity verified by showing a biometric credential (e.g., face) to a camera and, after identification, perform operations in or interact with a specific airport system.

Blockchain technology – A digital database containing information (such as records of financial transactions) that can be simultaneously used and shared within a large decentralized, publicly accessible network.

Border search exception – A Supreme Court–recognized exception to the Fourth Amendment in U.S. law that allows searches and seizures at international borders without a warrant or probable cause.

Bypass – When biometric data are tampered with post-capture, or when the camera is bypassed altogether.

Cashierless – The use of a self-service checkout kiosk (as in a store) to handle monetary transactions.

Central processing unit (CPU) – The component of a computer system that performs the basic operations (such as processing data) of the system, that exchanges data with the system's memory or peripherals, and that manages the system's other components.

Credential – An object or data structure that authoritatively binds an identity—via an identifier or multiple identifiers—to at least one authenticator possessed and controlled by a subscriber (passenger or airport staff).

Credential Authentication Technology – A TSA technology that ensures ID authentication, reservation verification, and Secure Flight pre-screening status at airport security checkpoints.

Cybercrime – Criminal activity (such as fraud, theft, or distribution of child pornography) committed using a computer, especially to illegally access, transmit, or manipulate data.

Cybersecurity – Measures taken to protect a computer or computer system (as on the Internet) against unauthorized access or attacks.

Dactyloscopic identification – Identification by comparison of fingerprints.

Database – A large collection of data organized especially for rapid search and retrieval (as by a computer).

Data packet switching – A method of transmitting data over a digital network by grouping it into smaller packets of data.

Designated aviation channeling (DAC) – A service in the aviation industry that supports conducting required background checks of employees in the sector, thereby ensuring safety and security.

Digital identity – A digital version of one's identity and the distinguishing character or personality of an individual.

Digital Trust in Places and Routines (DTPR) – Formerly Digital Transparency in the Public Realm, DTPR is an open-source initiative by Helpful Places that aims to promote an open-source communication standard to enable transparency, accountability, and control for people.

Distributed ledger (blockchain technology) – A mechanism for storing data, replicated numerous times and in a synchronized manner, geographically spread across multiple computers in different locations. A consensus of all the computers is kept current, making unwarranted mutations (e.g., a cyberattack) of the data generally impossible.

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E-passport – A formal document issued by an authorized official of a country to one of its citizens that is usually necessary for exit from and reentry to the country, allows the citizen to travel to a foreign country in accordance with visa requirements, and requests protection for the citizen while abroad.

EU-U.S. Privacy Shield program – A framework designed by the U.S. Department of Commerce and the European Commission to provide companies on both sides of the Atlantic with a mechanism to comply with data protection requirements when transferring personal data from the European Union to the United States in support of transatlantic commerce.

Fair Information Practice Principles (FIPPs) – FIPPs are five main guidelines published by the FTC that represent widely accepted concepts concerning fair information practice for privacy protection in the electronic marketplace.

Global Entry – A CBP program that, on arrival in the United States, allows expedited clearance for enrolled members who are pre-approved, low-risk travelers.

Health passport – Similar to a passport, a health passport contains biographic (and biometric) data as well as important health information such as the results of certain tests, vaccinations, and specific medical conditions.

Interoperability – The ability of a system (such as a biometric system) to work with or use the parts or equipment of another system.

Known Traveler Number – A number issued by the TSA, DHS, or DoD that indicates that a passenger has undergone a pre-flight background check or other screening before checking in for a flight.

Machine learning – The use and development of computer systems that are able to learn and adapt without following explicit instructions by using algorithms and statistical models to analyze and draw inferences from patterns in data.

Mass surveillance – A form of indiscriminate surveillance that uses systems that collect and analyze surveillance footage with the intent of identifying people for a specific purpose or range of purposes.

NEXUS – A voluntary program designed to speed up border crossings for low-risk, pre-approved Canadian travelers into the United States.

OneID – A concept by IATA that aims to promote robust, integrated identity management across the end-to-end passenger process and allows individuals to assert their identity online or in person.

Opt in – To voluntarily choose to participate.

Passenger journey – In this primer, the trip from an origin to a final destination.

PreCheck – A TSA program that lets eligible, low-risk travelers enjoy expedited security screening.

Preclearance – A program that allows passengers and their bags to be inspected and cleared at a foreign airport by U.S. CBP officers prior to travel to the United States.

Privacy by design – A set of seven principles in system or process design that advance the view that privacy assurance should ideally become an organization's default mode of operation and be incorporated from the start of the project.

Rap back (FBI) – Refers to the notification returned to an organization when an employee who has undergone a fingerprint-based state or federal criminal history record information check has a subsequent state or federal criminal history event.

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REAL ID – Pursuant to the REAL ID Act, Pub. L. 109-13, Congress mandated minimum security standards for drivers' licenses and identity documents and prohibits federal agencies from accepting "for any official purpose" state driver's licenses or identification cards unless those documents meet the standards set forth therein. Pub. L. 109-13 Title II Sec. 201 (3); REAL ID Act, Pub. L. 109-13, Sec. 202(a)(1).

Reasonable expectation of privacy – A person's reasonable expectation of privacy generally implies that someone who unreasonably and seriously compromises another's interest in keeping his or her affairs from being known can be held liable for that exposure or intrusion.

Risk profiling – In the aviation world, the screening of passengers within a specific framework and aiming to address security risks.

Seamless – Having no awkward transitions, interruptions, or indications of disparity.

Self-sovereign identity (SSI)– Recognition of the requirement that individuals own and control their identity without the intervention of an administrative authority.

Special needs exception – An exception to the Fourth Amendment requirement that government searches be supported by a warrant and probable cause that allows government searches that are primarily aimed at advancing some special need other than criminal law enforcement (e.g., security checkpoint searches) if the search program is reasonable given the balance of public and private interests.

Single sign on – An authentication scheme that allows a user to log in with a single set of login credentials, such as a username and password, to access multiple independent applications.

Spoof – A nonliving object that exhibits human traits (an artifact) that is presented to a camera or biometric sensor.

Supremacy Clause – A provision in the U.S. Constitution that establishes that the Constitution and federal laws take priority over any conflicting state laws.

Touchpoints – Points of contact or engagement between a passenger and a business (e.g., airline, TSA, concessionaire) that can occur at any time throughout a passenger's journey.

Transportation network companies – Businesses that provide prearranged transportation for compensation using an online-enabled application or platform (such as smartphone apps) to connect drivers using their personal vehicles with passengers.

Trusted traveler – A passenger who is a member of a risk-based program (e.g., Global Entry, TSA PreCheck, SENTRI, NEXUS, and FAST) to facilitate the entry of pre-approved travelers.

VeriFLY – Allows a passenger to verify the required credentials necessary to obtain a valid pass that will enable access to a service where a person must be physically present.

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APPENDIX A

Case Study: Amazon Go Cashierless Retail Experience

Summary

Initiated by Amazon, the Amazon Go retail experience (see Figure A-1 and Figure A-2) uses a combination of cameras, sensors, computer vision techniques, machine learning, and artificial intelligence to create a cashierless retail experience; the first airport location will be at Newark Liberty International Airport (EWR) (Amazon Go 2020). This technology, called "Just Walk Out," requires customers to identify themselves when they enter the store (Amazon 2020). The customer's identity is verified either by scanning the Amazon Go mobile app (in the case of an Amazon Go store) or their credit card (for other retailers equipped with the Just Walk Out technology). A biometric variant also exists, where instead of a QR code or credit card, customers enroll at the storefront enrollment kiosk by scanning their 3D hand palm biometric and linking that to their account.

The Just Walk Out technology leverages computer vision and machine learning to distinguish between customers and the items picked out and added to their virtual cart. The system does not rely on facial recognition but relies on movement tracking.

Each customer is associated with a name, an account, and a consumer profile that reflects all interactions with items on the shelves (which items are stared at, picked up, or bought). Future use of the customer's data is still undefined. However, the initial patent application included some examples where the customer's purchase history could be used to confirm which items were picked by the user.

The main benefits are significant time savings for customers and financial savings for retailers (reduced to no cashiering cost) (Bishop 2020b). The system architecture relies on Amazon's



Source: Adapted from Amazon Go 2020.

Figure A-1. Amazon Go shopping experience.

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Table A-1. Key facts for the Amazon Go case study.

What?	• Cashierless retail technology • Project implemented by Amazon
Where?	• 26 Amazon Go stores across the United States (e.g., Chicago, New York, Seattle, San Francisco) • Deployment at a few airports across the United States in stores of retail chain OTG (On-the-Go), including at EWR (OTG Management 2020)
Customer process steps	• To access Amazon Go stores: – Create an Amazon account – Download Amazon Go app – The Amazon Go mobile app interface will generate a QR code, or the user may add a hand palm scan to the account at an enrollment kiosk. – Customer enters the store by scanning the QR or hand palm (or credit card for other retail stores with Just Walk Out technology) – Just Walk Out technology tracks the customer and the interaction with store items. – Virtual cart validation – Exit
Who?	• Amazon patented the Just Walk Out technology. It provides the hardware (camera and sensors) along with the software.
Why?	• Amazon's goal is to create a seamless shopping experience where customers do not have to wait in line.
How?	• The concept relies on the combined use of cameras, sensor fusion, computer vision, and deep-learning algorithms to track customers during their time in the store, noting each item picked up, put back, or added to their virtual carts. For the biometric hand palm variant, 3D hand palm scanners and a palm recognition software are used.
Enrollment/digital identity creation and verification	• An Amazon account can be made online, and enrollment is done in the storefront at the enrollment kiosk. The online Amazon account can be linked to the hand palm biometric by scanning one's hand at the kiosk.
Verification of identity how?	• Customers are identified by either a QR code generated by the Amazon Go app in Amazon Go stores or their credit card in third-party retail stores where the Just Walk Out environment is available.
For?	• Amazon account holders

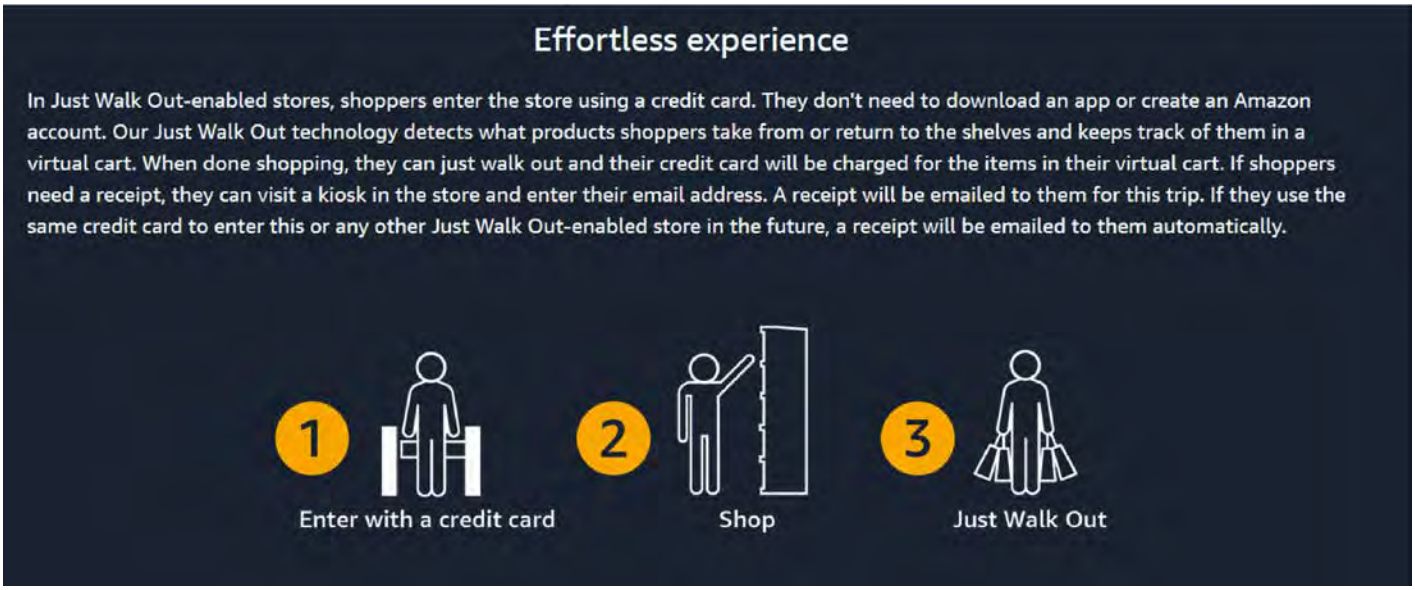
Just Walk Out technology and includes the camera and sensor hardware as well as the software system. Table A-1 contains a summary of the Amazon Go case study.

How Does It Work?

Before the Customer Journey

- Amazon Go relies on sensor fusion (analyzing and aggregating data from multiple sensors, including weight and movement sensors), advanced data hosting services through Amazon Web Services, and advanced computer vision–based machine learning (Amazon 2015).
- The hardware includes electronic shelves, cameras, fixtures, and a facility management system.
- The inventory management involves data storing and item identification.
- In the case of Amazon Go stores, customers create an Amazon account and download the Amazon Go app.
- In the case of Just Walk Out–enabled stores, customers access the store with their credit card.

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Source: Amazon 2020.

Figure A-2. Just Walk Out enables seamless store experiences for other retailers as well.

The Customer Journey

The customer journey can be described by the following:

- Each shopper (and that shopper's party) enters and is identified with a QR code generated by the Amazon Go app, biometric hand palm, or a credit card in the case of stores equipped with the Just Walk Out technology.
- The technology tracks the customer's movements and interactions with the different store items. As customers remove items from the shelves, those items are added to their virtual cart (Kumar et al. 2013).
- The customer receives a receipt and is charged when exiting the store.

Retention and Storage

Account data are saved on Amazon's own servers.

- Cameras track throughout the store in real time using the video feed streaming from a ubiquitous network of cameras (Bacco and Hiatt 2010).
- Amazon Go stores retain data on the interactions of individual customers with the items on the shelves. These data are used to further train the artificial intelligence running the architecture.
- Account data are retained and stored at Amazon until the passenger opts for the deletion of the account.
- Customers can elect to delete their biometric data.

System Architecture Flow Diagram

The flow diagram for this case study can be found in the Amazon Go Cashierless Retail Experience case study of Chapter 2.

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System Specifications

The system relies on an artificial intelligence (see Figure A-3) to track the customer in the store. The artificial intelligence architecture relies on Amazon Web Services for streaming services as well as advanced computer vision–based machine learning.

Computer Vision–Based Machine Learning

To associate the right items with the right customers, the technology relies on aggregating data from different sensors linked with the location information of the sensors. The artificial intelligence driving the architecture tracks the customers in the store by aggregating data from different sensors through sensor fusion and solving different identification and linking problems to associate customers, their location, and their interactions with items in the store.

Person Detection

The Just Walk Out technology does not use facial-recognition technology but relies on red-green-blue cameras equipped with depth- and distance-sensing capabilities. Each customer is associated with a general profile and an anonymized 3D point cloud. During the customer's time in the store, a deep-learning algorithm predicts the customer's location and associates that location with actions and interactions with store items (see Figure A-4).

System Architecture, Pre-Existing Systems, and Databases

The following explain the buildup of the system and its processes:

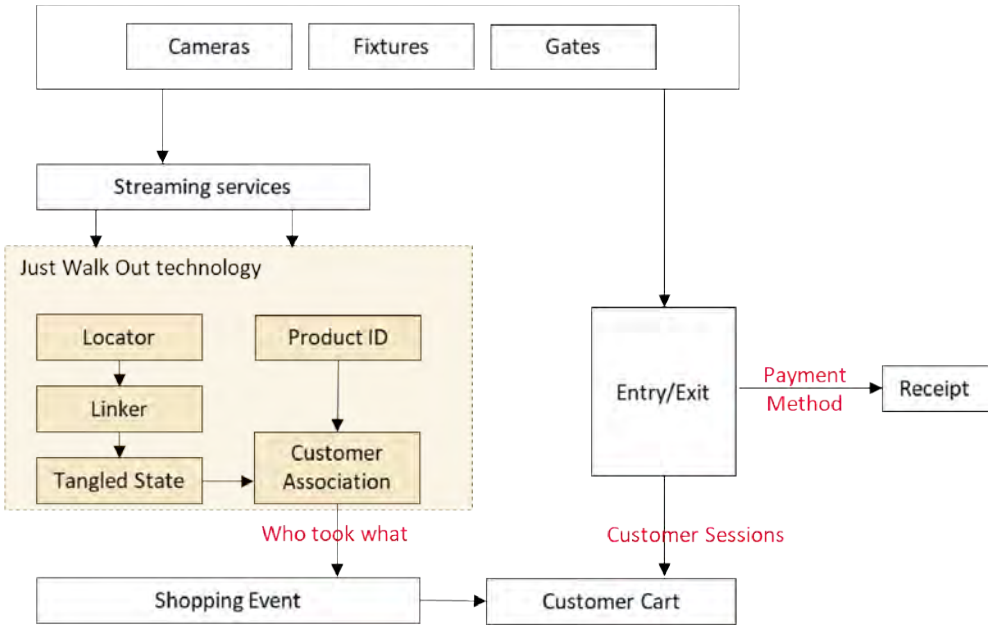
- Customer information and biometrics are collected at the entry gate.
- The identification of distinctive features on the hand(s) of the customer relies on a proprietary algorithm, with distinctive features of the hand specific to this proprietary technology architecture (see Figures A-5 through A-7).
- Hand biometrics data that are collected are either stored, if the customer has an Amazon account, or are deleted once the customer exits the store.
- The customer holds his or her palm over the device to opt in.



Source: Amazon 2019.

Figure A-3. Amazon artificial intelligence for the Amazon Go retail experience.

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Source: Amazon 2019.

Figure A-4. Logic structure of Amazon Go and technology components.

Stakeholders and Responsibilities

Stakeholders

The main stakeholders of Amazon Go concept are Amazon and the third-party retailers who have purchased Just Walk Out technology services.

Responsibilities

Amazon owns the Just Walk Out concept, which builds on a series of patents submitted by Amazon Technologies since 2013 (Puerini et al. 2014).

Case Study Review

Benefits

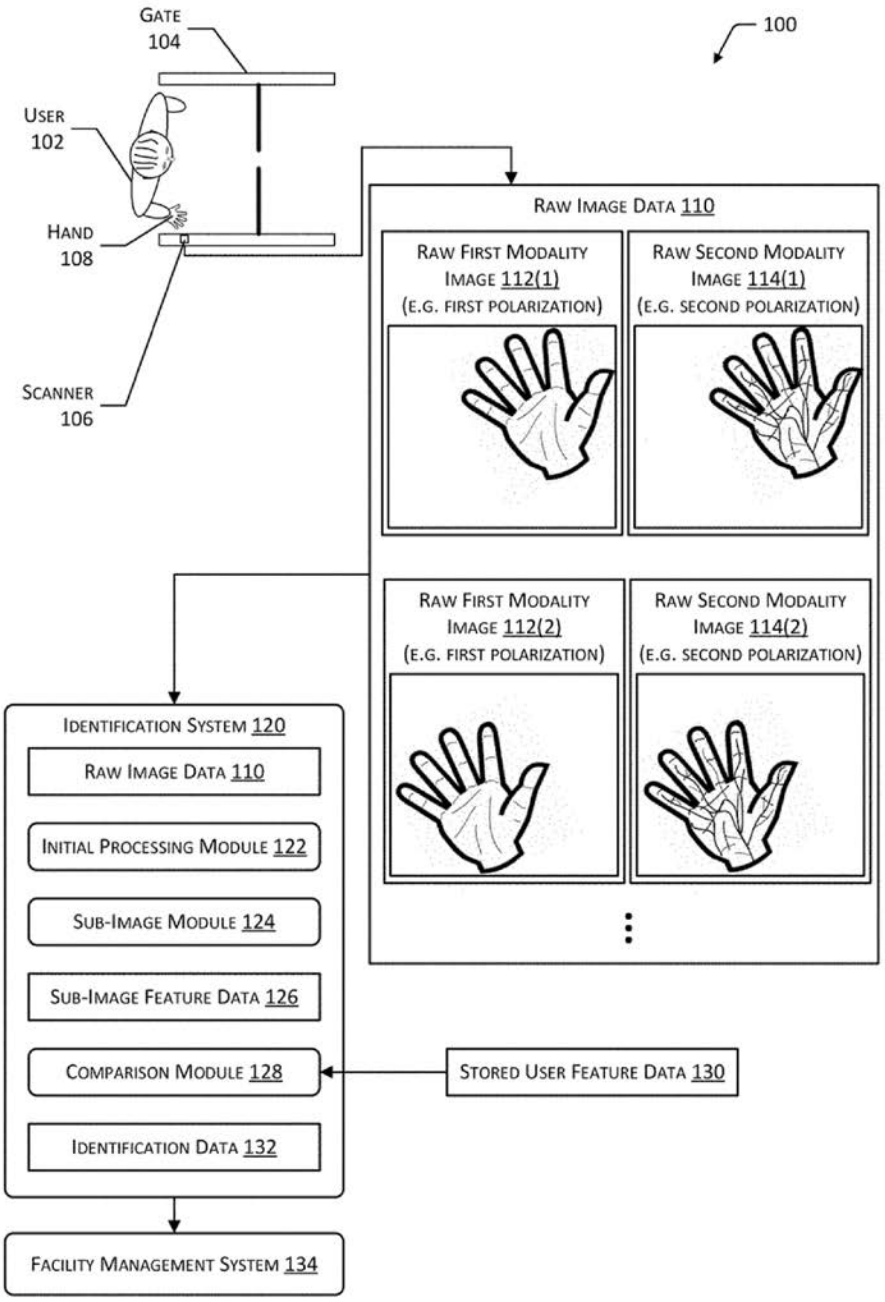
The Just Walk Out technology presents benefits both for the customer and the retailer. Benefits to customers include:

- Improved customer experience over time with deep learning allowing for faster tracking and more accurate identification of interaction with store items,
- Limited contact and interaction with store employees,
- Reduced time spent in the store, and
- No use of facial recognition.

Benefits for the retailers providing the Just Walk Out–enabled experience include:

- Adaptability of the Just Walk Out technology to different store layouts,
- Financial savings due to reduced staff cost related to cashiering or inventory,
- No use of facial recognition,
- Improved customer experience, and
- Detailed consumer profile data.

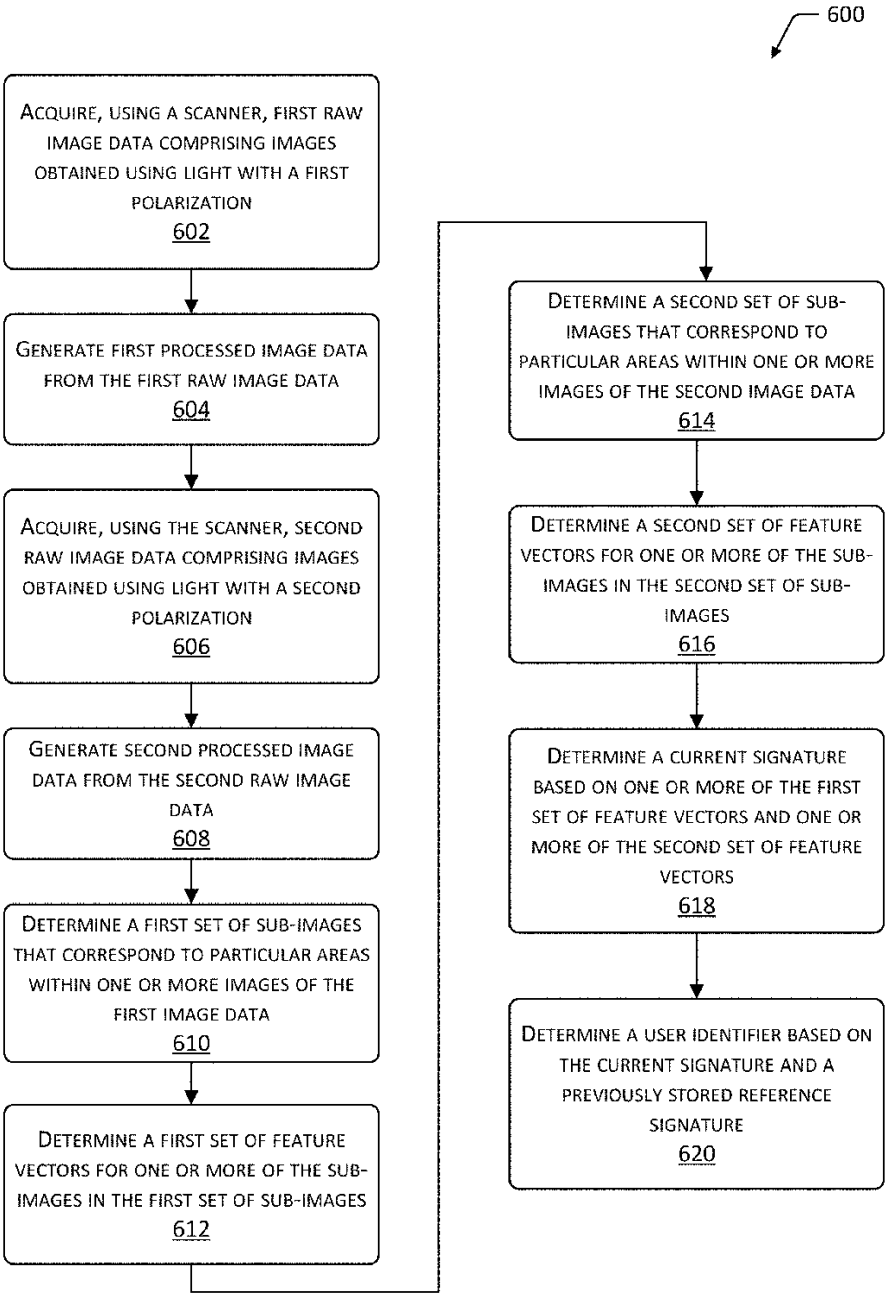
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Source: Kumar et al. 2018.
Note: Numbers in figure are part of the patent application and are not relevant to this discussion.

Figure A-5. Hand biometrics identification process.

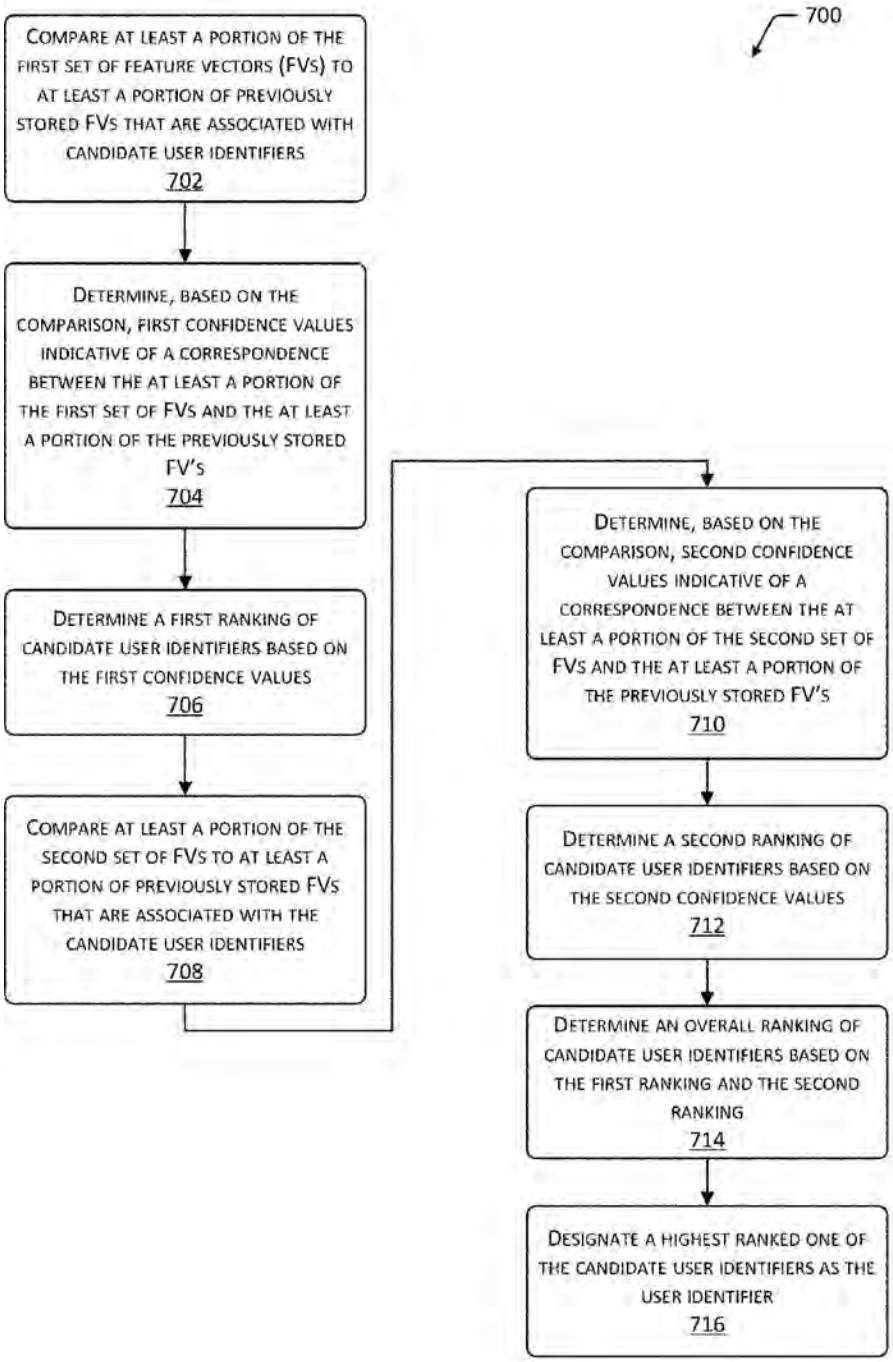
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Source: Kumar et al. 2018.
Note: Numbers in figure are part of the patent application and are not relevant to this discussion.

Figure A-6. Hand biometrics characteristics identification.

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Source: Kumar et al. 2018.
Note: Numbers in figure are part of the patent application and are not relevant to this discussion.

Figure A-7. Person detection based on hand biometrics identity.

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Responses From Customers

Crowd-sourced reviews for individual stores rate on average more than 4 stars out of 5. A stated choice survey conducted by the Shorr group in 2018 found that "84% of respondents to a survey said that they see Amazon Go as a 'type of grocery shopping experience' they'd enjoy more than traditional grocery shopping" and "over 25% of respondents said that they would pay more for grocery products if it meant they didn't have to wait in line at checkout" (Shorr 2018).

System Performance and Specifications Review

The Just Walk Out technology requires an approximately $1 million investment in hardware. Customers spending less time in the store allow for higher customer throughput per hour compared to a traditional store (Cheng 2019). The system is designed for 99% accuracy, including during the busiest periods. On average, Amazon Go stores process 550 customers per day and as many as 90 people at any given time.

Fall-Back Options

- Despite the absence of cashiers, a few store staff members are present to assist customers with any technical issues, answer questions, and process cash transactions.
- If customers are incorrectly charged for an item, they can contest the charge and be reimbursed.

Concerns

- Privacy and use of collected data: There is a lack of transparency on how the collected data factor into the overall Amazon personalized marketing strategy. The architecture relies on deep learning, and data mining of the customer's activity in the store would improve the experience and accuracy over time.
- Lack of transparency around the use of data slowed the adoption of the technology in Europe because of confrontation with GDPR regarding consent to the processing of personal data. In the context of the regulation, every consent request must state the precise purpose for which the data will be processed (Walters 2020).

Lessons Learned

The Just Walk Out technology initially relied on motion tracking and an anonymized cloud of data to identify individual customers. However, in December 2019, Amazon submitted a patent for a non-contact biometric identification system to identify customers by their hands (Kumar et al. 2018). Introducing hand biometrics as one of the identifying features of each customer is expected to simplify the person-detection component of the Just Walk Out architecture.

Findings and Trends

Findings

The Just Walk Out technology is a potential template for future models of touchless and cashierless retail at airports and other shopping locations. The use of biometrics for identification, camera tracking devices, image analysis software, and a program that automates product billing has proven to create a new user-friendly experience that is more efficient and is touch free and seamless. Amazon now has several versions of this retail technology in operation, and future revisions of the concept will only improve accessibility. Enrollment challenges may be resolved with biometric palm geometry recognition, which is also being trailed, or with a simple

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two-step authentication process. Another option is the use of not just one biometric, but two or more, thereby providing options for customers to choose from.

The main benefits behind using this type of software are significant time savings for customers and financial savings for retailers through lower operational expenditures on employees (e.g., labor wages, benefits). Some enrollment challenges may be resolved with simple two-step palm enrollment, which is interesting because it may seem less invasive when compared to a picture of one's face. Furthermore, this type of software presents a potential alternative to current and future models for touchless retail for airports and airlines.

Future Situation and Broader Implementation

Amazon submitted a patent application for a touchless hand scanning system named "Amazon One" and started implementing it in Amazon Go stores in October 2020. Customers scan their hands at the entry gate to enter the store.

Trends Identified

A trend identified in the Amazon case study is that the use of hand biometrics is aimed to facilitate a more efficient, user-friendly (retail) experience. In the cashierless system, time is saved, shopping requires less fumbling of personal items during payment, and simplicity for the user can bring satisfaction. In this specific case, it is not only the biometric technologies enabling this, but also the camera tracking system and software, which allow for the automatic charging of customers to their accounts. This is a trend that will likely expand to other sectors outside of retail and airports.

Similarly, the retail experience removes the need for contact points and human interactions, which has the added benefit of reducing the risk of transmissible diseases, which is especially beneficial in the COVID-19 era. The camera system and the 3D hand geometry scanners allow for tracking and identification without customers needing to touch any surfaces.

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APPENDIX B

Case Study: Denver–Daon Biometric Partnership

Summary

COVID-19 has changed passengers' needs at airports, and there is now a greater focus placed on health. Denver International Airport (DEN), as part of its goal to make the passenger journey safer and more predictable, is partnering with Daon, an innovator in biometric authentication with experience developing the TSA PreCheck program, to develop a series of pilots at the airport. The technology solutions are based on Daon's Glide HealthPass platform, which is a digital health passport that allows businesses to build an authorization process to meet their unique and evolving needs (Daon 2020). The first pilot is the VeriFLY reservation system, which is inspired by a self-identified population of health-conscious travelers who want to maximize social distancing while using touchless technologies (DEN Talks 2020). Using a reservation system similar to OpenTable, travelers make a reservation for access to a dedicated VeriFLY security lane; access to the security lane is via an e-gate. Once through screening, travelers would have access to a reserved, lower-capacity train car on the automated guideway transit system (AGTS) to access the concourses. VeriFLY is free to passengers, but they must download an app and register with some basic personal information and a photograph, which is stored by Daon. Benefits of the program include building confidence and certainty throughout the passenger journey, flattening the arrival peaks with the reservation system, promoting a touchless and document-free travel experience by incorporating biometrics, and reducing exposure for TSA officers. A second pilot applies the same technology to a remote screening facility to allow the families of employees to proceed through security outside the terminal area. See Table B-1 for more information on the Denver–Daon partnership.

Introduction

To address passenger health and safety, DEN has partnered with public health stakeholders and biometric technology firm Daon to develop a biometric authentication system known as VeriFLY, which is aimed at making passenger journeys safer and more predictable. A second pilot for remote screening takes the security checkpoint off of airport premises.

How Does It Work?

Before the Passenger Journey

VeriFLY is free to passengers, but they must download an app and register with some basic personal information (non-PII) and a photograph. Daon's IdentityX is a digital identity platform that delivers seamless identity establishment and verification, multi-factor authentication, and recovery. The technology solutions are based on Daon's Glide HealthPass platform, which is

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Table B-1. Key facts of case study on Denver–Daon partnership.

What?	• Biometric authentication system for security lane reservations and off-airport premises security screening
Where?	• DEN
Customer process steps	• Mobile app sign-up to enroll and make a reservation for the security checkpoint • On day of travel, passenger self-certifies health status in app and receives QR code • Arrive at airport for designated slot, verify body temperature, and use QR code to access dedicated security checkpoint lane
Who?	• Daon • DEN • Passengers • TSA
Why?	• Speed up passenger processing of passengers • Relieve pressure on security checkpoints
How? Technology used?	• VeriFLY: facial recognition to verify identity in the app to bring up a QR code for airport touchpoints • Remote screening: bring security checkpoint off of airport premises
Enrollment/digital identity creation and verification	• Voluntary program with enrollment completed on the app. Appointments on the app can be booked 2 weeks in advance.
Verification of identity how?	• Initial rollout using QR code, with future rollout of facial recognition via the mobile app
For?	• Health-conscious departing travelers who want to maximize social distancing and use of touchless technology while being able to get through checkpoint at a known time and access the concourse on a dedicated AGTS car • Remote screening for those passengers without bags traveling with airport employees

a digital health passport that allows businesses to build an authorization process to meet their unique and evolving needs. The database is housed and accessed entirely by Daon.

The Passenger Journey

Within 2 weeks prior to travel, the passenger uses the app to create a reservation for a checkpoint timeslot (FlyDenver 2020). On the day of travel, passengers self-certify their health status and receive a QR code to access the e-gate. The QR code verifies identity and reservation time, and a scanner verifies body temperature before access is granted.

Retention and Storage

Daon stores the facial biometric and personal information in its system, but customers can purge their information when they want to leave the program. The airport does not store any of the personal information.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Denver–Daon Biometric Partnership case study of Chapter 2.

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Stakeholders and Responsibilities

The primary stakeholder is Daon, which is responsible for the software solution and storage of biometrics. It is also funding the current VeriFLY pilot. DEN is another main stakeholder. It is responsible for advertising, staffing, and ensuring that the terminal layout and AGTS platform can accommodate the program (McLaughlin 2020). Passengers are the third main stakeholder since passenger participation is required to make the program successful.

An indirect stakeholder is TSA since the VeriFLY program involves use of a dedicated security checkpoint lane. Early involvement was required to confirm that a dedicated lane and screening staff would be provided for the trial. Future coordination with TSA will be required to adapt the program to changes such as the use of biometric credential authentication technology, which will remove the role of the travel document checker.

Future stakeholders are expected to include airlines and medical labs as the program is expanded to provide additional functionality.

Case Study Review

Benefits

Benefits of the VeriFLY passenger screening program include building confidence and certainty throughout the passenger journey, flattening the arrival peaks in the reservation system, promoting a touchless and document-free travel experience by incorporating biometrics, and reducing exposure for TSA officers.

Benefits of the remote screening pilot include relieving pressure on the terminal checkpoints and being able to provide an enhanced, touchless experience. There is also a financial benefit of deferring construction of capacity-driven projects.

Responses from Customers

The initial response from customers has been overwhelmingly positive. More than 90% of users said they were likely to use it again and would recommend it to friends. New passenger profiles are being created every day, which is a sign that VeriFLY is gaining in popularity (Merrit 2021).

System Specifications Review

There has not been a system specification review by a third party because this deployment is still in the early stages.

Fall-Back Options

VeriFLY is a voluntary program, so passengers do not have to participate. Regular security access in the terminal is still provided.

Concerns

The program is still in its early phases, so one of the operational concerns is whether there will be enough passenger demand to warrant a dedicated security screening lane and AGTS train car for passengers who participate in the program. The capacity reduction from the general population would need to be sufficiently offset by passengers using VeriFLY. Other concerns

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include appointment blocking or other attempts by participants to create a more private experience by reserving multiple timeslots.

Lessons Learned

In the early days of the VeriFLY program, there was a perception that there was an unintended consequence of packing more people into fewer train cars because some VeriFLY-dedicated train cars were going out empty. Through increased messaging and an increase in program popularity through the busy holiday period, this misconception was quickly corrected. Additionally, some passengers, on seeing the success of VeriFLY, tried to download the app and join the process midstream. This meant that their temperatures were not checked at the start of the process.

Findings and Trends

Findings

The VeriFLY program has grown at a steady pace since initial implementation and has become a program that state and local health departments have encouraged. Since the airport has been able to prove the concept, VeriFLY will expand to a remote screening facility to allow badged employees and their co-travelers to all clear security protocols based on their specific risk factors.

Future Situation and Broader Implementation

With the initial implementation success, future phases will include incorporation of the facial biometric and ability to access third-party health information to further automate the process. At higher levels on the ladder of confidence, the facial biometric can also eventually be tied into TSA biometric credential authentication technology to eliminate the traditional role of the travel document checker. This would create a document-free travel experience through the checkpoint.

Trends Identified

The Daon VeriFLY program aligns with the trend of deployment of integrated and multi-stakeholder biometric solutions. While VeriFLY currently provides access only to a security checkpoint lane and the AGTS train, the system is designed to be expandable to other areas of the airport. This could include areas such as concessions and ground transportation. Since VeriFLY is in early deployment, the additional functionality layers can be added as more people enroll and as the process at security is further refined and improved.

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APPENDIX C

Case Study: CBP Trusted Traveler Programs at U.S. Airports

Summary

Global Entry is a CBP program that allows expedited clearance for pre-approved, low-risk travelers on arrival in the United States. The program uses fingerprints plus passport or, more recently, facial biometric template matching. The program is open to U.S. citizens, U.S. nationals, U.S. lawful permanent residents, and foreign nationals from 15 countries. Additionally, participants of NEXUS and SENTRI trusted traveler programs are allowed use of the Global Entry kiosks (U.S. Department of Homeland Security 2012). As of August 2020, automatic kiosks are available at 75 U.S. POEs, including 15 preclearance airports.

At airports, program members proceed to Global Entry kiosks to have their fingerprints and/or picture taken and are matched against a gallery pre-compiled from a CBP database. The kiosk then issues the traveler a transaction receipt and directs the traveler to baggage claim and the exit.

Travelers must be pre-approved for the Global Entry program. Applications are reviewed by CBP, and information is processed through various government databases as part of a rigorous background check. Once conditionally approved, the individual goes to one of over 100 enrollment centers to provide fingerprints, a facial photograph, and identity verification documents (e.g., passport, proof of residency) for review by CBP. The biometrics collected at enrollment are used for facial matching each time a passenger uses a Global Entry kiosk. The Global Entry trusted traveler program benefits passengers, CBP, and airports by providing shorter wait times, faster transactions, and the ability to focus government resources on higher-risk targets. See Table C-1 for more information.

Introduction

Designed to provide quick, easy, and convenient processing, Global Entry is a CBP program that allows for expedited clearance for pre-approved, low-risk travelers on arrival to the United States. Global Entry is designed to reduce the time required for travelers entering the United States through the passenger verification process. SENTRI and NEXUS are similar programs.

How Does It Work?

Before the Passenger Journey

With this voluntary program, individuals submit an application on CBP's cloud-based Trusted Traveler Program (TTP; formerly the Global Online Enrollment System) website and provide personal information such as name, address, date of birth, place of birth, country of

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Table C-1. Key facts of CBP Trusted Traveler Program case study.

What?	• Expedited clearance for pre-approved, low-risk travelers entering the United States • Fingerprint and photograph used to confirm identify • Biometric matching 1:few as a gallery of photos of inbound passengers each day are used
Where?	• 75 U.S. POEs, including 15 preclearance airports
Customer process steps:	• Passengers enroll online and are approved after CBP reviews information, conducts interview, and validates biometrics. • At a kiosk in Federal Inspection Station (FIS), enrolled passengers have their fingerprints and/or picture taken. • A transaction receipt is presented to CBP officer at exit control area.
Who?	• CBP and DHS are responsible for program development and implementation. • Participating airport partnerships required for facility implementation, including enrollment centers, signage, and other marketing support • Enrolled passengers
Why?	• To fast-track border crossing procedures for individual travelers who are deemed low risk
How? Technology used?	• Identity as a service – centralized – third-party model • Global Entry kiosks contain camera and fingerprint scanners to obtain biometric tokens to compare against. • Applicant biometrics (fingerprints) are stored in the DHS Automated Biometric Identification System (IDENT); facial photographs are maintained by Global Enrollment System (GES).
Enrollment/digital Identity creation and verification	• Voluntary program with a multistep application process including an in-person interview
Verification of identity how?	• Facial and fingerprint biometrics are collected when completing in-person interview at Global Entry enrollment center. • Fingerprints taken at Global Entry kiosk are matched against those on file. (The kiosks use four fingerprints from one hand to validate identify.) • The facial biometric is matched against a gallery compiled from CBP's Automated Targeting System (ATS) or Unified Passenger Module (UPAX) system.
For?	• Pre-approved, low-risk air travelers to the United States who are U.S. citizens and lawful permanent residents; foreign nationals of certain countries

citizenship, travel history, and employment (U.S. Department of Homeland Security 2008b). Applicants also pay a nonrefundable fee. Information collected through the application is deposited into the Global Enrollment System (GES) as the system of record for CBP TTP. Personal information is maintained in a Privacy Act of 1974 system of records.

Applications are reviewed by CBP, and information is run through criminal and other government databases. Once conditionally approved, the individual goes to one of over 100 enrollment centers to provide fingerprints and a facial photograph along with native documents (e.g., passport, proof of residency) for review by CBP. Applicant biometrics are stored in the DHS Automated Biometric Identification System (IDENT). Alternatively, passengers are encouraged to complete Enrollment on Arrival, launched in July 2017 and offered at 41 airports, to reduce the burden on enrollment centers. Enrollment in Global Entry is valid for 5 years.

Applicants whose applications are rejected may seek redress under one of three methods: via the enrollment center, the DHS Traveler Redress Inquiry Program, or the CBP Trusted Traveler Ombudsman. Individuals may request information about their records in the GES/TTP

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system. Redress is available for U.S. citizens and lawful permanent residents through requests made under the Privacy Act of 1974.

Each trusted traveler is vetted through a "24-hour vetting" process against various databases, including Treasury Enforcement Communications and terrorist data sets (U.S. Government Accountability Office 2014). Treasury Enforcement Communications are designed to be a comprehensive enforcement and communications system that enables CBP and other agencies to create or access lookout data when (1) processing persons and vehicles entering the United States; (2) communicating with other computer systems, such as the FBI's National Crime Information Center; and (3) storing case data and other enforcement reports.

The Passenger Journey

When a passenger who is enrolled in Global Entry books an international trip, the passenger adds the trusted traveler membership number to the reservation. On reaching CBP immigration either on arrival to the United States or at a preclearance facility abroad, passengers are directed to a dedicated Global Entry area. Automated kiosks are located in the Federal Inspection Station (FIS) area of each participating airport.

In the legacy process, a passport or U.S. permanent resident card is inserted into the document reader (U.S. Department of Homeland Security 2009). Fingerprints are scanned at the kiosk and compared to the fingerprint biometrics on file to validate identify and confirm program membership. (Four fingerprints are used for validation.) Individuals are also prompted for a digital photograph and must answer several customs declaration questions. A transaction receipt is printed and presented to a CBP officer at the exit control area.

Starting in June 2018 and active at over 15 airports, a digital photograph is taken as the only means to confirm identity (Boyd 2020). No passport scan is required, and no customs questions are presented to the traveler. The facial biometric is matched against a gallery compiled from CBP's Automated Targeting System (ATS) or Unified Passenger Module (UPAX) system.

Retention and Storage

The digital photographs taken at the kiosk are saved to DHS IDENT, which the agency is in the process of transferring to a new, cloud-based Homeland Advanced Recognition Technology (HART) system (Boyd 2019). Image data are deleted 3 years after the participant abandons their membership or 3 years after their 5-year Global Entry membership expires without being renewed.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the CBP Trusted Traveler Programs at U.S. Airports case study of Chapter 2.

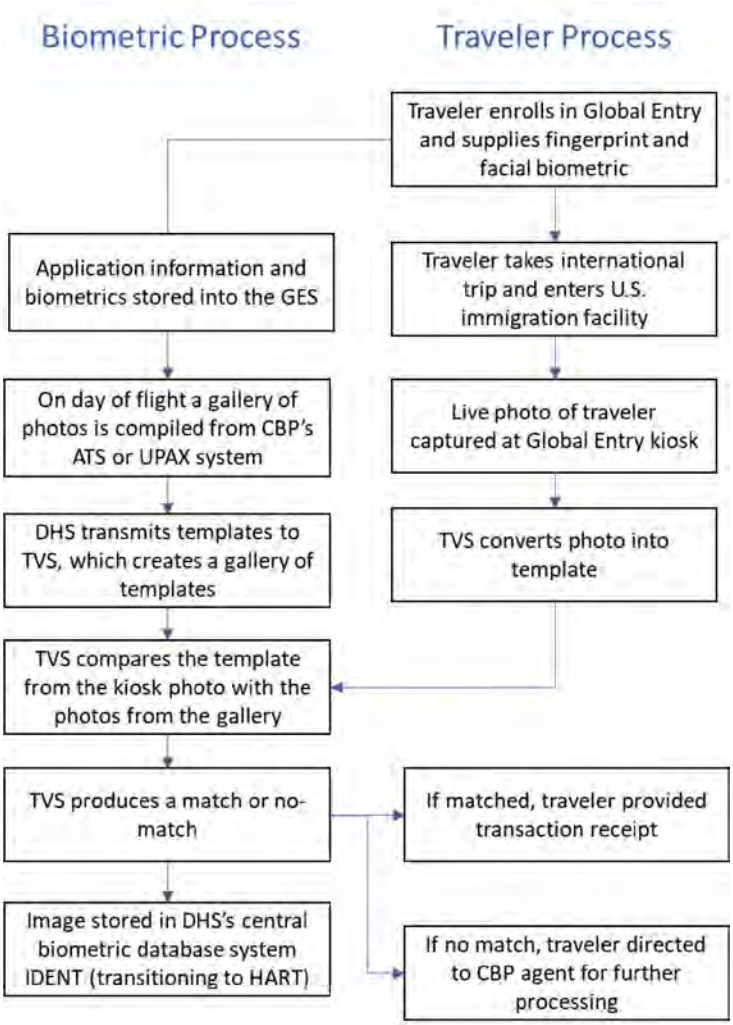
System Specifications

With 1:N matching, the TVS compares a live photo of a traveler against a gallery of other traveler photos (U.S. Government Accountability Office 2020). The stepwise process aligning the traveler process with the TVS matching process is shown in Figure C-1.

Stakeholders and Responsibilities

CBP is the largest stakeholder since the agency is primarily responsible for program development and implementation. Airports and passengers are also stakeholders. Passenger participation

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Source: Adapted from U.S. Government Accountability Office 2020.

Figure C-1. Stepwise process of TVS enrollment and at an airport.

is required to make the program successful, and airport coordination with CBP is required to ensure that the FIS facilities are designed and operated efficiently. Airport coordination efforts include facility modifications and electrical connections to accommodate the Global Entry process.

Case Study Review

Benefits

Global Entry benefits passengers, CBP, and airports. A 2018 CBP study found that Global Entry reduced wait times by more than 70%, with more than 75% of Global Entry passengers being processed in under 5 minutes (U.S. Customs and Border Protection 2018). This translates to an average savings of about 7 minutes. Feedback from passengers has been overwhelmingly positive and, on the average day, 10% of international air travelers use Global Entry kiosks (U.S. Customs and Border Protection 2018). In the first 10 years of Global Entry, membership grew to over 5.4 million members, with another 2 million enrolled in either NEXUS or SENTRI. The successful growth in Global Entry enrollments can be attributed to third-party participation

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by airlines and credit card companies, among others. While most airlines do not offer this benefit directly to their top-tier passengers anymore, airline co-branded credit cards as well as other credit cards include reimbursement of the Global Entry application fee as card perk.

For CBP, Global Entry allows the agency to identify low-risk travelers so that the focus can be placed on unknown and higher-risk travelers. It also accomplishes CBP's strategic goal of facilitating legitimate trade and travel while securing the homeland and works toward the Title 8 U.S.C. Section 1365b mandate, which requires that DHS operate a biometric entry and exit system and that it integrate registered traveler programs into this system. From an airport perspective, Global Entry allows for higher throughputs to be achieved in the same space and provides a customer service boost to passengers.

Responses from Customers

Global Entry generates "overwhelmingly positive comments," John Wagner, the acting deputy assistant commissioner for field operations [CBP], stated publicly (Sharkey 2013). Positive responses from passengers revolved around shorter lines and speed of processing. "Another reader, Eric Evans, said, 'No lines, and in less than two minutes I'm through Immigration.' And Greeley Koch, the executive director of the Association of Corporate Travel Executives, wrote, 'The longest I ever waited in line was 30 seconds'" (Sharkey 2013).

System Specifications Review

Biometric tokens have an extremely high match rate, but occasional issues do arise. About 3% of passengers do not have good readable fingerprints, so a match cannot be made at the kiosk. However, these failed matches can be processed with front-of-line privileges with a live CBP agent. Additionally, the fingerprint readers and cameras on the kiosks occasionally fail, which also requires manual processing by a CBP agent.

Fall-Back Options

Enrollees are not required to use the facial-recognition program and can instead opt to use the passport and fingerprint method, which will remain available. Any technical or processing issues at the Global Entry kiosk can be solved by giving the passenger front-of-line privileges to the next available CBP officer. Additionally, since this is a voluntary program, passengers do not have to participate. The traditional immigration process requiring a physical document check with a CBP officer is still provided in the inspection area.

Concerns

Privacy concerns over the storing and sharing of biometric data have been expressed. To address privacy concerns, a privacy impact assessment (PIA) update was completed in December 2019 (Mongin and Dansiek 2019) to assess the privacy impacts of (1) updates to the existing Global Entry kiosks to use cameras and facial-recognition technology provided by CBP's TVS, and (2) the use of photographs collected at the Global Entry kiosk (rather than the collection of travel documents and fingerprints) to verify traveler identity. The update addressed the risks discussed in the following:

- The risk that the biometrics CBP uses will be used for purposes inconsistent with the original mission is mitigated by the fact that associated photos of Global Entry travelers are stored in IDENT and may be available to authorized IDENT users for a variety of purposes, which DHS reviews and approves prior to granting access to the system, and that CBP owns and operates all Global Entry kiosks.

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- The risk that individuals may not know how CBP will use the photos is mitigated by privacy notices on the kiosk home screen, Global Entry website, visible signage at the FIS, and tear sheets at the FIS.
- The risk that TVS retains user information longer than necessary is mitigated by the fact that TVS does not retain images of travelers once their identities are verified by TVS.

In 2018, an Office of Inspector General report noted an additional concern that there could be a potential exploitation of Global Entry receipts at the exit control area due to failed authentication by the CBP officer (TSA 2018c). This was determined not to be a significant concern.

Lessons Learned

In talking to stakeholders involved in Global Entry implementations, the biggest hurdle that affected the success of a Global Entry implementation was the placement of the kiosks. Kiosk placement was important to ensure that access and use of the technology were self-intuitive to the user. It also required stakeholder ambassadors present to assist as necessary and to resolve kiosk issues so that the technology was properly used by the customer and system issues were resolved or reported in a timely manner. Some airports, for example, initially placed kiosks on the jet bridge or in the sterile corridor leading to the FIS. Other airports placed kiosks in entry passages to the FIS. Each terminal design represented a different challenge regarding correct placement and flow, even with new construction.

Additionally, the back end of any technological solution needs to tie into the CBP vendor to exchange data. The initial deployment was biometric and photo capture. CBP has since progressed with facial-recognition technology and one-to-few matching of photograph and biometrics against the APIS manifest.

While most of the challenges were associated with facility constraints, stakeholder willingness to invest in Global Entry as a process was an unforeseen challenge. Some airports initially declined the opportunity to serve as a pilot site, while others quickly agreed. These initial partners helped identify and solve some of the challenges that were experienced during the initial implementations.

Findings and Trends

Findings

The Global Entry program provides the ability for airports to safely and securely facilitate international air travel for citizens of select countries. Vetting pre-approved travelers in advance of their trips is a low-risk solution that streamlines the passenger verification and screening processes. By pre-approving low-risk travelers, CBP can focus its resources on higher-risk targets. The biometric implementation also reduces touchpoints and increases the throughput of facilities, which can be beneficial to capacity-constrained airports.

Future Situation and Broader Implementation

Global Entry is in the process of upgrading the passenger verification process from fingerprint biometrics to photograph biometrics at all 75 U.S. POEs. Global Entry is open to U.S. citizens, U.S. nationals, U.S. lawful permanent residents, and foreign nationals from 15 countries. Additionally, participants of NEXUS and SENTRI trusted traveler programs are allowed use of the Global Entry kiosks. The program is expected to continue to expand to include additional foreign nationals and passenger subsets in order to increase the population base of low-risk

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travelers. Finally, Global Entry can be expanded to include additional POEs, including general aviation facilities.

Trends Identified

This case study illustrates two of the key trends in biometrics: (1) identity verification solutions, and (2) emerging global standards.

Global Entry relies on the TVS to compare biometric images of travelers using kiosks with the database of biometrics provided at enrollment. TVS is an identity verification solution and not mass surveillance of passengers entering the United States This is especially important since these trusted travelers volunteered to provide their biometrics in exchange for travel and processing privileges.

Global standards are changing, and passport-free travel is a key example. Global Entry contributes to the emerging global trend of passport-free travel because the immigration process at the kiosks is transitioning to a system where facial biometrics replace the need to scan the passenger's passport. As Global Entry continues to evolve, it is anticipated that there will be no need to show a passport during the immigration process.

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APPENDIX D

Case Study: Seattle–Tacoma International Airport and Designated Aviation Channeling

Summary

Pursuant to the Aviation Transportation Security Act and implementing regulations,[112] TSA requires all employees of airport authorities and airline carriers as well as other airport stakeholder employees who require unescorted access to secure areas of an airport to submit an application and be approved for a SIDA badge. The vetting process screens an applicant's information against federal criminal and immigration databases to determine whether the applicant is a threat to transportation or national security. Starting in 2012, TSA authorized the use of DAC services by airport operators and aircraft carriers through which applicants submit their applications [which include both biographic and biometric data (i.e., fingerprint records)] to TSA (and the FBI) (Department of Homeland Security Inspector General 2013; Pilli 2020). Telos ID is one of two DAC service providers authorized by TSA and offers its DAC services at Seattle–Tacoma International Airport (SEA) (Airport Technology 2020). See Table D-1 for more information.

Introduction

In the case of an airport using DAC vendor services, the process for a SIDA badge begins when the airport employee submits the application, supported by authorized signers (or endorsers), to the DAC vendor. Telos, as a DAC service provider, takes and includes the applicant's fingerprints in the application with, in some cases, additional documentation (e.g., I-9 document) (Telos ID 2020b). The DAC vendor makes sure that the application is complete and properly formatted and verifies the employment and education information provided. The application is accompanied by fingerprints (and sometimes a photo), which are electronically submitted to TSA for a security threat assessment, which TSA in turn transmits to the FBI for a CHRC (Badgley 2020). The background check compares the applicant's information against federal criminal and immigration databases to determine whether the applicant is a threat to transportation or national security. If the application is approved by TSA, the results are provided to the airport operator, and a badge is issued, which not only authorizes access to secured areas of an airport but, depending on the biometric technology in use at the airport and by the employer, could be scanned to record reporting for duty for timekeeping purposes or to physically enter secured areas.

How Does It Work?

Before the Passenger Journey

The DAC service consists of various programs and systems used to collect applicant information. SEA primarily uses the DAC interface, which is a secure, web-based application hosted on servers managed by Telos. The back-end system is a database used to house collected

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Table D-1. Key facts on designated aviation channeling at SEA.

What?	• DAC services are used by airport operators nationwide. • Telos ID is one of two competitive vendors of services to airport and aircraft operators and stakeholders of DAC services for vetting of employees seeking unescorted access to secured areas of the airport (TSA requirement) (Ayers and Lucini 2012). • The application compiled by Telos includes fingerprints (straight or rolled) (matching is 1:1); if a facial image is provided the matching for facial image is 1:many. Fingerprints are required, and occasionally photos are provided to verify identity (e.g., in the case of a name change).
Where?	• Telos DAC services are currently at 90 airports in the United States.
Customer process steps	• At the request of airport stakeholder, applicant submits biographic and biometric data (fingerprints and sometimes a photo) to Telos (Telos ID 2020a). • Telos verifies the application information (e.g., checking employment and education references), ensuring completeness and proper format. • Application is electronically transmitted in an encrypted format to TSA. • TSA conducts a review of various government databases as a security threat assessment, and the FBI runs the fingerprints for a criminal history check [otherwise known as a Next Generation Identification (NGI) check]. • Results of approval or denial are provided to Telos, but more detailed information is only shared with airport operator or aircraft carrier. • Badge pickup is by applicant (airport employee), but if negative results are returned, airport or aircraft operator manually adjudicates the case.
Who?	• Telos, under contract with SEA, provides DAC services to airport staff (airlines, TSA, ground operations, airport management/operations, other airport stakeholders/suppliers) in accordance with airport security directive and pursuant to TSA requirements. • In the near future, Telos DAC services will also be used for CBP e-badge applicants (Burriesci 2020; Hamilton 2020).
Why?	• SIDA badge is required for all employees whose duties require unescorted access to secured areas of the airport. • Use of DAC biographic and biometric process creates a more efficient and secure vetting procedure, saving airport stakeholders time and money and standardizing the TSA/FBI–mandated procedure and minimizing TSA's administrative oversight to two DAC providers.
How? Technology used?	• The main DAC service used by SEA is a secure web-based application that allows SEA to collect applicant information and package it according to TSA and FBI specifications. • SEA uses the DAC services by integrating them with its identity management system (IdMS) software. SEA IdMS software is integrated with mobile/static fingerprint scanners, cameras, and software provided by Telos (Curtis 2020). • Telos electronically transmits the application and the fingerprints in an encrypted format. No specific technology is required for transmission, but fingerprint quality must meet FBI specifications.
Enrollment/digital Identity creation and verification?	• DAC services, while not mandatory, are provided in support of the SIDA badge application required by TSA.
Verification of identity how?	• Applicant provides two forms of government-issued identification and additional government-issued authorization (e.g., immigration forms such as an I-9), if warranted. Telos conducts checks for application data pertaining to employment and education history.
For?	• DAC processing is for issuance of SIDA badge for airport employees seeking unescorted access to secured areas within the airport. (Exceptions exist for federal, state, or local government employees and for certain other individuals, previously vetted through TSA/FAA and authorized by the airport operator due to continuous employment.) • In the near future, applicants seeking unescorted access to CBP's FIS area can apply for the e-badge using Telos DAC services.

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information for the purpose of transmitting to TSA for STA vetting and CHRC submissions to the FBI.

SEA uses the DAC service by integrating it with its identity management system (IdMS). SEA IdMS software is integrated with mobile/static fingerprint scanners, cameras, and software provided by Telos.

It is important to note that the DAC systems are subject to TSA MD 1400.3 IT Security, as well as Attachment 1: *TSA Information Assurance Handbook* (TSA 2018c). The DACs must also comply with DHS specifications regarding the safe handling of SSI, PII, and sensitive personally identifiable information (SPII).

Given that the DAC system contains SPII, the information systems and devices on which they are installed must be approved and granted authority to operate by DHS and TSA.

The Passenger Journey

At request of the airport, the applicant submits the application form with biographic and biometric data (fingerprints and, if relevant, a photo) to Telos for electronic transmission to TSA for SIDA badge for access to secured areas. Telos performs the following functions:

- Ensures biographic data within the application are complete and accurate and accompanied by FBI-required fingerprints.
- Verifies biographic information for employment and education history (including identifying unexplained gaps) by contacting listed references, employers, educational institutes, and so forth.
- Verifies identity, as part of the application, through a driver's license or other identity document (e.g., passport).
- Ensures data integrity by eliminating duplicate or erroneous data.
- Automates fingerprint web capture at a designated badging office, airport operator human relations office, or a mobile location. Provides capture devices (fingerprint scanners, cameras).
- Provides capability to attach documentation, such as copies of immigration records (e.g., Permanent Resident Card, employment authorization document, Form I-94).
- Via software, allows all the information to be packaged into a format compliant with the FBI Electronic Biometric Transmission Specification.
- Encrypts the transmission and sends to TSA for its STA and background check and to FBI for a criminal history record check via Next Generation Identification (NGI). Telos transmits the application via secure connections using the Secure Sockets Layer over the Simple Object Access Protocol or HTTPS (Hypertext Transfer Protocol Secure).
- Transmits the fee to TSA.
- TSA conducts a review of various government databases for the STA, and the FBI runs the fingerprints for an NGI check. The results of approval or denial are provided to Telos, but more detailed information is only shared with the airport operator or aircraft carrier. If approved, the employee picks up a SIDA badge, but if negative results returned, the airport operator manually adjudicates the case.

Retention and Storage

Telos saves biometric and biographic data in the application in accordance with TSA Privacy Act System of Records Notice (SORN) requirements for STA records. It is important to note that the DAC system stores this data in accordance with TSA MD 1400.3 IT Security, as well as Attachment 1: TSA Information Assurance Handbook. In addition, the DAC system must comply with other federal and DHS data retention policies.

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The data storage process and database structure are Telos ID proprietary system designs. TSA only holds the DAC to its requirements for the input to and output from the DAC system. These requirements are embodied in the previously mentioned TSA/DHS handbooks and policies.

Telos adheres to privacy requirements set by DHS/TSA for DAC service providers in support of TSA requirements of the Privacy Act of 1974 governing the collection, storage, and retention of TSA records. TSA has indicated that for SEA, state privacy laws may apply as well.

With respect to TSA, it retains information (fingerprints and biographic information) for 1 year after an individual's SIDA privilege is no longer valid. In addition, for those individuals who may originally have appeared to be a match to a government watch list but are subsequently cleared as not posing a threat to transportation or national security, information will be deleted or destroyed 7 years after completion of the STA or 1 year after any credential or access privilege granted based on the STA is no longer valid, whichever is longer. Information contained in the subject database on individuals who are actual matches to a government watch list or who otherwise pose a threat to transportation or national security will be deleted or destroyed 99 years after completion of the STA or 7 years after TSA learns that the individual is deceased, whichever is earlier.

For airport operators, per TSA regulation 49 CFR 1540, each operator must retain the STA application and supporting documents (verifying identity and work authorization) for 180 days following the end of the applicant's service to the operator.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Seattle–Tacoma International Airport and Designated Aviation Channeling case study in Chapter 2.

System Specifications

There are no specifications for the DAC system design for hardware, software, encryption, cameras, and so forth, but fingerprint quality must meet FBI requirements.

Stakeholders

Airport and aircraft operators are the main stakeholders, although TSA oversees the vetting process to ensure compliance with the law. In 2020, at some 78 airports nationwide, CBP began receiving from TSA those applications from applicants seeking unescorted access to the CBP FIS area, which CBP vetted through its UPAX system (a multi-record system with law enforcement information, including criminal, civil, and immigration data) (Calixte 2020).

Case Study Review

Benefits

According to the Telos ID website, Telos ID's DAC services improve data integrity, increase the efficiency of credentialing operations, and reduce costs (Telos ID n.d.). For background checks, DAC services enable submissions of workers' biographic and biometric data, including subscriptions to the FBI Record of Arrest and Prosecution Background (Rap Back) program, for individuals working in secured areas of U.S. commercial airports (TSA 2018a). Stakeholders have experienced a reduced administrative burden associated with operating a badging office.

Those stakeholders that sign up for the Rap Back subscription service achieve significant savings since they no longer are required to pay for and obtain an FBI criminal history check

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every 2 years. DAC services also assist in managing insider threats with FBI's Rap Back program, which provides employers with notice of an employee's criminal and arrest records by sending alerts in real time if there are any additions to an employee's FBI records.

Some benefits of using the DAC services are:

- DAC can also integrate with other users' workforce systems to meet additional badging, physical security, and personnel administrative needs.
- DAC Rap Back subscription service allows stakeholders to manage insider threats and provides employers with notice of an employee's criminal and arrest records by sending alerts in real time from the FBI if there are any additions to an employee's FBI records.
- DAC services can expedite employee on-boarding with real-time and combined CHRC and STA submissions.
- DAC eliminates spreadsheets that can expose PII.
- DAC frees up federal agency staff from data entry and accelerates processing of applications.

Responses from Airport Operators, Airlines, and Stakeholders

In an interview conducted on October 5, 2020, with a SEA representative, benefits include more efficient processing of some 12,000 applications annually and greater protections for PII.

System Specifications Review (Third Party or Other)

Telos provides capture devices (fingerprint scanners, cameras) and transmits application via a secure web portal. The fingerprints must meet FBI specifications for quality. If approved, the applicant picks up the SIDA badge, but if there are negative results, the airport operator receives the details from TSA and manually adjudicates the case.

Fall-Back Options

Applications may be submitted as a hard copy, either manually or by mail.

Concerns

With respect to privacy protections extended to Telos' processing of airport employee applications and related data, Telos maintains biometric and biographic data and yes/no results of vetting but does not receive the actual rap sheets or details regarding a finding that employees may have disqualifying events/offenses in their background. According to Telos, all information is stored in compliance with TSA requirements.

Lessons Learned

According to TSA sources, expanding the use of DAC services to multiple categories of airports has facilitated TSA's oversight role and benefited larger and smaller airports with greater efficiencies.

Findings and Trends

Findings

The DAC system promotes greater efficiency for pre-processing of SIDA badge applications for a large volume of workers.

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"As an encrypted, web-based solution, Telos ID's DAC services . . . [and] [i]ts modular design supports each airport's and air carrier's needs, and users can perform multiple functions on one platform" (Aviation Pros 2020).

TSA and CBP acknowledge the increased efficiencies and improved speed of application processing through the use of DAC services.

During Phase 1, CBP realized additional efficiencies in the airport employee vetting process for e-badges by piggybacking on the DAC services in cooperation with TSA. In the future, CBP intends to collect biometrics by requesting and receiving the set of fingerprints included in the application transmitted to TSA and the FBI.

Future Situation and Broader Implementation

Telos has indicated that its services could extend to Secure Flight vetting of passengers (but since Secure Flight is an assessment based on biographic data, it would not likely involve biometric data collection).

In the future, CBP intends to collect biometrics by requesting and receiving the set of fingerprints included in the application transmitted to TSA and the FBI.

Trends Identified

One emerging trend is the consolidation of the many types of background checks required by the U.S. government into a single application and process. The result is greater efficiencies in the vetting process for airport employees requiring access to secured areas at commercial, federal, and joint-use facilities.

According to Telos sources, it is exploring and implementing quality-integrated solutions, such as integrating customers' workforce systems to meet additional badging, physical security, and personnel administrative needs.

As of the time of writing, TSA PreCheck enrollment was conducted by IDEMIA. Starting in late 2020, Alclear, LLC, and Telos will also conduct TSA PreCheck enrollments for travelers. One of the challenges to expanded use of DAC services to airport-related functions is that most of the programs where vetting is a part use an established enrollment entity.

Endnote

112. Aviation Transportation Security Act, 49 U.S.C. §§ 44936 et. seq. (Pub. L. 107-71); 49 CFR part 1542; TSA Directive 1542-04-08G. *See also*, 49 U.S.C. §106(l)(6), stating "the Administrator is authorized to enter into and perform such contracts, leases, cooperative agreements, or other transactions as may be necessary to carry out the functions of the Administrator and the Administration."

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APPENDIX E

Case Study: Curb-to-Gate Program by CBP and Delta Air Lines at Hartsfield–Jackson Atlanta International Airport

Summary

CBP's Curb-to-Gate program is an application of facial recognition with the intent of reducing the number of times passengers must present their form of identification and boarding pass and of increasing the reliability and efficiency of airport border and security operations.

The Curb-to-Gate program at ATL as led by CBP and Delta Air Lines allows passengers to verify their identity in a reliable and more efficient manner through leveraging CBP's TVS:

- At check-in/bag-drop, a real-time photo is sent securely and encrypted to TVS.
- The incoming photo is matched to a biometric template in a pre-compiled library based on the airline's passenger manifests.
- In case of a positive match, the passenger's ID is verified, and the use of facial recognition for the rest of the journey is enabled.
- At bag drop, security, and boarding, the passenger's identity and right to entry/passage is verified using facial recognition.
- For border crossings, an entry/exit record is made in the ADIS.

The benefits of using the Curb-to-Gate program are time savings for passengers and airport personnel, lower operational expenditures for airports, optimization of the use of space throughout the airport, and an opportunity for airports to revolutionize the passenger travel experience. Additionally, Curb-to-Gate significantly reduces the number of physical interactions among airport stakeholders, thereby lowering the probability of transmitting diseases and hazardous pathogens (e.g., COVID-19). Furthermore, this represents a significant milestone in terms of the power of end-to-end facial recognition and demonstrates the capabilities of biometric software (A. W. 2020). See Table E-1 for more information.

Introduction

In the journey through an airport to an international flight, a passenger has to present passport and boarding pass multiple times before getting on the plane. Burdened with carrying cabin luggage, worrying about making their flight, and navigating the airport, this requirement has passengers fumbling with important documents and only adds to the often stressful experience.

This application of biometrics aims to take away the need to repeatedly present IDs and boarding passes, limiting it to one time at the start of the journey at the airport (Harper 2018). It also aims to greatly increase the security and efficiency of border and security operations. More recently, this application has also provided for a largely touchless journey, which is safer for all stakeholders traveling through and working at the airport.

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Table E-1. Key facts of case study on the Curb-to-Gate program by Delta.

What?	● Facial biometric template matching ● 1:few biometric matching ● Pilot project led by CBP
Where?	● Maynard H. Jackson Jr. International Terminal and Concourse F at ATL Atlanta ● Leased by Delta
Passenger process steps	● Curb-to-Gate program (to be expanded later) ● Departures ● Check-in ● Bag drop ● Security (TSA) ● 15 departure gates ● Arrivals ● CBP immigration checkpoint
Who?	● CBP led the trials in partnership with ATL, Delta Air Lines, and TSA. The DHS was involved since it manages and oversees access to the biometric information used in CBP's system. NEC was selected as technology hardware and software provider when it was the first vendor to meet specific technology requirements. NEC products include the NeoFace Express and NEC Enhanced Video Analytics (Beroukhim and Exum 2018).
Why?	● CBP's initial goal was to speed up processing using facial data of passengers who had been previously vetted by various systems. It was important to not re-vet everyone but to use an accurate and reliable system as a verification process for passenger identification and boarding. Stakeholder goals included being able to use the biometrics verification process for non-CBP passenger processing steps (e.g., check-in and replacement of paper boarding passes). The main goals of CBP were to: – Secure – Provide increased certainty as to the identity of travelers at multiple points in the travel continuum, – Simple – Eliminate the need for physical document and boarding-pass checks as well as the collection of fingerprints, – Facilitative – Establish a clear and easily understood process to reduce the potential for major bottlenecks within the air travel process, and – Compliant – Employ a high-integrity biometric entry and exit system that not only increases CBP's certainty as to the identity of travelers, but also more ably holds accountable those violating terms of admittance (U.S. Department of Homeland Security 2019).
How?	● Cameras: tablets, mounted, on-a-stick ● Matching live pictures taken at the checkpoint with preloaded biometric templates ● No collection of biographic data through TVS; pictures kept up to 12 hours ● Systems, databases, and so forth used: – TVS on virtual private cloud (VPC) – APIS – ADIS – DHS Office of Biometric Identity Management – DHS IDENT – ATS-UPAX
Enrollment/digital identity creation and verification	● No formal enrollment; TVS already has access to a large database of foreign nationals' pictures who have traveled to the United States or applied for visa or other permits. ● DHS Office of Biometric Identity Management

(continued on next page)

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Table E-1. (Continued).

Verification of identity how?	• Facial biometric is collected when applying for a passport/visa or during entry/exit of United States; verification matches biometric template of unique IDs of the pictures on file with a picture taken at the passenger checkpoint, which is sent encrypted to TVS.
For?	• International flights with Delta or partners Air France-KLM, Aeromexico, and Virgin Atlantic • To be expanded to U.S. domestic flights in subsequent steps in the program

This is achieved by CBP, which leverages its TVS to provide secure, efficient, automated biometric identity verification at the checkpoints for Delta (Future Travel Experience 2019a) and the airport as well as for PreCheck-eligible passengers at the TSA checkpoint (TSA 2020a). ATL also hosted one of the earlier versions of TSA's biometric authentication technology trials, but this is not covered in this specific use case review. Note that an actual passport is still necessary for international travel for identification at the destination countries.

TVS is a cloud-based facial biometric matching service that uses U.S. government databases of pictures on file of U.S. citizens as well as foreign nationals who have in the past traveled to the United States. For new travelers, enrollment for Delta's service is done at the check-in kiosk or airline counter.

Storage and retention of pictures sent to the TVS system are minimal, although the TVS system builds on pre-existing U.S. government databases, which keep records of foreigners for longer periods of time. For U.S. citizens, pictures are retained for no more than 12 hours in the TVS system before being purged. CBP temporarily retains facial images of non-immigrant aliens and lawful permanent residents for no more than 14 days. For certain other foreign nationals, pictures after matching are sent to CBP's IDENT database to update pictures on file and kept there for a maximum of 75 years. For both U.S. citizens and foreign nationals, an entry or exit record is stored, but it contains only the biographic information found in a passport; no pictures are in those records.

Furthermore, no pictures are stored on the local camera devices themselves by NEC. Airlines and airports are prohibited from keeping pictures of their passengers. The accuracy of the matching is high (99%), and the sending of data is regarded protected due to the secured connection between the NEC systems and TVS and due to the encryption of the pictures before they are sent.

The main benefits are significant time savings for passengers, and space and resource savings for airports and airlines. The system architecture is technology agnostic and, thus, allows for multiple hardware suppliers to develop specific solutions for airports.

How Does It Work?

Before the Passenger Journey

Several U.S. government databases are essential elements of the system behind the Curb-to-Gate program at ATL by CBP. CBP's TVS and its API are the systems that interact with private third parties' devices. The most important databases are briefly introduced here:

- HART (follow-up system for IDENT): The HART system is a centralized, DHS-wide biometric database that contains limited biographic and encounter-history information. The system will replace IDENT for the storage and processing of biometric and associated biographic information for national security, law enforcement, immigration and border

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management, intelligence, background investigations for national security positions, and other administrative uses.
- IDENT: The DHS database for the Automated Biometric Identification System that has, since 2003, collected pictures, fingerprints, and basic biographic data (e.g., passport information) of non–U.S. citizens who have traveled to the United States (Thales 2021).
- DOS database of passport photos and visa applications.
- APIS: A collection information, supplied by the airlines, of all passengers flying from U.S. airports.
- ADIS: CBP's system that keeps track of all entries/exits of U.S. citizens as well as foreign nationals; relating to visas.
- FBI NGI: The FBI's system for fingerprints, formerly known as the Integrated Automated Fingerprint Identification System.
- ATS-UPAX: This module system is a decision-support tool that is operated by DHS CBP; it combines data from numerous U.S. databases for passenger vetting.
- VPC: The firewalled virtual network on which the TVS system runs.

The TVS database and software are hosted in the United States on Amazon Web Services (dubbed the Gov Cloud), a version of Amazon Web Services that is designed to be able to host sensitive data and regulated workloads, and to address the most stringent U.S. government security and compliance requirements.

Prior to the passenger's journey, CBP's TVS system uses information from passenger booking information in the APIS database to create manifests (lists) of passengers that are expected to travel that day in its VPC. The TVS system preloads verified biometric templates of the passengers' pictures into the traveler manifests.

A biometric template is a digital representation of a biometric trait of an individual generated from a picture and processed by an algorithm. The template is usually represented as a sequence of characters and numbers comparable to a long string of letters and numbers.

The source for the biometric templates is DHS's decision-support tool called the ATS-UPAX, from which existing pictures are downloaded and converted into biometric templates. The pictures downloaded from ATS-UPAX are not stored in the TVS system.

ATS-UPAX sources pictures from multiple databases, such as the DHS's database IDENT and the Department of State's database of passport and photos of U.S. citizens and foreign nationals, but potentially from several other intelligence databases.

The Passenger Journey

1. At check-in/bag-drop, a real-time photo is made by NEC hardware (NEC 2020) with integrated cameras and sent securely and encrypted to the TVS matching service through the API.
2. In the TVS cloud service (VPC), the incoming encrypted photo is matched to the biometric templates in the relevant traveler manifest.
3. If a positive match is made, the passenger is verified, and the use of facial recognition for the remainder of the passenger journey is enabled (Today.com 2020). With a positive biometric match within the right manifest, the passenger record in the manifest is updated from "reported" to "confirmed."
4. At subsequent passenger process touchpoints (bag drop, security, and boarding), the passenger's identity and right to entry/passage are verified using facial recognition and matched with the corresponding specific traveler manifest (Baran 2020).
5. In the case of a border crossing, an entry or exit record consisting of only the biographic information from the passenger's passport is created or added to the pre-existing record and retained in the ADIS.

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6. In case of no match or if the passenger wishes to opt out of this process, a manual verification of the identity is done at each checkpoint. Pursuant to CBP inspection, the manual ID check is followed by a check of eligibility to enter/exit the United States (Murph 2018).

Retention and Storage

Storage and retention of pictures sent to the TVS system is minimal:

- No pictures are stored on any of the airports' or airlines' devices supplied by NEC. These only take the picture, encrypt the file, and immediately send it to the TVS through the API over a secured connection. Airlines and airports are prohibited from keeping pictures of passengers.
- For U.S. citizens, pictures are kept up to 12 hours in the TVS system (Patterson and Russell 2018).
- During the initial trial period, pictures of U.S. citizens and foreign nationals were kept in the ATS-UPAX for up to 14 days for the evaluation of the technology. This is no longer the case.
- For foreign nationals, pictures are sent to CBP's IDENT database after matching to keep the records' pictures up-to-date and current. These records are kept for a maximum of 75 years.
- For U.S. citizens and foreign nationals, an entry or exit record is stored in ADIS. This record contains only the biographic information found in a passport; no pictures are in this record.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Curb-to-Gate Program by CBP and Delta Air Lines at Hartsfield–Jackson Atlanta International Airport case study in Chapter 2.

Stakeholders and Responsibilities

CBP took leadership of the (pilot) project, partnering with Delta, ATL, TSA (TSA 2020b), and NEC to trial the TVS system on broader application use over multiple passenger checkpoints. This curb-to-gate trial followed several earlier pilot programs and built on CBP's earlier Departure Information System Test pilot and Departure Verification System (both in 2016).

Stakeholders

The main stakeholders of the Curb-to-Gate program are Delta Air Lines, ATL, TSA, NEC, and other SkyTeam partner airlines (AF-KLM, Aeromexico, Virgin Atlantic) (Mutzabaugh 2018). In an interview, CBP reiterated the importance of carrier buy-in from the onset and throughout the development and entire period of the program (personal communication).

Responsibilities and Governance

CBP was the main driver of the program, and the main requirements of the pilot were dictated by the CBP. After various collaborative discussions with several organizations, CBP took the lead on moving the project forward to establish a strategic pilot project. As the task and goals were outlined and agreed on, subject-matter work groups were formed (e.g., in data systems and information technology) and decided on protocols. For CBP, Delta Air Lines, and ATL, officials were designated and provided guidelines and project goals.

Timeline and Planning

Interviews with CBP revealed that multiple timelines were based on individual workgroup task capabilities. Weekly updates were established as the working groups progressed with the pilot projects, implementations, evaluations, and next steps.

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Procurement and Vendor Selection: NEC

NEC was the first vendor to demonstrate the capability to meet the biometric facial technical requirements for all the vested partners. Delta volunteered to pilot the biometric process at its ATL operational hub.

Case Study Review

Benefits

For passengers, an average reduction of 7 to 9 minutes was accomplished when adding up all the time savings at the passenger checkpoints that used the facial-recognition technology (Burt 2020; Linder 2020). Passengers also do not have to reach for and hand over their passports and boarding passes on several occasions, making the passenger checkpoints much less of a hassle and preventing possible transmission of communicable diseases through closer contact and interaction.

For the airport, airline, TSA, and CBP, the faster processing times through checkpoints mean a possible reduction of staffing, space, and other resources required for a similar throughput. This allows for expansion of throughput in the current assigned areas or freeing up of space for other uses. More specifically, CBP reported the benefits shown in Table E-2 in its August Report of 2019 (U.S. Department of Homeland Security 2019).

TVS is technology agnostic and comes with a range of tools (the API) for integration with stakeholder/partner technology for capturing facial biometrics. This means that other suppliers of the hardware and software (NEC) can be chosen, or the software can be developed in parallel.

The 1:few matching greatly improves matching speed and accuracy over the use of other large U.S. databases.

With respect to privacy, the pictures are encrypted before uploading to the cloud, where they are transformed to a biometric template: only the unique IDs are stored, and only for a short period of time. Pictures uploaded by the airline/airport are compared to the biometric templates, and only a positive/negative match is returned. This means no actual pictures are stored in the TVS database.

The process and flow of information are well defined for sharing of pictures for traveler verification as well as other information flows, because they have been analyzed by the Office of

Table E-2. CBP strategic benefits.

Improved business process	• An enhanced entry/exit business process that integrates within existing government and stakeholder business models
Stronger relationships	• An environment that allows CBP and stakeholders to work together and that allows for further airline modernization
A positive impact on inbound security and throughput	• Enhanced inbound security and more efficient throughput
Improved traveler experience	• An overall enhanced traveler experience
Improved data integrity	• Use DHS enterprise biometric repositories provided to ensure accurate biometric identity records
Enhanced Visa overstay enforcement	• Support the ID and tracking of Visa overstays by closing information gaps associated with current exit reporting capabilities, allowing for improved enforcement action

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Biometric Identity Management in its PIA. These PIA reports are available to the general public to read, greatly benefiting the digital transparency of the system.

For the CBP, the automated process lets the CBP officers focus on high-risk travelers, or even undocumented travelers, which aids the CBP with its border protection mandate.

Responses from Passengers

Feedback from passengers has been positive; according to passenger surveys performed during the pilot project, only 2% opted out of the process; 93% of customers had no issue using the technology at boarding (Steele 2019), and 72% found it more preferential than previous processes (Yamanouchi 2019a; Yamanouchi 2019b).

System Performance and Specifications Review

According to a CBP report of August 2019, attained online (U.S. Department of Homeland Security 2020), the accuracy of the facial-recognition matching was in the high 90s, and improvements in the technology have led to much lower false positives and false negatives. The matching rate was running at 97% to 99% according to CBP, as can be seen in the results from the August 2019 Report (U.S. Department of Homeland Security 2019) shown in Table E-3.

The estimated false positive rate based on the internal CBP analysis is 0.0103%, which is within the established key performance indicator (KPI) target of less than 0.1%.

Fall-Back Options

For those passengers who do not wish to take part in the Curb-to-Gate program [at the time of writing about 2% (Pitrelli 2020)], there are opt-out possibilities. This will mean having staff do manual verifications of identity as well as boarding-pass/passport checks.

Power cuts or system integrity failures can seriously hamper the operation of the passenger identity vetting system, as is the case with any computer system. Luckily, with the main system running in the cloud, this process can be hosted in a multitude of physical locations, thus protecting it from interruption. Similarly, mobile devices in the airport used at the passenger checkpoints can run software independently and on battery power, maintaining an encrypted connection, so that in the event of a loss of power, these devices can continue operation. The latest update from the traveler manifest should allow for continued operation, at least for those who are positively matched in the TVS system. Manual ID verification in the case of a negative match is more difficult at CBP since this requires connection to DHS/DOS for formal ID query.

Table E-3. Facial-recognition performance report, August 2019 (U.S. Department of Homeland Security 2019).

Modality	Number of Locations	Flight Count	Number of Travelers	Technical Match Rate
Air entry	11	446	34,716	99.2%
Air exit	16	92	11,545	97.6%
Air preclearance	4	45	6,559	99.4%

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Concerns

Media attention has raised concerns on the implementation of biometrics in four main categories:

1. Use of technology for (mass) surveillance,
2. Privacy,
3. Transparency, and
4. Technology standards.

Use of Technology for (Mass) Surveillance

The fear that facial-recognition technology will allow for mass surveillance of the public has seen media attention around the world, especially as some countries have implemented just that. The goal of the Curb-to-Gate program and its (system) design prevent this capability. CBP specifically states that TSA is not evaluating the use of facial comparison for law enforcement purposes; it is assessing its use for traveler identity verification as part of its mission to protect the nation's transportation systems to ensure freedom of movement for people and commerce. In the execution of the system, there are no means that allow, in this case, federal authorities to access the camera systems, and all the system does is generate automatic versions of the entry/exit reporting for which CBP was already keeping records.

Privacy

Privacy concerns (Hautala 2019) for any system that handles biographic or biometric personal information include misuse or loss of that information when collected, used, or stored. To counter privacy concerns, in the design of the TVS, CBP has implemented a privacy-by-design approach to ensure that privacy protections are embedded into its use of facial comparison technology. CBP employs four primary safeguards to secure the data: secure storage, brief retention periods, irreversible biometric templates, and strong encryption during data storage and transfer.

On top of that, CBP has submitted a number of PIAs (Kaplan 2018) on related pilots and programs to the DHS Privacy Office for adjudication and publication. These use the FIPPs to assess and mitigate any impact on an individual's privacy. These principles are rooted in the Privacy Act of 1974 and govern the use of personal information.

One example of the principles of privacy by design is showcased in the case of a data breach. If the TVS database were to be hacked, only the biometric templates that are stored in the database would be susceptible to theft. These are worthless because they cannot be recreated to biometric facial pictures without the software and codes used to make the template in the first place.

Transparency

When interviewed, CBP stressed the need to mitigate privacy concerns with outreach and openness. CBP provides transparency and general notification to the public through program information, such as frequently asked questions, available on the CBP website and the TVS PIAs and SORNs published on the Department of Homeland Security website. The PIAs and SORNs for the TVS and its predecessor projects explain all aspects of CBP's biometric entry/exit programs, including policies and procedures for the collection, storage, analysis, use, dissemination, retention, and deletion of data.

Technology Standards

System performance depends highly on adequate hardware and software roll out, in which case standards for the technology can protect against malfunctioning, misinterpretation, or other issues that may arise. One issue that has gained attention in the media is the reported bias in some demographic groups, as per NIST in 2018 (Grother et al. 2018). Another worry with

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facial-recognition technologies is that camera images are not of high enough quality, which can have an impact on the accuracy of the match.

Because data privacy, protection, and mitigation of algorithmic or operational bias are prime concerns, CBP actively makes improvements while seeking to ensure that there are no signs of bias and also engages in a robust public dialogue on appropriate standards. CBP also conducts outreach to privacy advocates, the NIST, and the DHS Science and Technology Directorate (S&T) to monitor performance and progress.

Lessons Learned

According to an interview with CBP, several lessons learned were important in the development of the program and its continuation to the next phases:

- Initially, there was skepticism and concern from potential end-users and stakeholders about the cost of equipment, programming, and training. Through discussions and joint planning, all parties were able to establish and implement mutually beneficial goals.
- Initially, there were various levels of government and private-entity scrutiny regarding government biometric and facial data collection and storage. CBP emphasized the importance of not replicating the vetting process and, for privacy reasons, not storing data. This proved beneficial in further privacy assessments.
- It was important to have airline and airport approval to access relevant company data and systems.
- It was important to allow subject-matter working groups to leverage expertise and focus on the immediate needs by not being too rigid in timelines and processes. At times, there was trouble visioning the end result among stakeholders, which resulted in some inflexibility. By including willing partners at the operational decision-making table, continued progress was possible, and a collaborative effort for a continuous focus on future enhancements was established.
- It was important to ensure that acquisition groups were in sync with implementation requirements. This expectation management was key for tracking progress and updating timelines for pilot programs.

Findings and Trends

Findings

The development of (federal) national standards to govern the use of biometrics in facial recognition for verification/identification is considered beneficial for:

- Picture quality and method of sharing, to further improve matching accuracy and speed;
- Method of encryption, and secure connection, to ensure further protection from digital threats such as hacks;
- Use of biometric (templates) regarding capture, use, and storage;
- Synthesis of lessons learned/recommendations from PIA to further privacy commitments enacted (e.g., storage and retention durations); and
- Information and transparency: Informing users and passengers of system functions and how these work in a campaign to gain trust and dispel negative media attention.

Starting in 2018, DHS S&T has performed independent biometric analyses using a sample of operational TVS probe and gallery facial images. Based on these analyses, DHS S&T made specific recommendations to CBP:

- Ensure that only ticketed travelers are allowed to use TVS for boarding or to increase match thresholds used for biometric exit; and

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- Carry out an exhaustive "Virtual Red Team" analysis to calculate the risk of false matches based on the demographics (age, country of origin, gender) of travelers on individual flights (U.S. Department of Homeland Security 2019).

Future Situation and Broader Implementation

The success of the Delta Air Lines ATL Curb-to-Gate program gave rise to implementation of TVS at numerous other airports in the United States (Steele 2020). In 2018, the entry process using facial comparison was reengineered and deployed in the air entry environment at 15 airports, including four preclearance locations, with plans to expand further in 2019. From recent reports by CBP regarding the next steps for further pilot programs/implementation of the TVS system, the following can be expected in the (near) future:

- Expansion of the program to domestic flights in the United States.
- Expansion of the trial to transferring passengers.
- By 2022, deployment of biometric exit using TVS to the top 20 airports, which account for more than 97% of commercial air travelers departing the United States.
- Full certification/integration with simplified arrivals and biometric entry/exit.
- Mass implementation: All passenger processes, including all transfers (rescreening and CBP immigration), will have the option for use of facial recognition.
- Development of standard process for CBP and TSA.
- Guidance related to required opt-out possibilities for passengers.
- Development of guidance on how TVS is potentially an enabler for the Flexible Facilitation Model, where mixing of, for instance, domestic and international passengers is possible without physical separation of the two passenger flows with terminal partitions (U.S. Department of Homeland Security 2019).

Besides these, CBP is actively working to expand stakeholder partnerships and adoption and is prioritizing the highest volume of international airports and carriers to achieve the biometric air exit implementation goal. CBP continues efforts to find innovative ways to use TVS with mobile phones, tablets, and watches. CBP will look to expand partnerships with international airports and governments and to further expand capabilities in preclearance locations to continually improve security and facilitation of traveler processes.

Trends Identified

This specific use case has shown the possible success of a seamless passenger journey through several checkpoints manned by different airport stakeholders (airline, customs, security). Trends identified relate to:

- Touchless touchpoints reduce the risk of transmission of communicable diseases as well as increase speed and ease of travel.
- There will be end-to-end integration of digital identity management throughout the passenger journey. This will continue even beyond the scope of the airport. All journey touchpoints will use biometrics, from booking to arriving at a hotel/destination, across different modes of transport, and using different suppliers of services (taxi, baggage pickup, airport, airline, security, taxi, hotels, etc.).
- The system provides incentives for passengers to use it because it allows for more ease of use, faster processing, and less hassle with official documents.

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APPENDIX F

Case Study: Known Traveller Digital Identity at Aéroport International Montreal–Pierre Elliot Trudeau, Montreal, Canada

Summary

To respond to the increasing demand for a more efficient and streamlined passenger process, the WEF and several airport and industry partners launched the KTDI program for the development of a trusted digital identity based on biometric technology. The goal of this initiative is to make it possible for all industry partners to use the digital identity securely and easily while protecting the passenger's privacy.

The KTDI program is still in the early development phases and relies on the self-sovereign concept, where passengers are the owners of their personal data and can elect to share them (or not) with different parties (e.g., airport, airlines, border authority). In its current design, the program identifies that the digital identity should be created by an identity-issuing authority, either a third party or a government authority. This authority would create a biometric template of the passenger as well as a unique security feature. With the use of distributed-ledger technology on a blockchain, the immutability of the digital identity can be secured.

The KTDI concept would use the distributed ledger to register every successful identity claim at various authentication touchpoints in order to build up trust through verifiable claims, which is a type of smart contract in the blockchain. In 2018, the governments of Canada and the Netherlands established a pilot group to drive the KTDI efforts. After the successful pilot, efforts for implementation were, at the time of writing, stalled due to the COVID-19 pandemic. See Table F-1 for more information on KTDI.

Introduction

After evaluation of the Cathay pacific trial, it was concluded by AMS that the passenger process could be further optimized. Biometric enrollment with the use of an e-passport is still a time-consuming procedure. If it would be possible to enroll once and then be able to fly every time regardless of airline without additional enrollment at the airport, this would save a processing step and create a more seamless passenger experience. A potential solution for such a process was initiated by the WEF: the KTDI (Known Traveller Digital Identity 2020). The KTDI concept (Accenture 2018b) shifts from conventional travel documentation verification to a digital format. The concept is piloted by cross-border collaboration between Canadian and Dutch public and private partners. In 2018, the governments of Canada and the Netherlands established a pilot group to drive the KTDI efforts and determine the potential of the concept for international connections. The pilot group includes the public authorities of both countries, the airports

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Table F-1. Key facts of case study on KTDI.

What biometric technique is used?	• Facial recognition • 1:1 and 1:few matching of biometrics
Where?	• Initial pilot at three international airports: – Aéroport International Montreal–Pierre Elliott Trudeau (YUL) – Toronto Pearson International Airport (YYZ) – Amsterdam Airport Schiphol (AMS)
Passenger process steps	• Enrollment first, prior to booking a flight – KTDI enrollment with a third party or a government authority • Departure at airport – Passenger and bag check-in follow the conventional check-in process – Enrollment by forwarding passenger flight details to the KTDI app on a mobile device or by enrolling at biometric kiosk in the departure hall – Entry to security filter with biometric lane for enrolled passengers – Border control emigration checkpoint with biometric lane – Boarding gate with biometric lane • Arrivals – Enrollment by forwarding passenger details through mobile phone (prior to departure or from the plane) or by enrolling at biometric kiosk in the arrival hall – Border control immigration checkpoint with biometric lane
Who?	• The KTDI facilitation was developed by the governments of Canada and the Netherlands, in partnership with WEF and several other partnering organizations. The complete pilot group consists of: – WEF – Government of Canada (Transport Canada) – Government of the Netherlands – YUL – Toronto Pearson International Airport – Amsterdam Airport Schiphol – KLM Royal Dutch Airlines – Air Canada – Accenture (Accenture 2018a) – Vision-Box – Idemia – Customs and Border Service Authority CBSA (Canada) – Immigration Refugees and Citizenship Canada (IRCC) – Koninklijke Marechaussee (Dutch Border Police) – Rijksdienst voor identiteitsgegevens (Dutch Identity authority) • The workgroup was initiated in 2018; pilot implementation in terminal was originally planned for 2020 but was postponed and is expected to commence in the near future.
Why?	• The KTDI program is based on a platform that enables a traveler-managed digital identity for international paperless travel to create a seamless passenger flow that is faster and more enjoyable. KTDI protects the passengers' privacy by giving them control over their data and who they share that with.
How? Technology used?	• KTDI is based on an interoperable digital identity, linked directly to government-issued identity documents (e.g., e-passports) of which the verifiable claim is stored on a blockchain. The platform allows for identity verification to airports, airlines, and border authorities using cryptography, distributed-ledger technology, and biometrics (facial recognition).

(continued on next page)

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Table F-1. (Continued).

Enrollment/digital identity creation and verification	• During the pilot, a one-time enrollment in the KTDI program was achieved, as follows: – Airline invites a select group of passengers to participate – Passenger creates and certifies digital ID with appropriate official issuing authority: local government digital identity-issuing office hosted at Schiphol airport or the Canada Border Services Agency (CBSA) in Canada – Passenger downloads mobile phone application and creates KTDI profile – The local identity-issuing authority makes a photograph of the passenger with the passenger's capturing device in a controlled environment. The passenger stores the image on the KTDI app, where it is encrypted, which the authority then verifies. – The passenger then tests this verified digital ID by making a selfie, which is compared to the digital ID, and if a positive match is made, it is verified by the local authority and becomes the first verifiable claim of a traveler's identity data stored on the KTDI blockchain. The photos are not stored anywhere other than on the passenger's own device. The blockchain only stores the verifiable claim. – In the mobile phone application, the passenger can then add information regarding upcoming flight, such as digital boarding cards. • Per-trip online check-in for each flight: – Prior to a flight, the passenger can push biometric details, passport details, and boarding card to the seamless system of the airport, where they are temporarily stored (maximum 24 hours) in a passenger data envelope (PDE). This can be repeated for each flight without having to enroll again. *Note:* The exact method of enrollment may be subject to change due to the fact the project is still in development.
Verification of identity how?	• The Digital ID and other details of the passenger are temporarily stored in the PDE. Directly after forwarding the data to the airport of departure or arrival, the system can perform all checks of whether the passenger is authorized to pass each touchpoint. At the airport on arrival at each biometric lane of the seamless process, the passenger's face is matched with the template stored in the system and processed according to the authorizations.
For which user groups?	• Pilot group of passengers between the Netherlands and Canada with a Dutch or Canadian Passport traveling with KLM or Air Canada between AMS and YUL and AMS and YYZ in either direction. The system could potentially be used for any nationality (with an e-passport) as long as the respective authority of the country provides the required digital ID verification and KTDI support.

and airlines offering a direct connection, and the technology companies involved with KTDI's development. As part of the initiative, the concept is planned to be implemented at AMS for KLM and Air Canada flights from and to Montreal and Toronto.

Note that as the pilot has not started, this case study description is limited to the main concept of operation, functions, and technical aspects.

How Does It Work?

Before the Passenger Journey

The KTDI system relies on two main platforms. The first platform is the KTDI-supported airport system hosted by an airport; the other is the KTDI app on the mobile device of the passenger, which holds the passenger's digital ID and flight information.

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The airport's orchestration platform that supports KTDI can accept passenger data when shared via the app prior to departure or arrival. The orchestration platform creates a passenger data envelope (PDE) for those data and will store it for the duration of the trip. The platform is typically hosted locally or on a cloud dedicated to the airport systems.

The KTDI app on a passenger's mobile device stores the passenger's biographic data (passport data) as well as the passenger's biometric data: the photographs of the passenger's face. These data are stored only locally on the mobile device and can be shared with airport platforms and country authorities if the passenger chooses to do so.

The touchpoints at the airports operate on a system provided by Vision-Box and are based on IATA's OneID concept. The system is made up of the following components for departures:

- KTDI enrollment kiosk for post check-in,
- Biometric lane with digital ID verification at security filter access,
- Biometric lane with digital ID verification at border control filter, and
- Biometric lane with digital ID verification at boarding gate.

The following components are installed for arrivals:

- KTDI enrollment kiosk at the arrival gate, and
- Biometric lane with digital ID verification at border control filter.

The Passenger Journey

Passengers that travel with the KLM or Air Canada flight from AMS to Aéroport International Montreal–Pierre Elliott Trudeau (YUL) or Toronto Pearson International Airport (YYZ) and vice versa can be invited to participate in the pilot. During the pilot, participation is for invitees only.

KTDI Enrollment and Preparation

The KTDI concept relies on a KTDI profile and a verified digital ID that is kept in a digital wallet on the mobile device of the passenger in the KTDI app. The digital ID contains biometric and biographic details of the passenger. These details are verified by an identity-issuing authority and are encrypted by a security certificate.

In the Netherlands, the verification of the KTDI profile on the mobile device takes place in the municipality office near the place of residence. The municipality office is designated as the organization to issue identification documents on behalf of the Rijksdienst voor Identiteitsgegevens (translates as National Office for Identity Data). Verification is done at the desk of the municipality office with the passenger being present.

In Canada, passports are issued by Immigration, Refugees, and Citizenship Canada (IRCC) at dedicated passport offices. Application for a digital ID and verification of the KTDI profile takes place at these IRCC offices and takes a few processing days as the Canada Border Services Agency (CBSA) checks the application. Passengers enroll at a kiosk, after which the passenger data are vetted by the office authorities.

Once set up and verified, the KTDI profile is ready for use with the mobile device and includes the first verifiable claim of a traveler's identity data. The digital identity, being from a trusted traveler (verifiable digital identity), is on the mobile device and can be expanded with travel data (e.g., boarding card) but also immigration data such as visas or electronic traveler authorization. The passenger has self-sovereignty over the data and can push data to the seamless

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airport/airline platform. The data remain with the passenger on the mobile device and are not centrally stored.

Airline Check-In and Airport Enrollment Per Trip

To prepare for a trip, the passenger can check in online or at the airport to receive a mobile boarding card. The boarding card can be added to the digital KTDI app, after which the passenger can forward all the relevant data to the seamless platform of the airport. This includes passport details, a biometric template, and boarding card details for the airline.

The seamless platform then creates a PDE that contains all the data required for the journey of the passenger by the airport, airline, and border control authorities, being either the Koninklijke Marechaussee (Dutch border police) or the CBSA. The platform provides the interfaces with the systems of the stakeholders and organizes the data flows. It also manages the business rules and organizes the authorizations that are received from the stakeholders.

The data that are provided by the passenger are pre-staged in the seamless platform. No passenger data are exchanged or stored at the blockchain; only transactions with pointers are provided. As long as all transactions (i.e., the verifiable claims) are without complication, the required level of trust is present, and the passenger's credibility is established.

Arrivals with the KTDI App

On arrival at the destination airport, a dedicated area with registration kiosks is present at the gate for the invited and enrolled passengers. These kiosks register the passengers' ID as well as their biometrics to create a PDE in the same way as is done at the departure airport. Again, the passenger's details will be verified against the systems of the border control agency.

At immigration, dedicated lanes for KTDI with a mantrap layout are equipped with a camera to recognize and confirm the passengers' ID and are also equipped with a near-field communication reader for the mobile phone with the digital ID or the e-passport, when required (for redundancy). When the passenger is authorized to proceed, the exit barrier opens.

Retention and Storage

Although the process at the airport is based on a per-trip model, enrollment in KTDI can be considered a for-life model. The digital identity that can be stored on the passenger's mobile phone is created for the lifetime of the passenger's e-passport and can be used for multiple journeys unless the passenger decides to opt out by deleting the KTDI app and its data from his/her mobile device.

The passenger's identity data are not stored on the blockchain. The blockchain holds the pointer to the verifiable claims (interactions) of a traveler's identity data with trusted entities. The record is expanded with every claim from every time the traveler uses the profile and thereby re-establishes the credibility of its identity. As the record expands, the credibility increases.

The mobile phone application can contain additional data related to the passenger. Prior to a trip, the passenger can push this information to the seamless flow system of the airport. The passenger manages the KTDI profile and digital ID with self-control over the permissions and data pushed to the respective airport, airline, and authority per trip. The identity data itself stay with the passenger and are not saved centrally at the airport or authority. The data that are made available by the passenger to the airport, airline, and authorities are encrypted and stored in the local databases for 24 hours and are only available for the trip (World Economic Forum 2020b).

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System Architecture Flow Diagram

The flow diagram of this case study can be found in the Known Traveller Digital Identity at Aéroport International Montreal–Pierre Elliot Trudeau, Montreal, Canada, case study in Chapter 2.

System Specifications – Digital ID and Self-Sovereign Identity

Instead of simply replacing the hardcopy ID with a softcopy digital ID derivative or alternative, the concept uses verifiable claims (interactions) of a traveler's identity data with trusted entities. The use of such claims ensures that the credibility of the traveler and his/her identity are established and authenticated. Blockchain technology is used to provide information for these verifiable claims as they are essentially transactions between the digital ID of the passenger and verifying entity. The use of a digital format enables optimization of the passenger process and improves risk management.

Stakeholders and Responsibilities

The KTDI program was brought together under the WEF (World Economic Forum 2020a) with many partners. The three airports, the two national governments, Accenture, and WEF led the program through its initial forming, setting its direction with innovation and the drive to organize pilot programs.

Stakeholders

The main stakeholders of the KTDI program were the YUL, YYZ, and AMS airports, together with the two airlines, KLM Royal Dutch Airlines and Air Canada. Also, three private companies were involved: Accenture, Vision-Box, and Idemia. As partners during the development of the KTDI program and thus stakeholders, the governments of Canada (Transport Canada, CBSA) and the Netherlands [the Koninklijke Marechaussee (Dutch border police)] were closely involved to facilitate the program's first pilot. Most importantly, the stakeholder group most affected by the program was the passengers who were selected flying KLM or Air Canada to take part in the pilot.

Responsibilities and Governance

The consortium between the stakeholders led, at the start of the project, to the formation of multiple working groups, all made up of members of the different stakeholders to maximize working together on multiple fronts. The working groups were guided by a project management committee and a steering committee, which oversaw the progress as well as guarded the project goals and timelines. Working groups were formed for tasks related to legal topics, communications, the system architecture and technology, implementation and performance measurement, and the delivery of pilot projects leading up to larger implementation. In this format, responsibilities among the different stakeholders were divided equally, with experts in each of the groups taking the lead and informing the other members.

Case Study Review

Benefits

Benefits from the implementation of KTDI are numerous and are different for each stakeholder. The main benefits relate to the passenger experience. KTDI allows for faster touchpoint processing, better privacy protection, and a touchless journey through the airport, which in

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the COVID-19 period is a key focus. The complete process allows passengers to only use their mobile device, relieving them of having to fumble with a passport and boarding pass. Faster processing through touchpoints at the airport will smoothen the journey and make it less stressful, allowing the passengers to spend more time relaxing, visiting retail shops, or selecting offerings such as food and beverages.

Furthermore, for government agencies/subcontractors, such as border protection, immigration, and airside security, the digitized process allows for a more efficient resource allocation, a smaller footprint in the airport, and, notably, a safer and more secure airport operation.

Airlines will benefit from efficiency increases as automation makes room for more personal service, where needed, and more streamlined processes. Passengers will predominantly interact with the airline through the application on their phones.

For airports, the biometric solutions provide means to increase automation, decrease the necessary footprint of touchpoints, decrease staff numbers, and process more passengers in less space. Especially with regard to COVID-19, the focus lies on creating a touchless and seamless travel experience, which KTDI paired with the right infrastructure offers.

Another benefit for airlines, airports, and passengers is the forward verification and country entry eligibility that is possible with the system, which prevents passengers being rejected at the country of destination and having to fly back. Arriving at the destination, passengers present their KTDI digital identity and, being vetted prior to arrival, they only have to wait for a positive match of the picture made at the e-gate.

Because the system architecture includes a blockchain component that aims to store the verifiable claims of confirmed identity rather than the digital identity itself, passengers can choose exactly what data they share with which stakeholders in the passenger journey (Kelleher 2019). The complete identity document or credential is only stored on the passenger's mobile device. KTDI is designed to comply with Canadian and European privacy-protection rules and is designed according to privacy-by-design principles.

Responses from Passengers

At the time of writing, with the biometric solution still being under development, no passenger feedback had been collected. Internal feedback by members of the working group and employees of the consortium had been used in the development of the mobile application, which had progressed to a second beta phase. Trials in a laboratory environment will aim to test all components together and link the systems of all the different stakeholders.

Fall-Back Options

The KTDI program is designed to replace the current touchpoint infrastructure at the airport, but a non-biometric, non-KTDI option will still be required for the near future for passengers that wish to opt out. The initial pilot program that was successfully rolled out only included a limited number of selected passengers per flight; thus, the standard processes were still available for passengers wishing not to use the KTDI system.

Concerns

Two points that were a concern for the consortium were (1) who would host the mobile application, and (2) the legal frameworks that would need to be agreed on. On the first concern, discussion was ongoing on whether the WEF would host the application or if a separate

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entity would need to be created. Development of the application by Transport Canada or CBSA (with experience in application development) or the Dutch government authority would not necessarily mean that party would host the application.

On the second concern, the consortium had knowledge of multiple industry providers of technology and hardware infrastructure that demonstrated the individual use and capabilities of the components; much more difficult would be the drafting of legal agreements between the multiple stakeholders regarding their roles in data processing, data control, data exchange, and how these legal agreements would be affected by privacy impact assessments that several stakeholders would go through.

Lessons Learned

One of the main lessons learned by the working groups of the consortium was that the creation of the legal framework was underestimated at first (Gendreau 2020). Expertise needed on a wide range of matters, from privacy protection for the two countries as well as the European GDPR, to data management, matters of intellectual property, and legal arrangements, is typical for collaborations between public and private companies. The drafting of these legal frameworks started slowly, and expertise had to be drawn from outside the consortium, affecting budgets and project timelines.

A second lesson learned related to the switching from a detailed project approach description and plan to a more flexible, phase-by-phase approach. This dynamic approach provided clearer short-term goals and deliverables that were easier to manage and attain by the project working group.

Findings and Trends

Findings

- There was ongoing development of the digital credential, with international standards developing simultaneously, with the ultimate goal of being interoperable with other (biometric) systems.
- The multi-stakeholder (multi-country) system proves a much more complex project, although the potential benefits in the long term (interoperability and scalability) can only be attained through this route.
- Lessons were learned on privacy of information and transmission and collaboration with foreign governments, especially regarding legal frameworks between public and private project partners.

Future Situation and Broader Implementation

The goals are to implement the KTDI system at all airports that offer flights between the Netherlands and Canada and to offer the seamless flow option to all passengers flying with the project stakeholder airlines (Air Canada and KLM). Proof of concept might attract more interested countries, airlines, and airports.

Until that time, a pilot in a lab environment aims to test the complete system for a small number of simulated passengers. Pilots at actual airports with actual passengers would follow before further implementation at the partner airports.

Scalability is another major focus for future implementation, especially as the blockchain component needs to be well designed to handle larger amounts of data traffic. Similarly, the

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hosting of the application on the passengers' mobile devices needs to be able to handle the data load as well as the interoperability and integration with multiple (new) airlines.

As many digital and biometric solutions are being developed around the world, the consortium is in constant contact with IATA and ICAO so as to follow developments in DTC and OneID. The consortium aims to align the KTDI system with these international standards and developments to increase the likelihood of successful interoperability with other systems.

Trends Identified

For this specific case study, all five trends identified in the second chapter make their appearance. Key in this biometric solution for passengers is the use of a mobile device (smartphone) for storage of their digital identity and biometrics. Looking at the system architecture design shows a trend toward a complex, multi-stakeholder system that aims to maximize long-term benefits of interoperability and scalability by aligning to global standards for digital identity credentials as well as standards for biometrics with regard to privacy, security, and ethical concerns. It is no surprise that the consortium in this case study has applied the design principles of privacy by design and is taking significant time preparing legal frameworks to mitigate potential risks.

Finally, this case study provides another example of a solution that leans heavily on biometric technologies for identity verification and takes care to distinguish itself from systems that might be used for mass surveillance.

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APPENDIX G

Case Study: The Seamless Passenger Journey at London Heathrow

Summary

The vision for a seamless passenger journey consists of the deployment of biometrically enabled touchpoints equipped with cameras on a pole and at check-in kiosks, self-service bag drops, ticket presentation gates, and self-boarding gates. This initiative targets both international and domestic customers and encompasses all terminals of the airport, with an initial focus on Terminal 5 (T5) and Terminal 2 (T2). At a later stage, other processing points will be added, such as for immigration and security screening and the ability to enroll and check in using a mobile app. Another parallel initiative consists of biometrically enabled preclearance of passengers traveling from the UK to the United States on selected flights.

On arrival at the first touchpoint, the departing passenger consents to the digital process, scans his/her boarding pass, and captures a photo of his/her face (i.e., takes an on-the-day image). The automated passenger identification involves matching this on-the-day image with the image stored in the passenger's e-passport chip. If there is a positive match, the passenger's ID is verified. Biometric matching and boarding-pass scanning will be used at subsequent touchpoints to recognize the passenger throughout the remaining journey.

For a passenger traveling to the United States, the process also involves matching the on-the-day image against the CBP's TVS. For the validation to work, it is a prerequisite that the passenger had previously visited the country or that his/her photo was otherwise accessible from a U.S. government database such as for a visa application (Heathrow 2020b).

At the heart of the process is the Passenger Authentication Scanning System (PASS2), which was developed by Atkins on behalf of LHR and manages passenger identity throughout the airport. The system incorporates Atkins' ID management platform (IMP) and is augmented for various tests and trials of different technologies, including those from Yoti, ICM, CEIA, Dorma Kaba, Rockwell Collins, and Aurora AI (Stewart 2019a).

Passengers' biometric data are stored and retained on a per-trip basis and will be cleared at the end of the operational day or after 24 hours (World Travel and Tourism Council 2019) [and within 30 days during the live trials, about which passengers were notified (Heathrow 2020a)]. LHR only keeps anonymized transaction data to monitor and improve the biometric technology. Due to privacy issues and the risk of data breach, LHR prefers not to hold passengers' biographic data in its IMP and is testing a third-party provider for the verification function. Booking details are not stored either since the passenger is still required to scan a boarding pass.

The main benefits are an enhanced experience with faster processing and reduced queuing times for passengers, which allows the airport and airlines to save and optimize their resources and minimize costly terminal expansions. From a government perspective, the automated process is much more accurate, and thus more secured, than manual ID verification.

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Key Facts

Table G-1 summaries the key aspects of the seamless passenger journey at LHR.

Introduction

LHR has been using facial biometrics for the departing journey for more than 10 years (Wilcox 2019b), with the original system designed to capture the domestic passengers' faces at ticket presentation gates and later at boarding gates. This was primarily set up to allow international and domestic passengers to mix in the combined departure lounge, while ensuring that they would board the correct flights.

Table G-1. Key facts of case study on the seamless passenger journey.

What?	• The seamless passenger journey • Facial recognition • 1:1 and 1:few biometric matching
Where?	• LHR
Passenger process steps	• All the processing points were tested in the lab or with live trials, but full deployment will happen either in the short term (a date has not been set yet due to COVID-19) or at a later stage. • Online and off-airport check-in (at a later stage) • Self-service kiosk and self-service bag drop (in the short term) • Improved security (risk-based; at a later stage) • Ticket presentation gate (in the short term) • Security risk-based screening (at a later stage) • Self-boarding gates (in the short term) • Border (at a later stage) • U.S. bound flights: cross-validation with CBP's TVS (live on some flights and planned for the remaining flights in the short term) • Cross-border with transit and destination airports (at a later stage)
Who?	• LHR was set to launch its first end-to-end biometric deployment in the summer of 2019, bringing facial recognition to the key processing points of the departing passenger's journey. Atkins' PASS2 system, which incorporates Atkins' IMP and various technologies, including those provided by Yoti, ICM, CEIA, Dorma Kaba, Rockwell Collins, and Aurora, were tested.
Why?	• The £50 million project is part of a wider program of investments to increase the speed, efficiency, and security of passengers' journeys, reduce congestion in LHR's terminals, and increase capacity in the existing facilities, thus postponing costly expansions (Heathrow 2018). Biometrics on the outbound passenger journey were first implemented at T5 to enable the terminal to co-locate domestic and international passengers. A face capture of the domestic passenger is performed at the ticket presentation gates when the passenger presents his/her boarding pass and then again at the boarding gates. This ensures that domestic passengers would not board an international flight and that international passengers would not board a domestic flight. The program was then extended to T2. After extensive testing, the program was ready for full deployment to include all passengers and additional touchpoints.
How?	• At the time of writing, manual authentication meant passengers needed to present IDs and travel documents such as boarding passes and passports many times throughout the outbound journey to show that they were authorized to fly. By offering passengers the option to use facial-recognition technology, the process becomes more streamlined and faster.

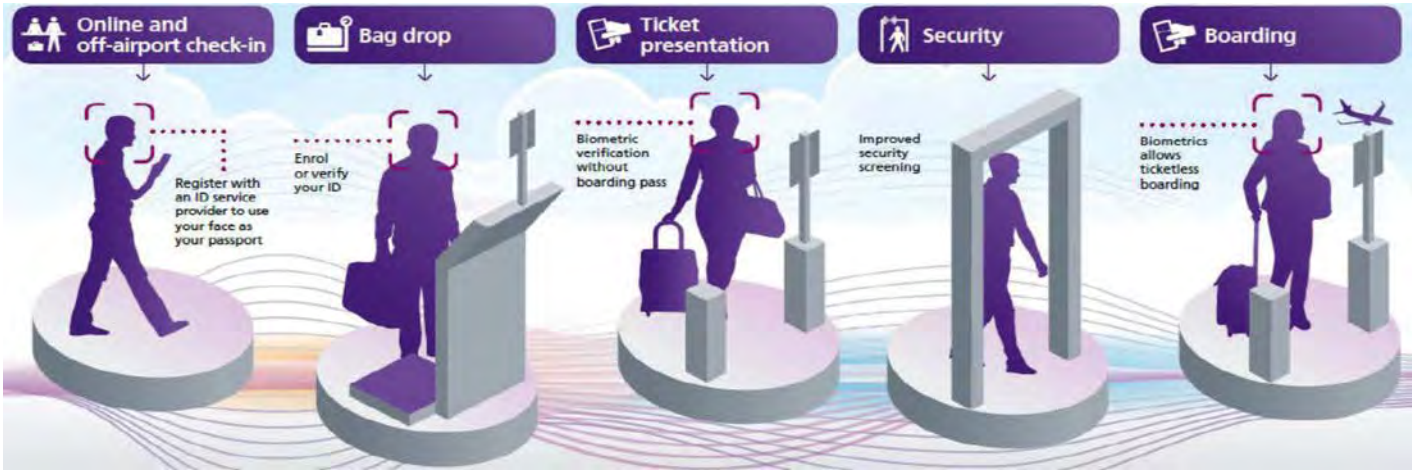
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Table G-1. (Continued).

Enrollment/digital identity creation and verification	• In-person enrollment: The passenger can enroll at a biometrically enabled self-service check-in kiosk, at a self-service bag drop, or at the ticket presentation gate; the passenger scans his/her e-passport [the machine-readable zone (MRZ) contains the passenger's biographic and biometric data], takes an on-the-day image that is then matched to the passport's image, and scans or retrieves his/her travel booking; the verified biometric data are then stored in the IMP. • Mobile enrollment (at a later stage): The passenger uses a mobile device ID verification app, provided by Yoti, to scan his/her passport's MRZ, gaining access to the passport's chip and biometric/biographic data. The passenger takes a selfie to authenticate the image on the passport. The verified biometric data are then stored in the IMP. This process was tested in principle, but it will be fully implemented at a later stage. • Because of privacy issues and the risk of data breach, LHR would prefer not to hold passengers' biographic data in its IMP; ideally, the verification function should be provided by a third party, and the only data stored in the local IMP would be the biometric data; the booking details would not be not stored either as the passenger is still asked to scan his/her boarding pass at each touchpoint.
Verification of identity how?	• Identity verification involves matching an on-the-day image taken of the passenger at one of the biometrically enabled self-service touchpoints to the image recorded on the passport's chip. A single PDE is created and stored in the local IMP. Biometric matching will then be used at subsequent touchpoints to prove it is still the same passenger without the requirement to again provide documentation (i.e., by matching subsequent scans of the passenger's facial biometrics to those extracted from the first on-the-day image).
For?	• The initiative will be extended to all domestic and international flights out of T5 (British Airways), T2, and eventually to all terminals.

In 2016, LHR developed a new business case to set out the future pathway for automation, innovation, and technology at the airport. In this process, LHR quickly realized that passenger identification was key to creating a seamless passenger journey for all customers, and the biometrics technology is the right one to help achieve this objective (Wilcox 2020). See Figure G-1 for an overview of the seamless journey.

LHR began a 3-year program of investments to implement automated self-services and augment the existing system for passenger identification, with the aim to increase the speed, efficiency,



Source: Beunardeau et al. 2019.

Figure G-1. Seamless journey vision.

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and security of passengers' journeys and reduce congestion inside the airport's terminals. After a series of successful trials, LHR was set to launch its first end-to-end biometric deployment in 2019, bringing facial recognition to the key processing points of the departing passenger's journey. The deployment was postponed due to the COVID-19 pandemic.

How Does It Work?

Before the Passenger Journey

More than 10 years ago, biometrics were adopted as a means of passenger identification, first at T5 and then T2. Therefore, when it was decided to extend the program to additional touchpoints and to all international and domestic passengers, the most cost-efficient and viable option was to augment the capability of the existing IMP, which was developed by Atkins.

The platform stores the passenger's name and biometric data captured at the first touchpoint (self-service check-in kiosk, self-service bag-drop, or ticket presentation gate) by reading the e-passport chip and taking a photo (an on-the-day image). The data stored are referred to as the PDE.

The biometric data stored in the e-passport chip are compared to the on-the-day image and, if there is a positive match, the passenger's ID is verified, and from that point onward, the on-the-day image is used to recognize the passenger throughout the remaining journey, without the need for additional paper-based identification.

While it is technically feasible to store additional passenger data, such as the PNR, in the IMP to eliminate the requirement to scan the boarding pass at each touchpoint, LHR staff made the decision to not integrate this step and, therefore, to not store passenger booking details in the IMP. The thinking behind this decision was that passengers will continue to hold a boarding pass (printed or on their mobile app) for the foreseeable future, as confirmed by a passenger survey. This decision eliminated the need to integrate the IMP with the airline DCS where passenger information such as the PNR is kept, significantly simplifying the IMP capabilities and shortening the validation and matching times.

The CBP TVS initiative relies on the CBP's own capabilities to almost instantaneously validate a passenger's identity by comparing his/her biometrics data (an on-the-day image captured at the boarding gate) against a pre-existing CBP-held database. (For the validation to work, the passenger has to be a returning passenger to the United States with his/her biometric and biographic data already captured by CBP in a previous visit, or his/her photo must otherwise be accessible from a U.S. government database such as for a visa application.)

Mobile enrollment was tested to prove that the technology worked and that the IMP could support the integration with a third-party identity app. However, this mobile enrollment capability had not gone live at the time of writing because LHR staff believe that airlines should integrate this capability within their own booking app and, therefore, they should take the lead in pushing this initiative forward. Considering that 60% to 70% of passengers at LHR travel for leisure, and a significant number of these passengers travel only once a year, this capability would have limited value overall.

The trial was conducted in partnership with Yoti and consisted of the passenger checking in on the Yoti webpage or using a mobile app. The passenger was asked to share biometric and biographic data by having the app access the passport MRZ and by taking a photo of the day. The data were then stored in a secure Yoti-controlled database that was set up specifically for this trial. At the following processing points, the passenger's face would be captured by the biometric-enabled touchpoint, translated into a specific readable format, and sent to Yoti's IMP for verification of identity. Using this approach, referred to as a "self-sovereign ID management"

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approach, LHR IMP would not store passport details and would therefore be less exposed to privacy concerns or data breeches.

The Passenger Journey

At the first touchpoint (self-service kiosk, self-service bag drop, or ticket presentation gate), the passenger's consent is given to use his/her face biometrics and biographic information; the passenger scans his/her boarding pass (printed on paper or on his/her mobile app) and his/her passport; the kiosk camera captures the passenger's face (on-the-day image); the boarding pass is validated directly with the airline DCS; the passenger's biometric data are accessed via the e-passport chip and verified against the on-the-day image; and if there is a match, the passenger's face biometrics, name, flight number, and PNR information, known as PDE, are stored in the airport IMP and will be used for identification at the other biometric-enabled processing points at the airport. The process and technology are capable of handling group enrollments versus one-on-one enrollments.

At other biometric-enabled processing points, the passenger is identified by a comparison between the camera capture at the processing point and the on-the-day image stored in the IMP.

In case of no match or if the passenger wishes to opt out, he/she can proceed to staffed processing points.

In the case of selected U.S.-bound flights and for returning passengers to the United States, the passenger scans his/her boarding pass at the boarding e-gate, the passenger's identity is validated, and the biometric data are verified against the CBP TVS database.

Retention and Storage

The PDE storage and retention are per trip, and all information is deleted at the end of the operational day or after 24 hours. After that, only anonymized transaction data are retained to monitor and improve the biometric technology. During live trials, the data were kept for 30 days.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Seamless Passenger Journey at London Heathrow case study in Chapter 2.

Stakeholders and Responsibilities

Stakeholders

In the early stages, airline partners (including British Airways) and suppliers (including the airport system and the hardware and IMP vendors) worked collaboratively to map the journey, define the concept of operations, and test the technology.

UK government bodies were involved in different stages of the process. For example, LHR was collaborating with the UK Department for Transport to test the prospect of risk-based differentiated security. In addition, LHR worked with lawyers and the Information Commissioner's Office (ICO) to look at the process from a privacy perspective.

The compatibility of the PDA data format and its integration with the government database was tested in principle on both inbound and outbound processes, but no live trials were implemented due to the regulatory and policy changes required to make the implementation possible.

As previously mentioned, the U.S. government, in particular CBP, also collaborated with LHR.

To gain further stakeholders' buy-in, demonstration days were held throughout the initiative. LHR also actively shared its experiences with other airports and aviation industry bodies, such as IATA and ACI.

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Responsibilities and Governance

The seamless travel initiative's governance can be summarized as follows:

- Heathrow staff led the initiative and secured the necessary funding for lab testing, live trials, and any other related activities.
- Several working groups and sub–working groups were established to include all stakeholders and focus on specific subject matters such as technology, processes, and regulation.

This governance structure developed at the beginning of the initiative in 2016 was kept in place.

Timeline and Planning

The program was initiated by Heathrow staff in a series of workshops to discuss the future of automation and passenger identification and how technology and innovation could shape the passenger journey. A business case to support the initiative was developed, and the initiative was officially started in 2016. Since then, LHR has embarked on a series of lab tests and live trials, which have provided invaluable insights on each aspect of the journey, including human interaction and perception. Those tests included the following:

- Testing individual and group enrollment and biometric matching at various airport touchpoints, such as kiosks, bag drops, and entrance to security;
- Enabling off-airport enrollment using an app;
- Allowing passengers' automated access at subsequent touchpoints once they have enrolled at the first touchpoint, thus expediting their journey;
- Looking at the prospect of enabling risk-based differentiated security in collaboration with the UK government;
- Testing a cross-border solution in collaboration with CBP's TVS system (Heathrow 2020b); and
- Testing a third-party provider of verification function.

The approach taken by LHR was to retrofit existing or new self-service equipment with cameras on a pole rather than have cameras integrated in the equipment itself. This decision was made to provide a sense of consistency across the various touchpoints and the terminals so that passengers would know what to expect and where to look.

Procurement and Vendor Selection

By early 2019, lab tests and trials were successfully completed, vendor selections were finalized, and an initial investment of £50 million was secured. The initial phase of the rollout at T2 and T5 was planned to start in mid-2019. Once completed, this would be the largest rollout of biometric technology, with more than 5000 touchpoints. Due to the global pandemic and the drastic decrease in passenger numbers, the deployment was put temporarily on hold.

Use Case Review

Benefits

The benefits for the various stakeholders are highlighted in the Table G-2.

Responses from Passengers

During the lab tests and live trials, it was shown that biometric verification without a boarding pass was technically feasible. However, passengers' feedback indicated that they prefer the combination of facial biometric capturing and boarding-pass scanning at different gates and

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Table G-2. Seamless journey stakeholder benefits.

Stakeholder	Benefits
Passengers	Enhanced experience with a more seamless journey
	Faster processing and reduced queuing times
Airport	Increased passenger processing throughputs postponing the need of costly capital expansion programs
	More efficient use of existing infrastructure
	Co-location of international and domestic passengers in the same departure lounge with significant increase in efficiency
Government	Enhanced security: ID validation is much more accurate than manual validation (almost double the matching rate)
	Reduced staffing costs as fewer passengers have to be manually processed
Airlines	The biggest beneficiaries
	Staff reduction or redeployment where most needed
	Operational improvements such as on-time performance

touchpoints. This approach seems more natural and reassures passengers that the process is secured and that they are correctly completing the departing journey. LHR therefore assumed that the boarding pass would not be disappearing in the short to middle term.

It is noteworthy that passengers' feedback was always positive when they were asked about identity management as a means to deliver a secured and seamless journey. However, they were quite skeptical and negative when hearing the term "biometrics."

In addition, LHR established a consultation group of 3,000 people to gauge the public's sentiment on privacy. The vast majority of the respondents said they were okay with sharing their personal data if they received transparent information on how their data would be used and if they had control over their data (who to share it with and why).

System Performance and Specification Review

To overcome the problem of poor ambient lighting, a decision was made early on to adopt cameras augmented with infrared technology. This technology consistently provides better quality images for matching purposes.

Some of the biggest challenges had to do with additional functions having to be performed at some of the processing points, such as visa checks (when visas are not electronic) and excess baggage payments.

Fall-Back Options

The intention was never to completely eliminate manned positions. Therefore, in the case of no matching, lack of consent, or if the technology failed, staffed counters or roaming agents would be always available at each touchpoint.

Concerns

Data privacy was addressed from the beginning, and the principles of privacy by design were incorporated in the system architecture where access to data was provided only on a need-to-know and authorized-to-know basis.

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LHR engaged ICO to perform a full review of the seamless passenger journey initiative. Specifically, the ICO reviewed the role and the relationships between the different business partners (airport, airlines, suppliers) involved in the processing of biometric personal data, confirmed that data management met GDPR/General Data Protection Act of 2018 (DPA18) provisions, and assessed LHR's proposed method for obtaining explicit passengers' consent to use their biometric data in a live environment.

Lessons Learned

The key lesson learned was that the seamless passenger journey initiative was not about technology, but rather was about human interaction with the technology. The technology exists, and lab tests proved that it worked. However, if the technology is not intuitive and user friendly, ultimately it will not serve its purpose, and this was proved over and over during live tests.

Apparently simple things, such as the distance from the camera during image capture, where the passenger was looking during image capture, crowding around cameras, and tight queuing in proximity of boarding gates, would cause poor image quality and no matching. Human interaction was carefully analyzed, and appropriate solutions were identified throughout the trials.

Findings and Trends

Findings

Biometrics, combined with self-service processing, deliver tangible efficiencies in processing times, reduction in staffing requirements, and a significant improvement in safety and security with a much higher matching rate than that measured in manual processes.

These benefits are significant when the deployment of biometrics and automation is on a large scale, as it will be in the case of LHR.

Future Situation and Broader Implementation

Other initiatives, such as mobile enrollment, risk-based security, and the integration of immigration in the seamless journey, are on hold as of the time of writing.

The cross-border initiative (between three airports, three governments, and two airlines) was initiated by Heathrow with involvement from IATA. It is in the technical testing phase, an identity management mobile app is in the process of being developed, and the function of the third-party broker (providing a platform to verify identity, which in the case of the United States is CBP, but is different in other countries) has been discussed and could hypothetically be provided by IATA. This solution minimizes risk to airports as they will not be responsible for holding biographic passenger data.

Trends Identified

A self-sovereign approach is the most promising form of passenger data management. Under this model, the passenger would control what personal data he/she wants to share and with whom. This model could be applied more broadly to the entire passenger journey. The model has successfully been adopted in other industries, such as financial services, but the adoption of such a model in aviation is proving more challenging.

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APPENDIX H

Case Study: Risk Management During COVID-19 Using Biometrics at Carrasco International Airport, Montevideo, Uruguay

Summary

Carrasco International Airport (MVD) has successfully implemented biometrics at its border control and boarding checkpoints at arrivals and departures, significantly increasing processing capacity at those points. With only international flights, all of the passengers go through border control—since 2016 choosing the new e-gates or queuing for the control podium. At the e-gates, passengers automatically enroll by scanning their passport, selecting their flight on a touch screen, and having their picture taken. If the picture matches the biometric information on the e-passport and no other issues are present, the passenger is allowed to exit or enter the country. With successful enrollment, for the departures process, the passenger can then complete boarding by simply approaching the e-gate at the boarding gate and having his/her face scanned. A positive match opens the gate, and all personal information is purged 30 minutes after departure. Passengers wishing not to use the biometric-enabled checkpoints may opt out and use the legacy facilities.

Processing times at the border control gates have been reduced from 40 or 50 seconds to approximately 15 seconds, and for those passengers with eligible e-passports, boarding times have also decreased. Additionally, the implementation of the border gates has reduced passenger and ground staff interaction and contributed to a more seamless and touchless passenger journey, improving health safety at the airport.

The use of biometric technology has also enabled the border control authority to initiate risk profiling (related to COVID-19) during the arrivals process. Passengers arriving from low-risk countries were able to pass through the border e-gate quickly, while passengers from higher-risk countries were easily directed to a secondary screening or additional health check. This greatly reduced the need for increased staff and other resources otherwise needed to check all passengers by their origins. This was all made possible by integrating data defined as advanced passenger information and PNR into the system platform. See Table H-1 for more information.

Introduction

At the international-only (no domestic services) airport of Montevideo in Uruguay, named Carrasco International Airport (Montevideo Carrasco Airport 2020), the airport operator Corporacion America has made significant progress in piloting and subsequently implementing in full a biometric system that automates most of the border crossing and boarding steps in the passenger journey. The airport has taken a leadership role in South America as the first to implement biometric solutions (Vision-Box 2018b).

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Table H-1. Key facts of case study on risk management during COVID-19 with biometrics.

What?	• Border and boarding biometrics enabled and implemented for risk management for COVID-19 • 1:1 matching at border control • 1:few matching at boarding
Where?	• Carrasco International Airport, Montevideo Uruguay
Passenger process steps	• Departures: ○ Border control ○ Boarding • Arrivals: ○ Immigration
Who?	• Lead: Corporacion America (Centre of Excellence) – Easy Airport Consulting Division • Stakeholders: Airport authorities: border control (immigration), Vision-Box (global vendor), Air Europe, and LATAM airlines
Why?	• Improve passenger processing speed though airport • Improve security as border officers can focus on high-risk passengers • Allow risk-based profiling for COVID-19 testing
How?	• Two biometrically enabled e-gates by Vision-Box; operating system a collaboration between Vision-Box and Corporacion America. • E-gates at border control as well as e-gates at boarding. The e-gates at border control verify the passport that is scanned (1:1 matching) and let the passenger select his/her flight, and then a picture is taken of the passenger's face for the temporary biometric token. • At boarding, the scanning of the passenger's face and subsequent successful matching (1:few) allows passage through the e-gate for boarding the aircraft.
Enrollment/digital identity creation and verification	• No prior enrollment is needed. Everything is done at the border control checkpoint. • Identity is verified by reading the e-passport and comparing to the stored biometric on the e-passport to the live picture taken of the passenger's face. • Trip model token: the biographic and biometric information used for the token is only used for one trip.
Verification of identity how?	• Matching of the e-passport's stored facial biometric with the image taken of the passenger's face at the e-gate (1:1 matching)
For?	• International passengers only (MVD airport only has international connections.) • Currently limited to passengers with: ○ Uruguayan e-passport or ID card ○ E-passport from other countries that do not have visa requirements

How Does It Work?

System Architecture, Pre-Existing Systems, and Databases

The biometric solutions at MVD are found at the border control e-gates and the boarding e-gates. A collaboration between Vision-Box and the Centre of Excellence team of Corporacion America—specifically the Easy Airport Consultation Division (Easy Airport 2020)—resulted in the creation of an in-house operating platform based on Vision-Box's Orchestra architecture. Although the system is connected to the airline DCS systems, it does not have direct access to API and PNR information. Rather, the Easy Airport system is returned a

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Ministry of Homeland Security risk scoring, which Carrasco International Airport uses for its risk-based assessments. This tailor-made system enables the operation of the e-gates, facilitates the temporary storage of the recorded biometrics, and provides an EasyWeb portal for its users and stakeholders.

The EasyWeb portal is a unique feature and required most of the custom programming to be done in-house. The portal allows airlines to track passengers through different checkpoints during their airport journey. This allows the airlines to have a complete picture of the passengers who are ready for boarding or who might still be on the way to the gate.

The EasyWeb portal also offers the airport Operational Control Centre (OCC) and third-party stakeholders some additional features, such as:

- Dashboards of useful information such as waiting times and passenger flow statistics; and
- Anonymized statistics of passenger flow, wait times, movements, and dwell times through the airport passenger journey.

The Passenger Journey

For departing passengers, enrollment using biometrics prior to the arrival at the airport is not needed. With a confirmed flight booking, the passenger can check in online or at the airport, drop off baggage, and then continue to the pre-security e-gate. At this point, only a valid boarding pass is scanned, allowing the passenger to continue to airport security.

After clearing security, biometric enrollment is done at the border control checkpoint since all flights at MVD are international. At this e-gate, the passport is scanned, and the passenger selects his/her flight on the touchscreen, after which a live picture is taken of the passenger. The passenger's identity is verified by comparing the passport's stored biometric information to the live picture and, if verified, passage is granted through the e-gate (Vision-Box 2018a).

The facial biometric is then stored temporarily in the system and passed to the e-gate at the correct boarding gate. When the passenger arrives at the boarding gate at the designated time, he/she will be granted passage in case of a successful match (1:few).

For the arriving passenger, he/she will only encounter the biometric-enabled border control gates that verify one's identity through facial recognition and comparison to the stored biometric information in the e-passport or ID card, which is similar to the departure process.

Retention and Storage

The government of Uruguay has a fairly modern approach to personal private information (PPI) and has laid down strict rules for the capture, use, retention, and deletion of this information. The goals are to keep the information safe, delete it when possible, and make sure that if other parties want to use it, the information benefits the whole community.

At MVD, the stored biographic and biometric data that are linked to flight information are purged 30 minutes after departure. What remains are the non-identifiable statistical data points that benefit, for example, the airport OCC, the airlines, or third parties.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Risk Management During COVID-19 Using Biometrics, Carrasco Airport, Montevideo, Uruguay, case study in Chapter 2.

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Stakeholders and Responsibilities

Stakeholders

The project was led by Corporacion America in collaboration with the airport (government) authorities at Montevideo and the vendor Vision-Box. The main stakeholders also include the two main airlines at the airport, Air Europa and LATAM, and several airport departments.

Responsibilities and Governance

Corporacion America, the airport authority, has taken the lead for this project and has put in place multiple working groups to make this effort a reality.

Use Case Review

Benefits

The main benefits of the implementation of the biometric-enabled e-gates at MVD have proven to be:

- Improved security,
- Improved health safety,
- Reduced staffing requirements,
- Faster border crossing and boarding times, and
- COVID-19–related risk profiling of passengers.

The first benefit relates to the improved security because the border control officers have more availability to focus their time on high-risk passengers. Instead of having to process all the passengers and focus the majority of their time on checking low-risk passengers, the border control officers can now invest most of their time on more high-value activities. This improved the border control officers' performance as well as job satisfaction. The security of flights has also increased. When boarding 200 passengers in 20 minutes and with airline ground staff often being stressed or in a rush, the time needed to accurately check each passport picture adds pressure.

The reduced number of interactions with ground staff and touchpoints are a more recent focus and benefit. The biometric-enabled e-gates have made the passenger journey at MVD move toward a fully touchless, seamless journey. Fewer touchpoints and less interaction decrease the risk of transmission of communicable diseases such as, for example, COVID-19.

Overall, the Easy Airport system, together with the EasyWeb portal, have allowed the airport and the airlines to adopt a much more data-intensive operation and to learn to better allocate resources and plan ahead. Data-driven solutions have, for example, reduced some staffing requirements at checkpoints or reallocated the priority of jobs. Also, with the portal, airlines have a complete picture of the passengers who are ready for boarding or who might still be on the way to the gate, which improves the effectiveness of the services that the ground staff can offer. This was a major incentive for the airlines to join the program.

For airlines and passengers, the faster border crossing and boarding of aircraft were a big benefit. Traditional border crossing checks would typically take 40 or 50 seconds but were replaced by a 15-second e-gate checkpoint. For airport capacity, this is a huge improvement. For boarding, in some examples, double or even triple the boarding speed was achieved since larger percentages of passengers could use the facial biometric–enabled e-gates. LATAM shared that its ground staff could manage the boarding process much better and provide a better passenger experience.

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Early in 2020, the use of the technology also enabled the border control authority to initiate risk profiling (related to COVID-19) during the arrivals process (Cerri 2020). Passengers arriving from low-risk countries were able to pass through the border e-gate quickly, while passengers from higher-risk countries were easily directed to a secondary screening or additional health check. This greatly reduced the need for increased staff and other resources otherwise needed to check all passengers by their origin. This was all made possible by integrating data defined as advanced passenger information and PNR into the system platform.

Responses from Passengers

Passenger reviews on the new systems have been positive, even if their knowledge of how the systems worked was basic. Since the use of biometrics began in 2016, passenger satisfaction has been high (Mujica et al. 2020).

Fall-Back Options

For the two biometric-enabled checkpoints, the normal legacy checkpoints are still available throughout the airport. After the implementation of the new systems in 2016, there has been a steady rise in adoption due to the phased-in availability to different nationalities and government ID cards. As adoption keeps rising, resource allocation for the legacy checkpoints will likely decrease.

Concerns

The primary concerns of the government authorities were the correct and proper use of data and minding the privacy of passengers. The design of the system followed stringent requirements to protect private information (biographic and biometric), and for stakeholders that wanted to use the EasyWeb portal, information had to be provided about how they would use the anonymized data and for what purpose. For several third-party stakeholders, the data were used for making the airport better in terms of passenger services, shorter waiting times, and improvements across the entire airport system. If stakeholders shared their intended use of the non-PPI information, then they were allowed to join and have access to the portal. In this instance, the government agencies maintained control over the admissions.

No other concerns were voiced by the airport authority, either regarding the communication or education of passengers or violations of their privacy rights.

Lessons Learned

The main lessons learned were related to the following:

- It proved beneficial to be able to leverage security improvements in the business case with stakeholders for collaboration.
- Several iterations were needed to improve the capturing of facial biometrics: influenced by lighting, time of day, face masks, and user feedback.
- Not all airlines were initially rushing to implement and collaborate, but COVID-19 and the new focus on health measures changed that.
- Limited eligibility of passports caused a slow increase in passenger adoption.

A lesson learned from discussions with potential airline stakeholders was that, rather than just the ROI, the improved security also proved of high importance in encouraging participation in the collaboration (Cabrera 2020). Government agencies noted that the improvement in security was a major driver.

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The capturing of facial biometrics with cameras installed at the e-gates was improved several times after learning the influence of lighting (time of day, light conditions outside, added lighting in the e-gates) as well as the sudden mandatory use of masks to prevent the spread of COVID-19. In the latter case, the actual algorithm for comparing and matching the biometrics was changed to meet the required standards. In addition, due to communication with airport staff and passengers, the e-gate went through several iterations of design. This changed the method of capture of the facial images as well as the interaction of passengers with the e-gate. Some examples of design changes were changes to the messaging, the layout of the touchscreen, and the overall lighting conditions at the e-gate.

Initially, not all airlines were in a hurry to implement the program. Air Europa and LATAM wanted to introduce the new systems primarily to reduce staff, increase efficiencies, and improve safety and security. These airlines already had a strong digital and online presence, so facilitating online check-in and such was already part of their processes. For smaller and more local airlines, participation in the project would cause them to have to upgrade their back-end systems. This meant these airlines would first have to invest in their own systems. With developments around COVID-19 and the move toward touchless passenger facilitation, many more airlines have joined the project.

The use of the e-gates was first limited to those with Uruguayan e-passports, accounting for about 15% to 20% of all passengers, followed by those with Brazilian and Argentinian e-passports, which accounted for another 10% to 15%. In time, Uruguayan ID cards were also allowed since travel within South America does not require passengers to have a passport. Thereafter, passengers with e-passports from countries not requiring visas were also allowed to use the e-gates. This caused a slow increase in technology adoption. Theoretical coverage of users is 97%, but a large percentage of the Brazilian population traveling to Uruguay does not have a passport, only an ID card, and these are not updated. In this case, the policy of a different country to not have up-to-date identification documents influenced the adoption rate in Montevideo. Real user adoption rates were closer to 50% at the time of writing.

Findings and Trends

Findings

Easy Airport presents the ability to meet different airline business cases and strategic objectives such as resource and time savings. It also presents the opportunity to install state-of-the-art technology and infrastructure to meet the airport's future demand. The added data portals for airlines and other stakeholders is a bonus and delivers anonymized data streams. Easy Airport was used during the COVID-19 pandemic in risk profiling passengers into specific categories automatically and notifying border agents of passengers entering the country from specific origins. Those passengers underwent a different health screening process.

Some key lessons learned for the airport authority and implementing teams were around the protection of passenger privacy and information (as well as transmission of such information to foreign governments) especially regarding legal frameworks between the project stakeholders.

Future Situation and Broader Implementation

According to MVD, one future consideration is looking into including children in the program since, at the time of writing, e-passports were only available for citizens 18 year or older. This would greatly improve the passenger experience because travelers with children have to use the legacy system and verify their identity with regular identity cards or passports.

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The use of facial recognition has also been considered for the pre-security check. This could shift the location for the enrollment to earlier in the passenger journey. The aim is to allow all touchpoints in the passenger journey to be used with the Easy Airport biometric solutions.

Trends Identified

The main trend identified in this case study is the increased use of biometric solutions to improve the passenger experience, create a seamless journey, and allow for less physical interaction with ground staff and touchpoints. The Easy Airport passenger journey in Montevideo is moving in the direction of becoming completely touchless, although several points of interaction still exist. Integration with other airport authorities and possibly the increased use of mobile devices will see this trend continue.

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APPENDIX I

Case Study: Digi Yatra and the Seamless Passenger Journey at Kempegowda International Airport, Bengaluru, India

Summary

In 2016, as part of a new terminal development program, Kempegowda International Airport (BLR) staff conducted a series of workshops to discuss the future of the passenger journey and how this could be improved by use of a single biometric token. The BLR Innovation Lab further developed the concept of operations and explored different types of biometrics (e.g., facial, fingerprints, and iris). In January 2017, BLR rolled out its first Digi Yatra trial, which went live on a Vistara Airlines flight (Airport Technology 2019; ET Bureau 2019). The initiative was then quickly expanded to AirAsia India and Spice Jet and is now set to include all domestic airlines. International flights will be involved at a later stage (Future Travel Experience 2019b).

At the heart of the initiative is the Common Digi Yatra ID Platform, an interface platform between the countrywide biometrics database owned by the Unique Identification Authority of India (UIDAI) and the local airport IMP.

To partake in the Digi Yatra seamless journey, the departing passenger can create a Digi Yatra ID any time prior to his or her trip. This is a one-time process that can be done online or in person at one of the airport's registration kiosks using a government-approved ID number (including a unique identifier if the passenger has previously been registered in UIDAI database) or a manual ID card. In case a manual ID card is used, an additional physical check by an officer is required at the registration kiosk. Once completed, the Digi Yatra ID is activated and will be permanently valid.

The passenger can enter his/her Digi Yatra ID number at the time of ticket booking, and it is then transmitted by the airline DCS to the airport IMP.

On the day of the travel, the passenger arrives at the airport registration kiosk, scans his/her boarding pass and has his/her iris and face captured by the kiosk's camera. The passenger's consent is taken to use his/her biometrics for the journey. The Digi Yatra ID platform performs boarding-pass validation with the airline DCS and a real-time biometric validation to verify the passenger's ID. At this point, a digital template of the passenger's face biometrics is activated in the airport IMP, and a single token combining this digital template with the PNR is created. This single token, also called the "passenger dataset," grants the passenger seamless and paper-free accessibility through the rest of the biometric-enabled processing points. In case of no match or if the passenger wishes to opt out, a manual verification of the travel document and identity is performed at each processing point.

A passenger's data storage and retention vary by each database. On the one hand, the Digi Yatra ID will exist indefinitely after activation. The passenger can update his/her profile if

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necessary and can choose to opt out at any time. On the other hand, the passenger dataset stored and retained in the airport IMP will be deleted 1 hour after the corresponding flight departs.

The Digi Yatra trial at BLR showed promising results as passenger processing times at biometric-enabled processing points considerably decreased compared to the manual process. The airport is aiming toward having 60% to 70% of passengers using the biometric-enabled journey. Once this goal is reached, the true magnitude of the reduction in processing times and staff count will be fully visible and measurable.

At the time of writing, due to the COVID-19 pandemic, the Digi Yatra ID platform's deployment has been delayed, and a manual validation process is in place. BLR is exploring the possibility of trialing another option, referred as a "self-sovereign" approach. Under this model, the passenger would control what personal data he/she wants to share and with whom. The passenger would complete the enrollment process using a government-issued validation number and would then take a selfie, unlock the e-passport chip (via the same app), and combine this information into a digital token that he/she can share with the airline and with the airport.

Key Facts

Table I-1 summarizes the key aspects of the seamless passenger journey at BLR.

Table I-1. Key facts of the case study on Digi Yatra.

What?	• Digi Yatra: seamless, biometric-enabled, paperless journey
Where?	• BLR, Bengaluru, India
Passenger process steps	• Outbound domestic passenger journey (curb to gate) (to be expanded to international flights at a later stage) • Enrollment kiosk • Airport entry e-gate • Check-in/bag drop • Security entry • Emigration (at a later stage) • Boarding e-gates • In the aircraft (tablets with biometric identification) (at a later stage) • Immigration (at a later stage)
Who?	• Digi Yatra is an initiative of the Ministry of Civil Aviation in partnership with the major privatized Indian airports and the International Airports Authority of India (IAAI). The common Digi Yatra ID platform, once operational, would enable the automated validation of the passenger's identity. The platform has not been implemented yet (due to COVID-19), but BLR has biometrically enabled several of the touchpoints listed previously and linked them into its own localized ID management platform. The ID validation process will be performed manually until the Digi Yatra platform is operational or a different approach, such as the self-sovereign ID management approach, is implemented. • The initial BLR trial was launched in partnership with Vistara Airlines. The selected vendor was Vision-Box, which provided the airport ID biometric platform technology (Orchestra). After Vistara Airlines, AirAsia India and SpiceJet were added. The program is in the rollout stage (albeit temporarily on hold due to COVID-19) and will include all domestic airlines. (IndiGo was the last remaining domestic carrier to be included.) The program will be expanded to international flights at a later stage.

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Table I-1. (Continued).

Why?	• The Ministry of Civil Aviation set several goals for Digi Yatra: ○ Delay costly airport capital expansions by looking at innovation and digitalization to improve efficiencies and reduce costs ○ Provide a better and easier passenger experience ○ Enhance security standards ○ Back up the program with a strong, verifiable government-issued identity ○ Provide a common identity management system that would work across all domestic airports in India and that could be used by all domestic and international passengers ○ Use Digi Yatra as a mean of communication with all enrolled passengers
How?	• The Common Digi Yatra ID platform was conceptualized by a joint venture between the government (with majority stake), the Airports Authority of India (AAI; with minority stake), and all private airport operators. The Common Digi Yatra ID platform offers core passenger services such as enrollment, authentication, and consented profile sharing. Passengers have to enroll once using a valid government-issued ID. The local airport identity management system uses the facial biometric data provided by the Digi Yatra ID platform to identify the passenger for all the processes inside the airport. The facial biometric data are purged from the airport system 1 hour after the passenger's flight takes off. • Until the Digi Yatra platform becomes operational, the identity check is manually performed.
Enrollment/digital identity creation and verification	• Passengers can enroll in Digi Yatra online by providing a valid government ID (preferably the Aadhaar number, which is a 12-digit unique identifier that the government issues to all its citizens after biometrically enrolling them into this countrywide program). Passengers obtain a personal Digi Yatra ID that has permanent validity. The activation (validation) process is performed once online or at an airport registration kiosk (where a photo-of-the-day is taken and the identity is validated against the government database accessible through the Aadhaar number). In all subsequent trips, this step is not needed if the passenger, at the time of booking, includes his/her personal Digi Yatra number to his/her reservation.
Verification of identity how?	• If the passenger uses his/her government-issued Aadhaar number, the verification and activation are performed once by matching the photo-of-the-day with the government-held biometric data, which is accessible through the Digi Yatra ID platform. If another form of government-issued ID is used (other than the Aadhaar number), the ID has to be manually checked at the airport to complete the activation process.
For?	• All domestic flights on Vistara Airlines, AirAsia India, and SpiceJet. To be expanded to IndiGo and all international flights (ET Government 2020; Business Traveller India 2020; Press Trust of India 2020).

Introduction

Aviation in India has been growing at a considerable pace, and India is expected to become the world's largest domestic civil aviation market in the next 10 to 15 years. More emphasis has been placed on investing in innovation and digitalization to keep up with this growth.

To this effect, the Ministry of Civil Aviation has started Digi Yatra in partnership with the privatized Indian airports and the Airports Authority of India (AAI). This initiative aims to transform the passenger experience at all Indian airports by reinventing airport terminals and explicitly rejecting the conventional approach of building bigger airports to manage more passengers.

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Specifically, Digi Yatra makes use of biometrics and identity management technologies to allow a seamless, paper-free journey (Awtaney 2019). As one of Digi Yatra's stakeholders, BLR became one of the first Indian airports to trial the concept. Passengers only need to show their face to gain access to all terminal processing checkpoints, which eliminates the requirement to repeatedly show travel and identification documents (Devaiah 2019).

How Does It Work?

Before the Passenger Journey

The Common Digi Yatra ID platform is an interface platform between the countrywide biometrics database owned by the Unique Identification Authority of India (UIDAI), under the Ministry of Electronics and Information Technology and the local airport ID management platform (Bhavan 2018b).

Specifically, the three platforms are:

- UIDAI platform: Countrywide database where Indian residents' demographic and biometric data are held; UIDAI is responsible for enrollment, updates, and authentication; any Indian resident can enroll by providing proof of identity and proof of address; as part of the enrollment process, at the enrollment center, a photo, fingerprints, and iris scans are taken; once the authentication is performed (within 60 to 90 days), a unique number, the Aadhaar number, is issued; the number has indefinite validity and can be used to validate identity; obtaining an Aadhaar number is not mandatory.
- Digi Yatra ID platform: Countrywide airport system database where the demographic and biometric identity data are linked to travel bookings; the enrollment is completed online (valid indefinitely); the activation is done once at an airport registration kiosk (if the Aadhaar number is used, that constitutes a valid proof of identity; if there is no Aadhaar number, the activation process is performed by a manual validation of a government-issued ID document); the validation is valid for 5 years; at the time of booking, the booking reservation number is linked to the Digi Yatra number and to the demographic and biometric data in a single ID token; this linkage is possible because of the integration of the airline's DCS with the Digi Yatra ID platform (Government of India 2020).
- Airport ID management platform: BLR uses Vision-Box's Orchestra ID management system to validate identity at the biometric-enabled passenger processing points (entry e-gates and all other downstream processing points); only the photo capture (biometric data) and the PNR are shared between Digi Yatra and the airport platform (Burt 2019).

Due to COVID-19, the Digi Yatra ID platform is not operational, and therefore automated identification validation is not possible, and the PNR is not automatically shared with the airport platform. Therefore, at the time of writing, identity validation was performed manually at the registration kiosk, and the PNR data capture was done at the airport e-gate by having the passenger scan his/her boarding pass.

BLR is exploring another option that would empower the passenger to digitally share his/her biographical and biometric data when required via an app. This approach is broadly known as the self-sovereign ID management approach. The passenger would complete the enrollment process using his/her Aarhaad number and would then take a selfie, unlock the e-passport chip (via the same app), and combine this information in a digital token that he/she can share with the airline (during the booking process) and with the airport (to access the biometric-enabled processing points). BLR is in the process of exploring and potentially trialing this option.

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Passengers complete the online Digi Yatra enrollment process and are issued a Digi Yatra number. The enrollment has to be completed only once. At the time of booking, passengers are asked to include their personal Digi Yatra number in their booking reservation, which is then transmitted by the airline DCS to the airport ID management platform; the PNR number on the boarding pass is used as the key to extract the passenger's Digi Yatra ID number from the airport ID Management platform at the time of validation at the airport entry e-gate (Khadakbhavi 2020). See Figure I-1 for a description of the Digi Yatra concept.

The Passenger Journey

The registration kiosk: The passenger scans his/her boarding pass (printed on paper or from his/her mobile app); the kiosk camera captures the passenger's biometrics (face and iris for Aadhaar validation); the passenger's consent is given to use his/her facial biometrics for other airport processing points; the Digi Yatra ID platform validates the boarding pass with the airline DCS and verifies the identity of the passenger with real-time biometric validation within the Digi Yatra ID platform (or, if this is the first time, the Digi Yatra ID platform performs a one-time validation of the passenger with the Aadhaar system using facial and iris biometric data); the passenger's facial biometric digital template is activated in the airport ID management platform with the creation of the passenger dataset, which includes the facial biometric digital template and the PNR for identification at the other biometric-enabled processing points at the airport; the passenger dataset is associated with a unique identifier; the facial biometric digital template is uploaded to the Digi Yatra ID platform.

Airport entry e-gate: Having completed the registration process, the passenger can now access the airport entry e-gate; the passenger is identified by the passenger dataset, and the e-gate opens.



Notes: PAX = passengers; DY = Digi Yatra; PESC = pre-embarkation security check. The immigration and in-the aircraft processing points will be incorporated in the journey at a future stage.
Source: Bhavan 2019.

Figure I-1. The Digit Yatra journey concept.

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Other biometric-enabled processing points: The passenger is identified by the passenger dataset.

In the case of no match or if the passenger wishes to opt out, a manual verification of the travel document and identity is performed at each processing point.

Retention and Storage

Storage and retention of the passenger data varies by database:

- Digi Yatra: The passenger data are retained indefinitely or until the passenger elects otherwise; the passenger can update and modify the information in the Digi Yatra profile if needed.
- The passenger dataset (photo and PNR) is stored and retained in the airport ID management platform only until 1 hour after the flight departs. After that, all data has to be deleted.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Digi Yatra and the Seamless Passenger Journey at Kempegowda International Airport, Bengaluru, India, case study in Chapter 2.

Stakeholders and Responsibilities

Stakeholders

This initiative is a partnership among the Ministry of Civil Aviation, the privatized Indian airports, and the Indian Airports Authority. BLR was one of the first airports to trial the concept (in 2017). The launch airline was Vistara Airlines, but the program was quickly expanded to include other domestic airlines, such as AirAsia India and SpiceJet. Other airlines, including international carriers, will join the program.

Responsibilities and Governance

The Digi Yatra program's governance can be summarized as follows (Khadakbhavi 2019):

- The Secretary of Ministry of Civil Aviation is in charge of the overall initiative.
- A steering committee (Digital Cell) composed of the CEOs (including BLR executives) of major Indian public–private partnership airports, AAI, the Bureau of Civil Aviation and Security, and the Central Industrial Security Force.
- A technical working committee (TWC) that is made up of subject-matter experts from the participating airports, including staff from BLR and the BLR Innovation Lab.
- The TWC conducted a series of workshops with all stakeholders, including airlines, online travel agents, regulators, airports, and the UIDAI, to explain the concept of Digi Yatra and gather consensus among all stakeholders.

Timeline and Planning

In 2016, as part of a new terminal development program, BLR staff conducted a series of workshops to discuss the future of the passenger journey and how this could be improved by use of a single biometric token. The BLR Innovation Lab further developed the concept and explored different types of biometrics (e.g., facial, fingerprints, and iris).

Procurement and Vendor Selection

Once BLR developed the concept for operation, a selection process involving several leading biometric service providers was conducted, and the preferred vendor, Vision-Box, was selected.

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The trial went live on a Vistara Airlines flight in January 2017 (Mayhew 2018; Vision-Box 2018a; Vision-Box 2019).

Case Study Review

Benefits

The benefits for the various stakeholders are highlighted in Table I-2 (Khan 2020).

Responses from Passengers

The feedback from passengers was positive. The majority found the process easy to understand, and they appreciated the convenience of not having to show the same documents multiple times. Passengers also noticed that queues were shorter and that processing times faster. Non–tech-savvy passengers could also successfully use the system because of its simplicity.

System Performance and Specifications Review

One of the main challenges had to do with the ambient lighting that affected the cameras and resulted in poor-quality images at some of the processing points. These images could not be used for matching purposes. This problem was addressed with manual intervention. As image capture technology and image enhancement algorithms continue to improve, this challenge should be overcome.

Fall-Back Options

If a passenger is not enrolled in the Digi Yatra program or if the authentication process for some reason does not work (no match), the passenger is processed manually at each processing

Table I-2. Digi Yatra and the seamless passenger journey: stakeholder benefits.

Stakeholder	Benefits
Passengers	Enhanced experience with a more seamless journey; no need to show boarding pass or ID at multiple processing points Faster processing and reduced queue times Same program across Indian airports
Airport	Increased passenger processing throughputs postponing the need of costly capital expansion programs Enhanced passenger experience and simplified processes accessible to all, including less tech-savvy passengers Better resource planning by knowing passenger loads and locations Reduced staffing costs as fewer passengers needing to be manually processed
Government	Enhanced security: minimize ID fraud, reduce the risk of allowing non-travelers into the terminal Scalability of the program to all airports across India Leveraging of the countrywide UIDAI Aadhaar database to make identification more secured and streamlined Reduced staffing costs as fewer passengers have to be manually processed
Airlines	Better resource planning Improved on-time departures through knowing where passengers are within the terminal Redeployment of staff where most needed

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point (valid government-issued ID and boarding pass are checked at each checkpoint). The passenger is still processed through the airport entry e-gates, albeit manually. Some of the terminal entry points do not have entry e-gates, and passengers with disabilities or others requiring assistance can use staffed entry points.

Concerns

To respond to privacy concerns, privacy considerations were addressed early on and incorporated into the concept development process. The techniques implemented to achieve data privacy by design included authentication protocols, verification of credentials, encryption, data minimization, privacy enablers in databases, data masking, secure data storage, and transparency-enhancing techniques.

The system is fully GDPR compliant, and all passenger data and travel records are purged from the airport ID management platform 1 hour after the flight departs.

In addition, Digi Yatra policy prevents airports from creating a profile of users and using that for marketing purposes without passenger's consent. Once enrolled, Digi Yatra users can opt out of the system, and, they can delete their profile at any time.

Lessons Learned

Understanding how passengers interact with the cameras and the equipment is fundamental to achieving positive outcomes. For example, passengers traveling in groups could crowd around the device, which would prevent the system from properly working. At BLR, the processing of passengers, even if traveling in groups, is done one at a time and, therefore, having more than one person standing in front of the camera creates matching issues. This was resolved by signage and by deploying adequate staff to coach the passengers along the journey. As passengers become more familiar with the technology, some of these challenges will be overcome.

The COVID-19 pandemic has accelerated the need to implement touchless technology and minimize the manual handling of IDs and travel documents. Touchless technology had not been the focus of the effort because some of the touchpoints still require some interaction (e.g., registration kiosk, bag-tag printer). Going forward, touchless solutions at all processing points will be further explored and implemented.

Findings and Trends

Findings

The decrease in passenger processing times is considerable, especially at the airport entry e-gates where there was a decrease of approximately 50% compared to the manual process. The boarding e-gates are also registering faster processing times. The airport goal is to have 60% to 70% of passengers using the biometric-enabled journey. Once this goal is reached, the true magnitude of the reduction in processing times and staff count will be fully visible and measurable.

Future Situation and Broader Implementation

It was anticipated that by the end of 2020, all passengers traveling on domestic flights would be able to use all the biometric-enabled touchpoints. The program will be extended to international flights, airline by airline, starting in 2021, assuming the inherent complications associated with immigration processes can be resolved (Lakshman 2020; OT Staff 2019).

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The Digi Yatra common ID platform implementation was delayed due to the global COVID pandemic. As Indian airports reprioritized their investments during these uncertain times, it was unclear as to when the program would resume. Before the crisis hit, the design phase was completed and the RFP to select the vendor had been prepared. Therefore, the reactivation of the program, once the decision is made, should be quite straightforward.

In the meantime, BLR has been exploring another identity verification solution that would not require the Digi Yatra platform and is based on the concept of self-management of digital personal data to empower each passenger with regard to what personal data to share and with whom. The passenger would enroll by using an identity app on his/her phone to take a selfie to verify biometric and biographic information by reading the chip on the e-passport or by using the Aadhaar number and the user-specific password generated by the UIDAI. At the time of booking, this app interfaces with the airline app to combine the biometric and biographic data with the PNR and generate a travel credential that can be shared with other parties, such as the airport and the government (if, in the future, border processing is included in the seamless journey). At the airport entry e-gate, the passenger, by showing his/her face and scanning his/her boarding pass, gains access if there is a positive match in the local airport IMP (i.e., if the PNR is valid and the passenger is who they say they are). The remaining processes are the same as before. This concept is still in an exploratory phase, and more work has to be done to make it a reality.

Trends Identified

BLR staff identified the following trends during the planning and implementation of the seamless journey:

- The vast majority of passengers have embraced this technology and appreciate the benefits that it provides; this trend is in line with the latest IATA Global Passenger Survey (IATA 2019).
- The success of this initiative is due to the adoption of a collaborative approach since all key stakeholders were invited to participate from the early conceptual stages.

Self-sovereign ID management is probably the best way forward since it empowers individuals to decide with whom and how to share their personal data.

Note: Source used throughout this appendix: Future Travel Experience 2019b.

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APPENDIX J

Case Study: Happy Flow at Aruba's Queen Beatrix International Airport

Summary

Happy Flow is aimed at improving the departure process at Aruba's Queen Beatrix International Airport (AUA) by smoothing the flow of passengers and creating a unique and seamless passenger experience. The program adopted protocols and standards that would benefit authorities, the airport, and airlines.

Using facial recognition at multiple passenger touchpoints, passenger identity and right to travel are verified more securely. Plans are underway to include off-airport elements in the passenger journey (e.g., hotel check-in and car rental). In the initial launch of Happy Flow, there were also ideas to preclear into the European Schengen area.

Travel documents are only required at the enrollment station. There, the passenger's identity is securely checked, and a virtual identity is created. After enrollment, the passenger goes through self-service passenger touchpoints (bag drop, security, border control, and boarding) where the passenger's face is matched against a secured database, and only authorized passengers are allowed to pass.

As with other facial-recognition initiatives and programs, the intent of Happy Flow was to promote a more streamlined departure process while improving overall security. Enrollment is the most time-consuming step, while processing is efficient after the biometric token is created. The system can be further optimized by introducing one-time enrollment, such as DTC, as well as measures to counter identity fraud from off-site/mobile enrollment. See Table J-1 for more information.

Introduction

In 2012, the government of Aruba, the Ministry of Justice in the Netherlands, AUA, KLM, and the Schiphol Group agreed on starting an innovation project aimed at improving the passengers' experience, smoothing passenger flow, and improving processes at AUA. The program was to adopt protocols and standards that would benefit the authorities, the airport, and the airline, and potentially even bring relief to the immigration process at Amsterdam Airport Schiphol for arriving flights from Aruba, but most of all would create a unique and seamless passenger experience.

The improved passenger experience has been the shared driver of the workgroup throughout the years and turned out to be the vital element in innovation and realization. This shared goal, independent from individual stakes and interests, created a shared sense of responsibility and motivation. Along the passenger process, each stakeholder benefited directly or indirectly from

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Table J-1. Key facts of case study on Happy Flow in Aruba.

What?	• Seamless passenger process using facial biometric matching
Where?	• AUA, Aruba
Customer process steps	• Process is the predecessor of IATA Seamless flow (OneID) • Departures: – Biometric enrollment per flight (verify passenger ID + boarding card) – Baggage drop-off with biometric assistance (counter with camera for walking-pace facial recognition that pulls up PNR – Biometric self-service border control checkpoint (exit check) combined functionality with airside check (reservation and travel document) – Duty free and airside terminal area (conventional, no applications of Happy Flow) – Boarding gate with separate Happy Flow lane (with camera for facial recognition)
Who?	• Aruba Airport Authority N.V. with KLM Royal Dutch Airlines and the government of Aruba. Supported by the Ministry of Justice in the Netherlands, KMAR (Dutch border police), Royal Schiphol Group, and Vision-Box.
Why?	• To improve the passenger experience, save time, and make the passenger process more efficient for the airline and airport
How? Technology used	• Verification of the ID and facial imagery takes place at the biometric kiosk; therefore, front loading and splitting the border control process • Identity verification at the kiosk by authentication of the e-passport and biometric verification of the holder and the biometric on the e-passport under government control • Biometric ID verification and exit control at border control • Privacy-by-design principles/GDPR were adhered to from the start of the project.
Enrollment/digital identity creation and verification	• Enrollment per departing flight at the self-service biometric kiosks when entering the terminal • Check-in through airline's systems conducted online or via CUSS kiosks • At the seamless flow kiosk: – Step 1: verify identity with biometric data on passport and authenticate passport – Step 2: scan boarding card at kiosk – Step 3: create a digital ID and temporarily store in a passenger data envelope with boarding card details
Verification of identity how?	• First by machine check of the authenticity of the passport in accordance with Frontex automated border control (ABC) standards, then by comparing the image stored on the passport with the person in front of the enrollment kiosk
For?	• KLM departing passengers to AMS (16+) with e-passport valid for participation [EU/European Entrepreneurial Region (EER) nationals, U.S. and Canadian nationals]

these improvements due to improved bilateral ties, increased passenger satisfaction, reduced queues and inefficiencies, and relieved pressure from authorities' operations. This resulted in the Happy Flow process, the name of which hints at the island's slogan, "One Happy Island," and the natural flow that was created. With the introduction of Happy Flow, the world's first biometric seamless passenger process was introduced. In the future, the concept is to be broadened to include mobile enrollment, the arrivals process of more airlines, and off-airport services such as for hotels and car rentals.

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How Does It Work?

Before the Passenger Journey

On arrival at the airport, the passenger can decide to participate in the Happy Flow program (Aruba Happy Flow 2020). On enrollment at the airport at a biometric kiosk, the biometric data of the passenger, including PNR and boarding card details, are stored in the PDE. All PDEs are stored in a database located at the airport. The Aruban border police receive all passport data for verification. The airline's DCS will receive the boarding card details to register boarding.

The Passenger Journey

Passengers that are over 16 years old, holding an e-passport, and traveling on the daily KLM flight from Aruba to Amsterdam are allowed to use the biometric systems provided at AUA in the non-U.S. departures area. Participation is optional; passengers can also follow the conventional process.

Biometric Enrollment

On entering the main terminal building, passengers on the KLM flight get the choice to opt for the Happy Flow process. Provided that the passenger has passed online check-in, the passenger can enroll at the Happy Flow biometric kiosks for that day's flight.

Passengers insert their e-passport in the Happy Flow kiosk, where the authenticity of the passport is verified, and the digital image of the passport holder is read from the passport chip by the machine. The built-in camera of the kiosk captures an image of the person in front of the camera and confirms his/her identity by matching it with the image from the passport. The verification of the passenger's identity is done in the same manner as at an automated border control gate in accordance with the European Frontex ABC standards. This requirement is the foundation for the federated identity management system and is an important part of the arrangement between the stakeholders.

After confirmation of the passenger's ID, its details are matched against the law enforcement databases to verify if the passenger is eligible to leave the country. This is checked within seconds after enrollment, and the status is recorded in the system. In the passenger process, this is well in advance of the actual border control checkpoint, leaving ample time for border control agents to respond to passengers of interest. In fact, the border control procedure is now divided in two stages, which creates potential new use cases for border control operations.

At the enrollment kiosk (see Figure J-1), the passenger also has to scan the boarding card. When the time and date (of the flight reservation) are correct, the boarding card details are stored in the PDE of the seamless flow database together with the biometric template. When both verifications are positive and the document is authenticated, the biometric image that is captured of the passenger's face is stored in the seamless flow database and serves as the single token for the successive steps in the Happy Flow process.

Facial-recognition cameras are used for all the next steps in the passenger process, essentially digitalizing and replacing the need for hardcopy documents (ID and boarding card). The passenger's face serves as a single token. At every biometric checkpoint, according to privacy-by-design principles, only the minimum information from the PDE that is required for that specific process step will be shared with the system and stakeholder. As a result, passengers are no longer required to show their travel documents as is the case with the conventional passenger process. This leads to a quick and hassle-free process where at every step the passenger only needs to show his/her face.

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Source: Aruba Happy Flow 2017.

Figure J-1. Image of a biometric enrollment kiosk for Happy Flow.

The passenger's data stored in the seamless flow database include the passport details, boarding card, and biometric template of the passenger's face. The data required for the passenger journey are stored by the seamless platform for less than 24 hours and are then deleted. The passenger needs to enroll for each new trip. The passport details and facial images are saved in a separate database of the immigration authorities for a longer term.

Baggage Drop-off Based on Biometric Match

After enrollment, the passenger proceeds to one of the separate hold baggage counters dedicated to Happy Flow. A camera integrated in the desk is directed to the passenger approach flow situated in front of the desk. When a passenger comes in view, facial-recognition software will recognize the passenger's face and confirm the corresponding digital ID.

As the passenger will be recognized while approaching the desk, the staff at the desk already get the PNR presented with the required travel documents at the workstation. Once passengers are at the desk, they simply hand over their baggage and can continue to the next step in the process. The airline staff at the desk can insert the number of bags, print the label(s), and tag the bags.

Border Control Checkpoint (Emigration/Exit Check)

As passengers make their way through the terminal, they arrive at the border control checkpoint, where separate Happy Flow lanes are present. These lanes are designed as mantraps, with barriers on entry and exit. Inside, the lanes are equipped with cameras situated at the end of the lane that can move vertically inside a console to find the right angle to recognize and confirm the passenger's ID. When a passenger is recognized and authorized, the e-gate will open (see Figure J-2). The lanes are fully automated and able to process passengers within

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Source: Aruba Happy Flow 2017.

Figure J-2. Biometric e-gates in the Happy Flow passenger journey.

approximately 3 seconds. Ultimately, the lanes at AUA should be able to speed up the biometric identification and authentication process to the time required for the passenger to pass through the lane at walking speed. This would eliminate the static moment between the barriers (therefore avoiding queues and retaining a seamless flow).

In case the camera is unable to recognize and confirm the passenger's ID, the passenger is requested to step out and follow the regular facilitation process next to the Happy Flow lanes. In case a passenger was flagged with a law enforcement or immigration issue, the immigration office will have been notified through its Happy Flow dashboard. The immigration officers will intercept this passenger at the border control exit checkpoint and take action appropriate to the issue. If the issue is resolved, the border police will change the flagged status of the passenger, and the passenger can continue.

For this step, the border control checkpoint is also combined with the entry to the security filter. To enter the security filter (or airside access since the actual security filter is not directly annexed but is a short walk away), a valid boarding card is required, which was already confirmed on enrollment.

Boarding Gate with Biometric Lane

At the gate serving the KLM flight to Amsterdam, a dedicated Happy Flow biometric lane is provided with similar and recognizable features to those in the previous checkpoints and much like the border control checkpoint but without the mantrap feature. The lane is fully automated and equipped with a moveable camera and access barrier at the end. As with the security checkpoint, the recognition and identification processing take place within a few seconds, after which the barrier opens and the passenger can board. The processing speed of the biometric lane is deliberately reduced to avoid queues in the boarding bridge.

In addition to these features, the e-gate has an interface with the airline's DCS in order to register the actual boarding of the passenger. The automated boarding lane is considerably faster compared to using conventional manual actions, which keeps queuing and standstills to a minimum.

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Retention and Storage

Because KLM is a European company and EU citizens' data are processed, the EU privacy regulations (GDPR rules) need to be adhered to. This means that the entire platform was developed based using the concept of privacy by design. The digital ID that is created during enrollment is encrypted and stored in a local database for temporary use by the airport, airline, and authorities. The data are stored for less than 24 hours and are deleted directly after. When the data are shared between stakeholders, each stakeholder only receives the fields of data that it is authorized to have and that are required for the purpose of that specific task:

- The Aruban Immigration Authority receives all data that are incorporated in the embedded chip of the e-passport for the task of the border control process.
- The airline receives only the data that are required to be registered in the flight manifest via the DCS or to pull up the PNR at baggage check-in.
- The airport receives only data that are de-personalized as it is not allowed to register any personally identifiable information (GDPR).

The data relevant to the Aruban Immigration Authority and law enforcement agencies (facial biometric details captured and passport data) are encrypted and saved permanently on a dedicated database.

System Architecture Flow Diagram

The flow diagram of this case study can be found in the Happy Flow at Aruba's Queen Beatrix International Airport case study in Chapter 2.

System Specifications

The pilot with KLM Royal Dutch Airlines is based on a per-trip model. The system as installed at AUA is built up of the following components:

- Biometric enrollment kiosk,
- Biometric baggage drop-off desk,
- Biometric border control lane,
- Biometric boarding control lane,
- Biometric template database,
- (Biometric) data-management system orchestration via business rules,
- Facial data recognition software,
- External database interfaces (like the connection with authorities' databases that have long-term storage of person's details), and
- Local database for temporary storage (less than 24 hours).

The overall system consists of physical passenger processing devices that are connected with the databases and systems through a dedicated and secure IP network. The operating system and hardware are supplied by Vision-Box. The operating system is at the center of the hardware and software components and manages all data flows. The operating system of Vision-Box also manages the business rules for decision making at the various checkpoints.

The operating system uses cameras at each part of the process (check-in/enrollment, baggage drop-off, border checkpoint, boarding) where identification of the passenger is required. Wayfinding and routing for Happy Flow throughout the terminal between these parts is passive and is done via signage.

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Biometric Boarding Lane and DCS Interface

The biometric lane at the gate operates as a standard common-use terminal equipment (CUTE) e-gate and has a data connection with the biometric database. As soon as the passenger's identity is recognized, the boarding card details are sent to the airline host DCS. When the passenger and boarding card are verified and approved, the airline host sends a signal to the gate that it can open, and the passenger is registered as boarded.

The lane also contains a standard boarding card reading device and, through the link with the DCS, non-enrolled passengers can also be processed.

Stakeholders and Responsibilities

Aruba Airport Authority N.V. is responsible for the biometric facilities at the airport and the availability and reliability of the equipment.

KLM Royal Dutch Airlines is responsible for the use and operation of the enrollment kiosks, the biometric check-in system, and the biometric boarding lanes.

The Aruban Immigration Authority is responsible for the use and operation of the biometric immigration lanes.

Case Study Review

Benefits

The Happy Flow pilot was introduced on the daily KLM flight with an Airbus A330 that uses two circle routes. The first is a round trip from Amsterdam to Bonaire to Aruba and back. The other is opposite and flies to Aruba first and back to Amsterdam via Bonaire. Consequently, about half of the passengers originate from Aruba, making up 150 passengers on average per day.

On this daily flight, between 100 and 130 passengers enroll and use Happy Flow, with an average participation of approximately 80%. Enrollment was open to all age groups, but after implementation was limited to adolescents and adults (16+ years of age) due to the fact that Aruban law requires verification by immigration authorities that minors are leaving the country with the permission of both parents.

The ultimate goal to improve the passenger's experience was met and, in the process, the implementation of Happy Flow using biometrics improved passenger satisfaction, reduced queues and inefficiencies, and relieved pressure on authorities' operations.

Responses from Customers

During passenger feedback research conducted by KLM, passengers indicated that they:

- Experienced less stress;
- Liked to be in control of their journey (rather than having to join the queue);
- Enjoyed the usability and decreased passenger processing times; and
- Still felt some concern over privacy, data storage, and data ownership.

KLM indicated that, although passengers were concerned about their privacy, they perceived KLM as a reliable airline and also expected it would deal with their data in a reliable way.

Some passengers indicated that they perceived the biometric process as extra secure and felt that only people who had something to hide would not want to enroll. Passengers from countries with stronger privacy concerns were more reluctant to enroll.

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Systems Specifications Review

The system is designed according to the Frontex ABC standards (EU). No actual information is available regarding the performance of the system (false rejection rates). The processing speed of gates at boarding is approximately 6 seconds.

Fall-Back Options

Passengers who did not enroll in Happy Flow traveled only with hand baggage (and were checked in online) or were families with children under 16 years old. The first group would simply pass by the enrollment kiosk as they headed directly to the first touchpoint in their journey: border control. Therefore, they passed through the terminal as quickly as possible. Families with children under 16 years old simply could not enroll and needed to check in together.

There are also passengers who travel with considerable baggage. They spend more time at the baggage drop-off and check-in desks, and the efficiency savings with Happy Flow would be more limited. The conventional passenger processing facilities are still in place, so passengers who do not enroll in Happy Flow are able to use the conventional processes.

Concerns

One of the obstacles experienced during the pilot was that the passport image quality stored on the e-passport chip varies per nationality. Some of these are of inadequate quality to perform a reliable match with the person at the enrollment kiosk. As a result, identity verification of these passengers cannot be performed at the required level of accuracy, and these passengers had to use the conventional passenger process.

Another challenge is the sensitivity and calibration of the touchpoint equipment at enrollment, baggage drop-off, border control, and boarding. The cameras for facial recognition require consistent lighting (with certain margins) on the passenger's face across the day. This may pose challenges at some gates due to the terminal design in terms of natural light. This can be solved with artificial lighting and adjustments of the camera view angle.

Lessons Learned

Many lessons were learned from the Aruba Happy Flow project, but the most important was that a cross-border public–private data sharing endeavor is possible even when including border authorities. This would not have been accomplished without the trust that the international stakeholders placed in each other and the openness of sharing each other's business interests. This led to a joint definition of the requirements and operational goals for the pilot and the development of a trust framework between parties. Ultimately this resulted in a system that was designed in accordance with privacy-by-design, EU GDPR, and Frontex requirements. This paved the way for cross-border interoperability and developments of international program such as OneID and ICAO DTC (Steenbergen 2020).

The project proved that it involved much more than just access control at the airport. Multiple stakeholders can benefit from efficiency gains and offer more and better service to passengers with the same system. KLM, for example, likes to have its passenger check-in process via its own user interface so that it can still make last-minute adjustments or ancillary sales (Knoppers 2020). At the time of the pilot, it was not possible to incorporate check-in with biometric enrollment at one kiosk. It is expected this could be incorporated in the next generation of the system. Passenger-centric thinking also sparked initiatives to include hotels and car rental companies in the program in the next generation of the system. Along the same line, online health registration of test results for COVID-19 or proof of vaccination could well be incorporated.

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Findings and Trends

Happy Flow catalyzed developments for automation and the use of biometrics in airport passenger processes. These developments range from international standards and recommended practices, professional associations and working groups, legislation, politics, technology, and research and innovation. AUA has not stood still since and has moved to the forefront of these developments and is planning to continue to do so in the future.

Future Situation and Broader Implementation

The Happy Flow concept has developed into a 2.0 version that is ready to be rolled out in the near future. This new version aims to process all airlines and be able to interface with any airline booking and departure control system. It is also the intention to add non-air stakeholders like hotels and rental car companies to the Happy Flow platform. The use of a (self-sovereign) trusted digital identity will be key to these future developments that will be in line with international standards such as ICAO DTC and IATA OneID. This would enable home enrollment and would reduce the required enrollment time at the airport.

AUA was one of the first airports to establish U.S. preclearance facilities. With large volumes of traffic to the United States and CBP opting in to biometric passenger processing, AUA also has ambitions to facilitate biometric processing for U.S.-bound traffic.

Trends Identified

Process facilitation: The airline and airport are seeking to have single (multiple-trip) enrollment that supports international interoperability such as with ICAO DTC or KTDI. This to avoid the need for (time-consuming) enrollment facilities at the airport and to increase overall efficiency.

Commercial services: The airport and airline are also seeking to introduce more stakeholders to the program that can provide services with the use of the passenger's biometric token. The trends are to provide more biometric-enabled services during the passenger journey and improve the passenger experience and non-aero revenues.

Fraud and risk management: The protection of the passengers' data is considered a basic requirement. It is expected by the passenger, and passengers trust the data with the airline in relation to the brand image. This means that secure data protection is important to help establish the brand value. In addition, passengers feel more secure knowing that a trusted system is in place that verifies the background of all passengers on a flight.

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APPENDIX K

Legal, Policy, and Privacy Review

The Supreme Court has recognized the common law root of the right to privacy.[113] There are four well-established common law privacy torts: (1) unreasonable intrusion upon someone's seclusion, (2) appropriation of a person's name or likeness, (3) unreasonable disclosure of private facts, and (4) publicity that unreasonably places another in a false light.[114] The factual scenario most likely implicated is the unreasonable intrusion upon someone's privacy, but even this is unlikely unless facial-recognition technology is deployed in circumstances where the intrusion is "especially private" (e.g., a bathroom).[115]

The Supreme Court has decided that while not expressly set out in the Constitution, privacy protections are implicit or within the "penumbra" of several provisions of the Constitution, to include the prohibition of unreasonable searches and seizures under the Fourth Amendment, and other amendments under the Bill of Rights.[116] In addition, while beyond the scope of this report, use of facial recognition combined with public surveillance can implicate First Amendment considerations (Zick 2007).

With respect to the issue of identification, the Self-Incrimination Clause in particular is implicated, but it does not protect against the compelled collection of physical evidence, such as blood, DNA, or fingerprints, or its introduction at trial.[117] Rather, the evidence sought must be testimonial and incriminating, and the Supreme Court has held that the disclosure of one's identity "is likely to be so insignificant in the scheme of things as to be incriminating only in unusual circumstances."[118] Thus, even without an individual's consent, the collection and disclosure of biometric data, which is neither testimonial nor incriminating, arguably does not violate the Self-Incrimination Clause, but recently lower courts have been split on the issue.[119]

Although the Supreme Court has not considered whether the specific use of facial-recognition technology might violate provisions of the Constitution, it has rendered decisions on the use of technology.[120] For example, in *Kyllo v. United States*, the Supreme Court held that the use of thermal imaging technology to determine the amount of heat within a person's house (presumed to be associated with marijuana cultivation) was an unreasonable search, given the heightened privacy expectation associated with one's home.[121] It may be argued, however, that the Court was more concerned with where the search occurred rather than the technology used to conduct the search (Haas 2019).

In subsequent decisions where the facts focused on the use of technology for the collection of location data or cell phone information, the Supreme Court rendered decisions that eroded in part the doctrine that Fourth Amendment protections did not extend to public spaces or information received from third parties. For example, in 2012, the Court decided that the installation of a GPS tracking device without a warrant violated the defendant's Fourth Amendment rights, holding that the physical installation of the device constituted an unlawful interference with the suspect's property interests in the vehicle.[122] Six years later, the Supreme Court

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decided that the government's acquisition of information from a third party on the defendant's cell-site location was a search under the Fourth Amendment for which a warrant supported by probable cause was required.[123]

Private Parties

As noted in Chapter 3, where a private party acts as an agent of the government, that conduct may implicate Fourth Amendment principles pertaining to searches. Examples where the courts have found that a search by a private party was conducted as an instrument of the government include where an airline employee, who had been an informant in the past and financially rewarded for his assistance, performed a luggage search,[124] and where an airline employee conducted a search of luggage pursuant to airline and FAA procedures.[125] The fact that the federal government has not compelled a private party to take a particular action does not by itself establish that such action is a private one.[126] For example, encouragement or endorsement of a certain action by authorizing it and removing all legal barriers to it "suffice[s] to implicate the [Constitution]."[127]

The issue of whether a company that sells facial-recognition technology to the government can be considered a state actor (and subject to Fourth Amendment rules to protect privacy) has arisen in a number of cases brought against Clearview AI.[128] In essence, the complaints filed against Clearview AI contend the company was acting as an agent of the federal government for:

- Allegedly scraping billions of facial images from the Internet,
- Performing facial scans of those images, and
- Creating a biometric database that allowed users of the database to immediately identify a member of the public merely by uploading a person's image to the database.[129]

DNA Identification as a Case Study

Although courts have not yet had the opportunity to examine the governmental use of facial-recognition technology,[130] recent case law on the collection of DNA and creation of DNA profiles provide a useful framework for understanding how courts will likely address such conduct in the future. Ultimately, the similarities and differences between DNA profiling and the use of facial-recognition technology arguably demonstrate the latter's legality.

This case law is the product of disputes arising from the DNA Analysis Backlog Elimination Act of 2000 (DNA Act), which created the Combined DNA Index System (CODIS).[131] CODIS, which is supervised by the FBI, connects DNA laboratories at the local, state, and national level; collects DNA profiles from arrestees, convicted offenders, and forensic evidence found at crime scenes; and sets uniform standards for DNA matching.[132] Although the DNA used for such matching is often referred to as a "genetic fingerprint," it does not have any association with a genetic disease or any other genetic disposition.[133] Thus, while it is functionally equivalent to a fingerprint, the information in the database is only useful for "human identity testing."[134]

However, that is not to say that the Supreme Court has not been divided over the government's asserted purpose for DNA identification. In *Maryland v. King*, a close 5-4 decision from 2013, the Court held that the use of a cheek swab to obtain an arrestee's DNA sample and DNA identification of the same arrestee as the perpetrator of an unsolved crime from 6 years earlier was reasonable under the Fourth Amendment.[135] Justice Kennedy reasoned that the fact that the DNA identification search was part of a routine booking procedure precluded any need for a warrant.[136]

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Next, Justice Kennedy balanced the government's legitimate interests against the degree to which the search intrudes upon an individual's privacy, finding that the former's interest in verifying the identity of a person in detention; determining the risks such person presents to facility staff, the existing detainee population, and new detainees; ensuring the accused person's availability for trial; deciding whether the individual can safely be released on bail; and preventing the detention of innocent people outweigh privacy intrusions from DNA identification.[137] Specifically, Justice Kennedy noted that the intrusion of a cheek swab to obtain a DNA sample is a minimal one because its context, police custody, diminishes privacy expectations; it is brief, painless, and does not break the skin; and the sample only reveals an individual's identity, not any genetic traits.[138] Thus, he concluded that the search in this case, unsupported by any individualized suspicion beyond that which supported the arrest, was reasonable.[139]

In his dissent, Justice Scalia responded the Court's assertion that "DNA is being taken, not to solve crimes, but to identify those in the state's custody, "taxes the credulity of the credulous."[140] He maintained that the majority uses "identify" to mean finding out what unsolved crimes the arrestee has committed, since, in this case, what the DNA match identified was a previously taken sample from an earlier crime.[141] In fact, King's identity was already known when he was charged—before the DNA database returned a match.[142] Consequently, because suspicionless searches have only been permitted when the primary purpose of the search was not to detect evidence of ordinary criminal wrongdoing, DNA identification searches must, Justice Scalia contended, require some level of individualized suspicion.[143]

It is well established that suspicionless searches of all persons—such as fingerprinting—for ordinary law-enforcement purposes would violate the Fourth Amendment.[144]

Extrapolating from both *King* opinions, arguably the use of facial-recognition technology to identify air passengers is even more safely within the ambit of Fourth Amendment reasonableness than DNA identification. Unlike DNA identification, identification via facial recognition would be for non–law enforcement investigatory purposes, including verifying that the passenger may lawfully enter or exit the country and ensuring the safety and security of other passengers, the airplane crew, and society at large. Moreover, if suspicionless searches of passengers' persons and properties can already be lawfully conducted at airports as lawful border searches or administrative searches, a search that is more limited in scope, not physically intrusive, and functionally equivalent to the current practice of examining a passenger's passport is consistent with the Fourth Amendment.

Survey of Federal Privacy Laws Relevant to Airport Operators and Stakeholders

Generally, federal privacy laws (1) govern a federal agency's collection and use of biometric data, or (2) protect consumers' privacy with respect to the collection and use of their biometric data and provide them a cause of action. There are a number of federal privacy laws applicable to the collection and use of biometric data by federal agencies as well as specific activities by various industries in distinct business sectors, some of which are relevant to airport operators and stakeholders. Interestingly, a 2015 GAO report observes that there are no federal laws restricting the capture of facial images except with respect to matters associated with minors.[145]

With respect to legal protections pertaining to commercial uses of biometric data, including facial-recognition technology, federal laws can be divided into three broad categories that address privacy and consumer protection for "(1) the capture of facial images; (2) the collection,

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use, and sharing of personal data; and (3) unfair or deceptive acts or practices, such as failure to comply with a company's stated privacy policies."[146]

The overarching federal law governing federal agencies' collection, use, and disclosure of PII such as biometric data is the Privacy Act of 1974.[147]

Privacy Act of 1974

The Privacy Act of 1974 governs what information is collected, maintained, and used by federal agencies.[148] Where a federal agency creates records retrievable by personally identifying information to satisfy requirements required by law,[149] legal requirements governing the collection, use, retention, dissemination, disclosure, and maintenance of individuals' personal information are set out in the Privacy Act of 1974. When seeking personal information from an individual, the act requires agencies to provide a Privacy Act Statement that informs the person about the authority for the collection, the purpose, the routine uses of such information, and the consequences should the individual decline to provide the information.[150] In addition to controls imposed on federal agencies, the act creates rights for persons on whom the records are maintained.[151]

Generally, an agency must publish public notice in the Federal Register about the collection, maintenance, use, and dissemination of information about individuals maintained in a system of records (referred to as a System of Records Notice or SORN) and must conduct a PIA analyzing how the data are used and determining whether that use meets applicable legal and policy privacy requirements.[152] Further, agencies may not disclose records on individuals without their consent unless one of 12 exceptions apply (such as for law enforcement and national security purposes).[153] The act imposes recordkeeping, retention, and destruction requirements as well.[154] There are civil and criminal penalties applicable to violations of the act.[155]

In addition to the application of the Privacy Act, federal agencies' collection of biometric data, particularly facial-recognition data, is authorized or governed by laws, regulations, and policies pertaining to:

- Requirements on airport operators to establish airport security programs;[156]
- Requirements for anyone, including airport and air carrier employees, requiring unescorted access to aircraft and secure areas to obtain a SIDA badge;[157]
- The issuance of airman certificates;[158]
- The collection of biometric data from travelers for border control purposes;[159]
- The establishment of an improved visa issuance system, ESTA;[160]
- The creation of trusted traveler programs;[161] and
- Travel security, defense, counterterrorism, and law enforcement.[162]

Even where federal agencies are using biometric or facial-recognition technology linked to sector-specific activities, there are gaps in the coverage of the law for the regulation of substantive and procedural protections for subjects' data.

For example, CBP implemented a biometric exit system required by law, but it has not mandated cooperation from carriers or other authorities.[163] After almost a decade of collecting biometrics (fingerprints) from arriving aliens,[164] CBP launched the TVS in 2017 to capture facial images from passengers departing the United States.[165] Under TVS, information from a passenger's check-in for a flight is used to compile a gallery of pre-existing photographs of the passenger, such as from visas or passports, and may include photographs from previous encounters with CBP or other DHS components. Prior to boarding, the camera takes a photograph

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of the passenger, and TVS compares it to the photos in the gallery to verify the passenger's identity. In essence, TVS uses an algorithm to "perform both 1:N and 1:1 facial-recognition matching."[166] Once confirmed, after a matter of seconds, the passenger is free to board, and CBP creates an exit record for the passenger.[167]

Photographs of U.S. citizens and exempt aliens are deleted by CBP within 12 hours of verification.[168] CBP has indicated that U.S. citizens entering or departing the United States may opt out of having their picture taken and that the verification will be done manually by a CBP officer or airline/airport representative.[169] Further, according to a 2020 GAO report, foreign nationals may also opt out if an air carrier or third party conducts the facial-recognition verification.[170] CBP, however, stores photographs of non-immigrant aliens and lawful permanent residents for up to 14 days in a database, and photos of "in-scope" travelers[171] are stored for 75 years in another database. CBP stores biographic exit records for every traveler, regardless of their citizenship or status. CBP stores biographic exit records of U.S. citizens and lawful permanent residents for 15 years and exit records of non-immigrant aliens for 75 years.[172] No photos are shared with travel stakeholders, only the results of the biometric match (or no match).[173]

In an effort to assess and address CBP's use of TVS, the DHS Privacy Office sought a review and recommendations from the DHS Data Privacy and Integrity Advisory Committee (DPIAC).[174] The DPIAC made a number of recommendations to improve transparency with respect to notice (both substantive and procedural), consult with the scientific community to determine length of years of reliability/accuracy of facial images, data minimization, steps to ensure greater data accuracy (including training partner airlines on use of the technology), and accountability and auditing.[175]

While CBP's TVS has not been free from criticism, as of July 2020, only two lawsuits had been filed in connection with the use of its facial-recognition technology for border entry/exit purposes. Both complaints were filed in connection with requests filed under the Freedom of Information Act for records pertaining to CBP's TVS.[176]

Similarly, TSA has been developing biometric solutions to perform travel document checks for passengers proceeding through security checkpoints. In 2018, the agency released its TSA Biometrics Roadmap for Aviation Security & the Passenger Experience. The Biometrics Roadmap outlines four goals:

- Partner with CBP on biometrics for international travelers,
- Operationalize biometrics for TSA PreCheck travelers,
- Expand biometrics to additional domestic travelers, and
- Develop support infrastructure for biometric solutions (TSA 2018b).

In collaboration with CBP, TSA is deploying Credential Authentication Technology, which is to be used to authenticate the ID credential presented by the passenger (TSA 2020c).

Lastly, the authors note that even in the absence of any lawsuit challenging FBI's direct use of facial-recognition technology, there are numerous cases reporting challenges to the use, reliability, or introduction into evidence of the results of the agency's matches during criminal proceedings.[177]

The Federal Trade Commission Act and Deceptive and Unfair Practices

A core mission of the FTC is consumer protection. The Federal Trade Commission Act of 1914 (FTCA), as amended, authorizes the FTC to, among other things:

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- Prevent and address unfair or deceptive acts or practices;
- Seek monetary damages and other relief for conduct injurious to consumers;
- Prescribe rules defining with specificity acts or practices that are unfair or deceptive, including the ability to establish requirements designed to prevent such acts or practices;
- Conduct investigations relating to the organization, business, practices, and management of entities engaged in commerce; and
- Make reports and recommendations to Congress and the public.[178]

Within the scope of the FTCA is the authority to combat unfair and deceptive practices regarding the collection and use of biometric data.[179] The FTC's authority, however, is limited. For example, it may not require retailers to have privacy policies, only that if they do, they may not engage in unfair or deceptive practices (Zimmerman 2018; Voss and Houser 2019). Under the FTCA, the FTC can determine that an act is "unfair" if it includes such acts or practices involving foreign commerce that:

- Cause or are likely to cause reasonably foreseeable injury within the United States, or
- Involve material conduct occurring within the United States.[180]

The FTCA further provides that:

> The Commission shall have no authority . . . to declare unlawful an act or practice on the grounds that such act or practice is unfair unless the act or practice causes or is likely to cause substantial injury to consumers which is not reasonably avoidable by consumers themselves and not outweighed by countervailing benefits to consumers or to competition."[181]

As an example of the exercise of its enforcement authority, the FTC recently amended a 2012 settlement order involving Facebook's alleged deceptive practice concerning consumers' control over the privacy of their personal information (i.e., sharing of users' personal data with third parties without their knowledge or consent). On April 28, 2020, the FTC formally approved an order to revise the settlement to where Facebook agreed to pay $5 billion in penalties, make changes in Facebook's approach to privacy management, and be subject to FTC monitoring (FTC 2020).[182]

Finally, as discussed *infra,* while the FTC has not issued specific regulations governing the use of biometric data technology (Stewart 2019b; Wright 2019), it has undertaken to issue guidance, interpretive statements, and other similar public pronouncements to attempt to advise employers on privacy protections and best practices (Stewart 2019b; Wright 2019).

Authorities to Control Travel During a Pandemic

Federal law authorizes the Center for Disease Control and Prevention (CDC) to exercise broad powers to conduct or require cooperative efforts to prevent the introduction, transmission, and spread of quarantinable and serious communicable diseases such as COVID-19.[183] The CDC may require the assistance of Customs and Coast Guard officers as well.[184]

Generally, this authority extends to screening, isolating, and quarantining, and "plac[ing] a person under surveillance" if the CDC director "has reason to believe" that the "arriving person" is infected with or has been exposed to a communicable disease listed under Executive Order 13295.[185] Subject to certain limitations precluding the denial of entry of U.S. citizens and legal permanent residents, the scope of CDC's authority and ability to screen arriving persons at ports of entry and engaged in interstate travel is clear (see CDC 2020).

CDC surveils or quarantines persons believed to have been exposed to a disease but who are not yet ill and isolates persons infected with a communicable disease.[186] Of note, TSA may assist

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CDC by screening passengers in the interest of aviation safety, maintaining a do-not-board list (identifying persons identified by CDC as public health threats),[187] and canceling flights (if the CDC has determined that a person aboard the aircraft has been exposed or infected with a pandemic disease).[188] Similarly, airlines may refuse to board passengers infected with communicable diseases in accordance with DOT regulations (i.e., when the decision is based on "reasonable judgment that relies on current medical knowledge or on the best available evidence" that the person poses a "direct threat" to the health and safety of others).[189]

The law is less clear with respect to exit screening for outbound international flights. Although untested in the courts, arguably the authority to prevent the introduction, transmission, and spread of quarantinable and serious communicable diseases would support screening of departing persons, including contact tracing of infected travelers who have been in contact with other persons in the United States or who might be on a return trip to the United States shortly or to comply with WHO regulations.[190]

In late May of 2019, the CDC issued a checklist for states' health departments to design a contact tracing plan (CDC 2019a). Also, the CDC website on contact tracing is frequently updated to provide links to resources, tools, and guidance for public health staff and others (CDC 2019b). Note that health data are heavily regulated under federal and state law with extensive privacy protections.[191]

U.S. Congress Legislative Activity

In its 116th session, Congress took up a number of bills proposing to legislate the use of biometric data, and there was every expectation that the next Congress would entertain similar legislation (see Figure K-1). Two bills merit monitoring. The bill that most clearly implicates the use of facial recognition and interests of airport stakeholders is the Commercial Facial Recognition Privacy Act (S. 847). The bill would, subject to certain important exceptions, generally prohibit the commercial use of facial-recognition technology to identify and track consumers without the end users' consent. As proposed, the bill would place limitations on the third-party sharing of collected facial biometric data, would require certain entities to meet minimum data security standards (to be published by FTC in consultation with NIST), and would treat violations as unfair or deceptive acts or practices under the FTCA. The bill, which includes several exemptions, would except "security applications" that use the technology for loss prevention or to detect and prevent criminal activity.[192]

Another significant development is Senator Sherrod Brown's announcement on June 18, 2020 that he intends to introduce the Data Accountability and Transparency Act of 2020, a comprehensive privacy and data protection bill. The bill would not only create a privacy agency but would prohibit private companies and government agencies from collecting personal data unless it is "strictly necessary" to carry out one of a few specified purposes. More significantly, it would ban the use of facial surveillance technology (EPIC 2020). The impacts on the use of facial-recognition technology could be enormous. This broad, sweeping bill would appear to prohibit any data aggregator (which includes federal agencies) from using facial-recognition technology or collecting, using, or sharing any personal data obtained from facial-recognition technology.

The topic of facial recognition has also received attention in several Congressional hearings. The House Committee on Oversight and Reform has held three hearings on facial recognition.[193] Three key concerns were identified: accuracy, transparency, and protection of civil liberties.[194]

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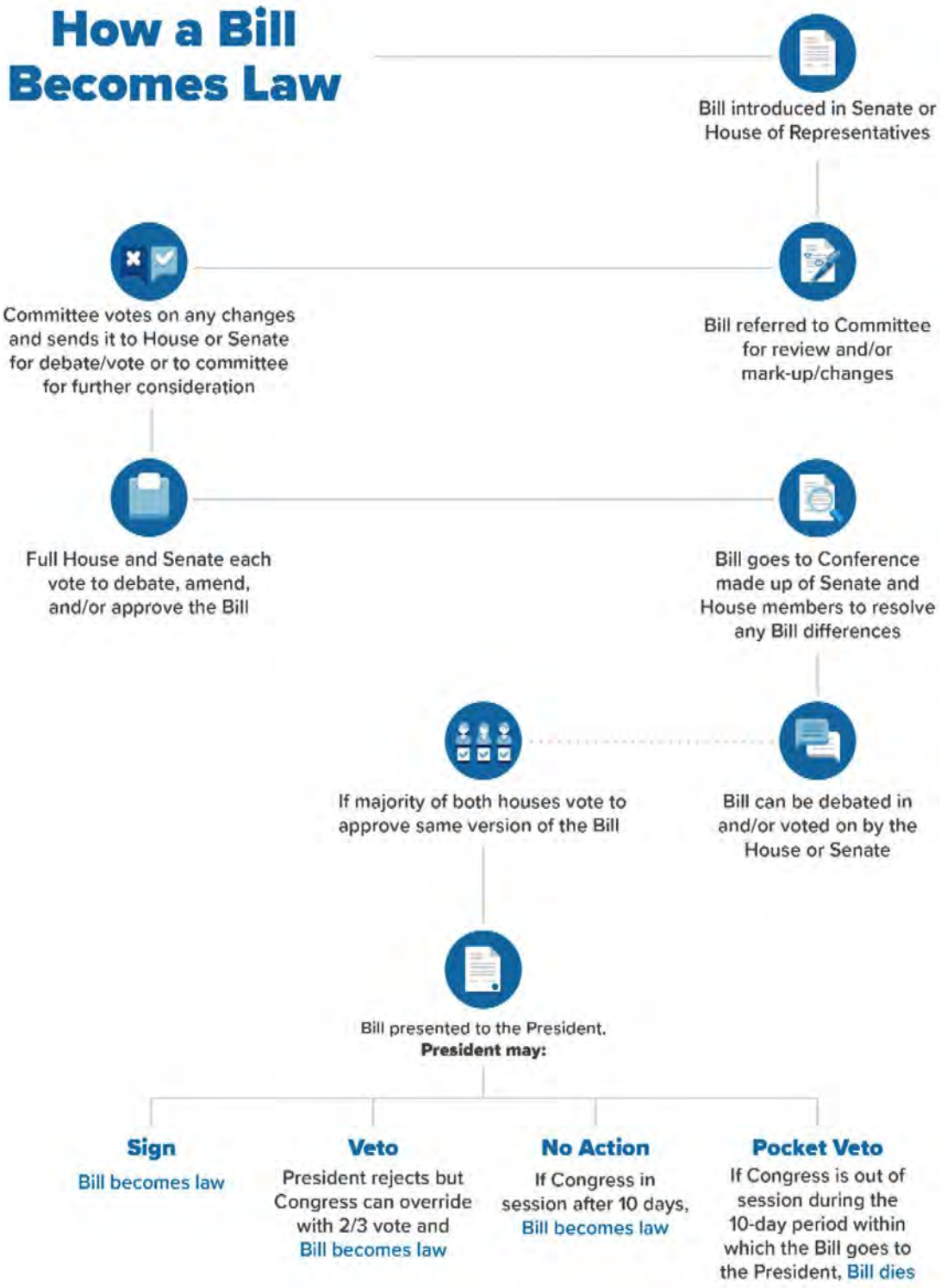


Figure K-1. How a bill becomes law.

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Survey of State Laws

This section contains a survey and description of state laws that specifically govern the collection and use of biometric data, including facial-recognition technology (see Figure K-2).

Illinois

Over a 2-year period (2018 to 2019), over 200 lawsuits were filed for alleged violations of BIPA, and reportedly this is on the rise (Prescott 2020). In 2019, the Illinois Supreme Court issued a decision in *Rosenbach v. Six Flags Entertainment Corporation* that a plaintiff need not allege an actual injury to establish that he qualified as an "aggrieved" person under specific provisions of BIPA and allowed the suit for penalties and damages to proceed.[195] In this case, Six Flags theme park took a minor's thumbprint, without his consent and without informing him about the purpose of data collection, for a season's pass and future reentry to the theme park.[196] The reason this decision is significant is not only because it recognizes technical violations as sufficient to show harm under BIPA, but also potentially lowers the bar for litigants to bring lawsuits in federal courts that rely on satisfying a "case or controversy" showing to establish standing to bring a lawsuit. [197]

In 2016, a plaintiff sued Google claiming in its photo-tagging that the company scanned her facial features from a photograph and created a facial template from the photograph in violation of BIPA.[198] The court held that biometric information obtained from a photograph qualified as a biometric identifier under BIPA, rejecting Google's argument that only face scans taken in person would qualify as biometric identifiers under the statute.[199] Subsequently, the case was dismissed on grounds relating to a lack of proof of concrete injuries.[200]

Following on the heels of *Rosenbach*, users of Facebook living in Illinois brought suit in California for violations of BIPA associated with Facebook's use of facial-recognition

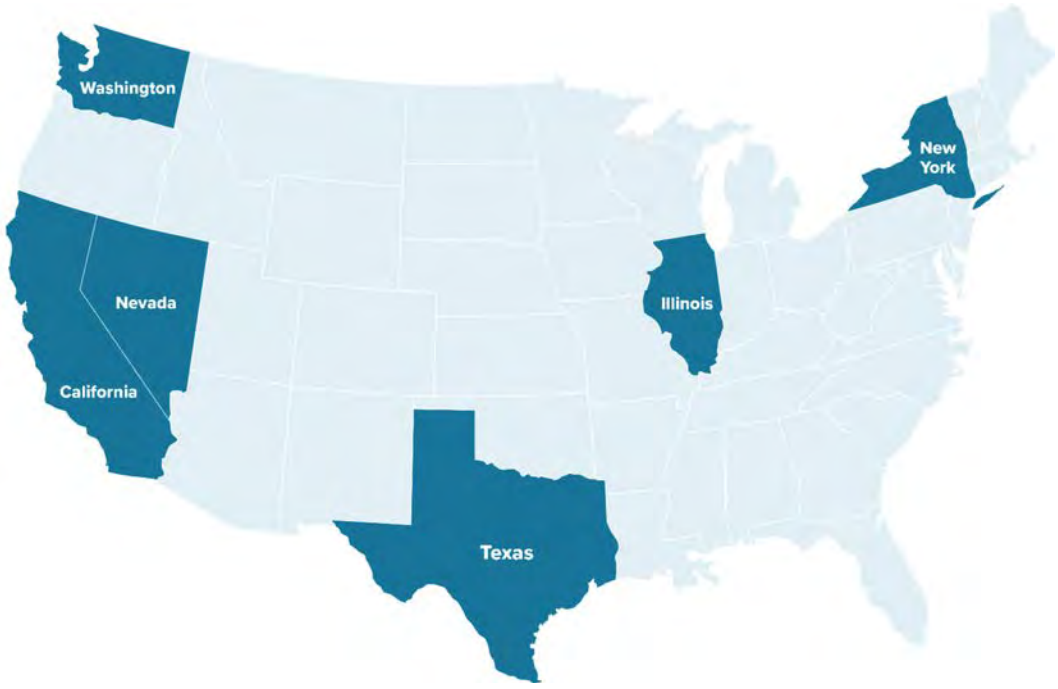


Figure K-2. States with state laws surveyed.

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technology to "tag" persons in users' pictures.[201] The Ninth Circuit Court of Appeals determined that Facebook's collection, use, and storage of biometric identifiers without a written release violated the procedural protections provided under BIPA, emphasizing that this was the type of harm targeted by the act, and thus found the plaintiffs had articulated "a concrete and particularized harm, sufficient to confer . . . standing."[202] Even potentially more important was the Appellate Court's rejection of Facebook's argument that there should be no extraterritorial application of BIPA for actions Facebook failed to undertake in another state. The Court opined: "it is reasonable to infer that the [Illinois] General Assembly contemplated BIPA's application to individuals who are located in Illinois, even if some relevant activities occur outside the state."[203]

The general rule with respect to application of a state's law outside the state is that, absent a clear intent expressed in the provisions of the statute, courts will generally not find extraterritorial application of the law and generally limit the scope of the law to things or persons within the state. [204] Not only are the courts wrestling with the extraterritorial application of BIPA, they are considering lawsuits against third parties that have derived facial "geometries" from photographs, despite BIPA specifically excluding photographs from the definition of a biometric identifier subject to the law's provisions.[205]

In addition to litigation over the specific terms in BIPA, two decisions subsequent to *Rosenbach* are worth mentioning as well as bearing on prerequisites to the ability to bring suit: *Miller v. Southwest Airlines*[206] and *Crooms v. Southwest Airlines Co*.[207] In both cases, airline ramp agents from Midway Airport alleged that Southwest Airlines' practice of scanning and using their fingerprints for timekeeping purposes violated BIPA. The Seventh Circuit in *Miller* and the District Court in *Crooms* held that unions represented the plaintiffs' interests under a collective bargaining agreement, Illinois could not divest the union of its role. Ultimately, the courts decided that federal law[208] required the plaintiffs to bring their claims before an adjustment board (and not in federal court), and thus affirmed dismissal of their claims.[209]

As these cases illustrate, there are many unsettled questions, particularly with respect to the interplay of federal and state law, to include the degree of harm a plaintiff needs to show to have standing to pursue a case in federal courts versus technical violations under state law to bring a case under BIPA. As one court recently noted, federal courts and state courts define injury differently for purposes of determining standing to bring a suit in the respective courts.[210]

To date, most of the cases under BIPA have been class actions targeting employers' use of biometric technology at work. These lawsuits are not only on the rise but are expensive and difficult to defend against. For example, in 2020, Facebook offered to settle a class action lawsuit (in California) for alleged violations of BIPA for $650 million (after the trial judge initially rejected the proposed $550 million settlement) (Shibu 2020). In May, plaintiffs filed lawsuits against Facebook in Texas and Arizona, adding to the many other suits pending nationwide, and the company is reportedly facing billions in damages (Justia n.d.).

There is also uncertainty with respect to application of BIPA (and other states' biometric privacy laws) under circumstances where airport stakeholders are partnering with government agencies, such as CBP under the TVS program. Factors bearing on applicability, compliance, and potential liability include whether the state law addresses only commercial uses of biometric data and excludes government agencies from its scope, as do all of the biometric state laws discussed in this section (with the exception of the recent Washington law), and whether the airport stakeholder's partnership can be viewed as acting as an agent of the government by collecting facial images and transmitting them to CBP (and not retaining them), but this becomes complicated if the stakeholder collects the data for a dual business purpose as well, such as for facilitating the boarding process.

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Texas

The Texas biometric privacy law, like that of Illinois, requires persons or entities that collect biometric data to inform individuals before capturing the biometric data and to obtain the individual's consent.[211] Unlike the Illinois law, the Texas statute does not require a written release. The Texas law, however, like the Illinois law, does prohibit the sale of biometric information, and it similarly sets restrictions on the storage of such information.[212] Lastly, although modeled after BIPA, it lacks any private cause of action, relying on the state's attorney general to enforce and, as appropriate, impose sanctions.[213]

Washington

Washington state's biometric privacy statute entered into effect in 2017 and regulates commercial uses of biometric data to require notice, consent, and a limited ability to sell the data.[214] Washington's law also does not include photographs, video or audio recordings, or facial geometry as biometric identifiers.[215] Unlike the Texas law, it does not require that consent to collection of biometric data be in writing. Further, in a significant departure from its Illinois and Texas counterparts, the Washington state law carves out an exemption to biometric data collection and storage; businesses may collect and store such information without providing notice and obtaining consent so long as the information is collection for "security purposes,"[216] including collection, storage, and use of the information for purposes of preventing shoplifting, fraud, and theft.[217] As with Texas, there is no private cause of action, and enforcement is delegated to the state attorney general.[218]

On March 12, 2020, the governor signed a new facial biometric law, backed by Microsoft, the purpose of which was to limit state and local government authorities' use of facial-recognition technology.[219] Reportedly, Microsoft, Amazon, and Comcast met with state legislative members to address their concerns from earlier versions of the bill, and Microsoft issued a statement endorsing the bill.[220] The act will go into effect in 2021.

California

California's California Consumer Privacy Act (CCPA), which entered into effect on January 1, 2020,[221] provides consumer and employee rights and regulates commercial uses of biometric data by including it in the definition of personal information.[222] California's law applies on a somewhat more limited scale than the Illinois, Texas, and Washington laws. The California law targets any company that both (1) operates in California and (2) either makes at least $25 million in annual revenue, gathers data on more than 50,000 users, or makes more than half its money from user data.[223] The law treats biometric information, including images of one's face, as personal information, and provides rights to consumers to protect their personal information.

The CCPA provides the right to know what the company is collecting and why, including information about the sale of their personal information, the right to opt out of the sale of their personal information to third parties, the right to have their data deleted, with some exceptions, and the right to equal service and pricing even if they exercise their privacy rights (Shank 2019). The law provides a limited private cause of action against businesses that fail to "implement and maintain reasonable security procedures and practices appropriate to the nature of the information."[224] The CCPA provides a unique provision that requires consumers to provide notice to the company before initiating any legal action. This period of 30 days is to allow the company time to cure the violation and, if this is done, no action may be pursued.[225]

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In sum, with respect to privacy laws enacted in Texas, Washington, and California, there are variances with respect to:

- The definition of biometric data (e.g., application to the image/data collected, but not to the analysis of the data),[226] and
- The scope of coverage (commercial purpose but not extending to data collected for security purposes, arguably excluding timekeeping uses).[227]

New York

New York amended its existing data-breach notification laws with its 2019 Stop Hacks and Improve Electronic Data Security (SHIELD) Act, which went into effect in March 2020. The SHIELD Act broadens the definition of private information to include biometric information to protect the private information of state residents by requiring businesses to implement and maintain information security protocols.[228] Previously, New York had passed a limited biometric legislation, §201-a, which applies specifically in the employment context. It prohibits fingerprinting "as a condition of securing employment or of continuing employment."[229] The SHIELD Act extends protections to any state resident and extends the obligation to provide notification of a data breach under New York's breach notification law beyond the more previously limited protected class of affected persons or businesses that conducted business in New York. The law, as amended, does not expressly provide for a private right of action but authorizes the state's attorney general to pursue a maximum fine for failing to notify those affected from $150,000 to $250,000 (*The National Law Review* 2020).

Nevada

Effective October 2019, Nevada enacted SB 220, which amended existing state law to require operators of websites and online services to post privacy notices on their websites and became the first state to provide consumers with the ability to opt out of the sale of their personal information.[230] Although similar to the CCPA, SB 220 is narrower in scope (Kohne et al. 2019).

Most recently, similar legislation has been introduced in several other states (e.g., Arizona, Florida, and Massachusetts).[231] States have also enacted laws to protect the privacy of minors and students.[232] It is anticipated that this trend will only continue as privacy concerns increase.

Municipalities

Recently, municipal bans on the use of facial-recognition technology have sprung up in cities across the country. Cities with such bans include San Francisco, Oakland, Boston, Somerville (MA), Cambridge City (MA), Northampton (MA), Portland (ME), and Brookline (MA) (Hudgins 2019). Other cities are considering similar measures, and California, New Hampshire, and Oregon have already passed laws that ban the use of facial-recognition technology in police body cameras, and New York and New Jersey appeared poised to do the same (Read 2020; Johnston 2019; Jarmanning 2020). As a general rule, municipal governments have cited concerns about potential abuses from the use of facial-recognition technology, particularly with respect to surveillance activities; impacts on minorities' freedom of association; and imperfections in, and invasiveness of, the technology, citing bias.[233]

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International Organization Activities

There are a number of international organizations working, coordinating, and collaborating with others to support initiatives to achieve a more seamless travel experience in the air environment. Significantly, most advocate for global standards in an obvious effort to incorporate uniformity in international aviation.[234]

Beginning in the late 1960s, ICAO begin work on developing a machine-readable travel document (MRTD) and in 1980 produced Doc 9303 (currently in its seventh edition: ICAO 2015) setting specifications for passports and other travel documents. Over the next couple of decades, ICAO adopted and incorporated facial-recognition technology as a core concept for MRTD and leading to the promotion of its use for e-passports (ICAO 2007). ICAO initiated an ongoing project operating the ICAO Public Key Directory (PKD), which is a central repository for exchanging the information required to authenticate e-passports. Overall, the objective is to advocate for, and assist in, creating a globally interoperable system [section 3.2 of Doc 9303 (ICAO 2015)]. Specifically, ICAO is focused on the DTC, which could serve as an e-passport (i.e., extracted data) or be issued in parallel to or in replacement of a physical e-passport (Cole 2019). Recently, ICAO published guidelines pertaining to DTC (ICAO 2020).

In an example of an alternate use of biometrics for an identity purpose, the UN High Commissioner for Refugees (UNHCR) in 2018 issued its Strategy on Digital Identity and Inclusion, a global effort to encourage and support countries "giving everybody access to a legal and digital identity" (UNHCR 2018). UNHCR indicated that refugees and other displaced persons represent a "marginalized" sector of a population in greatest need of an identity to benefit, for example, from the receipt of basic assistance, protection, and relief services. States in turn benefit from the registration of stateless persons, refugees, and displaced persons with a clearer picture of those residing in their territory. "UNHCR assists member states in ensuring that refugees and asylum seekers, stateless persons, and other forcibly displaced are—digitally speaking—not left behind" (UNHCR 2018). In furtherance of this effort, UNHCR began rolling out Population Registration and Identity Management EcoSystem (PRIMES), which, using biometrics, provides a platform for all UNHCR registration and identity management tools. A consolidated global database is in place. Many of the pilots testing PRIMES are being conducted in partnership with governments, academic institutions, international organizations, and private-sector companies for proof of concepts, establishing use cases and processes, and testing IT infrastructure.

In the 1990s, IATA and ACI launched a joint initiative, "Smart Security," focused on improving the passenger flow process, particularly through security checkpoints. Smart Security sought to address anticipated growth in passenger traffic by focusing on strengthening security, increasing operational efficiencies, and improving the passenger experience with a more seamless approach (ACI n.d.). Over the years, the initiative evolved to combine IATA's Checkpoint of the Future and ACI's Better Security parallel programs to seek to achieve a roadmap for streamlining the passenger flow process, a key component of which is the integration of innovative biometric technology. As of 2018, more than 100 airports worldwide had implemented a variation of Smart Security (Smiths Detection 2018).

In December 2006, the Simplifying Passenger Travel Interest Group, of which ACI is a founding member and which is made up of airlines, airports, and others, published its recommendations for the ideal process flow (IPF) for the travel of passengers through an airport. It proposed that many of the processes be automated using biometrics, incorporating ICAO standard biometrics for the passenger identification process in the IPF and at all airports (ACI 2006).

Building on earlier efforts such as Smart Security and digital identity management, in 2018 IATA announced the OneID initiative, which proposed the use of biometric technology to establish a "secure, seamless, and efficient journey" (ICAO 2019) to facilitate the passenger

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experience through security touchpoints by "sharing a single set of passenger identity information among authorized stakeholders in accordance with data privacy rules" (ICAO 2019). The initiative includes the concept of a DTC, the global policy and technical specifications of which are being shaped and developed by the ICAO. Generally, the concept has been described as a "virtual credential, derived from and linked to an issuing document, securely stored on a mobile device or in the cloud and accessed via biometric authentication, essentially enabling document-free travel" (Entrust Datacard 2019). Significantly, the initiative recognizes that a globally coordinated "horizontal" approach is needed to replace a disjointed, fragmented response to date (IATA 2018a). OneID expressly includes, within its desired end state, a plan to have OneID coincide with IATA's NEXTT vision, described in the following (IATA 2018b).

IATA and ACI are pursuing jointly the NEXTT initiative, the goal of which is to make the most of the latest technological advances to transform airports, improve the travel experience from check-in to boarding, including the cargo transport experience, and develop a "common vision to enhance the on-ground transport experience, guide industry investments, and help governments improve the regulatory framework" (Appleton 2019). In response to concerns about siloed sector developments, IATA and ACI broadened the focus of the initiative to "business and cultural change as well as how airports and airlines can develop their operations to achieve additional efficiencies and ultimately increase the capacity" (Appleton 2019). The initiative highlights the benefits of "off-airport" processes, use of city centers, and interconnected systems, and includes incorporation of biometric technology without necessarily advocating any single option or solution. Per IATA, "[t]he concept involves the use of a trusted digital identity, biometric recognition technology, and a collaborative identity management platform" (IATA 2020).

Similarly, the WEF has promoted the KTDI, and in March 2020, it published a white paper offering guidance and recommendations on specifications and a facial-recognition framework for industry (World Economic Forum 2020b). To promote "responsible use" of facial-recognition resources, WEF developed, among a diverse group of stakeholders, a set of principles (and best practices) designed to inform decision makers and product teams on issues pertaining to bias mitigation, proportional use of the technology, privacy, accountability, consent, right to accessibility, children's rights, and alternative options, to name a few (World Economic Forum 2020c). Specifically, the Netherlands and Canada are conducting an ongoing pilot to test KDTI aspects in a "real-life, cross-border context" (World Economic Forum 2020c) and to further inform future pilots and implementation.

The white paper is intended to document the standards, open specifications, capabilities, functionalities, and industry best practices captured by the initial pilot and "provide guiding principles for the KTDI concept and any related future pilots toward the end-state vision of global interoperability" (World Economic Forum 2020c).

This synopsis illustrates how various initiatives have evolved. More importantly, the organizations pursuing initiatives to design biometric identity and authentication and air transport process improvements have clearly made substantial efforts to produce coordinated pilots and strategies (e.g., NEXTT) integrating and incorporating efforts of multiple stakeholders, and it is expected that that trend will continue.

Endnotes

113. *See U.S. Dep't of Justice v. Reporters Comm. for Freedom of the Press,* 489 U.S. 749, 763 & n. 15 (1989) (recognizing the common law's protection of a privacy right).
114. *Restatement (Second) of Torts,* § 652A (Am. Law. Inst. 1977).
115. *Rivera v. Google, Inc.*, 366 F. Supp. 3d 998 (N.D. Ill. 2018); *Jacobson v. CBS Broad., Inc.,* 19 N.E.3d 1165, 1180 (Ill. App. Ct. 2014).

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116. *Griswold v. Connecticut,* 381 U.S. 479, 483-485 (1965).
117. *Schmerber v. California,* 384 U.S. 757, 764 (1966).
118. *Hiibel v. Sixth Judicial Dist. Court,* 542 U.S. 177, 191 (2004).
119. *Compare, e.g., United States v. Wright*, 431 F. Supp. 3d 1175 (Nev. D. Ct. 2020) (Forcing the defendant to unlock his phone with his face violated the Fifth Amendment) with *State v. Andrews*, 2020 N.J. LEXIS 898 (Aug. 10, 2020) (Court held compelled disclosure of cell phone passcode did not incriminate the defendant as the passcode was not substantive information or a clue to the crime).
120. *Kyllo v. United States*, 533 U.S. 27, 34 (2001).
121. *Id.,* at 40.
122. *United States v. Jones*, 565 U.S. 400 (2012).
123. *Carpenter v. United States,* 138 S. Ct. 2206 (2018); *see also Riley v. California,* 573 U.S. 373 (2014) (police officers could not search without a warrant defendants' cell phones under the Fourth Amendment exception for a search incident to an arrest).
124. *United States v. Walther*, 652 F.2d 788, 793 (9th Cir. 1981).
125. *People v. Owens,* 184 Cal. Rptr. 509 (Cal. Ct. App. 1982).
126. Skinner v. Ry. Labor Executives' Ass'n, 489 U.S. 602, 615 (1989).
127. *Id.,* at 616.
128. *Calderon v. Clearview AI, Inc.*, 20 civ. 1296 (CM), 20 civ. 2222 (CM), 20 civ. 3053 (CM), 20 Civ. 3104 (CM), 20 Civ. 3481 (CM), 20 Civ. 3705 (CM), 2020 WL 2792979, at *1 (S.D.N.Y. May 29, 2020); Class Action Complaint, *Mutnick v. Clearview AI, Inc*., No. 1:20-cv-00512, 2020 WL 378474 (N.D. Ill. Jan. 22, 2020).
129. *Id.*
130. In a review of case law to determine any suits filed against the FBI, CBP, or TSA regarding the use of facial recognition, the only cases located to date concerned actions filed under the Freedom of Information Act for agency records pertaining to CBP's use of facial recognition technology. *See, e.g.*, *ACLU v. DHS*, Case 1:20-cv-02213 (S.D.N.Y, March 12, 2020); *EPIC v. CBP*, No. 19-cv-689 (D.D.C. March 12, 2019) (settled on April 24, 2020).
131. 34 U.S.C. §§ 12592-93 (2017); *id.* §§ 40701-44 (2019).
132. *Maryland v. King*, 569 U.S. 435, 444-45 (2013). As Judge O'Scannlain has helpfully explained:

> CODIS can be used in two different ways. First, law enforcement can match one forensic crime scene sample to another forensic crime scene sample, thereby allowing officers to connect unsolved crimes through a common perpetrator. Second, and of perhaps greater significance, CODIS enables officials to match evidence obtained at the scene of a crime to a particular offender's profile. In this latter capacity, CODIS serves as a potent tool for monitoring the criminal activity of known offenders.

United States v. Kincade, 379 F.3d 813, 819-20 (9th Cir. 2004).
133. *Id.* at 818 n.4.
134. *King*, 569 U.S. at 445.
135. *Id.*
136. *King*, 569 U.S. at 448. "Seemingly" because, as Justice Scalia himself notes, "the opinion does not really contain what you would call a rule of decision," Justice Kennedy at various points relies on the search incident to arrest exception and the special needs exception to justify warrantless DNA identifications.
137. *Id.* at 465.
138. *Id.* at 461–64. Interestingly, Justice Kennedy also noted that "[t]he special needs cases, though in full accord with the result reached here, do not have a direct bearing on the issues presented in this case, because unlike the search of a citizen who has not been suspected of a wrong, a detainee has a reduced expectation of privacy." *Id.* at 463.
139. *Id*. at 465–66.
140. *Id*. at 466.
141. *Id*. at 474.
142. *Id*. at 474–75.
143. *Id*. at 468.
144. *United States v. Mitchell*, 652 F.3d 387, 411 (3d Cir. 2011) (citing *Hayes v. Florida*, 470 U.S. 811, 813–18 (1985); *Davis v. Mississippi*, 394 U.S. 721, 727 (1969)), cert. denied, 565 U.S. 1275 (2012); see also *Green v. Berge*, 354 F.3d 675, 680 (7th Cir. 2004) (Easterbrook, J., concurring) ("What is 'reasonable' under the fourth amendment for a person on conditional release, or a felon, may be unreasonable for the general population. Just as parolees' homes may be searched without a warrant or probable cause, while both are required to search a free person's home, so it may be that collection of DNA samples from the general population would require person-specific cause—or at least a 'special need'. . . .").
145. U.S. Government Accountability Office 2015, at 28 [referencing the Video Voyeurism Prevention Act of 2004, codified at 18 U.S.C. § 1801)], but also would include the Children's Online Privacy Protection Act in that list (15 U.S.C. §§ 6501-6506).
146. *Id.*

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147. 5 U.S.C. § 552a (2018).
148. *Id.*
149. Examples of systems or programs specifically related to border enforcement and aviation operations for which federal agencies collect biometric data include IDENT, DHS Automated Biometric Identification System (formerly U.S. VISIT), E-Verify, TWIC, CBP Airport Security Program, TVS, Global Entry/Pre√, and Passport.
150. 5 U.S.C. § 552a (e)(3).
151. Individuals have the right to access records containing information about themselves, to amend incorrect information, and to sue the agency for violations of the statute. 5 U.S.C. 552a(d) and (g).
152. E-Government Act of 2002 (Pub. L. 107-347), codified at 44 U.S.C. § 3601 et seq.
153. 5 U.S.C. § 552a (j) and (k).
154. *Id.* § 552a(e).
155. *Id.* § 552a(g) and (i).
156. 49 U.S.C. § 44903 (2018).
157. 49 CFR 1540.5; *see also* 19 CFR 122.181-188 (CBP Airport Security Program for unescorted access to CBP areas).
158. 49 U.S.C. § 44703 (2018).
159. 8 U.S.C. §§ 1181, 1185, and 1221 and 19 U.S.C. §1433 (IDENT, US-VISIT, Traveler Verification Service); Intelligence Reform and Terrorism Prevention Act of 2004 (Public Law 108–458; 118 Stat. 3638); Implementing Recommendations of the 9/11 Commission Act of 2007 (Public Law 110–53; 121 Stat. 266); *see also*, Pope, Carra, *Biometric Data Collection in an Unprotected World: Exploring the Need for Federal Legislation Protecting Biometric Data*, 26 J.L. & Pol'y 769 (2018) (comprehensive discussion of DHS agencies' use of biometric data).
160. Section 711 of the Implementing Recommendations of the 9/11 Commission Act of 2007, Pub. L. 110-53.
161. 8 U.S.C. § 1365b (2018), 49 U.S.C. §§ 114 note and 44919.
162. *See, e.g.*, USCIS Fact Sheet (June 2020), https://www.uscis.gov/sites/default/files/USCIS/Refugee%2C%20Asylum%2C%20and%20Int%27l%20Ops/Refugee_Screening_and_Vetting_Fact_Sheet.pdf.
163. *See* 8 U.S.C. § 1365b (2018); Biometric Exit Frequently Asked Questions (FAQs), https://www.cbp.gov/travel/biometrics/biometric-exit-faqs (Last modified: Friday, May 15, 2020).
164. DHS Office of Biometric Identification Management (OBIM) was created in March 2013, replacing the United States Visitor and Immigration Status Indicator Technology (US-VISIT) Program, the purpose of which is to use biometrics to establish and verify the identity of foreign nationals who apply for visas and seek to enter the United States. https://www.dhs.gov/obim.
165. Biometric Exit Frequently Asked Questions (FAQs), https://www.cbp.gov/travel/biometrics/biometric-exit-faqs (Last modified: Friday, May 15, 2020); *see also* DHS Data Privacy and Integrity Advisory Committee (DPIAC), Report 2019-01, *Privacy Recommendations in Connection with the Use of Facial Recognition Technology* (Feb. 2019).
166. GAO-20-568 at 14, Facial Recognition Technology (Sept. 2020).
167. S*upra* note 165, at 37.
168. *Id.*
169. *Id.*: *see also* Transportation Security Administration and U.S. Customs and Border Protection: Deployment of Biometric Technologies, Cong. Rept. (Aug. 2019), https://www.tsa.gov/sites/default/files/biometrics report.pdf.
170. *Supra* note 165.
171. *See* 8 CFR 235.1(f) (defining "in scope travelers" as "any person who may be required by law to provide biometrics upon entry into the United States pursuant to 8 CFR 235.1(f)(ii), or upon exit from the United States pursuant to 8 CFR 215.8. "In-scope" travelers include any alien other than those specifically exempt as outlined in the CFR. Exempt aliens include: Canadian citizens under Section 101(a)(15)(B) of the Immigration and Nationality Act who are not otherwise required to present a visa or be issued a form I-94 or Form I-95; aliens younger than 14 or older than 79 on the data of admission; aliens admitted A-1, A-2, C-3 (except for attendants, servants, or personal employees of accredited officials), G-1, G-2, G-3, G-4, NATO-1, NATO-2, NATO-3, NATO-4, NATO-5, or NATO-6 visas, and certain Taiwan officials who hold E-1 visas and members of their immediate families who hold E-1 visas unless the Secretary of State and the Secretary of Homeland Security jointly determine that a class of such aliens should be subject to the requirements of paragraph (d)(1)(ii); classes of aliens to whom the Secretary of Homeland Security and the Secretary of State jointly determine it shall not apply; or an individual alien to whom the Secretary of Homeland Security, the Secretary of State, or the Director of Central Intelligence determines it shall not apply." at 4 (fn. 9).
172. Privacy Impact Assessment for the Traveler Verification Service DHS/CBP/PIA-056 at 21, November 14, 2018, https://www.dhs.gov/sites/default/files/publications/privacy-pia-cbp030-tvs-november2018_2.pdf;

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Report 2019-XX of the DHS Data Privacy and Integrity Advisory Committee (DPIAC): Privacy Recommendations in Connection with the Use of Facial Recognition Technology As approved in Public Session, https://www.dhs.gov/sites/default/files/publications/DPIAC%20DRAFT%20Biometrics%20Recommendation%20Report%20v4_02.06.2018.pdf.

173. *Supra* note 165.
174. *Supra* note 172.
175. *Id*.
176. *See, e.g., ACLU v. DHS*, Case 1:20-cv-02213, (S.D.N.Y. Mar. 12, 2020); *EPIC v. CBP*, No. 19-cv-689 (D.D.C. Mar. 12, 2019) (settled on April 24, 2020).
177. *See, e.g., United States v. Martinez*, No. 04 CR 543, 2019 LEXIS 192708 (N.D. Ill. Nov. 6, 2019)
178. 15 U.S.C. §§ 41-58, as amended
179. Pub. L. No. 63-203, § 5 [codified at 15 U.S.C. § 45(a)].
180. Supra note 178, § 45(a)(4).
181. *Id.* § 45(n).
182. For a summary of Facebook's history with FTC, see Wilson 2019.
183. 42 U.S.C. §§ 264-65. *See also* 42 U.S.C. § 268(b) (requiring Customs and Coast Guard officers to aid in the enforcement of quarantine rules and regulations).
184. *See* 42 U.S.C. § 268 (b) (2018).
185. 42 CFR § 71.32(a).
186. *See* CDC 2020.
187. 49 U.S.C. §114(m) and §106(l)(4)
188. 49 U.S.C. § 44905(b).
189. 14 CFR 382.21 and 382.19 (c)(1).
190. The World Health Organization, of which the U.S. is a member, has issued binding regulations, the International Health Regulations (IHR), which require among many things that members detect, monitor, and respond effectively to disease outbreaks that could spread internationally. World Health Assembly Res. 58.3, arts. 6-7, 9,13, (May 23, 2005) available at https://www.who.int/ihr/publications/9789241580496/en/.
191. Health Insurance Portability and Accountability Act (HIPAA) Pub. L. 104-191 (1996), particularly 42 U.S.C. § 1320d–6 (penalties for wrongful disclosure); *see also* Determann 2020.
192. *See* Neuberger 2019; *See* www.congress.gov (last visited July 24, 2020). *See, e.g.,* Privacy Bill of Rights, S. 1214, 116th Cong. (2019); Consumer Data Privacy and Security Act, S. 3456, 116th Cong. (2020); American Data Dissemination Act, S. 142, 116th Cong. (2019); Data Care Act of 2019, S. 2961, 116th Cong. (2019); Digital Accountability and Transparency to Advance Privacy Act, S. 583, 116th Cong. (2019); H.R. 3900, 116th Cong. (2019); Privacy Score Act of 2020, H.R. 6227, 116th Cong. (2020); Office of Biometric Identity Management Authorization Act of 2019, H.R. 1729, 116th Cong. (2019).
193. Facial Recognition Technology (Part I): Its Impact on Our Civil Rights and Liberties (May 22, 2019); Facial Recognition Technology (Part II): Ensuring Transparency in Government Use (June 4, 2019); and Facial Recognition Technology (Part III): Ensuring Commercial Transparency & Accuracy (Jan. 15, 2020). https://www.youtube.com/watch?time_continue=7&v=2dpazLVUo_w&feature=emb_title.
194. *Id*.
195. *Rosenbach v. Six Flags Entm't Corp*., 129 N.E.3d 1197 (Ill. 2019).
196. *Id.* at 6-9.
197. *Patel v. Facebook,* Inc., 932 F.3d 1264 (9th Cir. 2019), *cert. denied*, 140 S. Ct. 937 (2020).
198. *Rivera v. Google, Inc*., 238 F. Supp. 3d 1088, 1091 (N.D. Ill. 2017).
199. *Id.,* at 1095.
200. *Rivera v. Google*., 366 F. Supp. 998, 1014 (N.D. Ill. 2018).
201. *Id.*
202. *Id.,* at 1276; *see also*, Jackson 2020.
203. *Patel*, 932 F.3d at 25. *But see Rivera v. Google*, 366 F. Supp. 3d 998 (N.D. Ill. 2018) (A showing of a concrete injury required to establish standing for a violation of BIPA).
204. BIPA May Apply to Clearview AI's Creation of Biometric Data, Law360 Expert Analysis (February 18, 2020).
205. 740 ILCS 10 ("Biometric identifiers do not include . . . photographs . . . "); *See, e.g., Mutnick v. Clearview AI, Inc. et al.*, No. 1:20-cv-00512 (N.D. Ill. January 22, 2020); *Vance v. IBM Corp*., No. 1:20-cv-00577 (N.D. Ill. January 24, 2020).
206. *Miller v. Southwest Airlines Co*., 926 F.3d 898 (7th Cir. 2019)
207. No. 19-cv-2149, 2020 WL 2404878, at *1 (N.D. Ill., May 12, 2020).
208. The Railway Labor Act, 45 U.S.C. § 151 et seq.
209. *Supra* notes 206 and 208; *Crooms*, 2020 WL 2404878.
210. *See, Bryant v. Compass Grp. USA, Inc*., 958 F. 3d 617 (7th Cir. May 5, 2020) for a discussion of standing requirements and decisions (Standing found in case where plaintiff alleged violation of BIPA for the

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failure to comply with statutory requirements associated with an account to biometrically access a vending machine).

211. Tex. Bus. & Com. Code Ann. § 503
212. *Id*.
213. *Id.*
214. Wash. Rev. Code § 19.375 (excludes physical or digital photographs, as well as video or audio recordings from the definition of "biometric identifier); Wash. H.B. 1493 (2017).
215. Wash. H.B. 1493 (2017).
216. *Id.* § 3(4).
217. *Id.* § 4
218. *Id.* § 5.
219. Wash. S.B. 6280 (2020).
220. *See, e.g.,* Crawford 2020 and Hofer 2020.
221. Cal. Civ. Code § 1798.100 et seq.
222. *Id.*; For detailed summary, see Zaller 2020.
223. Cal. Civ. Code § 1798.140(c).
224. *Id.* at § 1798.150: (a) (1) Any consumer whose nonencrypted and nonredacted personal information, as defined in subparagraph (A) of paragraph (1) of subdivision (d) of Section 1798.81.5, is subject to an unauthorized access and exfiltration, theft, or disclosure as a result of the business's violation of the duty to implement and maintain reasonable security procedures and practices appropriate to the nature of the information to protect the personal information may institute a civil action for any of the following:

    (A) To recover damages in an amount not less than one hundred dollars ($100) and not greater than seven hundred and fifty ($750) per consumer per incident or actual damages, whichever is greater.
    (B) Injunctive or declaratory relief.
    (C) Any other relief the court deems proper.

    (2) In assessing the amount of statutory damages, the court shall consider any one or more of the relevant circumstances presented by any of the parties to the case, including, but not limited to, the nature and seriousness of the misconduct, the number of violations, the persistence of the misconduct, the length of time over which the misconduct occurred, the willfulness of the defendant's misconduct, and the defendant's assets, liabilities, and net worth.

    See also Attorney General website at California Consumer Privacy Act (CCPA), https://oag.ca.gov/privacy/ccpa (last visited Aug. 4, 2020).
225. CCPA § 1798.150 (a)-(b).
226. Tex. Bus. & Com. Code Ann. § 503; Wash. Rev. Code § 19.375. (excludes physical or digital photographs, as well as video or audio recordings, from the definition of "biometric identifier); Cal. Civ. Code § 1798.100 et seq.
227. *Id.;* Wash. Rev. Code § 19.375.
228. S.5575A, 2019-2020 Leg. Sess. (N.Y. 2019), available at https://legislation.nysenate.gov/pdf/bills/2019/S5575A.
229. N.Y. Lab. Law §201-a (McKinney 2019).
230. S.B. 220, 2019 Leg., 80th Sess. (Nev. 2019), available at https://www.leg.state.nv.us/App/NELIS/REL/80th2019/Bill/6365/Text.
231. H.R. 2478, 50th Leg., Second Reg. Sess. (Ariz. 2012); H.R. 1153, 2019 Sess. (Fla. 2019); S. 120, 191st General Court (Mass. 2019).
232. *See, e.g.,* National Conference of State Legislatures 2020.
233. *See, e.g.,* Gandhi 2020 and Ringrose 2019.
234. *See, e.g.*, ICAO n.d.-a.

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APPENDIX L

Best Practices and Privacy by Design

Included in this appendix are additional examples of best practices, checklists, and guiding privacy principles.

The FIPPs core principles are:

1. The collection limitation principle. There should be limits on the collection of personal data, and any such data should be obtained by lawful and fair means and, where appropriate, with the knowledge or consent of the data subject.
2. The data quality principle. Personal data should be relevant to the purposes for which they are to be used and, to the extent necessary for those purposes, should be accurate, complete, and kept up-to-date.
3. The purpose specification principle. The purposes for which personal data are collected should be specified not later than at the time of data collection and the subsequent use limited to the fulfillment of those purposes or such others as are not incompatible with those purposes and as are specified on each occasion of change of purpose.
4. The use limitation principle. Personal data should not be disclosed, made available, or otherwise used for purposes other than those specified, except (a) with the consent of the data subject, or (b) by the authority of law.
5. The security safeguards principle. Personal data should be protected by reasonable security safeguards against such risks as loss or unauthorized access, destruction, use, modification, or disclosure.
6. The openness principle. There should be a general policy of openness about developments, practices, and policies with respect to personal data. Means should be readily available of establishing the existence and nature of personal data and the main purposes of their use, as well as the identity and usual residence of the data controller.
7. The individual participation principle. Individuals should have the right:
   a. To obtain from a data controller, or otherwise, confirmation of whether the data controller has data relating to them;
   b. To have data relating to them communicated to them, within a reasonable time, at a charge, if any, that is not excessive, in a reasonable manner, and in a form that is readily intelligible to them;
   c. To be given reasons if a request made under subparagraphs (a) and (b) is denied and to be able to challenge such denial; and
   d. To challenge data relating to them and, if the challenge is successful, to have the data erased, rectified, completed, or amended.
8. The accountability principle. A data controller should be accountable for complying with measures that give effect to the principles stated previously (IAPP n.d.).

An FTC report (FTC 2012a) elaborated on the recommended practices from a 2010 FTC preliminary report (FTC 2010). Building on its 2010 preliminary report that had recommended

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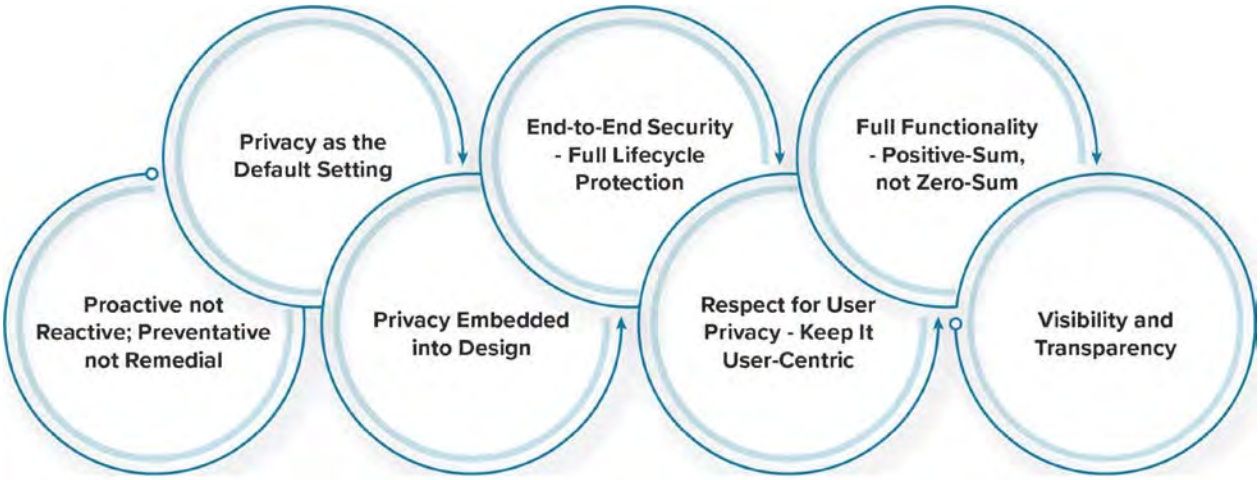
that businesses build privacy protections into their operations as described by privacy by design, the 2012 FTC report proposed best practices that offer simplified choices that give consumers more meaningful control, and would increase the transparency of data collection and use practices.

- Companies should incorporate into their practices substantive privacy protections, such as data security, reasonable collection limits, sound retention and disposal practices, and data accuracy.
- Companies should maintain comprehensive data-management procedures throughout the life cycle of their products and services.
- Companies do not need to provide choice before collecting and using consumer data for practices that are consistent with the context of the transaction or the company's relationship with the consumer or are required or specifically authorized by law.
- For practices requiring choice, companies should offer the choice at a time and in a context in which the consumer is making a decision about his or her data. Companies should obtain affirmative express consent before (1) using consumer data in a materially different manner than claimed when the data was collected, or (2) collecting sensitive data for certain purposes.
- Privacy notices should be clearer, shorter, and more standardized to enable better comprehension and comparison of privacy practices.
- Companies should provide reasonable access to the consumer data they maintain, and the extent of access should be proportionate to the sensitivity of the data and the nature of its use.
- All stakeholders should expand their efforts to educate consumers about commercial data privacy practices (FTC 2012a).[235]
- The bulk of these recommendations mirror provisions in BIPA and other state laws regulating the collection, use, retention, and sharing of biometric data (FTC 2012a).

Privacy-by-Design Checklist

For examples of other checklists, see Prescott 2020 and Information Commissioner's Office n.d. Privacy-by-design principles are shown in Figure L-1. Another useful checklist was compiled in connection with the U.K. 2019 legislation to implement Article 25 of the GDPR:

- We consider data protection issues as part of the design and implementation of systems, services, products, and business practices.



Source: Inter*VISTAS* Consulting, Inc.

Figure L-1. Privacy-by-design principles.

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- We make data protection an essential component of the core functionality of our processing systems and services.
- We anticipate risks and privacy-invasive events before they occur and take steps to prevent harm to individuals.
- We only process the personal data that we need for our purposes(s), and we only use the data for those purposes.
- We ensure that personal data are automatically protected in any IT system, service, product, or business practice, so that individuals should not have to take any specific action to protect their privacy.
- We provide the identity and contact information of those responsible for data protection both within our organization and to individuals.
- We adopt a plain-language policy for any public documents so that individuals easily understand what we are doing with their personal data.
- We provide individuals with tools so that they can determine how we are using their personal data and whether our policies are being properly enforced.
- We offer strong privacy defaults and user-friendly options and controls, and we respect user preferences.
- We only use data processors that provide sufficient guarantees of their technical and organizational measures for data protection by design.
- When we use other systems, services, or products in our processing activities, we make sure that we only use those whose designers and manufacturers take data protection issues into account.
- We use privacy-enhancing technologies to assist us in complying with our data protection by design obligations (Information Commissioner's Office n.d.).

Recent EU developments include the 2019 EU Data Protection Guidelines (4/2019) on the GDPR requirements for Data Protection by Design and by Default (European Data Protection Board 2019).[236] The draft guidelines (which were open for comment until January 2020) proposed measures and guidance addressing data protection by design, data protection by default, data subjects' rights, safeguard requirements, practical guidance on the application of the principles, and certification. Thus, the guidelines propose guidance on what data protection obligations mean in practice and how to implement the data protection principles effectively. Similarly, the EU published guidelines on consent describing the elements of valid consent [i.e., under GDPR article 4(11)] (European Data Protection Board 2020a).

The EU also published Guidelines (2/2020) on GDPR provisions on transfers of personal data between EU and non-EU public authorities and bodies (European Data Protection Board 2020b).

Endnotes

235. S*ee also* FTC 2012b.
236. *See also* European Data Protection Board 2020a and European Data Protection Board 2020b. For a comprehensive summary of the elements of data protection by design and by default, see Boardman et al. 2019. www.twobirds.com/en/news/articles/2019/global/edpb-publishes-guidelines-on-data-protection-by-design-and-by-default.

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APPENDIX M

IATA OneID and Seamless Flow

IATA OneID or Seamless Flow

The concept of this use case is that the passenger enrolls with his/her biometric details prior to each journey and is then able to pass several processing steps at the airport by just showing his/her face (e.g., for check-in, bag drop, security, border control, boarding, or even airline lounge entry and duty-free purchases) by using biometric verification touchpoints. The processing steps can also include the arrival process at the destination airport or the arrival process on return at the origin airport. In addition to smooth processing of the passenger, the system can also recognize and differentiate passenger personas and provide services tailored to the persona on the journey (for example, for priority or disabled passengers).

The biometric enrollment takes place at kiosks located at the airport that are capable of reading e-passports. As part of identity proofing, the passenger's identity is verified based on the e-passport or another biometric identity card, and the image of the passenger that is stored on the chip (1:1 matching). After positive identification, the system takes a new photo of the passenger and stores this as the biometric credential in the database of the biometric system for a set period. The database will not contain more data than the daily throughput at the airport; this way, the 1:1 search time to match the face of the passenger with the stored image will be limited. As opposed to TVS, the database will only be populated with images of passengers that actively enroll and give their consent to use their biometrics.

Before each trip, passengers enroll with their biometric details, or when DTC is used, push their biometric credentials from their mobile device to the database of the seamless system, where the data are stored in a PDE. Besides the passenger's digital identity credentials, the PDE can also contain personal information that can be of use for the process at the airport, such as loyalty program status or disabilities. Other information, such as age, sex, and nationality, may be of special interest for border authorities to use for targeting passengers that have a higher risk profile. In the post COVID-19 era, medical information related to the passenger, such as proof of vaccination, may be stored in the PDE and used for a frictionless journey.

After enrollment, based on the information in the PDE, the data-management platform verifies whether the passenger is authorized to pass through all steps of the passenger process. This authentication is performed based on business rules that are programmed in the data-management platform. As a result, just after enrollment, the system knows exactly what to do with the passenger for each touchpoint.

The biometric touchpoints are all connected with and managed by the data-management platform through a local area network. The data-management platform and biometric database can either run from a server located at the airport or from a server in the cloud. This data-management platform also manages the interfaces with third-party databases such as airline DCS;

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border police systems, databases, and watch lists; and potentially other airport systems such as point of sales. The business rules for decision making related to the identified person are determined by information retrieved from third-party systems. These business rules can potentially be dynamic for border control agencies for targeting specific groups of passengers.

After 15 years of driving this initiative, IATA hopes its solution can at first be adopted in markets that are struggling with capacity. OneID is going to be possible in the coming years in two or three countries; however, a rollout of the project and wider adoption will be limited until a digital ID specification is in place.

Stakeholders in the Biometric System

Because the Seamless Flow system covers multiple steps in the passenger process, there are multiple stakeholders. The border authorities are the responsible party for the border control process and, as such, are a core stakeholder, whereas the responsibility for the passenger checks at boarding lies with the airlines. The airport is responsible for access control to the security area, so the airport also has an important stake.

Required IT Infrastructure or Facilities

The Seamless Flow system extends to multiple verification touchpoints at airports. More extensive network infrastructure will be required compared to the earlier biometric systems, depending on the number of touchpoints that are connected and the scale. A seamless system typically contains the following components:

- Biometric enrollment stations;
- Biometric touchpoints that support self-service access;
- Biometric touchpoints for other processes, such as bag drop, dynamic wayfinding, and point of sale;
- A secure local area network;
- Data-management platform and face-recognition algorithm; and
- Biometric template database (either in the cloud or in a local database).

The concept of Seamless Flow requires interfaces with other systems as it attempts to ease the passenger journey between multiple stakeholders. The seamless data-management platform requires an interface with the airlines' DCS at check-in when a combined CUSS/biometric device is used, and again, an interface with the DCS upon boarding. For the border control checkpoint, an interface is required with the systems and search-list databases of the border authorities.

Note: When the biometric system needs to interface with CUTE/Common Use Passenger Processing System (CUPPS), the equipment will require certification—the vendor of the hardware needs to pay for this certification and will forward the cost to the owner(s). In case the CUTE/CUPPS is owned by multiple airlines in a so-called CUTE Local User Board (CLUB), this will require approval of the CLUB members.

Passenger Groups That Can Use the System

All passengers can use the system who are holding an e-passport and are traveling by airlines that participate in the seamless travel service that is provided at the airport.

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APPENDIX N

ICAO Digital Travel Credential

ICAO DTC Guiding Principles

ICAO recently published guiding principles for the use of digital travel credentials. These principles state that:

- The DTC must be at least as secure as an eMRTD.
- The information contained in the DTC must be derived from the travel document issuing authority's data from the eMRTD.
- Life-cycle management of the DTC may not be dependent on the life-cycle management of the eMRTD.
- Changes must not be needed in current eMRTD standards or issuing process for authorities not intending to issue DTCs.
- Revocation of a DTC must not result in the automatic withdrawal of the eMRTD associated with the DTC.
- Revocation of the eMRTD must automatically revoke all underlying DTCs.
- A DTC must always be issued by a travel document issuing authority (ICAO 2020).

Implementation of DTC

There are three types of DTC:

- Type 1 – eMRTD bound: consisting of a virtual credential only, with the eMRTD as a physical authenticator.
- Type 2 – eMRTD + physical credential bound: consists of a virtual credential and a physical credential in addition to the eMRTD.
- Type 3 – physical credential bound: consists of a virtual credential and a physical credential but no eMRTD.

Life Cycle of a DTC

The Guiding Core Principles for the Development of Digital Travel Credential (ICAO 2020) includes an overview of the creation of DTC as well as a discussion of the three types of DTC and of DTC validity, unique identifiers, use by a traveler, invalidation, and revocation.

Best Practices in the Use of DTC

As DTC and eMRTD are similar except for their form factor, verification of a DTC by a receiving entity requires the same procedures and same levels of scrutiny as for the eMRTD. The potential risks resulting from the use of the DTC are similar or identical to those of the

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eMRTD. Suggested best practices include preventing unauthorized access to the virtual component of the DTC during transmission or when stored, use of passive authentication and verifying the issuing authority is a trusted entity (state), and checking that the virtual component is not an unauthorized copy by verifying the physical component.

Risk Analysis

Many of the risks associated with DTCs are shared with eMRTDs. Some core risks are unique to the DTC. Guiding Core Principles for the Development of Digital Travel Credential differentiates between risks that are shared with the eMRTD and those that are unique to DTCs and suggests some mitigation strategies (ICAO 2020).

Other considerations, such as the use of diplomatic, official, and other passports, involve recognizing the potential impacts or lack of standardization. Because the DTC's Logical Data Structure 1 (identity and document information only) does not hold additional data such as visas and travel history, the use of DTCs could have an effect on traditional risk management activities performed by border control and other authorities when deciding whether to board or admit a traveler. Logical Data Structure 2 can support the additional information, but first-generation DTCs will not support this functionality. The storing of multiple DTCs on a single device needs to be carefully considered to ensure that all the potential exploitations of vulnerable groups are addressed.

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APPENDIX O

Accuracy of Facial Recognition

With the application of facial-recognition systems in aviation, it is important that all stakeholders are aware of how accurate these systems are. Inaccurate capturing equipment or facial-recognition technology causing a false positive match must not result in allowing a passenger to gain illegal passage at a border checkpoint or to illegally enter a plane. The algorithm should not be so strict that almost every person that approaches a touchpoint is rejected, nor should the time to render the decision be too long. There needs to be a balance between security and facilitation.

The matching algorithm analyzes the features stored in the biometric template on enrollment with the features derived from the image made of the live person on verification to produce a similarity score. If the score reaches a certain threshold, the algorithm decides that it is a match. A biometric system will never result in a 100% match. This is because it depends on the quality and amount of useful information that can be extracted from the image. The quality of useful information depends on varying conditions between enrollment and verification but also on the difference in appearance of the person that needs to be identified (e.g., aging, glasses, face mask).

Accuracy relates to the number of times that the system gives the correct response. For the response, two variables are most important in operation: the false acceptance rate (FAR) and the false rejection rate (FRR). The FAR is the relative number of passengers that can pass through to the system who should not be allowed to do so. The FRR is the relative amount of passengers that are rejected by the system when they had the right to pass. An increase in the size of the database will lead to a lower accuracy, with higher false acceptances because there will be more similar biometric templates. Raising the threshold too high will lead to a low FAR but may also lead to longer processing times.

Which rates are acceptable are to be determined by the authorities and other stakeholders. The performance of different face-recognition algorithms from different vendors is reported on a frequent basis by NIST. NIST has been testing the performance of face-match algorithms since 2017 by using a standard measuring method against large image databases. Although the accuracy depends on many other factors of the system as well, the performance of the facial-recognition algorithm is the basis of a reliable face-recognition system.

In the facial-recognition algorithm, the similarity threshold can be adjusted, and this will affect the FRR and FAR. After installation of the biometric verification touchpoints at the location, the system needs to be tested and commissioned. When the system is accepted and operational, it can be load tested by having live people enrolled and passing through the biometric touchpoints. Fine tuning may include, for example, adjusting the uniform illumination

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level of the touchpoint or protecting the camera view from backlight. After some time, the system can be checked for the FRR.

In the passenger facilitation process, it is also possible that the face that is presented for verification does not exist in the database. A passenger that did not enroll in the program will not have a biometric template in the database. The system will give a rejection, but it is not false. After a series of attempts, the system should raise an alarm, and manual intervention will be needed.

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Abbreviations and acronyms used without definitions in TRB publications:

A4A	Airlines for America
AAAE	American Association of Airport Executives
AASHO	American Association of State Highway Officials
AASHTO	American Association of State Highway and Transportation Officials
ACI–NA	Airports Council International–North America
ACRP	Airport Cooperative Research Program
ADA	Americans with Disabilities Act
APTA	American Public Transportation Association
ASCE	American Society of Civil Engineers
ASME	American Society of Mechanical Engineers
ASTM	American Society for Testing and Materials
ATA	American Trucking Associations
CTAA	Community Transportation Association of America
CTBSSP	Commercial Truck and Bus Safety Synthesis Program
DHS	Department of Homeland Security
DOE	Department of Energy
EPA	Environmental Protection Agency
FAA	Federal Aviation Administration
FAST	Fixing America's Surface Transportation Act (2015)
FHWA	Federal Highway Administration
FMCSA	Federal Motor Carrier Safety Administration
FRA	Federal Railroad Administration
FTA	Federal Transit Administration
GHSA	Governors Highway Safety Association
HMCRP	Hazardous Materials Cooperative Research Program
IEEE	Institute of Electrical and Electronics Engineers
ISTEA	Intermodal Surface Transportation Efficiency Act of 1991
ITE	Institute of Transportation Engineers
MAP-21	Moving Ahead for Progress in the 21st Century Act (2012)
NASA	National Aeronautics and Space Administration
NASAO	National Association of State Aviation Officials
NCFRP	National Cooperative Freight Research Program
NCHRP	National Cooperative Highway Research Program
NHTSA	National Highway Traffic Safety Administration
NTSB	National Transportation Safety Board
PHMSA	Pipeline and Hazardous Materials Safety Administration
RITA	Research and Innovative Technology Administration
SAE	Society of Automotive Engineers
SAFETEA-LU	Safe, Accountable, Flexible, Efficient Transportation Equity Act: A Legacy for Users (2005)
TCRP	Transit Cooperative Research Program
TDC	Transit Development Corporation
TEA-21	Transportation Equity Act for the 21st Century (1998)
TRB	Transportation Research Board
TSA	Transportation Security Administration
U.S. DOT	United States Department of Transportation

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